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Class 12 Physics Chapterwise PYQs

2026 – 2003 | All CBSE Board Papers

Chapterwise previous year questions, sorted by marks and year

Chapter 11: Dual Nature of Radiation and Matter

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HOW TO USE THIS BOOKLET

1. **Attempt a section** on your own. Questions run newest year first.
2. **Check the Answer Key,** then follow a question number to its solution.
3. **Study the solution.** Note each *Common Mistake,* then return via "Back to Q n".

SECTION A

1-Mark Questions

Multiple choice · 120 questions

> *Attempt these first, then check the Answer Key and open any solution.*

1. *Two statements are given - one labelled as Assertion (A) and the other labelled as Reason (R). Select the correct answer from the options below:*

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Both Assertion (A) and Reason (R) are false.

Assertion (A): If accelerated electrons are passed through a narrow slit, a diffraction pattern is observed.

Reason (R): Electrons behave as both particles and waves.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Both Assertion (A) and Reason (R) are false.

Check Solution → **[2026 • Set 55-1-1]**

2. From the graph, the work functions of A and B are (h is Planck's constant and e value of charge on an electron):

(A) ν_1 and ν_2

(B) V_1 and V_2

(C) $h\nu_1$ and $h\nu_2$

(D) $\frac{h\nu_1}{e}$ and $\frac{h\nu_2}{e}$

Check Solution → [2026 • Set 55-1-1]

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3. For radiation of frequency $\nu > \nu_2$ incident on the surfaces of A and B, the maximum kinetic energy of ejected electron is:

(A) greater for metal A because it has a smaller work function.

(B) greater for metal B because it has a larger work function.

(C) greater for metal B because it has higher threshold frequency.

(D) the same for both metal A and metal B because it is independent of work functions of metals.

Check Solution → [2026 • Set 55-1-1]

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4. If the intensity of the incident radiation for both metals A and B, is doubled keeping its frequency constant, then:

(A) the slope of the parallel lines will increase.

(B) the slope of the parallel lines will decrease.

(C) the threshold frequencies for both A and B will decrease.

(D) the slope of the parallel lines will not change but more electrons will be emitted per second.

Check Solution → [2026 • Set 55-1-1]

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5. **(OR variant for Q30(iv))**

The threshold frequency for a metal surface is ν_0. If the radiation of frequency $3\nu_0$ illuminates the surface, the maximum kinetic energy (KE) of photoelectrons is E_1. If the frequency were increased to $6\nu_0$, the maximum KE of the

photoelectrons becomes E_2. Then $\frac{E_1}{E_2}$ equals:

OR

Let m be the slope of the graph line for metal B. If e is the value of electron charge, then Planck's constant 'h' is given by:

(A) 1/3 (or me)

(B) 1/2 (or $\frac{1}{me}$)

(C) 2/5 (or $\frac{m}{e}$)

(D) 3/4 (or $\frac{e}{m}$)

Check Solution → **[2026 • Set 55-1-1]**

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6. **Assertion (A):** On increasing the intensity of incident light of frequency ν ($>\nu_0$) on a photosensitive surface, the photocurrent increases.

Reason (R): The stopping potential for a photosensitive surface increases with increase of frequency ν ($>\nu_0$) of incident light.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false and Reason (R) is also false.

Check Solution → **[2026 • Set 55-2-1]**

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7. The maximum kinetic energy of the electrons emitted from a photosensitive surface depends on:

(A) work function of the surface ϕ_0 only.

(B) frequency of the incident radiation ν only.

(C) intensity of the incident radiation I only.

(D) Both ϕ_0 and ν.

Check Solution → [2026 • Set 55-2-1]

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8. A photosensitive surface is illuminated by radiations of wavelength λ_1, λ_2 ($>\lambda_1$) and λ_3 one by one and photoemission is observed in each case. λ_1 and λ_2 lie in UV range and λ_3 in visible range. If V_1, V_2 and V_3 are stopping potentials in these cases respectively, then:

(A) $V_1 = V_2 = V_3$

(B) $V_1 > V_2 > V_3$

(C) $V_1 > V_3 > V_2$

(D) $V_3 > V_2 > V_1$

Check Solution → [2026 • Set 55-2-2]

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9. The equivalent mass of a photon of wavelength 100 nm is:

(A) $2{\cdot}21\times 10^{-35}$ kg

(B) $6{\cdot}63\times 10^{-33}$ kg

(C) $1{\cdot}11\times 10^{-35}$ kg

(D) $3{\cdot}31\times 10^{-27}$ kg

Check Solution → [2026 • Set 55-2-3]

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10. **Assertion (A):** Photoelectric current depends upon the intensity of the incident radiation.

Reason (R): Stopping potential is independent of the intensity of the incident radiation.

Select the correct answer from the codes (A), (B), (C) and (D) given below:

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Both Assertion (A) and Reason (R) are false.

Check Solution → [2026 • Set 55-3-1]

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11. Radiation of wavelength 200 nm is incident on a photosensitive surface of work function 4·2 eV. The kinetic energy of fastest photoelectrons emitted from this surface will be close to:

(A) 3·5 eV

(B) 3·0 eV

(C) 2·5 eV

(D) 2·0 eV

Check Solution → [2026 • Set 55-3-1]

---

12. A proton and an alpha particle have equal momentum. The ratio of their kinetic energies (E_p / E_α) and de Broglie wavelengths associated with them (λ_p / λ_α) respectively are:

(A) 2, 1

(B) 1, 2

(C) 4, 1

(D) 1, 4

Check Solution → [2026 • Set 55-3-1]

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13. The velocity of the photoelectrons emitted from a surface depends upon the:

(A) Wavelength of incident radiation only

(B) Intensity of incident radiation only

(C) Work function of the surface only

(D) Both wavelength of the incident radiation and work function of the surface

Check Solution → [2026 • Set 55-3-2]

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14. The kinetic energy of a charged particle is increased to four times of its initial value. The de Broglie wavelength associated with the particle will:

(A) increase by 100% of its initial value.

(B) increase by 50% of its initial value.

(C) decrease by 25% of its initial value.

(D) decrease by 50% of its initial value.

Check Solution → [2026 • Set 55-4-1]

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15. While studying photoelectric emission from a given surface, the wavelength of the incident radiation is changed from 600 nm to 400 nm, keeping the intensity of radiation the same. Then:

(A) cut-off potential will decrease.

(B) cut-off potential will increase.

(C) saturation current will decrease.

(D) saturation current will increase.

Check Solution → [2026 • Set 55-4-1]

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16. Radiation of wavelength 331 nm irradiates the following metals:MetalWork Function (eV)Na1·92K2·15Ca3·20Mo4·17Which of the following statements is correct?

(A) Only Na and K show photoelectric emission.

(B) Only Mo will not show photoelectric emission.

(C) All of the given metals show photoelectric emission.

(D) None of them show photoelectric emission.

Check Solution → [2026 • Set 55-4-1]

17. In an experiment on photoelectric emission, the intensity of incident radiation is halved keeping its frequency constant. As a result:

(A) cut-off potential will increase.

(B) maximum kinetic energy of photoelectrons will remain the same.

(C) photocurrent will increase.

(D) cut-off potential will decrease.

Check Solution → [2026 • Set 55-4-2]

18. Photons of frequency ν are incident on the surfaces of two metals A and B of threshold frequencies $\frac{\nu}{2}$ and $\frac{\nu}{3}$. The ratio of maximum kinetic energy of electrons emitted from metal A to that from metal B is:

(A) $\frac{1}{3}$

(B) $\frac{3}{4}$

(C) $\frac{2}{3}$

(D) $\frac{3}{2}$

Check Solution → [2026 • Set 55-4-3]

19. **Assertion (A):** Photoelectric effect is a spontaneous phenomenon.
Reason (R): According to the wave picture of radiation, an electron would take hours/days to absorb sufficient energy to overcome the work function and come out from a metal surface.
Select the correct answer from the codes given below:

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Both Assertion (A) and Reason (R) are false.

Check Solution → [2026 • Set 55-5-1]

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20. Light of which of the following colours will have the maximum energy in a photon associated with it?

(A) Red light

(B) Yellow light

(C) Green light

(D) Blue light

Check Solution → [2026 • Set 55-5-1]

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21. The momentum of a photon associated with a microwave of wavelength 4.00 cm is:

(A) 1.66×10^{-32} kg ms^{-1}

(B) 1.83×10^{-34} kg ms^{-1}

(C) 2.05×10^{-34} kg ms^{-1}

(D) 1.66×10^{-34} kg ms^{-1}

Check Solution → [2026 • Set 55-5-2]

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22. A 500 nm photon is incident normally on a perfectly reflecting surface and is reflected. The value of momentum transferred to the surface is:

(A) 3.87×10^{-43} kg ms^{-1}

(B) 2.5×10^{-30} kg ms^{-1}

(C) 2.65×10^{-27} kg ms^{-1}

(D) 1.33×10^{-27} kg ms^{-1}

Check Solution → [2026 • Set 55-5-3]

23. Let λ_e, λ_p and λ_d be the wavelengths associated with an electron, a proton and a deuteron, all moving with the same speed. Then the correct relation between them is:

(A) $\lambda_d > \lambda_p > \lambda_e$

(B) $\lambda_e > \lambda_p > \lambda_d$

(C) $\lambda_p > \lambda_e > \lambda_d$

(D) $\lambda_e = \lambda_p = \lambda_d$

Check Solution → [2025 • Set 55-1-1]

24. The stopping potential V_0 measured in a photoelectric experiment for a metal surface is plotted against frequency ν of the incident radiation. Let m be the slope of the straight line so obtained. Then the value of charge of an electron is given by (h is the Planck's constant.)

(A) mh

(B) $\frac{m}{h}$

(C) $\frac{h}{m}$

(D) $\frac{1}{\text{mh}}$

Check Solution → [2025 • Set 55-1-1]

25. A source produces monochromatic light of frequency 5.0×10^{14} Hz and the power emitted is 3.31 mW. The number of photons emitted per second by the source, on an average, is:

(A) 10^{16}

(B) 10^{12}

(C) 10^{10}

(D) 10^{20}

Check Solution → [2025 • Set 55-1-2]

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26. In a photoelectric experiment with a material of work function 2.1 eV, the stopping potential is found to be 2.5 V. The maximum kinetic energy of ejected photoelectrons is:

(A) 0.4 eV

(B) 2.1 eV

(C) 2.5 eV

(D) 4.6 eV

Check Solution → [2025 • Set 55-1-3]

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27. Atomic spectral emission lines of hydrogen atom are incident on a zinc surface. The lines which can emit photoelectrons from the surface are members of

(A) Balmer series

(B) Paschen series

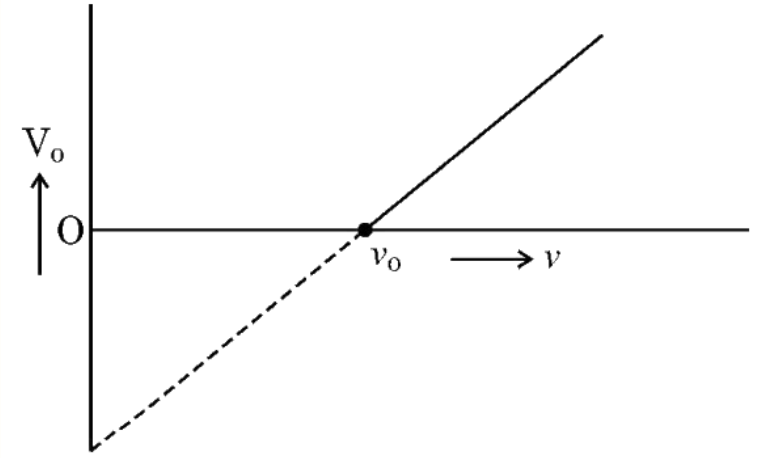
(C) Lyman series

(D) Neither Balmer, nor Paschen nor Lyman series

Check Solution → [2025 • Set 55-2-1]

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28. **Case study:** When a photon of suitable frequency is incident on a metal surface, photoelectron is emitted from it. If the frequency is below a threshold frequency (ν_0) for the surface, no photoelectron is emitted. For a photon of frequency $\nu(\nu > \nu_0)$, the kinetic energy of the emitted photoelectrons is $h(\nu-\nu_0)$. The photocurrent can be stopped by applying a potential V_0, called 'stopping potential' on the anode. Thus maximum kinetic energy of photoelectrons $K_m = eV_0 = h(\nu-\nu_0)$. The experimental graph between V_0 and ν for a metal is shown in figure. This is a straight line of slope m.



The straight line graphs obtained for two metals

(A) coincide each other.

(B) are parallel to each other.

(C) are not parallel to each other and cross at a point on y-axis.

(D) are not parallel to each other and do not cross at a point on y-axis.

Check Solution → [2025 • Set 55-2-1]

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29. The value of Planck's constant for this metal is

(A) $\frac{e}{m}$

(B) $\frac{1}{\text{me}}$

(C) me

(D) $\frac{m}{e}$

Check Solution → [2025 • Set 55-2-1]

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30. The intercepts on ν-axis and V_0-axis of the graph are respectively:

(A) $\nu_0, \frac{h\nu_0}{e}$

(B) $\nu_0, h\nu_0$

(C) $\frac{h\nu_0}{e}, \nu_0$

(D) $h\nu_0, \nu_0$

Check Solution → [2025 • Set 55-2-1]

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31. When the wavelength of a photon is doubled, how many times its wave number and frequency become, respectively?

(A) $2, \frac{1}{2}$

(B) $\frac{1}{2}, \frac{1}{2}$

(C) $\frac{1}{2}, 2$

(D) 2, 2

Check Solution → [2025 • Set 55-2-1]

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32. The momentum of a photon is 5.0×10^{-29} kg m/s. Ignoring relativistic effects (if any), the wavelength of the photon is

(A) 1.33 μm

(B) 3.3 μm

(C) 16.6 μm

(D) 13.3 μm

Check Solution → [2025 • Set 55-2-1]

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33. A beam of red light and a beam of blue light have equal intensities. Which of the following statements is true ?

(A) The blue beam has more number of photons than the red beam.

(B) The red beam has more number of photons than the blue beam.

(C) Wavelength of red light is lesser than wavelength of blue light.

(D) The blue light beam has lesser energy per photon than that in the red light beam.

Check Solution → [2025 • Set 55-4-1]

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34. **Assertion (A):** For monochromatic incident radiation, the emitted photoelectrons from a given metal have speed ranging from zero to a certain maximum value.

Reason (R): Each metal has a definite work function.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Both Assertion (A) and Reason (R) are false.

Check Solution → [2025 • Set 55-4-1]

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35. Choose the correct statement :

(A) Photons of light show diffraction whereas electrons do not show diffraction.

(B) Electrons have momentum whereas photons do not have momentum.

(C) Photons of light and electrons both exhibit dual nature.

(D) All electromagnetic radiations do not have photons.

Check Solution → [2025 • Set 55-4-1]

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36. The kinetic energy of an alpha particle is four times the kinetic energy of a proton. The ratio $\frac{\lambda_\alpha}{\lambda_p}$ of de Broglie wavelengths associated with them will be:

(A) 1

(B) $\frac{1}{2\sqrt{2}}$

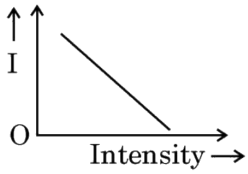
(C) $\frac{1}{4}$

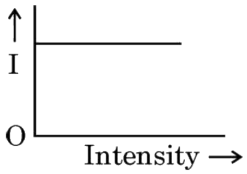
(D) $\frac{1}{2}$

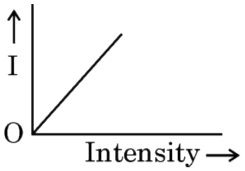
Check Solution → [2025 • Set 55-5-1]

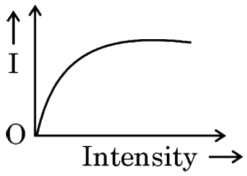
37. **Case Study:** Einstein explained photoelectric effect on the basis of Planck's quantum theory, where light travels in the form of small bundles of energy called photons. The energy of each photon is hν, where ν is the frequency of incident light and h is Planck's constant. The number of photons in a beam of light determines the intensity of the incident light. A photon incident on a metal surface transfers its total energy hν to a free electron in the metal. A part of this energy is used in ejecting the electron from the metal and is called its work function. The rest of the energy is carried by the ejected electron as its kinetic energy.

(i) Which of the following graphs shows the variation of photoelectric current I with the intensity of light?

(A) 

(B) 

(C) 

(D) 

(A) (A) Curve concave-down rising then flat

(B) (B) Straight line through origin with positive slope

(C) (C) Straight horizontal line

(D) (D) Curve concave-up rising

Check Solution → [2025 • Set 55-5-1]

38. **(ii)** When the frequency of the incident light is increased without changing its intensity, the saturation current:

(A) increases linearly

(B) decreases

(C) increases non-linearly

(D) remains the same

Check Solution → [2025 • Set 55-5-1]

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39. **(iii)** Which of the following graphs can be used to obtain the value of Planck's constant?

(A) Photocurrent versus Intensity of incident light

(B) Photocurrent versus Frequency of incident light

(C) Cut-off potential versus Frequency of incident light

(D) Cut-off potential versus Intensity of incident light

Check Solution → [2025 • Set 55-5-1]

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40. **(iv) (a)** Red light, yellow light and blue light of the same intensity are incident on a metal surface successively. K_R, K_Y and K_B represent the maximum kinetic energy of photoelectrons respectively, then:

OR

(b) Which of the following metals exhibits photoelectric effect with visible light?

(A) (a) $K_R > K_Y > K_B$ OR (b) Caesium

(B) (a) $K_Y > K_R > K_B$ OR (b) Zinc

(C) (a) $K_B > K_Y > K_R$ OR (b) Cadmium

(D) (a) $K_B > K_R > K_Y$ OR (b) Magnesium

Check Solution → [2025 • Set 55-5-1]

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41. Which of the following electromagnetic waves has photons of largest momentum?

(A) X-rays

(B) AM radio waves

(C) Microwaves

(D) TV waves

Check Solution → [2025 • Set 55-5-1]

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42. The work function of a material is 2·21 eV. Which of the following cannot produce photoelectrons from it?

(A) Red light

(B) Blue light

(C) Violet light

(D) Green light

Check Solution → [2025 • Set 55-6-1]

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43. The momentum (in kg m/s) of a photon of frequency $6{\cdot}0 \times 10^{14}$ Hz is:

(A) $6{\cdot}63 \times 10^{-22}$

(B) $1{\cdot}326 \times 10^{-27}$

(C) $2{\cdot}652 \times 10^{-26}$

(D) $3{\cdot}978 \times 10^{-24}$

Check Solution → [2025 • Set 55-6-1]

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44. The de Broglie wavelength associated with an electron moving with energy 5 eV is:

(A) 0·75 nm

(B) 1·2 nm

(C) 2·4 nm

(D) 0·55 nm

Check Solution → [2025 • Set 55-6-2]

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45. The wavelength of a photon is equal to the wavelength associated with an electron. Both will have the same value of:

(A) energy

(B) angular momentum

(C) velocity

(D) linear momentum

Check Solution → [2025 • Set 55-6-3]

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46. **Assertion (A):** The minimum negative potential applied to the anode in a photoelectric experiment at which photoelectric current becomes zero, is called cut-off voltage.

Reason (R): The threshold frequency for a metal is the minimum frequency of incident radiation below which emission of photoelectrons does not take place.

Select the correct answer:

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Both Assertion (A) and Reason (R) are false.

Check Solution → [2025 • Set 55-7-1]

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47. The frequency of a photon of energy 1.326 eV is:

(A) 1.18×10^{14} Hz

(B) 3.20×10^{14} Hz

(C) 4.20×10^{15} Hz

(D) 4.80×10^{15} Hz

Check Solution → [2025 • Set 55-7-1]

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48. Two beams, A and B whose photon energies are 3.3 eV and 11.3 eV respectively, illuminate a metallic surface (work function 2.3 eV) successively. The ratio of maximum speed of electrons emitted due to beam A to that due to beam B is:

(A) $\frac{1}{3}$

(B) $\frac{1}{9}$

(C) 3

(D) 9

Check Solution → [2024 • Set 55-1-1]

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49. The waves associated with a moving electron and a moving proton have the same wavelength λ. It implies that they have the same:

(A) momentum

(B) angular momentum

(C) speed

(D) energy

Check Solution → [2024 • Set 55-1-1]

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50. **Assertion (A):** In photoelectric effect, the kinetic energy of the emitted photoelectrons increases with increase in the intensity of the incident light.
Reason (R): Photoelectric current depends on the wavelength of the incident light.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false and Reason (R) is also false.

Check Solution → [2024 • Set 55-1-1]

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51. **Assertion (A):** Photoelectric current increases with an increase in intensity of incident radiation, for a given frequency of incident radiation and the accelerating potential.

Reason (R): Increase in the intensity of incident radiation results in an increase in the number of photoelectrons emitted per second and hence an increase in the photocurrent.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false and Reason (R) is also false.

Check Solution → **[2024 • Set 55-2-1]**

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52. The work function for a photosensitive surface is 3.315 eV. The cut-off wavelength for photoemission of electrons from this surface is:

(A) 150 nm

(B) 200 nm

(C) 375 nm

(D) 500 nm

Check Solution → **[2024 • Set 55-2-1]**

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53. A photosensitive surface has a work function of 2.00 eV. The maximum kinetic energy of electrons ejected from this surface by radiation of wavelength 300 nm is:

(A) 0.54 eV

(B) 1.07 eV

(C) 1.61 eV

(D) 2.14 eV

Check Solution → **[2024 • Set 55-2-2]**

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54. The momentum (in SI units) associated with a photon of energy 1.5 eV is:

(A) 410^{-27}

(B) 810^{-28}

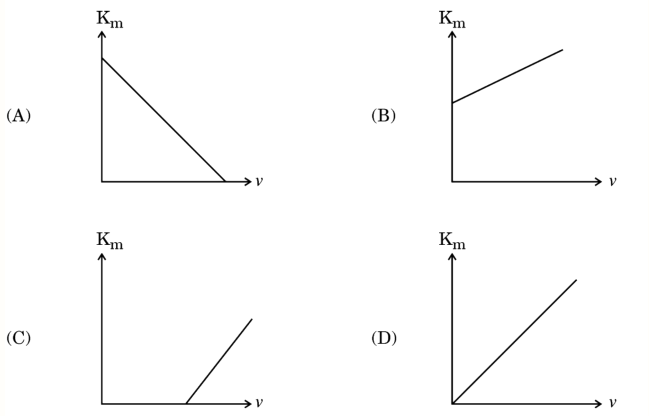
(C) 210^{-30}

(D) 610^{-29}

Check Solution → [2024 • Set 55-2-3]

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55. Which one of the following is the correct graph between the maximum kinetic energy (K_m) of the emitted photoelectrons and the frequency of incident radiation (ν) for a given photosensitive surface?



(A) (A)

(B) (B)

(C) (C)

(D) (D)

Check Solution → [2024 • Set 55-3-1]

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56. **Assertion (A):** Photoelectric effect demonstrates the particle nature of light.
Reason (R): Photoelectric current is proportional to frequency of incident radiation.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

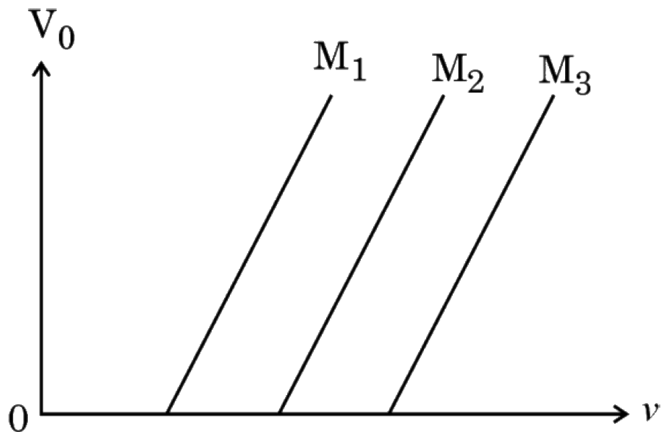
(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false and Reason (R) is also false.

Check Solution → [2024 • Set 55-3-1]

---

57. The figure shows the variation of stopping potential V_0 with frequency ν of incident radiation, for three materials M_1, M_2 and M_3 with work functions ϕ_1, ϕ_2 and ϕ_3 respectively. Then:



(A) $\phi_1 > \phi_2 > \phi_3$

(B) $\phi_2 > \phi_3 > \phi_1$

(C) $\phi_3 > \phi_2 > \phi_1$

(D) $\phi_2 > \phi_1 > \phi_3$

Check Solution → [2024 • Set 55-3-2]

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58. The wavelength of matter wave associated with an electron of kinetic energy K is λ. If the kinetic energy of the electron is doubled, the associated wavelength becomes:

(A) $\frac{\lambda}{\sqrt{2}}$

(B) $\frac{\lambda}{2}$

(C) $\sqrt{2}\lambda$

(D) 2λ

Check Solution → [2024 • Set 55-3-3]

---

59. **Assertion (A):** Electrons are ejected from the surface of zinc when it is irradiated by yellow light.

Reason (R): Energy associated with a photon of yellow light is more than the work function of zinc.

(A) If both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(B) If both Assertion (A) and Reason (R) are true and Reason (R) is not the correct explanation of Assertion (A).

(C) If Assertion (A) is true and Reason (R) is false.

(D) If both Assertion (A) and Reason (R) are false.

Check Solution → [2024 • Set 55-4-1]

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60. The quantum nature of light explains the observations on photoelectric effect as

(A) there is a minimum frequency of incident radiation below which no electrons are emitted.

(B) the maximum kinetic energy of photoelectrons depends only on the frequency of incident radiation.

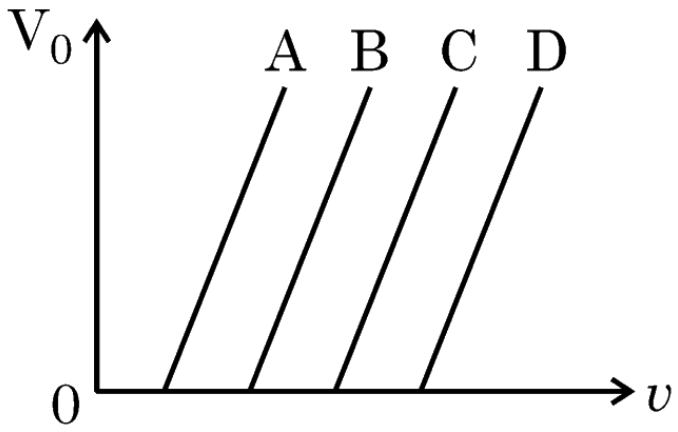
(C) when the metal surface is illuminated, electrons are ejected from the surface after sometime.

(D) the photoelectric current is independent of the intensity of incident radiation.

Check Solution → [2024 • Set 55-4-1]

---

61. The variation of the stopping potential (V_0) with the frequency (ν) of the incident radiation for four metals A, B, C and D is shown in the figure. For the same frequency of incident radiation producing photo-electrons in all metals, the kinetic energy of photo-electrons will be maximum for metal:



(A) A

(B) B

(C) C

(D) D

Check Solution → [2024 • Set 55-5-1]

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62. **Assertion (A):** Photoelectric effect demonstrates the particle nature of light.
Reason (R): Photoelectric current is proportional to intensity of incident radiation for frequencies more than the threshold frequency.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false and Reason (R) is also false.

Check Solution → [2023 • Set 55-1-1]

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63. A photon of energy 7 eV is incident on a metal surface having the work function of 3.75 eV. The stopping potential is:

(A) 1.75 eV

(B) 2.45 eV

(C) 3.25 eV

(D) 3.75 eV

Check Solution → [2023 • Set 55-1-2]

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64. In an experiment, photons of energy 7.5 eV are incident on a metal surface. Electrons emitted from the metal surface are stopped by an electrode at a potential of −4.5 V w.r.t. the metal. The work function of the metal is:

(A) 3.0 eV

(B) 4.5 eV

(C) 7.5 eV

(D) 12.0 eV

Check Solution → [2023 • Set 55-1-3]

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65. Which one of the following metals does not exhibit emission of electrons from its surface when irradiated by visible light?

(A) Rubidium

(B) Sodium

(C) Cadmium

(D) Caesium

Check Solution → [2023 • Set 55-2-1]

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66. A photon of wavelength 663 nm is incident on a metal surface. The work function of the metal is 1.50 eV. The maximum kinetic energy of the emitted photoelectrons is

(A) $3.0\times 10^{-20}\,J$

(B) $6.0\times 10^{-20}\,J$

(C) $4.5\times 10^{-20}\,J$

(D) $9.0\times 10^{-20}\,J$

Check Solution → [2023 • Set 55-2-2]

67. The energy of a photon of wavelength 663 nm is

(A) $6.64\times 10^{-20}\,J$

(B) $5.18\times 10^{-19}\,J$

(C) $3.0\times 10^{-19}\,J$

(D) $2.0\times 10^{-19}\,J$

Check Solution → [2023 • Set 55-2-3]

68. A graph is plotted between the stopping potential (on y-axis) and the frequency of incident radiation (on x-axis) for a metal. The product of the slope of the straight line obtained and the magnitude of charge on an electron is equal to:

(A) h (Planck's constant)

(B) $\frac{h}{e}$

(C) he

(D) $\frac{e}{h}$

Check Solution → [2023 • Set 55-3-1]

69. Light of frequency 6.4×10^{14} Hz is incident on a metal of work function 2.14 eV. The maximum kinetic energy of the emitted electrons is about:

(A) 0.25 eV

(B) 0.51 eV

(C) 1.02 eV

(D) 0.10 eV

Check Solution → [2023 • Set 55-3-1]

70. In an experiment on photoelectric effect, the intensity of incident radiation is increased, keeping the frequency ν the same. The number of photoelectrons

emitted will:

(A) increase

(B) decrease

(C) remain same

(D) depend on frequency

Check Solution → [2023 • Set 55-3-2]

---

71. A proton and an alpha particle have the same kinetic energy. The ratio of de Broglie wavelengths associated with the proton to that with the alpha particle is:

(A) 1

(B) 2

(C) $2\sqrt{2}$

(D) $\frac{1}{2}$

Check Solution → [2023 • Set 55-4-1]

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72. Photons of energy 3.2 eV are incident on a photosensitive surface. If the stopping potential for the emitted electrons is 1.5 V, the work function for the surface is:

(A) 1.5 eV

(B) 1.7 eV

(C) 3.2 eV

(D) 4.7 eV

Check Solution → [2023 • Set 55-4-1]

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73. Photons of energy 4.3 eV are incident on a photosensitive surface of work function 2.3 eV. The stopping potential for photoelectrons is:

(A) 2.0 eV

(B) 2.3 eV

(C) 4.3 eV

(D) 6.6 eV

Check Solution → [2023 • Set 55-4-2]

74. A light of frequency ν is incident on a metal surface whose work function is W_0. The kinetic energy of emitted electron is K. If the frequency of the incident light is doubled then the kinetic energy of emitted electron will be:

(A) 2K

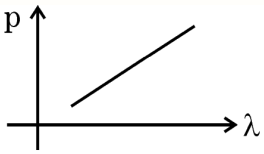
(B) more than 2K

(C) between K and 2K

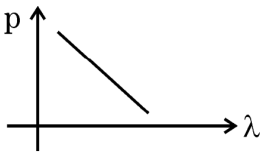
(D) less than K

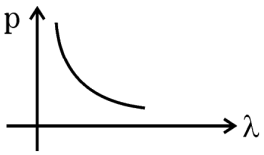
Check Solution → [2023 • Set 55-4-3]

75. Which of the following graphs correctly represents the variation of a particle momentum with its associated de-Broglie wavelength?

(A) 

(B) 

(C) 

(D) 

(A) Graph (A)

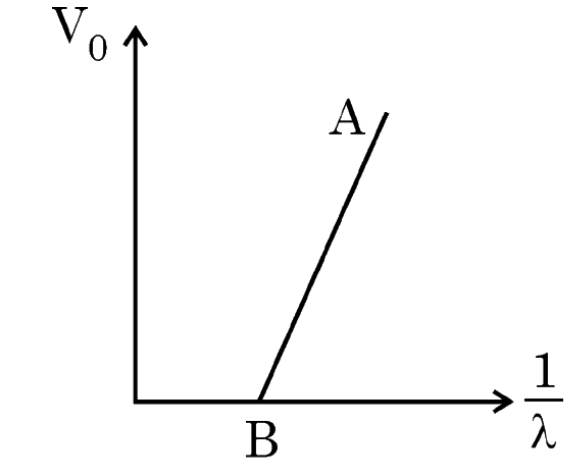
(B) Graph (B)

(C) Graph (C)

(D) Graph (D)

Check Solution → [2023 • Set 55-5-1]

76. Figure shows a plot of stopping potential (V_0) versus $\frac{1}{\lambda}$ where λ is the wavelength of the radiation causing photoelectric emission from a surface. The slope of the line is equal to



(A) $\frac{h}{e}$

(B) $\frac{hc}{e}$

(C) $\frac{h}{c}$

(D) $\frac{h^2 c}{e}$

Check Solution → [2023 • Set 55-5-2]

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77. E, c and v represent the energy, velocity and frequency of a photon. Which of the following represents its wavelength?

(A) $\frac{hv}{c^2}$

(B) hv

(C) $\frac{hc}{E}$

(D) $\frac{hv}{c}$

Check Solution → [2023 • Set 55-5-3]

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78. Two statements are given - one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer to this question from the codes (A), (B), (C) and (D) as given below:
(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(B) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false and Reason (R) is also false.
Assertion (A): In his study of photoelectric emission, Hallwachs connected a negatively charged zinc plate to an electroscope. He found that negatively charged particles were emitted from the zinc plate under the action of visible light.
Reason (R): An uncharged zinc plate becomes positively charged when it is irradiated by visible light.

Check Solution → [2021]

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79. If the frequency of the radiation incident on a photosensitive surface increases ($\nu > \nu_0$), how will the stopping potential change?

Check Solution → [2021]

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80. An electron is accelerated through a potential difference of 100 V. Calculate the de Broglie wavelength associated with it.

Check Solution → [2021]

---

81. A proton and a deuteron are moving with the same speed. Find the ratio of the de Broglie wavelength (λ_p / λ_d) associated with them.

Check Solution → [2021]

---

82. Photons of energies 1 eV and 2 eV are successively incident on a metallic surface of work function 0·5 eV. The ratio of kinetic energy of most energetic photoelectrons in the two cases will be

(A) 1 : 2

(B) 1 : 1

(C) 1 : 3

(D) 1 : 4

Check Solution → [2020 • Set 55-1-1]

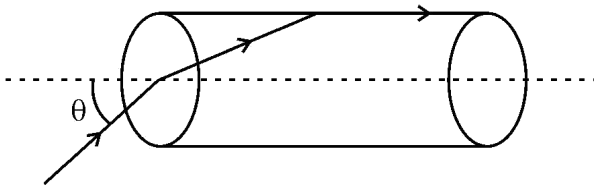
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83. A proton and an electron have equal speeds. Find the ratio of de Broglie wavelengths associated with them.

Check Solution → [2020 • Set 55-1-1]

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84. The variation of the stopping potential (V_0) with the frequency (ν) of the light incident on two different photosensitive surfaces M_1 and M_2 is shown in the figure. Identify the surface which has greater value of the work function.



Check Solution → [2020 • Set 55-1-1]

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85. What is the wavelength of a photon of energy 3.3×10^{-19} J?

Check Solution → [2020 • Set 55-2-1]

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86. The threshold wavelength for two photosensitive surfaces A and B are λ_1 and λ_2, respectively. What is the ratio of the work functions of the two surfaces?

Check Solution → [2020 • Set 55-2-2]

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87. The threshold frequencies of two photosensitive surfaces are ν_1 and ν_2, respectively. What is the ratio of the velocities of the photoelectrons emitted

from these surfaces when light of frequency ν is incident on them and photoemission occurs?

Check Solution → [2020 • Set 55-2-3]

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88. In photoelectric effect, the number of emitted photoelectrons is proportional to _____ of incident light.

OR

Light of frequency ν is incident on a photosensitive surface of threshold frequency ν_0 ($\nu > \nu_0$). The value of kinetic energy of the emitted photoelectrons will be _____.

Check Solution → [2020 • Set 55-3-1]

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89. The kinetic energy of a proton and that of an α-particle are 4 eV and 1 eV, respectively. The ratio of the de-Broglie wavelengths associated with them, will be

(A) 2:1

(B) 1:1

(C) 1:2

(D) 4:1

Check Solution → [2020 • Set 55-4-1]

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90. A photocell connected in an electrical circuit is placed at a distance d from a source of light. As a result, current I flows in the circuit. What will be the current in the circuit when the distance is reduced to $d/2$?

(A) I

(B) 2I

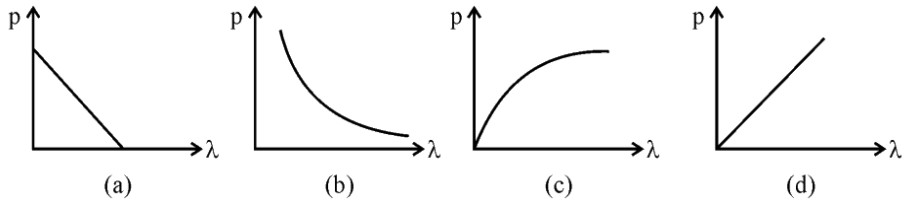
(C) 4I

(D) $\sqrt{2}\, I$

Check Solution → [2020 • Set 55-4-1]

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91. The graph showing the correct variation of linear momentum (p) of a charge particle with its de-Broglie wavelength (λ) is —



(A) (a)

(B) (b)

(C) (c)

(D) (d)

Check Solution → [2020 • Set 55-5-1]

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92. If photons of frequency ν are incident on the surfaces of metals A & B of threshold frequencies $\nu/2$ and $\nu/3$ respectively, the ratio of the maximum kinetic energy of electrons emitted from A to that from B is

(A) 2:3

(B) 3:4

(C) 1:3

(D) 3:2

Check Solution → [2020 • Set 55-5-1]

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93. Define the term "threshold frequency", in the context of photoelectric emission.

Check Solution → [2019 • Set 55-1-1]

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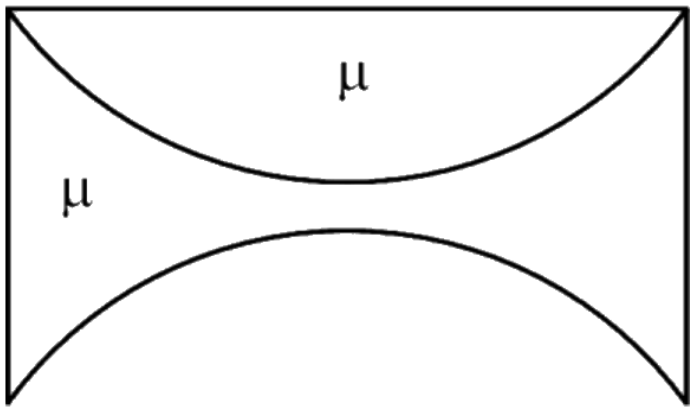
94. Define the term "Intensity" in photon picture of electromagnetic radiation.

Check Solution → [2019 • Set 55-1-1]

---

95. On the basis of the graphs shown in the figure, answer the following questions:
(a) Which physical parameter is kept constant for the three curves?

(b) Which is the highest frequency among ν_1, ν_2 and ν_3?



Check Solution → [2019 • Set 55-3-1]

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96. In photoelectric emission, when the frequency of incident radiation is doubled, will the maximum kinetic energy of photoelectrons also be doubled? Justify your answer.

Check Solution → [2019 • Set 55-3-2]

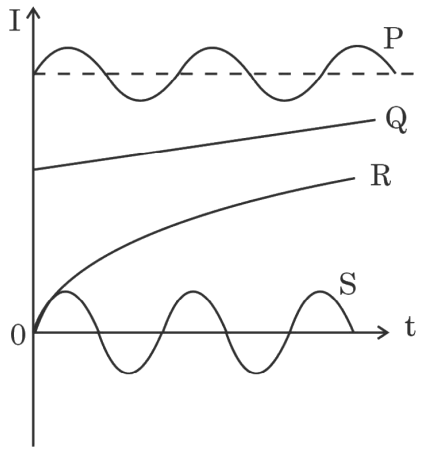
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97. Define intensity of radiation based on photon picture of light.

Check Solution → [2019 • Set 55-3-3]

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98. The figure shows the variation of stopping potential V_s with the frequency ν of the incident radiations for two photosensitive metals P and Q. Which metal has smaller threshold wavelength? Justify your answer.



Check Solution → [2019 • Set 55-4-1]

99. Plot a graph showing the variation of photoelectric current versus intensity of light.

Check Solution → [2019 • Set 55-4-2]

100. Plot a graph of stopping potential (V_s) versus the frequency (ν) of incident radiation in photoelectric emission.

Check Solution → [2019 • Set 55-4-3]

101. Plot a graph of the de-Broglie wavelength associated with a proton versus its momentum.

Check Solution → [2019 • Set 55-5-1]

102. Plot a graph of the de-Broglie wavelength associated with electron as a function of accelerating potential.

Check Solution → [2019 • Set 55-5-2]

103. Draw graphs showing variation of photoelectric current with applied voltage for two incident radiations of equal frequency and different intensities. Mark the graph for the radiation of higher intensity.

Check Solution → [2018]

104. A photosensitive surface emits photoelectrons when red light falls on it. Will the surface emit photoelectrons when blue light is incident on it? Give reason.

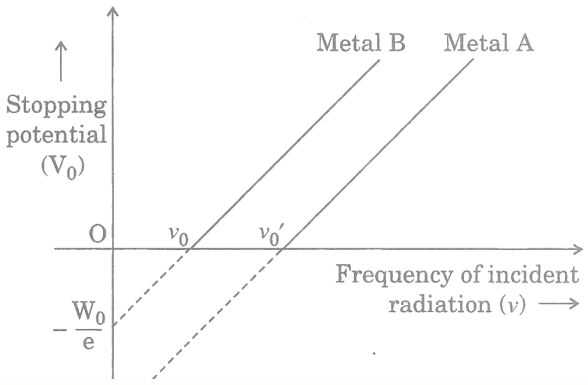
Check Solution → [2017]

105. Name the phenomenon which shows the quantum nature of electromagnetic radiation.

Check Solution → [2017]

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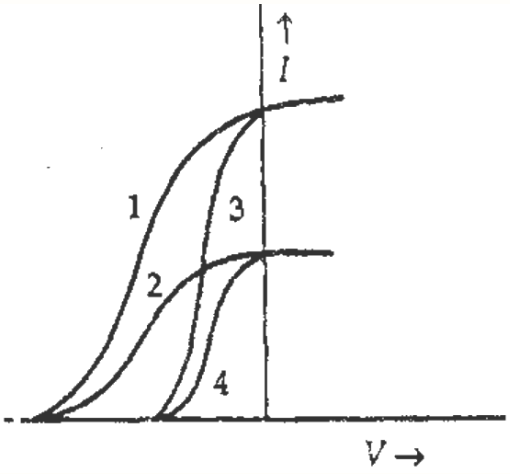
106. The graph shows variation of stopping potential V_0 versus frequency of incident radiation ν for two photosensitive metals A and B. Which of the two metals has higher threshold frequency and why ?



Check Solution → [2014]

---

107. The given graph shows the variation of photo-electric current (I) versus applied voltage (V) for two different photosensitive materials and for two different intensities of the incident radiation. Identify the pairs of curves that correspond to different materials but same intensity of incident radiation.



Check Solution → [2013]

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108. Does the 'stopping potential' in photoelectric emission depend upon
(i) the intensity of the incident radiation in a photocell?
(ii) the frequency of the incident radiation?

Check Solution → [2013]

---

109. A proton and an electron have same kinetic energy. Which one has greater de-Broglie wavelength and why?

Check Solution → [2012]

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110. Show on a graph the variation of the de Broglie wavelength (λ) associated with an electron, with the square root of accelerating potential (V).

Check Solution → [2012]

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111. Define the term 'stopping potential' in relation to photoelectric effect.

Check Solution → [2011]

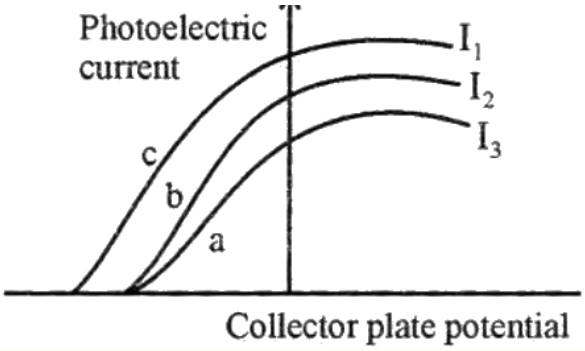
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112. Show graphically, the variation of the de-Broglie wavelength (λ) with the potential (V) through which an electron is accelerated from rest.

Check Solution → [2011 • Set 55-1-1]

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113. The figure shows a plot of three curves a, b, c showing the variation of photocurrent vs collector plate potential for three different intensities I_1, I_2 and I_3 having frequencies ν_1, ν_2 and ν_3, respectively incident on a photosensitive surface.
Point out the two curves for which the incident radiations have same frequency but different intensities.



Check Solution → [2009]

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114. The stopping potential in an experiment on photoelectric effect is 1.5 *V*. What is the maximum kinetic energy of the photoelectrons emitted?

Check Solution → [2009]

---

115. An electron and an alpha particle have the same de-Broglie wavelength associated with them. How are their kinetic energies related to each other?

Check Solution → [2008]

---

116. What is the stopping potential of a photocell, in which electrons with maximum kinetic energy of 6 eV are emitted?

Check Solution → [2008]

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117. An electron, an alpha-particle and a proton have the same kinetic energy. Which one of these particles has the largest de-Broglie wavelength?

Check Solution → [2007]

---

118. Ultraviolet radiations of different frequencies ν_1 and ν_2 are incident on two photosensitive materials having work functions W_1 and W_2 ($W_1 > W_2$) respectively. The kinetic energy of the emitted electrons is same in both the cases. Which one of the two radiations will be of higher frequency?

Check Solution → [2007]

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119. With what purpose was famous Davisson-Germer experiment with electrons performed?

Check Solution → [2006]

---

120. Electrons are emitted from a photosensitive surface when it is illuminated by green light but electron emission does not take place by yellow light. Will the electrons be emitted when the surface is illuminated by (i) red light, and (ii) blue light?

Check Solution → [2005]

SECTION B

2-Mark Questions

VSA · 60 questions

> *Section B of the paper, 2 marks each. Sorted newest year first.*

121. In a photoelectric experiment, the emitter plate is irradiated with radiation of 200 nm. The photocurrent becomes zero when the collector plate potential is −0.80 V. Calculate the work function (in eV) of the emitter.

Check Solution → **[2026 • Set 55-1-1]**

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122. Find the ratio $\frac{\lambda_\alpha}{\lambda_p}$ of de Broglie wavelength λ_α associated with an alpha particle to de Broglie wavelength λ_p associated with a proton if both are moving with the (a) same velocity (b) same kinetic energy.

Check Solution → **[2026 • Set 55-1-2]**

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123. In an electron microscope, accelerated electrons have wavelength of 0.011 nm. Calculate the voltage through which electrons were accelerated to attain this wavelength.
(Take $e = 1.6 \times 10^{-19}$ C, $m_e = 9 \times 10^{-31}$ kg, $h = 6.6 \times 10^{-34}$ J·s)

Check Solution → **[2026 • Set 55-1-3]**

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124. Find the ratio $(\lambda_\alpha/\lambda_p)$ of the de Broglie wavelengths λ_α and λ_p associated respectively with an alpha particle and a proton:
(i) if they are moving with the same kinetic energy.
(ii) just after they are accelerated through the same potential difference.

Check Solution → **[2026 • Set 55-2-1]**

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125. A particle of mass M at rest splits up into two particles of masses m_1 and m_2 having non-zero velocities. Calculate the ratio of the de Broglie wavelengths

associated with the two particles.

Check Solution → [2026 • Set 55-3-1]

126. Calculate the de Broglie wavelength of a neutron having kinetic energy of 150 eV.

Check Solution → [2026 • Set 55-3-2]

127. Light of frequency $5{\cdot}0 \times 10^{14}$ Hz is incident on a metal surface. If the maximum speed of photoelectrons emitted is $6{\cdot}63 \times 10^{5}\ m\ s^{-1}$, calculate the threshold frequency for the surface.

Check Solution → [2026 • Set 55-3-3]

128. A light of wavelength 400 nm is incident on metal surface whose work function is $3.0 \times 10^{-19}\ J$. Calculate the speed of the fastest photoelectrons emitted.

Check Solution → [2025 • Set 55-2-1]

129. Radiations of two frequencies are incident on a metal surface of work function 2.0 eV one by one. The energies of their photons are 2.5 eV and 4.5 eV respectively. Find the ratio of the maximum speed of the electrons emitted in the two cases.

Check Solution → [2025 • Set 55-2-2]

130. The threshold wavelength of a metal is 450 nm. Calculate (i) the work function of the metal in eV and (ii) the maximum energy of the ejected photoelectrons in eV by incident radiation of 250 nm.

Check Solution → [2025 • Set 55-2-3]

131. The threshold frequency for a given metal is 3.6×10^{14} Hz. If monochromatic radiations of frequency 6.8×10^{14} Hz are incident on this metal, find the cut-off potential for the photoelectrons.

Check Solution → [2025 • Set 55-5-1]

132. A laser beam of frequency $3{\cdot}0 \times 10^{14}$ Hz produces average power of 9 mW. Find (i) the energy of photon of the beam, and (ii) the number of photons emitted per second on an average by the source.

Check Solution → [2025 • Set 55-6-1]

133. A laser beam of wavelength 500 nm and power 5 mW strikes normally on a perfectly reflecting surface of area 1 mm^2 of a body. It rebounds back from the surface. Find the force exerted by the laser beam on the body.

Check Solution → [2025 • Set 55-6-2]

134. Consider a neutron (mass m) of kinetic energy E and a photon of the same energy. Let λ_n and λ_p be the de Broglie wavelength of neutron and the wavelength of photon respectively. Obtain an expression for $\frac{\lambda_n}{\lambda_p}$.

Check Solution → [2024 • Set 55-1-1]

135. The minimum intensity of white light that our eyes can perceive is about 0.1 nW m^{-2}. Calculate the number of photons of this light entering our pupil (area 0.4 cm^2) per second. (Take average wavelength of white light = 500 nm and Planck's constant = 6.6×10^{-34} J s)

Check Solution → [2024 • Set 55-3-1]

136. Light of wavelength 500 nm is incident on caesium metal (work function 2.14 eV) and photoemission of electrons occurs. Calculate the (i) kinetic energy (in eV) of the fastest electrons and (ii) stopping potential for this situation. (Take hc = 1240 eV·nm)

Check Solution → [2024 • Set 55-3-2]

137. Light of wavelength 600 nm is incident on potassium (work function 2.3 eV). Will photoemission of electrons occur? What is the longest wavelength that will cause photoemission of electrons? (Take hc = 1240 eV·nm)

Check Solution → [2024 • Set 55-3-3]

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138. A proton and α-particle are accelerated through different potentials V_1 and V_2 respectively so that they have the same de Broglie wavelengths. Find $\frac{V_1}{V_2}$.

Check Solution → [2024 • Set 55-4-1]

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139. What are matter waves? A proton, an electron and an α-particle have the same kinetic energy. Write the de Broglie wavelengths associated with them in increasing order.

Check Solution → [2024 • Set 55-4-2]

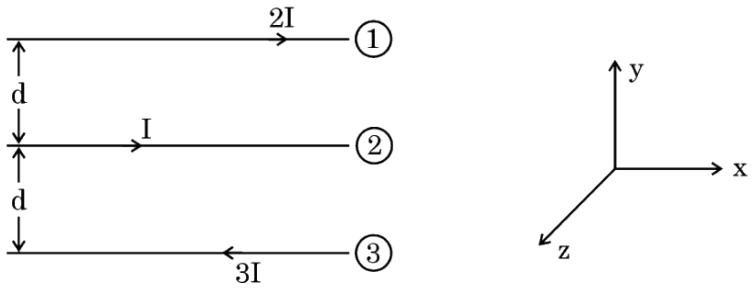
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140. The ratio of de Broglie wavelengths of a proton and a deuteron accelerated by potential V_p and V_d respectively, $(\frac{\lambda_p}{\lambda_d})$ is $\frac{1}{2}$. Find $\frac{V_p}{V_d}$.

Check Solution → [2024 • Set 55-4-3]

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141. The figure shows v_m^2 versus $\frac{1}{\lambda}$ graph for photoelectrons emitted from a surface where v_m is the maximum speed of electrons and λ is the wavelength of incident radiation. Using this graph and Einstein's photoelectric equation, obtain the expression for Planck's constant and work function of the surface.



Check Solution → [2023 • Set 55-4-1]

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142. (a) How will the De Broglie wavelength associated with an electron be affected when the (i) velocity of the electron decreases? and (ii) accelerating potential is increased? Justify your answer.

OR

(b) How would the stopping potential for a given photosensitive surface change if (i) the frequency of the incident radiation were increased? and (ii) the intensity of incident radiation were decreased? Justify your answer.

Check Solution → [2023 • Set 55-5-1]

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143. Plot a graph showing the variation of photo electric current, as a function of anode potential for two light beams having the same frequency but different intensities I_1 and I_2 ($I_2 > I_1$). Mention its important features.

Check Solution → [2023 • Set 55-5-3]

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144. Plot suitable graphs to show the variation of photoelectric current with the collector plate potential for the incident radiation of
(i) the same intensity but different frequencies ν_1, ν_2 and ν_3 ($\nu_1 < \nu_2 < \nu_3$)
(ii) the same frequency but different intensities I_1, I_2 and I_3 ($I_1 < I_2 < I_3$)

Check Solution → [2022 • Set 55-1-1]

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145. Photoelectric emission occurs when a surface is irradiated with the radiation of frequency (i) ν_1 and (ii) ν_2. The maximum kinetic energy of the electrons emitted in the two cases are *K* and 2K respectively. Obtain the expression for the threshold frequency for the surface.

Check Solution → [2022 • Set 55-2-1]

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146. (i) Name the factors on which photoelectric emission from a surface depends.
(ii) Define the term 'threshold frequency' for a photosensitive material.

Check Solution → [2022 • Set 55-3-1]

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147. What are matter waves? A proton and an alpha particle are accelerated through the same potential difference. Find the ratio of the de Broglie wavelength associated with the proton to that with the alpha particle.

Check Solution → [2022 • Set 55-4-1]

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148. Why it is the frequency and not the intensity of light source that determines whether emission of photoelectrons will occur or not? Explain.

Check Solution → [2022 • Set 55-5-1]

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149. For the light of wavelength 400 nm incident on the cathode of a photocell, the stopping potential is 6 *V*. If the wavelength of incident light is increased to 600 nm, calculate the new stopping potential. (Take $h = 4.14 \times 10^{-15}$ eV$\cdots$)

Check Solution → [2021]

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150. (a) Write two main observations of photoelectric effect experiment which could only be explained by Einstein's photoelectric equation.
(b) Draw a graph showing variation of photocurrent with the anode potential of a photocell.

Check Solution → [2020 • Set 55-1-1]

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151. Explain, how the process of emission of photoelectrons is different from the process of emission of β-particles.

Check Solution → [2020 • Set 55-1-3]

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152. Define the terms (a) threshold frequency, and (b) stopping potential. How were these terms incorporated in Einstein's photoelectric equation?

Check Solution → [2020 • Set 55-3-1]

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153. The de Broglie wavelengths associated with an electron and a proton are equal. Prove that the kinetic energy of the electron is greater than that of the proton.

Check Solution → [2020 • Set 55-3-2]

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154. (a) In what ways are matter waves different from electromagnetic waves?
(b) If the kinetic energy of a particle is reduced to one-fourth ($\frac{1}{4}$) of its initial value, how many times will the de Broglie wavelength associated with it become?

Check Solution → [2020 • Set 55-3-3]

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155. Light of same wavelength is incident on three photo-sensitive surfaces *A*, *B* and *C*. The following observations are recorded.
(i) From surface *A*, photo electrons are not emitted.
(ii) From surface *B*, photo electrons are just emitted.
(iii) From surface *C*, photo electrons with some kinetic energy are emitted.

Compare the threshold frequencies of the three surfaces and justify your answer.

OR

If the frequency of light incident on the cathode of a photo-cell is increased, how will the following be affected? Justify your answer.

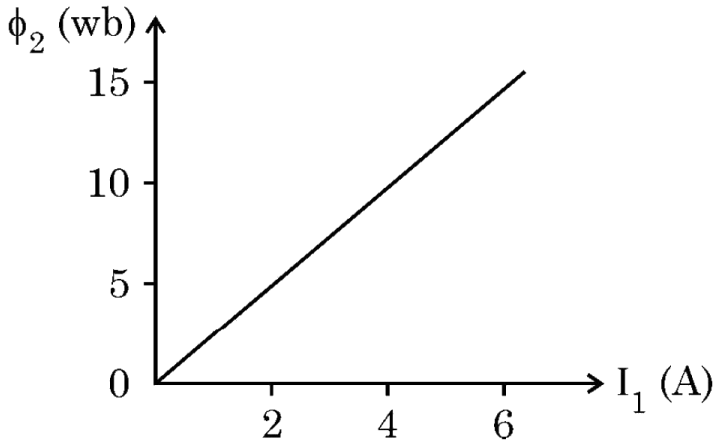
(i) Energy of the photo electrons.

(ii) Photo current.

Check Solution → [2020 • Set 55-4-1]

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156. Figure shows the stopping potential (V_0) for the photo electron versus ($1/\lambda$) graph, for two metals *A* and *B*, λ being the wavelength of incident light.



(a) How is the value of Planck's constant determined from the graph ?

(b) If the distance between the light source and the surface of metal *A* is increased, how will the stopping potential for the electrons emitted from it be effected ? Justify your answer.

Check Solution → [2020 • Set 55-5-1]

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157. In case of photo electric effect experiment, explain the following facts, giving reasons.

(a) The wave theory of light could not explain the existence of the threshold frequency.

(b) The photo electric current increases with increase of intensity of incident light.

Check Solution → [2020 • Set 55-5-2]

158. Why is wave theory of electromagnetic radiation not able to explain photoelectric effect? How does photon picture resolve this problem?

Check Solution → [2019 • Set 55-1-1]

159. Plot a graph showing variation of de Broglie wavelength (λ) associated with a charged particle of mass m, versus $\frac{1}{\sqrt{V}}$, where V is the potential difference through which the particle is accelerated. How does this graph give us the information regarding the magnitude of the charge of the particle?

Check Solution → [2019 • Set 55-1-1]

160. An electron, a proton and an alpha particle have the same kinetic energy. Which one has the shortest wavelength?

Check Solution → [2019 • Set 55-1-2]

161. Explain with the help of Einstein's photoelectric equation any two observed features in photoelectric effect which cannot be explained by wave theory.

Check Solution → [2019 • Set 55-1-2]

162. Define the terms, (i) threshold frequency and (ii) stopping potential in photoelectric effect.

Check Solution → [2019 • Set 55-1-3]

163. Plot a graph of photocurrent versus anode potential for a radiation of frequency ν and intensities I_1 and I_2 ($I_1 < I_2$).

Check Solution → [2019 • Set 55-1-3]

164. A photon and a proton have the same de-Broglie wavelength λ. Prove that the energy of the photon is $(2m\lambda c/h)$ times the kinetic energy of the proton.

Check Solution → [2019 • Set 55-2-1]

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165. The wavelength of light from the spectral emission line of sodium is 590 nm. Find the kinetic energy at which the electron would have the same de-Broglie wavelength.

Check Solution → [2019 • Set 55-2-3]

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166. If light of wavelength 412.5 nm is incident on each of the metals given below, which ones will show photoelectric emission and why ?

Metal	Work Function (eV)
Na	1·92
K	2·15
Ca	3·20
Mo	4·17

Check Solution → [2018]

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167. If light of wavelength 412.5 nm is incident on each of the metals given below, which ones will show photoelectric emission and why?

Metal — Work Function (eV)

Na — 1.92

K — 2.15

Ca — 3.20

Mo — 4.17

Check Solution → [2018]

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168. Calculate the de-Broglie wavelength of the electron orbitting in the $n = 2$ state of hydrogen atom.

Check Solution → [2016]

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169. A proton and an α particle are accelerated through the same potential difference. Which one of the two has (i) greater de-Broglie wavelength, and (ii) less kinetic energy? Justify your answer.

Check Solution → [2016]

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170. Write briefly the underlying principle used in Davison-Germer experiment to verify wave nature of electrons experimentally. What is the de-Broglie wavelength of an electron with kinetic energy (K.E.) 120 eV?

Check Solution → [2016]

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171. The equivalent wavelength of a moving electron has the same value as that of a photon of energy 6×10^{-17} J. Calculate the momentum of the electron.

Check Solution → [2015]

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172. A proton and an α-particle have the same de-Broglie wavelength. Determine the ratio of (i) their accelerating potentials (ii) their speeds.

Check Solution → [2015]

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173. A proton and a deuteron are accelerated through the same accelerating potential. Which one of the two has
(a) greater value of de-Broglie wavelength associated with it, and
(b) less momentum ?
Give reasons to justify your answer.

Check Solution → [2014]

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174. (i) Monochromatic light of frequency 6.0×10^{14} Hz is produced by a laser. The power emitted is 2.0×10^{-3} W. Estimate the number of photons emitted per second on an average by the source.
(ii) Draw a plot showing the variation of photoelectric current versus the intensity of incident radiation on a given photosensitive surface.

Check Solution → [2014]

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175. A deuteron and an alpha particle are accelerated with the same accelerating potential. Which one of the two has
(1) greater value of de-Broglie wavelength, associated with it and
(2) less kinetic energy ? Explain.

Check Solution → [2014]

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176. An α-particle and a proton are accelerated from rest by the same potential. Find the ratio of their de Broglie wavelengths.

Check Solution → [2012]

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177. Write Einstein's photoelectric equation. State clearly the three salient features observed in photoelectric effect, which can be explained on the basis of the above equation.

Check Solution → [2012]

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178. An electron is accelerated through a potential difference of 100 volts. What is the de-Broglie wavelength associated with it? To which part of the electromagnetic spectrum does this value of wavelength correspond?

Check Solution → [2010]

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179. Plot a graph showing the variation of stopping potential with the frequency of incident radiation for two different photosensitive materials having work

functions W_1 and W_2 ($W_1 > W_2$). On what factors does the (i) slope and (ii) intercept of the lines depend?

Check Solution → **[2010]**

180.

The work function of lithium is 2.3 eV. What does it mean? What is the relation between the work function W and threshold wavelength of a metal?

Check Solution → **[2003]**

SECTION C

3-Mark Questions

SA · 73 questions

> *Section C of the paper, 3 marks each. Sorted newest year first.*

181. (a) A particle of charge q and mass m is accelerated through a potential difference V. Deduce an expression for the de Broglie wavelength associated with the particle.

Check Solution → **[2026 • Set 55-2-3]**

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182. (b) Plot a graph for a particle of charge q and mass m showing variation of de Broglie wavelength λ, as $\frac{1}{\lambda}$ with $\sqrt{V}$, where V is the potential difference through which the particle is accelerated.

Check Solution → **[2026 • Set 55-2-3]**

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183. (a) Explain how the dual aspect of matter is evident in the de Broglie relation.
(b) Radiation of wavelength λ is incident on a photosensitive surface. Find the de Broglie wavelength of electrons emitted from the surface. Assume that the work function of the surface is negligible.

Check Solution → **[2026 • Set 55-4-1]**

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184. Photoemission of electrons occurs from a metal ($\phi_0 = 1.96$ eV) when light of frequency 6.4×10^{14} Hz is incident on it. Calculate:
(a) Energy of a photon in the incident light,
(b) The maximum kinetic energy of the emitted electrons, and
(c) The stopping potential.

Check Solution → **[2026 • Set 55-5-1]**

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185. What are de Broglie waves? Show that the wavelength of the electromagnetic radiation is equal to the de Broglie wavelength of its quantum (photon).

Check Solution → [2026 • Set 55-5-2]

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186. Answer the following giving reason:
(a) All the photo electrons do not eject with the same kinetic energy when monochromatic light is incident on a metal surface.
(b) The saturation current in case (a) is different for different intensity.
(c) If one goes on increasing the wavelength of light incident on a metal surface, keeping its intensity constant, emission of photo electrons stops at a certain wavelength for this metal.

Check Solution → [2025 • Set 55-1-1]

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187. **(a)** Draw a plot of frequency ν of incident radiations as a function of stopping potential V_0 for a given photo emissive material. What information can be obtained from the value of the intercept on the stopping potential axis?
(b) Calculate: (i) the momentum and (ii) de Broglie wavelength, of an electron with kinetic energy of 80 eV.

Check Solution → [2025 • Set 55-1-2]

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188. **(a)** Mention any three features of results of experiment on photoelectric effect which cannot be explained using the wave theory of light.
(b) In his experiment on photoelectric effect, Robert A. Millikan found the slope of the cut-off voltage versus frequency of incident light plot to be $4.12 \times 10^{-15}\ V\,s$. Calculate the value of Planck's constant from it.

Check Solution → [2025 • Set 55-1-3]

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189. Explain the following observations using Einstein's photoelectric equation :
(a) Photoelectric emission does not occur from a surface when the frequency of the light incident on it is less than a certain minimum value.
(b) It is the frequency, and not the intensity of the incident light which affects

the maximum kinetic energy of the photoelectrons.
(c) The cut-off voltage (V_0) versus frequency (ν) of the incident light curve is a straight line with a slope $\frac{h}{e}$.

Check Solution → [2025 • Set 55-4-1]

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190. **(a)** Define 'work function' of a metal. How can its value be determined from a graph between stopping potential and frequency of the incident radiation?
(b) The work function of a metal is 2.4 eV. A stopping potential of 0.6 V is required to reduce the photocurrent to zero, in a photoelectric experiment. Calculate the wavelength of light used.

Check Solution → [2025 • Set 55-7-1]

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191. **(a)** Briefly explain Einstein's photoelectric equation.
(b) Four metals with their work functions are listed below:
K = 2.3 eV, Na = 2.75 eV, Mo = 4.17 eV and Ni = 5.15 eV.
The radiation of wavelength 330 nm from a laser source placed 1 m away, falls on these metals. Which of these metals will not show photoelectric emission? What will happen if the laser source is brought closer to a distance of 50 cm?

Check Solution → [2025 • Set 55-7-2]

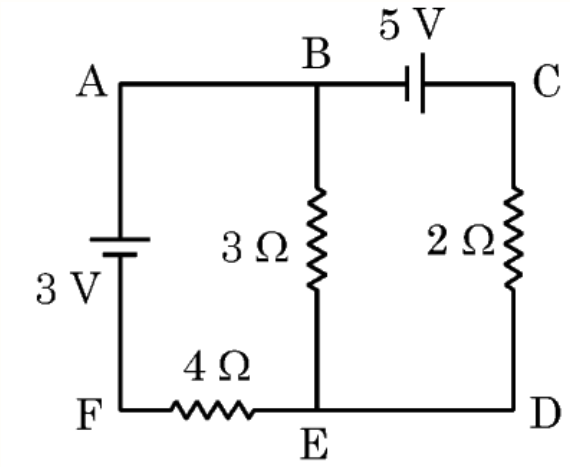
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192. Answer the following, giving reasons:
(a) The maximum kinetic energy of the photoelectrons is independent of the intensity of incident radiation.
(b) Photoelectric current increases with the increase in the intensity of the incident radiation.
(c) The stopping potential V_0 varies linearly with the frequency ν of the incident radiation for a given photosensitive surface.

Check Solution → [2025 • Set 55-7-3]

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193. de Broglie wavelength λ as a function of $\frac{1}{K}$ for two particles of masses m_1 and m_2 are shown in the figure. Here, K is the energy of the moving particles.



(a) What does the slope of a line represent?
(b) Which of the two particles is heavier?
(c) Is this graph also valid for a photon?
Justify your answer in each case.

Check Solution → [2024 • Set 55-2-1]

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194. (a) Briefly explain de Broglie hypothesis for wave nature of matter.
(b) Find the ratio of de Broglie wavelength associated with a proton and an alpha particle when both are
(i) accelerated from rest through the same potential difference, and
(ii) moving with the same kinetic energy.

Check Solution → [2024 • Set 55-2-2]

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195. (a) Write Einstein's photoelectric equation. How did Millikan prove the validity of this equation?
(b) Explain the existence of threshold frequency of incident radiation for photoelectric emission from a given surface.

Check Solution → [2024 • Set 55-5-1]

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196. A photosensitive surface of work function 2.1 eV is irradiated by radiation of wavelength 150 nm. Calculate (i) the threshold wavelength, (ii) energy (in eV) of an incident photon, and (iii) maximum kinetic energy of emitted photoelectron.

Check Solution → [2024 • Set 55-5-2]

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197. The threshold frequency for a metal is 3.0×10^{14} Hz. A beam of frequency 9.0×10^{14} Hz is incident on the metal. Calculate (i) the work function (in eV) of the metal and (ii) the maximum speed of photoelectrons.

Check Solution → [2024 • Set 55-5-3]

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198. How does Einstein's photoelectric equation explain the emission of electrons from a metal surface? Explain briefly. Plot the variation of photocurrent with: (a) collector plate potential for different intensity of incident radiation, and (b) intensity of incident radiation.

Check Solution → [2023 • Set 55-1-1]

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199. The threshold wavelength for a metal is 3315 Å. What should be the wavelength of light incident on the metal surface so that the maximum kinetic energy of the emitted photoelectrons be 1.25 eV?

Check Solution → [2023 • Set 55-1-2]

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200. A particle is associated with a de Broglie wave of wavelength 1.2 nm. The kinetic energy of the particle is made four times. Calculate the new value of the wavelength of the wave.

Check Solution → [2023 • Set 55-1-3]

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201. Calculate the wavelength of de Broglie waves associated with a proton having $\left(\frac{500}{1.673}\right)$ eV energy. How will the wavelength be affected for an alpha particle having the same energy?

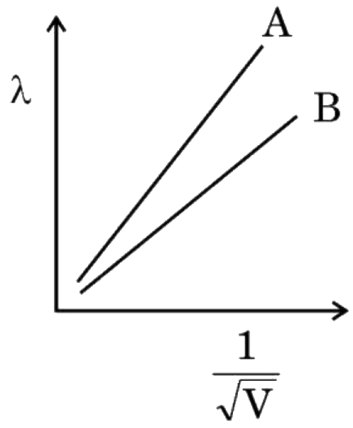
Check Solution → [2023 • Set 55-3-1]

202. (a) A particle of mass m and charge q is accelerated through a potential difference V. Plot a graph of de Broglie wavelength λ associated with it as a function of V.

(b) Calculate the energy acquired by, and de Broglie wavelength associated with, an electron accelerated through a potential difference of 400 V.

Check Solution → [2023 • Set 55-3-2]

203. (a) The figure shows de Broglie wavelength (λ) for two particles A and B having same charges but different masses as a function of $\frac{1}{\sqrt{V}}$, where V is the potential through which the particles are accelerated.



(i) Which of them is heavier and why?

(ii) What does the slope of the line represent?

(b) Calculate the momentum of an electron having de Broglie wavelength of 3.0 .

Check Solution → [2023 • Set 55-3-3]

204. Photoelectrons are emitted from a metal surface when illuminated with UV light of wavelength 330 nm. The minimum amount of energy required to emit the electrons from the surface is 3.5×10^{-19} J. Calculate:

(i) the energy of the incident radiation, and

(ii) the kinetic energy of the photoelectron.

Check Solution → [2022 • Set 55-1-1]

205. Ultra-violet light of wavelength 200 nm from a source is incident on a metal surface. If the stopping potential is −2.5 V, (a) calculate the work function of the metal, and (b) How would the surface respond to a high intensity red light of wavelength 6328 produced by a laser?

Check Solution → [2022 • Set 55-1-2]

206. The de-Broglie wavelength associated with an electron is 0.30 nm. Calculate: (a) the speed, and (b) the kinetic energy (in eV), of the electron.

Check Solution → [2022 • Set 55-1-3]

207. Find the ratio of the de Broglie wavelengths associated with an alpha particle and a proton, if both
(a) have the same speeds,
(b) have the same kinetic energy,
(c) are accelerated through the same potential difference.

Check Solution → [2022 • Set 55-2-1]

208. An electron is accelerated from rest through a potential difference of 100 V. Find:
(i) the wavelength associated with
(ii) the momentum of and
(iii) the velocity required by, the electron.

Check Solution → [2022 • Set 55-3-1]

209. (a) Calculate the frequency of a photon of energy 6.5×10^{-19} J.
(b) Can this photon cause emission of an electron from the surface of Cs of work function 2.14 eV? If yes, what will be maximum kinetic energy of the photoelectron?

Check Solution → [2022 • Set 55-3-2]

210. The work function of a metal is 2.31 eV. Photoelectric emission occurs when light of frequency 6.4×10^{14} Hz is incident on the metal surface. Calculate:
(i) the energy of the incident radiation,
(ii) the maximum kinetic energy of the emitted electron and
(iii) the stopping potential of the surface.

Check Solution → [2022 • Set 55-3-3]

211. (a) Use Einstein's photoelectric equation to depict the variation of the maximum kinetic energy (E_k) of electrons emitted, with the frequency (ν) of the incident radiation.
(b) A photosensitive surface is illuminated with a beam of (i) yellow light, and (ii) red light, both of the same intensity. In which case will
(I) photoelectrons have more E_k?
(II) more number of electrons be emitted?
Justify your answer in each case.

Check Solution → [2022 • Set 55-4-1]

212. The wavelength of the waves associated with a particle having kinetic energy E is λ. How and by what factor should its kinetic energy be changed so that the wavelength becomes $(\frac{\lambda}{3})$? Also, find the ratio of the final to the initial velocity of the particle.

Check Solution → [2022 • Set 55-4-2]

213. A proton is accelerated through a potential difference V. After acceleration, the de Broglie wavelength associated with it is λ. If the proton is replaced by an alpha particle, then find the de Broglie wavelength associated with it if it were accelerated through the same potential difference V. What will be the momentum of the alpha particle?

Check Solution → [2022 • Set 55-4-3]

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214. (a) Calculate the energy and momentum of a photon in a monochromatic beam of wavelength 331.5 nm.
(b) How fast should a hydrogen atom travel in order to have the same momentum as that of the photon in part (a)?

Check Solution → [2022 • Set 55-5-1]

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215. An alpha particle is accelerated through a potential difference of 100 V. Calculate:
(i) The speed acquired by the alpha particle, and
(ii) The de-Broglie wavelength associated with it.
(Take mass of alpha particle = 6.4×10^{-27} kg)

Check Solution → [2022 • Set 55-5-2]

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216. (a) Give an example each of a metal from which photoelectric emission takes place when irradiated by (i) UV light (ii) visible light.
(b) The work function of a metal is 4.50 eV. Find the frequency of light to be used to eject electrons from the metal surface with a maximum kinetic energy of $6.06\times 10^{-19}\ J$.

Check Solution → [2022 • Set 55-5-3]

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217. The maximum kinetic energy of the photoelectrons emitted is doubled when the wavelength of light incident on the photosensitive surface changes from λ_1 to λ_2. Deduce expressions for the threshold wavelength and work function for the metal surface in terms of λ_1 and λ_2.

Check Solution → [2020 • Set 55-2-1]

218. What do you mean by wave nature of an electron? How was quantisation of angular momentum of the orbiting electron in Bohr's model of hydrogen atom explained by de Broglie hypothesis?

Check Solution → [2020 • Set 55-3-1]

219. (a) Plot a graph to show the variation of stopping potential with frequency of incident radiation in relation to photoelectric effect.
(b) Use Einstein's photoelectric equation to show how from this graph, (i) Threshold frequency, and (ii) Planck's constant can be determined.

OR

(a) How does one explain the emission of electrons from a photosensitive surface with the help of Einstein's photoelectric equation?
(b) Work function of aluminium is 4.2 eV. If two photons each of energy 2.5 eV are incident on its surface, will the emission of electrons take place? Justify your answer.
(c) The stopping potential in an experiment on photoelectric effect is 1.5 V. What is the maximum kinetic energy of the photoelectrons emitted? Calculate in Joules.

Check Solution → [2019 • Set 55-3-1]

220. (a) State briefly, with what purpose was Davisson and Germer experiment performed and what inference was drawn from this.
(b) Obtain an expression for the ratio of the accelerating potentials required to accelerate a proton and an α-particle to have the same de-Broglie wavelength associated with them.

OR

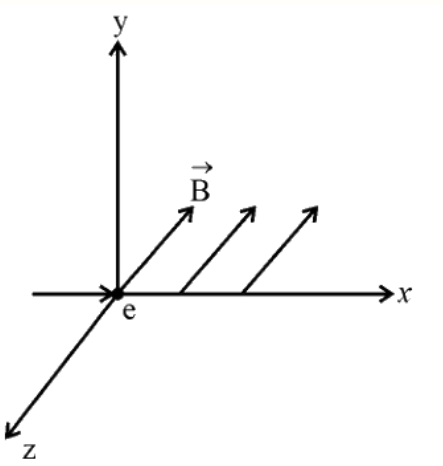
(a) An electron and a proton are accelerated through the same potential. Which one of the two has
(i) greater value of de-Broglie wavelength associated with it, and
(ii) lesser momentum?
Justify your answer in each case.

(b) How is the momentum of a particle related with its de-Broglie wavelength? Show the variation on a graph.

Check Solution → [2019 • Set 55-4-1]

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221. When a given photosensitive material is irradiated with light of frequency ν, the maximum speed of the emitted photoelectrons equals v_{max}. The graph shown in the figure gives a plot of v_{max}^2 varying with frequency ν.



Obtain an expression for
(a) Planck's constant, and
(b) The work function of the given photosensitive material in terms of the parameters '*l*', '*n*' and the mass '*m*' of the electron.
(c) How is threshold frequency determined from the plot?

Check Solution → [2019 • Set 55-5-1]

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222. When light of frequency ν_1 is incident on a photosensitive surface, the stopping potential is V_1. If the frequency of incident radiation becomes $\nu_1/2$, the stopping potential changes to V_2. Find out the expression for the threshold frequency for the surface in terms of V_1 and V_2.
If the frequency of incident radiation is doubled, will the maximum kinetic energy of the photoelectrons also be doubled? Give reason.

Check Solution → [2019 • Set 55-5-2]

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223. Write Einstein's photoelectric equation using the quantum picture of radiation. Explain briefly the three characteristic features observed in photoelectric effect which can be explained by Einstein's equation and not by the wave theory of light.

Check Solution → [2019 • Set 55-5-3]

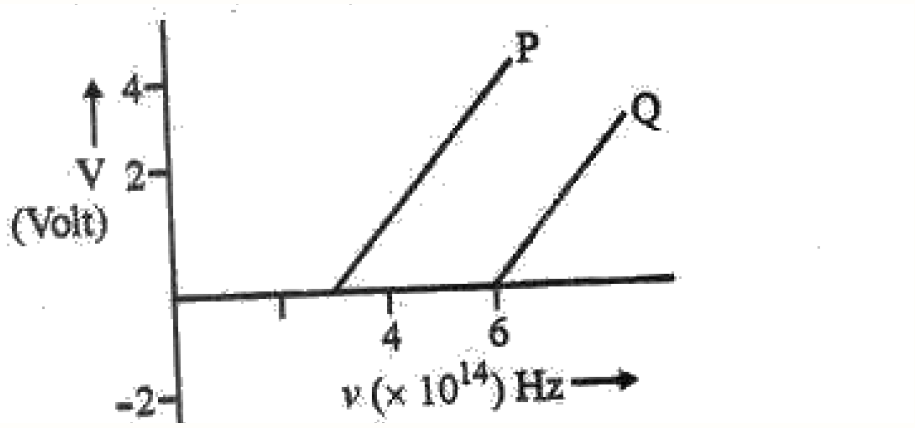
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224. (i) How does one explain the emission of electrons from a photosensitive surface with the help of Einstein's photoelectric equation?
(ii) The work function of the following metals is given: Na = 2.75 eV, K = 2.3 eV, Mo = 4.17 eV and Ni = 5.15 eV. Which of these metals will not cause photoelectric emission for radiation of wavelength 3300 A from a laser source placed 1 m away from these metals? What happens if the laser source is brought nearer and placed 50 cm away?

Check Solution → [2017]

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225. In the study of a photoelectric effect the graph between the stopping potential V and frequency ν of the incident radiation on two different metals P and Q is shown below:



(i) Which one of the two metals has higher threshold frequency?
(ii) Determine the work function of the metal which has greater value.
(iii) Find the maximum kinetic energy of electron emitted by light of frequency 8×10^{14} Hz for this metal.

Check Solution → [2017]

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226. (i) State two important features of Einstein's photoelectric equation.
(ii) Radiation of frequency 10^{15} Hz is incident on two photosensitive surfaces P and Q. There is no photoemission from surface P. Photoemission occurs from surface Q but photoelectrons have zero kinetic energy. Explain these observations and find the value of work function for surface Q.

Check Solution → [2017]

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227. (a) Draw a plot showing the variation of photoelectric current with collector potential for different frequencies but same intensity of incident radiation.
(b) Use Einstein's photoelectric equation to explain the observations from this graph.
(c) What change will you observe if intensity of incident radiation is changed but the frequency remains the same?

Check Solution → [2017]

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228. (a) Draw a graph showing variation of photocurrent with anode potential for a particular intensity of incident radiation. Mark saturation current and stopping potential.
(b) By how much would the stopping potential for a given photosensitive surface go up if the frequency of the incident radiations were to be increased from 4×10^{15} Hz to 8×10^{15} Hz?

Check Solution → [2017]

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229. Using photon picture of light, show how Einstein's photoelectric equation can be established. Write two features of photoelectric effect which cannot be explained by wave theory.

Check Solution → [2017]

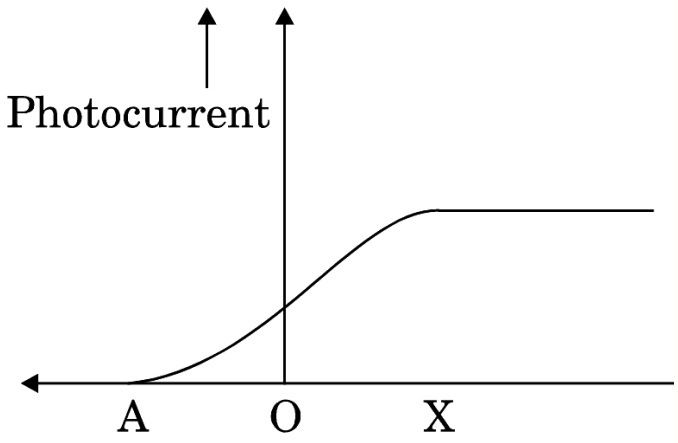
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230. Explain giving reasons for the following:
(a) Photoelectric current in a photocell increases with the increase in the intensity of the incident radiation.
(b) The stopping potential (V_0) varies linearly with the frequency (ν) of the incident radiation for a given photosensitive surface with the slope remaining the same for different surfaces.
(c) Maximum kinetic energy of the photoelectrons is independent of the intensity of incident radiation.

Check Solution → [2017]

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231. The following graph shows the variation of photocurrent for a photosensitive metal:



(a) Identify the variable *X* on the horizontal axis.
(b) What does the point *A* on the horizontal axis represent?
(c) Draw this graph for three different values of frequencies of incident radiation ν_1, ν_2 and ν_3 ($\nu_1 > \nu_2 > \nu_3$) for same intensity.
(d) Draw this graph for three different values of intensities of incident radiation I_1, I_2 and I_3 ($I_1 > I_2 > I_3$) having same frequency.

Check Solution → **[2017]**

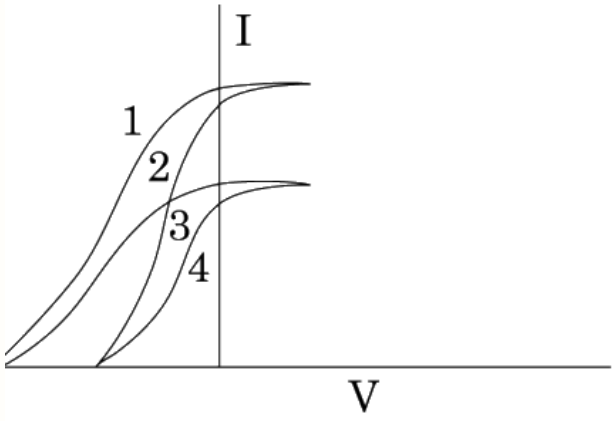
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232. Sketch the graphs showing variation of stopping potential with frequency of incident radiations for two photosensitive materials *A* and *B* having threshold frequencies $\nu_A > \nu_B$.
(i) In which case is the stopping potential more and why?
(ii) Does the slope of the graph depend on the nature of the material used? Explain.

Check Solution → **[2016]**

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233. The given graph shows the variation of photo-electric current (*I*) with the applied voltage (*V*) for two different materials and for two different intensities of the incident radiations. Identify and explain using Einstein's photo electric equation the pair of curves that correspond to (i) different materials but same intensity of incident radiation, (ii) different intensities but same materials.



Check Solution → [2016]

234. State two important properties of photon which are used to write Einstein's photoelectric equation. Define (i) stopping potential and (ii) threshold frequency, using Einstein's equation and drawing necessary plot between relevant quantities.

Check Solution → [2016]

235. (a) Define the term 'intensity of radiation' in photon picture.
(b) Plot a graph showing the variation of photo current vs collector potential for three different intensities $I_1 > I_2 > I_3$, two of which (I_1 and I_2) have the same frequency ν and the third has frequency $\nu_1 > \nu$.
(c) Explain the nature of the curves on the basis of Einstein's equation.

Check Solution → [2016]

236. (a) Define the term 'intensity of radiation' in terms of photon picture of light.
(b) Two monochromatic beams, one red and the other blue, have the same intensity. In which case (i) the number of photons per unit area per second is larger, (ii) the maximum kinetic energy of the photoelectrons is more? Justify your answer.

Check Solution → [2015]

237. Write Einstein's photoelectric equation and mention which important features in photoelectric effect can be explained with the help of this equation.
The maximum kinetic energy of the photoelectrons gets doubled when the wavelength of light incident on the surface changes from λ_1 to λ_2. Derive the expressions for the threshold wavelength λ_0 and work function for the metal surface.

Check Solution → [2015]

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238. An electron microscope uses electrons accelerated by a voltage of 50 kV. Determine the de-Broglie wavelength associated with the electrons. Taking other factors, such as numerical aperture etc. to be same, how does the resolving power of an electron microscope compare with that of an optical microscope which uses yellow light ?

Check Solution → [2014]

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239. (a) Why photoelectric effect can not be explained on the basis of wave nature of light? Give reasons.
(b) Write the basic features of photon picture of electromagnetic radiation on which Einstein's photoelectric equation is based.

Check Solution → [2013]

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240. Define the terms (i) 'cut-off voltage' and (ii) 'threshold frequency' in relation to the phenomenon of photoelectric effect.
Using Einstein's photoelectric equation show how the cut-off voltage and threshold frequency for a given photosensitive material can be determined with the help of a suitable plot/graph.

Check Solution → [2012]

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241. Write Einstein's photoelectric equation. State clearly how this equation is obtained using the photon picture of electromagnetic radiation.

Write the three salient features observed in photoelectric effect which can be explained using this equation.

Check Solution → [2012]

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242. Write two characteristic features observed in photoelectric effect which support the photon picture of electromagnetic radiation.
Draw a graph between the frequency of incident radiation (ν) and the maximum kinetic energy of the electrons emitted from the surface of a photosensitive material. State clearly how this graph can be used to determine (i) Planck's constant and (ii) work function of the material.

Check Solution → [2012]

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243. Draw a plot showing the variation of photoelectric current with collector plate potential for two different frequencies, $\nu_1 > \nu_2$, of incident radiation having the same intensity. In which case will the stopping potential be higher? Justify your answer.

Check Solution → [2011]

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244. An electron and a photon each have a wavelength 1.00 nm. Find
(i) their momenta,
(ii) the energy of the photon and
(iii) the kinetic energy of electron.

Check Solution → [2011 • Set 55-1-1]

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245. Light of wavelength 2000 Å falls on a metal surface of work function 4.2 eV. What is the kinetic energy (in eV) of the fastest electrons emitted from the surface?
(i) What will be the change in the energy of the emitted electrons if the intensity of light with same wavelength is doubled?
(ii) If the same light falls on another surface of work function 6.5 eV, what will be the energy of emitted electrons?

Check Solution → [2011 • Set 55-2-1]

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246. A proton and an alpha particle are accelerated through the same potential. Which one of the two has (i) greater value of de-Broglie wavelength associated with it, and (ii) less kinetic energy? Justify your answers.

Check Solution → [2009]

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247. An electromagnetic wave of wavelength λ is incident on a photosensitive surface of negligible work function. If the photo-electrons emitted from this surface have the de-Broglie wavelength λ_1, prove that $\lambda = \frac{2mc}{h}\lambda_1^2$.

Check Solution → [2008]

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248. In a plot of photoelectric current versus anode potential, how does
(i) the saturation current vary with anode potential for incident radiations of different frequencies but same intensity?
(ii) the stopping potential vary for incident radiations of different intensities but same frequency?
(iii) photoelectric current vary for different intensities but same frequency of incident radiations?
Justify your answer in each case.

Check Solution → [2007]

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249. Draw a schematic diagram of the experimental arrangement used by Davisson and Germer to establish the wave nature of electrons. Explain briefly how the de-Broglie relation was experimentally verified in case of electrons.

Check Solution → [2007]

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250. Define the terms threshold frequency and stopping potential in relation to the phenomenon of photoelectric effect. How is the photoelectric current affected on increasing the (i) frequency (ii) intensity of the incident radiations and why?

Check Solution → [2006]

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251. Ultraviolet light of wavelength 2271 from a 100 W mercury source radiates a photo cell made of molybdenum metal. If the stopping potential is 1.3 V, estimate the work function of the metal. How would the photo cell respond to high intensity (10^5 W m^{-2}) red light of wavelength 6328 produced by a He-Ne laser?

Plot a graph showing the variation of photoelectric current with anode potential for two light beams of same wavelength but different intensity.

Check Solution → [2005]

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252. Define the term 'work function' of a metal. The threshold frequency of a metal is f_0. When the light of frequency $2f_0$ is incident on the metal plate, the maximum velocity of electrons emitted is v_1. When the frequency of the incident radiation is increased to $5f_0$, the maximum velocity of electrons emitted is v_2. Find the ratio of v_1 to v_2.

Check Solution → [2004]

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253. Why are de-Broglie waves associated with a moving football not visible? The wavelength λ of a photon and the de-Broglie wavelength of an electron have the same value. Show that the energy of the photon is $\frac{2mc\lambda}{h}$ times the kinetic energy of the electron, where m, c and h have their usual meanings.

Check Solution → [2003]

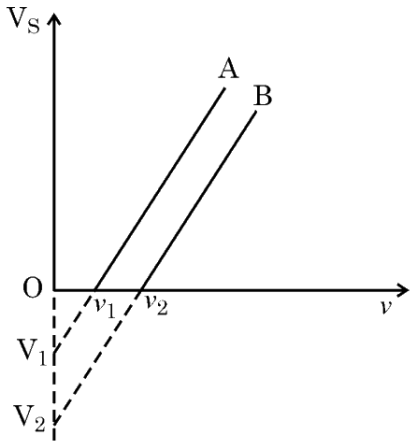
SECTION D

4-Mark Questions

Case Study · 3 questions

> *Section D of the paper, 4 marks each. Sorted newest year first.*

254. A researcher performs an experiment on photo-electric effect using two metals A and B with unknown work functions. She illuminates the surfaces of A and B with monochromatic radiation of various frequencies and records the value of corresponding stopping potentials (V_s). The graph shows the variation of stopping potential (V_s) with the frequency of incident radiation (ν) for metals A and B.



Check Solution → [2026 • Set 55-1-1]

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255. Figure shows the variation of photoelectric current measured in a photo cell circuit as a function of the potential difference between the plates of the photo cell when light beams A, B, C and D of different wavelengths are incident on the photo cell. Examine the given figure and answer the following questions:

(i) Which light beam has the highest frequency and why?
(ii) Which light beam has the longest wavelength and why?
(iii) Which light beam ejects photoelectrons with maximum momentum and why?

Check Solution → [2023 • Set 55-2-1]

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256. What is the effect on threshold frequency and stopping potential on increasing the frequency of incident beam of light? Justify your answer.

Check Solution → [2023 • Set 55-2-1]

SELF-CHECK

Answer Key

All MCQs at a glance

> *Attempt the MCQs first, then check here. Tap any question number to open its full worked solution.*

Q1 **A**	Q15 **B**	Q29 **C**	Q43 **B**	Q57 **C**	Q71 **B**
Q2 **C**	Q16 **B**	Q30 **A**	Q44 **D**	Q58 **A**	Q72 **B**
Q3 **A**	Q17 **B**	Q31 **B**	Q45 **D**	Q59 **D**	Q73 **A**
Q4 **D**	Q18 **B**	Q32 **D**	Q46 **B**	Q60 **A**	Q74 **B**
Q5 **C**	Q19 **A**	Q33 **B**	Q47 **B**	Q61 **A**	Q75 **D**
Q6 **B**	Q20 **D**	Q34 **B**	Q48 **A**	Q62 **B**	Q76 **B**
Q7 **D**	Q21 **A**	Q35 **C**	Q49 **A**	Q63 **C**	Q77 **C**
Q8 **B**	Q22 **C**	Q36 **C**	Q50 **D**	Q64 **A**	Q82 **C**
Q9 **A**	Q23 **B**	Q37 **B**	Q51 **A**	Q65 **C**	Q89 **B**
Q10 **B**	Q24 **C**	Q38 **D**	Q52 **C**	Q66 **B**	Q90 **C**
Q11 **D**	Q25 **A**	Q39 **C**	Q53 **D**	Q67 **C**	Q91 **?**
Q12 **C**	Q26 **C**	Q40 **C**	Q54 **B**	Q68 **A**	Q92 **B**
Q13 **D**	Q27 **C**	Q41 **A**	Q55 **C**	Q69 **B**	
Q14 **D**	Q28 **B**	Q42 **A**	Q56 **C**	Q70 **A**	

WORKED ANSWERS

Detailed Solutions

Every question worked step by step

Each card restates the question, shows the answer, then the full working. Use "Back to Q n" to return.

Q1

Two statements are given - one labelled as Assertion (A) and the other labelled as Reason (R). Select the correct answer from the options below:
(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).
(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).
(C) Assertion (A) is true, but Reason (R) is false.
(D) Both Assertion (A) and Reason (R) are false.
Assertion (A): *If accelerated electrons are passed through a narrow slit, a diffraction pattern is observed.*
Reason (R): *Electrons behave as both particles and waves.*

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of (A).

[2026 • Set 55-1-1]

SOLUTION

We test each statement on its own, then check whether the Reason explains the Assertion.

Assertion (A): When a beam of accelerated electrons is sent through a narrow slit, a diffraction pattern is obtained on the screen. This is true. Diffraction is a wave phenomenon, and the Davisson and Germer experiment, along with G. P. Thomson's electron diffraction experiments, confirmed that electrons produce diffraction patterns just like waves do.

Reason (R): Electrons behave both as particles and as waves. This is also true. This is the principle of wave-particle duality. The wave nature of a moving electron is described by the de Broglie relation.

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

Now we check the link. Diffraction can only happen if the electron has a wave character. So the wave-particle duality stated in (R) is exactly the reason why the diffraction in (A) is seen. Hence (R) correctly explains (A).

The other options fail: (B) wrongly says (R) does not explain (A); (C) and (D) wrongly mark one or both statements as false. So the correct choice is **(A) Both Assertion and Reason are true and Reason is the correct explanation**.

Final answer. Correct option is **(A)**: both A and R are true and R correctly explains A.

← *Back to Q1*

Q2

From the graph, the work functions of A and B are (h is Planck's constant and e value of charge on an electron):

(A) ν_1 and ν_2

(B) V_1 and V_2

(C) **$h\nu_1$ and $h\nu_2$**

(D) $\frac{h\nu_1}{e}$ and $\frac{h\nu_2}{e}$

[2026 • Set 55-1-1]

SOLUTION

Einstein's photoelectric equation, written for the stopping potential, is the key here.

$$eV_0 = h\nu - \phi_0$$

$$V_0 = \frac{h}{e}\nu - \frac{\phi_0}{e}$$

This is a straight line of V_0 against ν. The line cuts the ν-axis where $\mathbf{V_0 = 0}$, and at that point ν equals the threshold frequency ν_0. At threshold, the photon energy just equals the work function.

$$\phi_0 = h\nu_0$$

From the graph, metal A meets the ν-axis at ν_1 and metal B meets it at ν_2. So their threshold frequencies are ν_1 and ν_2.

$$\phi_A = h\nu_1 \; \phi_B = h\nu_2$$

Checking the options: ν_1,ν_2 are only frequencies, not energies; V_1,V_2 are potentials; the $\frac{h\nu}{e}$ form gives a voltage, not work function. Only $h\nu_1$ and $h\nu_2$ have units of energy and use the correct threshold values. So the **work functions are $h\nu_1$ and $h\nu_2$**.

Final answer. Correct option is **(C)**: the work functions of A and B are **$h\nu_1$** and **$h\nu_2$**.

← *Back to Q2*

Q3

For radiation of frequency $\nu > \nu_2$ incident on the surfaces of A and B, the maximum kinetic energy of ejected electron is:

(A) greater for metal A because it has a smaller work function.

(B) greater for metal B because it has a larger work function.

(C) greater for metal B because it has higher threshold frequency.

(D) the same for both metal A and metal B because it is independent of work functions of metals.

[2026 • Set 55-1-1]

SOLUTION

We compare the maximum kinetic energy of the two metals when the same radiation falls on both.

By Einstein's photoelectric equation, the maximum kinetic energy of an emitted electron is the photon energy left over after the work function is paid.

$$KE_{max} = h\nu - \phi_0$$

For a fixed incident frequency ν, the term $h\nu$ is the same for both surfaces. So the metal with the smaller work function ϕ_0 keeps more energy as kinetic energy.

From the graph the threshold frequencies are ν_1 for A and ν_2 for B, with $\boldsymbol{\nu_1 < \nu_2}$. Since $\boldsymbol{\phi_0 = h\nu_0}$, this means metal A has the smaller work function.

$$\phi_A = h\nu_1 < \phi_B = h\nu_2$$

Therefore, at the same frequency, metal A gives the larger maximum kinetic energy.

Ruling out the rest: (B) and (C) wrongly favour B (which has the larger work function and higher threshold, so less leftover energy); (D) is wrong because KE_{max} clearly depends on the work function. The correct answer is **greater for metal A, because A has the smaller work function**.

Final answer. Correct option is **(A)**: maximum kinetic energy is greater for metal A as it has the smaller work function.

← *Back to Q3*

Q4

If the intensity of the incident radiation for both metals A and B, is doubled keeping its frequency constant, then:

(A) the slope of the parallel lines will increase.

(B) the slope of the parallel lines will decrease.

(C) the threshold frequencies for both A and B will decrease.

(D) the slope of the parallel lines will not change but more electrons will be emitted per second.

[2026 • Set 55-1-1]

SOLUTION

The graph drawn is stopping potential V_0 against frequency ν. We must see what doubling the intensity does to it.

$$V_0 = \frac{h}{e}\nu - \frac{\phi_0}{e}$$

The slope of this line is fixed.

$$\text{slope} = \frac{h}{e}$$

Both h (Planck's constant) and e (electron charge) are universal constants. So the slope cannot change with intensity. This is why the two metal lines are parallel in the first place.

The threshold frequency depends only on the metal.

$$\nu_0 = \frac{\phi_0}{h}$$

Since ϕ_0 is a property of the surface, the threshold frequencies stay the same too. So options (A), (B) and (C) are all wrong.

Doubling the intensity at fixed frequency only doubles the number of photons arriving per second. Each photon (with $\nu > \nu_0$) ejects one electron, so the number of photoelectrons emitted per second, and hence the saturation current, doubles. The energy per electron is unchanged.

So the correct answer is **the slope does not change, but more electrons are emitted per second**.

Final answer. Correct option is **(D)**: the slope stays the same but more electrons are emitted per second.

← *Back to Q4*

Q5

(OR variant for Q30(iv))

The threshold frequency for a metal surface is ν_0. *If the radiation of frequency* $3\nu_0$ *illuminates the surface, the maximum kinetic energy (KE) of photoelectrons is* E_1. *If the frequency were increased to* $6\nu_0$, *the maximum KE of the photoelectrons becomes* E_2. *Then* $\frac{E_1}{E_2}$ *equals:*

OR

Let m be the slope of the graph line for metal B. If e is the value of electron charge, then Planck's constant 'h' is given by:

(A) 1/3 (or me)

(B) 1/2 (or $\frac{1}{me}$)

(C) **2/5 (or $\frac{m}{e}$)**

(D) 3/4 (or $\frac{e}{m}$)

[2026 • Set 55-1-1]

SOLUTION

This is an OR question. Both parts lead to the same option letter, so I solve both.

Part 1 (kinetic energy ratio): The maximum kinetic energy of a photoelectron is the photon energy above threshold.

$$E = h(\nu - \nu_0)$$

At $\nu = 3\nu_0$:

$$E_1 = h(3\nu_0 - \nu_0) = 2h\nu_0$$

At $\nu = 6\nu_0$:

$$E_2 = h(6\nu_0 - \nu_0) = 5h\nu_0$$

Taking the ratio:

$$\frac{E_1}{E_2} = \frac{2h\nu_0}{5h\nu_0} = \frac{2}{5}$$

Part 2 (finding h from the slope): The V_0 versus ν graph has slope m, and from Einstein's equation the slope equals $\frac{h}{e}$.

$$m = \frac{h}{e} \Rightarrow h = me$$

Note that the answer-key value $\frac{2}{5}$ from Part 1 and the expression me from Part 2 sit under the same option letter in this paper. The intended marked answer is **option (C)** with $\frac{E_1}{E_2} = \frac{2}{5}$.

COMMON MISTAKE

Watch the subtraction: KE_{max} *uses* $(\nu - \nu_0)$*, not* ν *alone. Forgetting the* $-\nu_0$ *would wrongly give the ratio* **3:6 = 1:2**.

Final answer. Correct option is **(C)**: $\frac{E_1}{E_2} = \frac{2}{5}$ (and for the OR part, $h = me$).

← *Back to Q5*

Q6

Assertion (A): *On increasing the intensity of incident light of frequency* ν $(>\nu_0)$ *on a photosensitive surface, the photocurrent increases.*
Reason (R): *The stopping potential for a photosensitive surface increases with increase of frequency* ν $(>\nu_0)$ *of incident light.*

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A).

[2026 • Set 55-2-1]

SOLUTION

We judge each statement, then test the explanation link.

Assertion (A): On raising the intensity of light (with $\nu > \nu_0$), the photocurrent increases. This is true. Intensity is the number of photons striking the surface per unit area per second. With $\nu > \nu_0$, each photon can free one electron, so more photons mean more photoelectrons per second, i.e. a larger photocurrent.

Reason (R): The stopping potential increases as the frequency ν $(>\nu_0)$ of the incident light increases. This is also true. From Einstein's equation:

$$eV_0 = h\nu - \phi_0$$

$$V_0 = \frac{h}{e}\nu - \frac{\phi_0}{e}$$

So V_0 rises linearly with ν.

Both statements are correct. But (A) is about how photocurrent changes with intensity, while (R) is about how stopping potential changes with frequency. These are two separate, independent facts. The stopping-potential law in (R) says nothing about why the current depends on intensity, so it does not explain (A).

Hence the correct choice is **(B) both true, but R is not the correct explanation of A**.

Final answer. Correct option is **(B)**: both A and R are true, but R does not explain A.

← *Back to Q6*

Q7

The maximum kinetic energy of the electrons emitted from a photosensitive surface depends on:

(A) work function of the surface ϕ_0 only.

(B) frequency of the incident radiation ν only.

(C) intensity of the incident radiation I only.

(D) Both ϕ_0 and ν.

[2026 • Set 55-2-1]

SOLUTION

The maximum kinetic energy of photoelectrons is read straight off Einstein's photoelectric equation.

$$KE_{max} = h\nu - \phi_0$$

Looking at this equation, KE_{max} changes with two things:

$$KE_{max} \text{ increases linearly with } \nu \text{ (slope } h)$$

$$KE_{max} \text{ decreases as the work function } \phi_0 \text{ increases}$$

So both the frequency of the incident radiation and the work function of the surface decide the maximum kinetic energy. That means option (A) alone and

option (B) alone are each incomplete.

Intensity does not appear anywhere in the equation. Raising the intensity only sends more photons per second, which increases the photocurrent, but it leaves the energy of each photon, and hence KE_{max}, unchanged. So option (C) is wrong.

Therefore the maximum kinetic energy depends on **both ϕ_0 and ν**, which is option (D).

Final answer. Correct option is **(D): KE_{max}** depends on both the work function ϕ_0 and the frequency ν.

← *Back to Q7*

Q8

A photosensitive surface is illuminated by radiations of wavelength λ_1, λ_2 ($>\lambda_1$) and λ_3 one by one and photoemission is observed in each case. λ_1 and λ_2 lie in UV range and λ_3 in visible range. If V_1, V_2 and V_3 are stopping potentials in these cases respectively, then:

(A) $V_1 = V_2 = V_3$

(B) $V_1 > V_2 > V_3$

(C) $V_1 > V_3 > V_2$

(D) $V_3 > V_2 > V_1$

[2026 • Set 55-2-2]

SOLUTION

Stopping potential depends on the energy of the incident photons, so we link it to wavelength.

From Einstein's equation, using $\nu = \frac{c}{\lambda}$:

$$eV = h\nu - \phi_0 = \frac{hc}{\lambda} - \phi_0$$

For one fixed surface, ϕ_0 is constant, so a smaller wavelength means a larger photon energy and therefore a larger stopping potential. In short, V rises as λ

falls.

We are told the ordering of wavelengths. Both λ_1 and λ_2 are in the UV range with $\boldsymbol{\lambda_2 > \lambda_1}$, and λ_3 is in the visible range, so λ_3 is the longest of all.

$$\lambda_1 < \lambda_2 < \lambda_3$$

Since stopping potential is highest for the shortest wavelength:

$$V_1 > V_2 > V_3$$

So λ_1 (shortest, highest energy) gives the largest stopping potential, and λ_3 (longest, lowest energy) gives the smallest. This rules out (A), (C) and (D). The correct order is $\mathbf{V_1 > V_2 > V_3}$.

Final answer. Correct option is **(B)**: $\mathbf{V_1 > V_2 > V_3}$.

← *Back to Q8*

Q9

The equivalent mass of a photon of wavelength 100 nm *is:*

(A) $\mathbf{2{\cdot}21\times 10^{-35}\ kg}$

(B) $6{\cdot}63\times 10^{-33}$ kg

(C) $1{\cdot}11\times 10^{-35}$ kg

(D) $3{\cdot}31\times 10^{-27}$ kg

[2026 • Set 55-2-3]

SOLUTION

A photon has zero rest mass, but it carries energy and momentum, so we can assign it an equivalent (or dynamic) mass.

The photon energy is

$$E = h\nu = \frac{hc}{\lambda}$$

Equating this to the mass-energy relation $\boldsymbol{E = m\,c^2}$ gives the equivalent mass:

$$m = \frac{E}{c^2} = \frac{hc}{\lambda c^2} = \frac{h}{c\lambda}$$

Now substitute the values, with $\lambda = 100\ \text{nm} = 110^{-7}\ m$, $h = 6.6310^{-34}\ J\,s$ and $c = 310^{8}\ m\,s^{-1}$.

$$m = \frac{6.6310^{-34}}{(310^{8})(110^{-7})}$$

$$m = \frac{6.6310^{-34}}{310^{1}}$$

$$m = 2.2110^{-35}\ \text{kg}$$

This matches option (A). The other options come from arithmetic slips in the powers of ten, so the equivalent mass is $\mathbf{2.2110^{-35}\ kg}$.

COMMON MISTAKE

Keep the powers of ten lined up: $c\lambda = (310^{8})(10^{-7}) = 310^{1}$. *A common error is to get* 310^{15}, *which throws the answer off by orders of magnitude.*

Final answer. Correct option is **(A)**: the equivalent mass of the photon is $\mathbf{2.2110^{-35}\ kg}$.

← Back to Q9

Q10

Assertion (A): *Photoelectric current depends upon the intensity of the incident radiation.*
Reason (R): *Stopping potential is independent of the intensity of the incident radiation.*
Select the correct answer from the codes (A), (B), (C) and (D) given below:

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

[2026 • Set 55-3-1]

SOLUTION

We judge both statements, then test whether one explains the other.

Assertion (A): Photoelectric current depends on the intensity of the incident radiation. This is true. Intensity fixes the number of photons reaching the surface per second. Each photon (above threshold) can release one electron, so a higher intensity means more photoelectrons per second and a larger saturation current.

Reason (R): Stopping potential is independent of the intensity. This is also true. The stopping potential comes from Einstein's equation:

$$eV_0 = h\nu - \phi_0$$

$$V_0 = \frac{h}{e}\nu - \frac{\phi_0}{e}$$

Here V_0 depends only on the frequency ν and the work function ϕ_0, with no intensity term. So changing the intensity leaves V_0 unchanged.

Both statements are correct, but they describe two different experimental facts. (A) is about photocurrent versus intensity; (R) is about stopping potential being intensity-free. The fact that V_0 is intensity-independent does not explain why the current depends on intensity. So (R) is not the explanation of (A).

Hence the correct choice is **(B) both true, but R is not the correct explanation of A**.

Final answer. Correct option is **(B)**: both A and R are true, but R does not explain A.

← *Back to Q10*

Q11

Radiation of wavelength 200 nm *is incident on a photosensitive surface of work function* 4·2 eV. *The kinetic energy of fastest photoelectrons emitted from this surface will be close to:*

(A) 3·5 eV

(B) 3·0 eV

(C) 2·5 eV

(D) **2·0 eV**

SOLUTION

By Einstein's photoelectric equation, the maximum kinetic energy of the emitted electrons equals the energy of the incident photon minus the work function of the surface.

$$\mathrm{KE}_{\max} = h\nu - \phi = \frac{hc}{\lambda} - \phi$$

Here $\lambda = 200$ **nm** and $\phi = 4.2$ **eV**. Using the handy result **hc = 1240 eV nm**, the photon energy is

$$E = \frac{hc}{\lambda} = \frac{1240\ \text{eV nm}}{200\ \text{nm}}$$

$$E = 6.2\ \text{eV}$$

Substituting into the photoelectric equation,

$$\mathrm{KE}_{\max} = 6.2\ \text{eV} - 4.2\ \text{eV}$$

$$\mathrm{KE}_{\max} = 2.0\ \text{eV}$$

So the fastest photoelectrons carry about **2.0 eV** of kinetic energy, which matches option (D). Option (A) 3.5 eV and (B) 3.0 eV would need a photon energy of 7.7 eV or 7.2 eV, and option (C) 2.5 eV would need 6.7 eV, none of which fit a 200 nm photon.

TIP

Remember **hc = 1240 eV nm**. It turns photon energy into a one-step division by the wavelength in nm.

Final answer. The kinetic energy of the fastest photoelectrons is 2.0 eV, option **(D)**.

← *Back to Q11*

Q12

A proton and an alpha particle have equal momentum. The ratio of their kinetic energies (E_p / E_α) *and de Broglie wavelengths associated with them* ($\lambda_p / \lambda_\alpha$) *respectively are:*

(A) 2, 1

(B) 1, 2

(C) **4, 1**

(D) 1, 4

[2026 • Set 55-3-1]

SOLUTION

Both particles have the same momentum p, so compare their kinetic energies and de Broglie wavelengths in terms of p.

Kinetic energy in terms of momentum is

$$KE = \frac{p^2}{2m}$$

For equal p, this gives $\mathbf{KE} \propto \frac{1}{m}$. The alpha particle has four times the proton mass, so $\mathbf{m_\alpha = 4\, m_p}$. Therefore

$$\frac{E_p}{E_\alpha} = \frac{m_\alpha}{m_p} = \frac{4\, m_p}{m_p} = 4$$

$$\frac{E_p}{E_\alpha} = 4 : 1$$

The de Broglie wavelength depends only on momentum,

$$\lambda = \frac{h}{p}$$

Since both particles share the same p, their wavelengths are equal,

$$\frac{\lambda_p}{\lambda_\alpha} = \frac{h/p}{h/p} = 1$$

So the kinetic energy ratio is **4** and the wavelength ratio is **1**, which is option (C). The other options mismatch one of the two ratios: the energy ratio cannot

be 1 or 2, and the wavelength ratio must be 1 because λ is fixed by momentum alone.

Final answer. The ratios are $\mathbf{E_p : E_\alpha = 4}$ and $\boldsymbol{\lambda_p : \lambda_\alpha = 1}$, option **(C)**.

← *Back to Q12*

Q13

The velocity of the photoelectrons emitted from a surface depends upon the:

(A) Wavelength of incident radiation only

(B) Intensity of incident radiation only

(C) Work function of the surface only

(D) **Both wavelength of the incident radiation and work function of the surface**

[2026 • Set 55-3-2]

SOLUTION

Apply Einstein's photoelectric equation to the fastest electrons, which leave with the maximum velocity v_{max}.

$$\frac{1}{2}mv_{max}^2 = h\nu - \phi = \frac{hc}{\lambda} - \phi$$

Solving for the maximum velocity,

$$v_{max} = \sqrt{\frac{2}{m}\left(\frac{hc}{\lambda} - \phi\right)}$$

This expression shows v_{max} depends on the **wavelength** λ of the incident radiation, through the photon energy $\frac{hc}{\lambda}$, and on the **work function** ϕ of the surface.

It does not depend on the intensity of the radiation. Intensity only sets how many photons arrive per second, so it controls the number of photoelectrons (the photocurrent), not their speed. This rules out options (A), (B) and (C), each of which names only a single factor or the intensity. Option (D) correctly names both factors.

Final answer. Photoelectron velocity depends on both the wavelength and the work function, option **(D)**.

← *Back to Q13*

Q14

The kinetic energy of a charged particle is increased to four times of its initial value. The de Broglie wavelength associated with the particle will:

(A) increase by 100% of its initial value.

(B) increase by 50% of its initial value.

(C) decrease by 25% of its initial value.

(D) decrease by 50% of its initial value.

[2026 • Set 55-4-1]

SOLUTION

The de Broglie wavelength of a particle of mass m and kinetic energy K is found by writing the momentum in terms of kinetic energy, $\boldsymbol{p = \sqrt{2mK}}$.

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mK}}$$

For a fixed particle, m and h are constant, so

$$\lambda \propto \frac{1}{\sqrt{K}}$$

The kinetic energy is raised to four times its value, $\boldsymbol{K' = 4K}$. Taking the ratio of new to old wavelength,

$$\frac{\lambda'}{\lambda} = \sqrt{\frac{K}{K'}} = \sqrt{\frac{K}{4K}}$$

$$\frac{\lambda'}{\lambda} = \sqrt{\frac{1}{4}} = \frac{1}{2}$$

So $\boldsymbol{\lambda' = \frac{\lambda}{2}}$, meaning the wavelength falls to half its starting value. The change is

$$\frac{\lambda' - \lambda}{\lambda} = \frac{\lambda/2 - \lambda}{\lambda} = -\frac{1}{2} = -50\%$$

The wavelength **decreases by 50%**, which is option (D). It cannot increase, so options (A) and (B) are wrong, and a 25% decrease (option C) would require $K' = \frac{16}{9}K$, not 4K.

Final answer. The de Broglie wavelength decreases by 50%, option **(D)**.

← *Back to Q14*

Q15

While studying photoelectric emission from a given surface, the wavelength of the incident radiation is changed from 600 nm to 400 nm, keeping the intensity of radiation the same. Then:

(A) cut-off potential will decrease.

(B) cut-off potential will increase.

(C) saturation current will decrease.

(D) saturation current will increase.

[2026 • Set 55-4-1]

SOLUTION

The cut-off (stopping) potential V_0 is fixed by the maximum kinetic energy of the photoelectrons, through $eV_0 = h\nu - \phi$.

$$V_0 = \frac{h\nu - \phi}{e} = \frac{1}{e}\left(\frac{hc}{\lambda} - \phi\right)$$

As λ decreases, the photon energy $\frac{hc}{\lambda}$ rises, so V_0 increases. Checking with numbers using $hc = 1240$ eV nm:

$$E_1 = \frac{1240}{600} = 2.07 \text{ eV} \Rightarrow eV_0^{(1)} = 2.07 - \phi$$

$$E_2 = \frac{1240}{400} = 3.10 \text{ eV} \Rightarrow eV_0^{(2)} = 3.10 - \phi$$

Since $E_2 > E_1$, we get $V_0^{(2)} > V_0^{(1)}$, so the **cut-off potential will increase**. This is option (B), and it rules out option (A).

The saturation current depends on the number of photoelectrons per second, which is set by the intensity. The intensity is held constant here, so the saturation current does not change. This rules out options (C) and (D).

Final answer. The cut-off potential will increase, option **(B)**.

← *Back to Q15*

Q16

Radiation of wavelength 331 nm irradiates the following metals:MetalWork Function (eV)Na1·92K2·15Ca3·20Mo4·17Which of the following statements is correct?

(A) Only Na and K show photoelectric emission.

(B) **Only Mo will not show photoelectric emission.**

(C) All of the given metals show photoelectric emission.

(D) None of them show photoelectric emission.

[2026 • Set 55-4-1]

SOLUTION

Photoelectric emission occurs only when the incident photon energy is at least equal to the work function, that is $E_{photon} \geq \phi$. First find the photon energy for **λ = 331 nm** using **hc = 1240 eV nm**.

$$E = \frac{hc}{\lambda} = \frac{1240 \text{ eV nm}}{331 \text{ nm}}$$

$$E \approx 3.75 \text{ eV}$$

Now compare this 3.75 eV photon energy with each work function:

For Na, **3.75 eV > 1.92 eV**, so emission occurs.

For K, **3.75 eV > 2.15 eV**, so emission occurs.

For Ca, **3.75 eV > 3.20 eV**, so emission occurs.

For Mo, **3.75 eV < 4.17 eV**, so no emission occurs.

The photon can free electrons from Na, K and Ca but not from Mo. So **only Mo will not show photoelectric emission**, which is option (B). Option (A) is wrong

because Ca also emits, option (C) is wrong because Mo does not, and option (D) is wrong because three metals do emit.

Final answer. Only Mo fails to emit, so the correct statement is option **(B)**.

← *Back to Q16*

Q17

In an experiment on photoelectric emission, the intensity of incident radiation is halved keeping its frequency constant. As a result:

(A) cut-off potential will increase.

(B) **maximum kinetic energy of photoelectrons will remain the same.**

(C) photocurrent will increase.

(D) cut-off potential will decrease.

[2026 • Set 55-4-2]

SOLUTION

Halving the intensity while keeping the frequency fixed changes how many photons arrive each second, but not the energy carried by each photon.

The photocurrent is proportional to the number of photoelectrons released per second, which tracks the number of incident photons per second, so it tracks intensity. Halving the intensity halves the photocurrent. This makes option (C) false, since the current decreases rather than increases.

The maximum kinetic energy and the stopping potential are set only by the photon energy and the work function.

$$KE_{max} = h\nu - \phi,\ V_0 = \frac{h\nu - \phi}{e}$$

Neither expression contains intensity, so with ν and ϕ unchanged, both KE_{max} and V_0 stay the same. This makes options (A) and (D) false, because the cut-off potential neither increases nor decreases.

Therefore the **maximum kinetic energy of photoelectrons remains the same**, which is option (B).

CONCEPT

Intensity is a number-of-photons knob, not an energy-per-photon knob. It moves the current, never the stopping potential.

Final answer. The maximum kinetic energy of photoelectrons remains unchanged, option **(B)**.

← *Back to Q17*

Q18

Photons of frequency ν are incident on the surfaces of two metals A and B of threshold frequencies $\frac{\nu}{2}$ and $\frac{\nu}{3}$. The ratio of maximum kinetic energy of electrons emitted from metal A to that from metal B is:

(A) $\frac{1}{3}$

(B) $\frac{3}{4}$

(C) $\frac{2}{3}$

(D) $\frac{3}{2}$

[2026 • Set 55-4-3]

SOLUTION

Using Einstein's photoelectric equation written with threshold frequency ν_0, the maximum kinetic energy is

$$KE_{max} = h(\nu - \nu_0)$$

For metal A, the threshold frequency is $\frac{\nu}{2}$, so

$$KE_A = h(\nu - \frac{\nu}{2}) = \frac{h\nu}{2}$$

For metal B, the threshold frequency is $\frac{\nu}{3}$, so

$$\mathrm{KE}_B = h(\nu - \frac{\nu}{3}) = \frac{2h\nu}{3}$$

Taking the ratio,

$$\frac{\mathrm{KE}_A}{\mathrm{KE}_B} = \frac{h\nu/2}{2h\nu/3} = \frac{1}{2} \times \frac{3}{2}$$

$$\frac{\mathrm{KE}_A}{\mathrm{KE}_B} = \frac{3}{4}$$

So the maximum kinetic energies are in the ratio **3 : 4**, which is option (B). The arithmetic clearly rules out $\frac{1}{3}$, $\frac{2}{3}$ and $\frac{3}{2}$.

Final answer. The ratio of maximum kinetic energies is $\frac{3}{4}$, option **(B)**.

← *Back to Q18*

Q19

Assertion (A): *Photoelectric effect is a spontaneous phenomenon.*
Reason (R): *According to the wave picture of radiation, an electron would take hours/days to absorb sufficient energy to overcome the work function and come out from a metal surface.*
Select the correct answer from the codes given below:

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

[2026 • Set 55-5-1]

SOLUTION

Examine the Assertion and the Reason separately, then decide whether the Reason explains the Assertion.

Assertion (A): the photoelectric effect is spontaneous. This is true. In experiments, photoelectrons appear almost instantly, within about 10^{-9} *s* of the light reaching the surface, with no measurable time lag even when the light is very faint.

Reason (R): on the wave picture, an electron would need hours or days to soak up enough energy to escape. This is also true. A classical wave spreads its energy over the whole surface, so any single electron, sitting on a tiny area, would gather energy very slowly and would need an impossibly long build-up time to reach the work function for weak light.

Now the link: this exact failure of the wave model, predicting a long delay that is never observed, is what forces the photon picture. In the photon model one photon hands all its energy to one electron in a single event, so emission is naturally instantaneous. So the wave-picture delay in R is precisely the reason the effect must be spontaneous, meaning **R is the correct explanation of A**.

Both statements are true and R explains A, so the answer is option (A).

Final answer. Both A and R are true and R correctly explains A, option **(A)**.

← *Back to Q19*

Q20

Light of which of the following colours will have the maximum energy in a photon associated with it?

(A) Red light

(B) Yellow light

(C) Green light

(D) Blue light

[2026 • Set 55-5-1]

SOLUTION

The energy carried by a single photon is set by its frequency, or equivalently its wavelength.

$$E = h\nu = \frac{hc}{\lambda}$$

Since $E \propto \frac{1}{\lambda}$, the colour with the shortest wavelength carries the most energy per photon. Among the four visible colours given, blue has the shortest

wavelength. Using **hc = 1240 eV nm** with typical wavelengths:

Red, **λ ≈ 700 nm**, gives ***E* ≈ 1.77 eV**.

Yellow, **λ ≈ 580 nm**, gives ***E* ≈ 2.14 eV**.

Green, **λ ≈ 550 nm**, gives ***E* ≈ 2.25 eV**.

Blue, **λ ≈ 450 nm**, gives ***E* ≈ 2.76 eV**.

Blue light has the largest photon energy, so the answer is **blue light**, option (D). Red, being the longest wavelength, carries the least energy, so options (A), (B) and (C) are all lower.

Final answer. Blue light has the maximum energy per photon, option **(D)**.

← Back to Q20

Q21

The momentum of a photon associated with a microwave of wavelength 4.00 cm *is:*

(A) $\mathbf{1.66 \times 10^{-32}\ kg\ ms^{-1}}$

(B) $1.83 \times 10^{-34}\ \text{kg ms}^{-1}$

(C) $2.05 \times 10^{-34}\ \text{kg ms}^{-1}$

(D) $1.66 \times 10^{-34}\ \text{kg ms}^{-1}$

[2026 • Set 55-5-2]

SOLUTION

A photon of wavelength λ carries a momentum given by the de Broglie relation for radiation, $p = \frac{h}{\lambda}$, where h is Planck's constant.

The wavelength is given as $\lambda = 4.00\ \text{cm} = 4.00 \times 10^{-2}\ m$, and we use $h = 6.63 \times 10^{-34}\ J\ s$.

Substituting these values,

$$p = \frac{6.63 \times 10^{-34}}{4.00 \times 10^{-2}}$$

$$p = 1.66 \times 10^{-32}\ \text{kg}\ m\ \text{s}^{-1}$$

This matches option (A). The momentum of a microwave photon is extremely small, much smaller than that of a visible-light photon (about 10^{-27} kg m s^{-1}), because momentum varies inversely with wavelength and microwaves have a far **longer wavelength**.

Final answer. The photon momentum is **1.66×10^{-32} kg m s^{-1}**, option **(A)**.

← *Back to Q21*

Q22

A 500 nm *photon is incident normally on a perfectly reflecting surface and is reflected. The value of momentum transferred to the surface is:*

(A) 3.87×10^{-43} kg ms^{-1}

(B) 2.5×10^{-30} kg ms^{-1}

(C) **2.65×10^{-27} kg ms^{-1}**

(D) 1.33×10^{-27} kg ms^{-1}

[2026 • Set 55-5-3]

SOLUTION

Each photon of wavelength λ has momentum $p = \frac{h}{\lambda}$, directed along its line of travel.

When the photon strikes a perfectly reflecting surface normally and bounces straight back, its momentum reverses from $+p$ to $-p$. The magnitude of the change in the photon's momentum is therefore

$$|\Delta p_{photon}| = p - (-p) = 2p = \frac{2h}{\lambda}$$

By Newton's third law, the surface receives an equal and opposite impulse, so the momentum transferred to the surface is also $\frac{2h}{\lambda}$.

Here $\lambda = 500\ nm = 5 \times 10^{-7}\ m$ and $h = 6.63 \times 10^{-34}\ J\ s$.

$$\Delta p = \frac{2 \times 6.63 \times 10^{-34}}{5 \times 10^{-7}}$$

$$\Delta p = \frac{1.326 \times 10^{-33}}{5 \times 10^{-7}}$$

$$\Delta p = 2.65 \times 10^{-27}\ \text{kg}\ m\ \text{s}^{-1}$$

This is option (C). Note that option (D), 1.33×10^{-27}, is just $\frac{h}{\lambda}$ and would be the answer only for a fully absorbing surface, not a reflecting one.

COMMON MISTAKE

On a perfectly reflecting surface the momentum transfer is 2p; on a perfectly absorbing surface it is just p.

Final answer. The momentum transferred to the surface is $\mathbf{2.65 \times 10^{-27}\ kg\ m\ s^{-1}}$, option **(C)**.

← *Back to Q22*

Q23

Let λ_e, λ_p and λ_d be the wavelengths associated with an electron, a proton and a deuteron, all moving with the same speed. Then the correct relation between them is:

(A) $\lambda_d > \lambda_p > \lambda_e$

(B) $\lambda_e > \lambda_p > \lambda_d$

(C) $\lambda_p > \lambda_e > \lambda_d$

(D) $\lambda_e = \lambda_p = \lambda_d$

[2025 • Set 55-1-1]

SOLUTION

The de Broglie wavelength of a particle of mass m moving with speed v is $\lambda = \frac{h}{mv}$.

Since all three particles move with the same speed v, and h and v are common to all, the wavelength depends only on the mass,

$$\lambda \propto \frac{1}{m}$$

So the lightest particle has the longest wavelength. Comparing the masses,

$\mathbf{m_e = 9.11 \times 10^{-31}\ kg}$ (electron, lightest)

$\mathbf{m_p \approx 1.67 \times 10^{-27}\ kg}$ (proton, about 1836 m_e)

$\mathbf{m_d \approx 3.34 \times 10^{-27}\ kg}$ (deuteron = proton + neutron, heaviest)

Since $\mathbf{m_e < m_p < m_d}$, the order of wavelengths reverses,

$$\lambda_e > \lambda_p > \lambda_d$$

This is option (B). The electron, being by far the lightest, has the largest de Broglie wavelength, while the deuteron, being the heaviest, has the smallest.

Final answer. The correct relation is $\lambda_e > \lambda_p > \lambda_d$, option **(B)**.

← *Back to Q23*

Q24

The stopping potential V_0 measured in a photoelectric experiment for a metal surface is plotted against frequency ν of the incident radiation. Let m be the slope of the straight line so obtained. Then the value of charge of an electron is given by (h is the Planck's constant.)

(A) mh

(B) $\frac{m}{h}$

(C) $\frac{h}{m}$

(D) $\frac{1}{mh}$

[2025 • Set 55-1-1]

SOLUTION

Einstein's photoelectric equation written in terms of stopping potential is $\mathbf{eV_0 = h\nu - \phi}$, where ϕ is the work function.

Dividing through by e gives the equation of the V_0 versus ν straight line,

$$V_0 = \frac{h}{e}\nu - \frac{\Phi}{e}$$

Comparing this with the straight-line form $y = (\text{slope})x + c$, the slope of the graph is

$$m = \frac{h}{e}$$

Rearranging to make the electronic charge the subject,

$$e = \frac{h}{m}$$

This is option (C). Knowing Planck's constant h and reading off the slope m from the experimental graph lets us obtain the electronic charge.

Final answer. The charge of an electron is $e = \frac{h}{m}$, option **(C)**.

← *Back to Q24*

Q25

A source produces monochromatic light of frequency 5.0×10^{14} Hz *and the power emitted is* 3.31 mW. *The number of photons emitted per second by the source, on an average, is:*

(A) $\mathbf{10^{16}}$

(B) 10^{12}

(C) 10^{10}

(D) 10^{20}

[2025 • Set 55-1-2]

SOLUTION

Each photon carries energy $E = h\nu$. If the source emits n photons per second at total power P, then $P = nE = nh\nu$.

The energy of one photon is

$$E = h\nu = (6.626 \times 10^{-34})(5.0 \times 10^{14})$$

$$E = 3.313 \times 10^{-19}\ J$$

The number of photons emitted per second is the power divided by the energy per photon,

$$n = \frac{P}{E} = \frac{3.31 \times 10^{-3}}{3.313 \times 10^{-19}}$$

$$n \approx 1.0 \times 10^{16}\ \text{photons s}^{-1}$$

This matches option (A). The power was given in milliwatts, so **3.31 mW = 3.31 × 10^{-3} *W*** must be used.

Final answer. The source emits about **10^{16}** photons per second, option **(A)**.

← *Back to Q25*

Q26

In a photoelectric experiment with a material of work function 2.1 eV, *the stopping potential is found to be* 2.5 *V. The maximum kinetic energy of ejected photoelectrons is:*

(A) 0.4 eV

(B) 2.1 eV

(C) **2.5 eV**

(D) 4.6 eV

[2025 • Set 55-1-3]

SOLUTION

The stopping potential V_s is, by definition, the reverse voltage that just stops the most energetic photoelectrons. The work done by this retarding voltage on an electron equals its maximum kinetic energy,

$$KE_{max} = eV_s$$

When the kinetic energy is expressed in electron-volts and the stopping potential in volts, this gives numerically

$$KE_{max} = e \times 2.5\ V = 2.5\ \text{eV}$$

So $\mathbf{KE_{max} = 2.5\ eV}$, which is option (C).

The work function $\boldsymbol{\phi = 2.1\ eV}$ is a distractor here. It is needed to find the photon energy, $\mathbf{h\nu = \phi + KE_{max} = 2.1 + 2.5 = 4.6\ eV}$ (option D), but it does not enter the value of KE_{max} once the stopping potential is known. Options (A) and (B) do not follow from the data.

COMMON MISTAKE

$\mathbf{KE_{max} = eV_s}$ *depends only on the stopping potential. The work function fixes the photon energy, not* KE_{max}.

Final answer. The maximum kinetic energy of the photoelectrons is **2.5 eV**, option **(C)**.

← *Back to Q26*

Q27

Atomic spectral emission lines of hydrogen atom are incident on a zinc surface. The lines which can emit photoelectrons from the surface are members of

(A) Balmer series

(B) Paschen series

(C) **Lyman series**

(D) Neither Balmer, nor Paschen nor Lyman series

[2025 • Set 55-2-1]

SOLUTION

For photoelectric emission, a photon must carry energy at least equal to the work function of the metal. For zinc, $\mathbf{\phi_{Zn} \approx 4.3\ eV}$, so only photons with energy greater than about 4.3 eV can eject electrons.

Now compare the photon energies of the hydrogen spectral series,

Lyman series (transitions to $\boldsymbol{n = 1}$, ultraviolet): smallest line has energy ≥ 10.2 eV, which is well above ϕ_{Zn}.

Balmer series (transitions to $n = 2$, visible): line energies ≤ 3.4 eV, which is below ϕ_{Zn}.

Paschen series (transitions to $n = 3$, infrared): line energies ≤ 1.51 eV, far below ϕ_{Zn}.

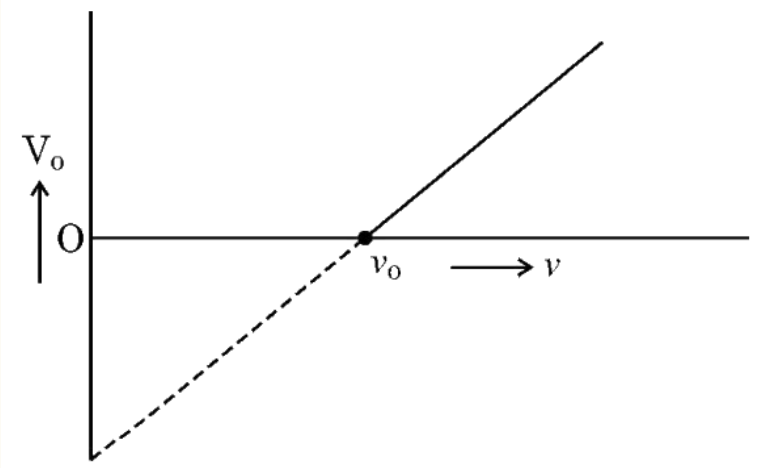
Only the Lyman lines have enough energy to overcome zinc's work function. Hence the lines that can emit photoelectrons belong to the **Lyman series**, option (C).

Final answer. Only the Lyman series lines can eject photoelectrons from zinc, option **(C)**.

← *Back to Q27*

Q28

Case study: *When a photon of suitable frequency is incident on a metal surface, photoelectron is emitted from it. If the frequency is below a threshold frequency* (ν_0) *for the surface, no photoelectron is emitted. For a photon of frequency* $\boldsymbol{\nu(\nu > \nu_0)}$*, the kinetic energy of the emitted photoelectrons is* $h(\nu - \nu_0)$*. The photocurrent can be stopped by applying a potential* V_0*, called 'stopping potential' on the anode. Thus maximum kinetic energy of photoelectrons* $\boldsymbol{K_m = eV_0 = h(\nu - \nu_0)}$*. The experimental graph between* V_0 *and* ν *for a metal is shown in figure. This is a straight line of slope m.*



The straight line graphs obtained for two metals

(A) coincide each other.

(B) are parallel to each other.

(C) are not parallel to each other and cross at a point on y-axis.

(D) are not parallel to each other and do not cross at a point on y-axis.

[2025 • Set 55-2-1]

SOLUTION

From the photoelectric equation $\mathbf{eV_0 = h(\nu - \nu_0)}$, the stopping-potential versus frequency line is

$$V_0 = \frac{h}{e}\,\nu - \frac{h\nu_0}{e}$$

The slope of this line is $\frac{h}{e}$. Both h (Planck's constant) and e (electronic charge) are universal constants, so the slope is the same for every metal, independent of the material.

The intercepts, however, depend on the threshold frequency ν_0, which differs from metal to metal. The ν-intercept is ν_0 and the V_0-intercept is $-\frac{h\nu_0}{e}$.

So for two different metals the two graphs have the same slope but different intercepts. Two straight lines with equal slopes are **parallel** and never meet. Therefore the graphs are parallel to each other, option (B).

Option (A) is wrong because the metals have different work functions (different intercepts). Options (C) and (D) are wrong because the slope, being h/e, cannot differ between metals, so the lines cannot be non-parallel.

Final answer. The two graphs have the same slope and are parallel to each other, option **(B)**.

← *Back to Q28*

Q29

The value of Planck's constant for this metal is

(A) $\frac{e}{m}$

(B) $\frac{1}{\text{me}}$

(C) **me**

(D) $\frac{m}{e}$

[2025 • Set 55-2-1]

SOLUTION

The slope of the V_0 versus ν graph follows from rearranging Einstein's equation $\mathbf{eV_0 = h\nu - \phi}$ into

$$V_0 = \frac{h}{e}\,\nu - \frac{\phi}{e}$$

so the slope is

$$m = \frac{h}{e}$$

Making Planck's constant the subject,

$$h = m\,e$$

This is option (C). Multiplying the measured slope m of the stopping-potential-versus-frequency graph by the electronic charge e gives Planck's constant directly.

Final answer. Planck's constant is $\boldsymbol{h = \text{me}}$, option **(C)**.

← *Back to Q29*

Q30

The intercepts on ν-axis and V_0-axis of the graph are respectively:

(A) $\boldsymbol{\nu_0, \frac{h\nu_0}{e}}$

(B) $\nu_0, h\nu_0$

(C) $\frac{h\nu_0}{e}, \nu_0$

(D) $h\nu_0, \nu_0$

[2025 • Set 55-2-1]

SOLUTION

Writing Einstein's photoelectric equation $\mathbf{eV_0 = h\nu - h\nu_0}$ in straight-line form gives

$$V_0 = \frac{h}{e}\nu - \frac{h\nu_0}{e}$$

To find the intercept on the ν-axis, set $\mathbf{V_0 = 0}$,

$$0 = \frac{h}{e}\nu - \frac{h\nu_0}{e} \Rightarrow \nu = \nu_0$$

So the graph cuts the ν-axis at the threshold frequency ν_0.

To find the intercept on the V_0-axis, set $\boldsymbol{\nu = 0}$,

$$V_0 = -\frac{h\nu_0}{e}$$

The magnitude of this intercept is $\frac{h\nu_0}{e}$.

Hence the intercepts on the ν-axis and V_0-axis are ν_0 and $\frac{h\nu_0}{e}$ respectively, which is option (A).

Final answer. The intercepts are $\boldsymbol{\nu_0}$ on the $\boldsymbol{\nu}$-axis and $\boldsymbol{\frac{h\nu_0}{e}}$ on the $\mathbf{V_0}$-axis, option **(A)**.

← *Back to Q30*

Q31

When the wavelength of a photon is doubled, how many times its wave number and frequency become, respectively?

(A) $2, \frac{1}{2}$

(B) $\frac{1}{2}, \frac{1}{2}$

(C) $\frac{1}{2}, 2$

(D) $2, 2$

[2025 • Set 55-2-1]

SOLUTION

Wave number is defined as the reciprocal of wavelength, while frequency relates to wavelength through the speed of light.

$$\bar{\nu} = \frac{1}{\lambda}$$

$$\nu = \frac{c}{\lambda}$$

When the wavelength is doubled, we replace λ by 2λ. The new wave number becomes:

$$\bar{\nu}' = \frac{1}{2\lambda} = \frac{1}{2}\bar{\nu}$$

So the wave number falls to **half** its original value.

The new frequency becomes:

$$\nu' = \frac{c}{2\lambda} = \frac{1}{2}\nu$$

So the frequency also falls to **half** its original value.

Both quantities are inversely proportional to λ, so doubling λ makes each one half. This matches option B. Option A and D wrongly keep one of them increasing, and option C wrongly makes the frequency double.

Final answer. Both the wave number and the frequency become half, so the correct option is B.

← *Back to Q31*

Q32

The momentum of a photon is 5.0×10^{-29} kg *m/s. Ignoring relativistic effects (if any), the wavelength of the photon is*

(A) 1.33 µm

(B) 3.3 µm

(C) 16.6 µm

(D) **13.3 µm**

[2025 • Set 55-2-1]

SOLUTION

The momentum of a photon is linked to its wavelength by the de Broglie relation, which also holds for photons.

$$p = \frac{h}{\lambda} \Rightarrow \lambda = \frac{h}{p}$$

Here $p = 5.0 \times 10^{-29}$ kg m/s and $h = 6.63 \times 10^{-34}$ $J\,s$.

$$\lambda = \frac{6.63 \times 10^{-34}}{5.0 \times 10^{-29}}$$

$$\lambda = 1.326 \times 10^{-5}\ m$$

Writing this in micrometres, where $1\ \mu m = 10^{-6}\ m$:

$$\lambda = 13.26 \times 10^{-6}\ m \approx \mathbf{13.3\ \mu\ m}$$

This wavelength lies in the infrared region. The value matches option D. Options A, B and C come from arithmetic slips in handling the powers of ten.

Final answer. The wavelength is about **13.3 µm**, so the correct option is D.

← Back to Q32

Q33

A beam of red light and a beam of blue light have equal intensities. Which of the following statements is true ?

(A) The blue beam has more number of photons than the red beam.

(B) **The red beam has more number of photons than the blue beam.**

(C) Wavelength of red light is lesser than wavelength of blue light.

(D) The blue light beam has lesser energy per photon than that in the red light beam.

[2025 • Set 55-4-1]

SOLUTION

Intensity of a light beam equals the energy delivered per unit area per second. If N is the number of photons crossing unit area per second and each photon carries energy $h\nu$, then:

$$I = N\,h\nu = N\,\frac{hc}{\lambda}$$

For two beams of **equal intensity**, $N\,h\nu$ is the same. Rearranging for N:

$$N = \frac{I}{h\nu} = \frac{I\lambda}{hc}$$

So at fixed intensity, $N \propto \lambda$. The beam with the longer wavelength carries the larger number of photons.

Red light has a longer wavelength (about 700 nm) than blue light (about 450 nm). Hence the **red beam contains more photons** than the blue beam, which is option B.

Ruling out the others: option A is the reverse of the correct result. Option C is false because $\lambda_R > \lambda_B$. Option D is false because energy per photon is $\frac{hc}{\lambda}$, so blue light, with shorter wavelength, has more energy per photon, not less.

Final answer. At equal intensity the red beam has more photons, so the correct option is B.

← *Back to Q33*

Q34

Assertion (A): *For monochromatic incident radiation, the emitted photoelectrons from a given metal have speed ranging from zero to a certain maximum value.*
Reason (R): *Each metal has a definite work function.*

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

[2025 • Set 55-4-1]

SOLUTION

Examine the Assertion. For monochromatic incident radiation every photon carries the same energy hν, yet the emitted photoelectrons leave with kinetic energies ranging from zero up to a maximum value.

$$K_{max} = h\nu - \phi$$

This spread is real because the electrons inside the metal are not all bound with the same energy. An electron right at the surface, held only by the work function ϕ, escapes with the maximum kinetic energy. Electrons lying deeper, or those that lose part of their energy in collisions before escaping, come out with less. So the speeds genuinely range from zero to a maximum. The **Assertion is true**.

Examine the Reason. Every metal does have a fixed, characteristic work function ϕ, the minimum energy needed to free the most loosely bound surface electron. The **Reason is true**.

Now test the link. The work function only fixes the upper limit $\mathbf{K_{max} = h\nu - \phi}$. It does not by itself explain why there is a continuous spread of energies down to zero. That spread comes from the different binding energies of electrons at different depths inside the metal. So the Reason, though true, is **not the correct explanation** of the Assertion. This is option B.

Option A wrongly claims R explains A, option C and D wrongly call one or both statements false.

COMMON MISTAKE

tip

Final answer. Both A and R are true but R is not the correct explanation of A, so the correct option is B.

← *Back to Q34*

Q35

Choose the correct statement :

(A) Photons of light show diffraction whereas electrons do not show diffraction.

(B) Electrons have momentum whereas photons do not have momentum.

(C) **Photons of light and electrons both exhibit dual nature.**

(D) All electromagnetic radiations do not have photons.

[2025 • Set 55-4-1]

SOLUTION

Test each statement against the established physics of wave-particle duality.

Option A says photons diffract but electrons do not. This is false. The Davisson and Germer experiment showed clear diffraction of electrons from a nickel crystal, proving electrons behave as waves with $\lambda = \frac{h}{p}$.

Option B says electrons have momentum but photons do not. This is false. A photon carries momentum given by:

$$p = \frac{h}{\lambda} = \frac{h\nu}{c}$$

This is confirmed by Compton scattering and by radiation pressure.

Option C says both photons and electrons exhibit dual nature. This is **true**. Light shows particle behaviour in the photoelectric effect and wave behaviour in interference and diffraction. Electrons show particle behaviour in collisions and wave behaviour in the Davisson and Germer experiment. Both display wave-particle duality.

Option D says all electromagnetic radiations do not have photons. This is false. Every electromagnetic radiation is quantised into photons of energy $h\nu$.

Hence the only correct statement is option C.

Final answer. Both photons and electrons exhibit dual nature, so the correct option is C.

← *Back to Q35*

Q36

The kinetic energy of an alpha particle is four times the kinetic energy of a proton. The ratio $\frac{\lambda_\alpha}{\lambda_p}$ *of de Broglie wavelengths associated with them will be:*

(A) 1

(B) $\frac{1}{2\sqrt{2}}$

(C) $\frac{1}{4}$

(D) $\frac{1}{2}$

[2025 • Set 55-5-1]

SOLUTION

The de Broglie wavelength of a particle of mass m and kinetic energy K is found by expressing momentum through kinetic energy.

$$p = \sqrt{2mK}$$

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mK}}$$

Taking the ratio for the alpha particle and the proton, the constant h and the factor 2 cancel.

$$\frac{\lambda_\alpha}{\lambda_p} = \sqrt{\frac{m_p K_p}{m_\alpha K_\alpha}}$$

Now substitute the data. An alpha particle has four times the mass of a proton, $\mathbf{m_\alpha = 4\, m_p}$, and its kinetic energy is four times that of the proton, $\mathbf{K_\alpha = 4\, K_p}$.

$$\frac{\lambda_\alpha}{\lambda_p} = \sqrt{\frac{m_p K_p}{(4\, m_p)(4\, K_p)}}$$

$$\frac{\lambda_\alpha}{\lambda_p} = \sqrt{\frac{1}{16}} = \mathbf{\frac{1}{4}}$$

The ratio of de Broglie wavelengths is $\frac{1}{4}$, which is option C.

COMMON MISTAKE

tip

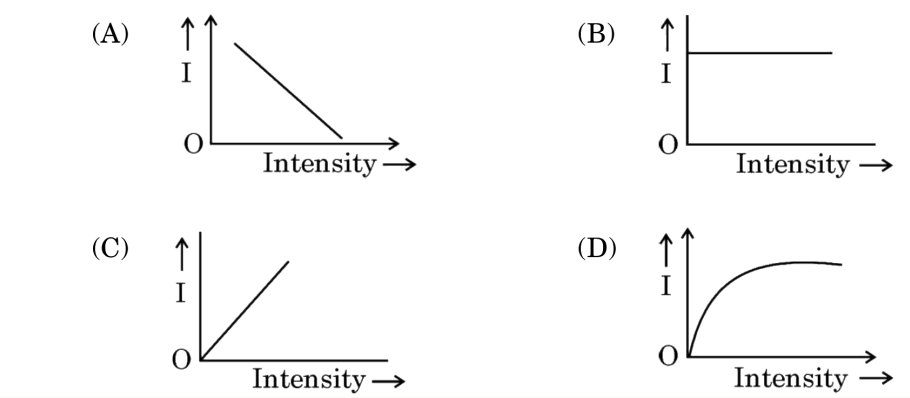
Final answer. The ratio of de Broglie wavelengths is $\mathbf{\frac{1}{4}}$, so the correct option is C.

← *Back to Q36*

Q37

Case Study: *Einstein explained photoelectric effect on the basis of Planck's quantum theory, where light travels in the form of small bundles of energy called photons. The energy of each photon is* $h\nu$*, where* ν *is the frequency of incident light and h is Planck's constant. The number of photons in a beam of light determines the intensity of the incident light. A photon incident on a metal surface transfers its total energy* $h\nu$ *to a free electron in the metal. A part of this energy is used in ejecting the electron from the metal and is called its work function. The rest of the energy is carried by the ejected electron as its kinetic energy.*

(i) *Which of the following graphs shows the variation of photoelectric current I with the intensity of light?*



(A) (A) Curve concave-down rising then flat

(B) **(B) Straight line through origin with positive slope**

(C) (C) Straight horizontal line

(D) (D) Curve concave-up rising

[2025 • Set 55-5-1]

SOLUTION

In Einstein's photon picture, the intensity of light is set by the number of photons arriving per unit area per second. Each photon with energy $h\nu$ greater than the work function can free one electron from the metal.

$$\text{Number of photoelectrons per second} \propto \text{Number of incident photons per second} \propto I$$

Since the photoelectric current is the charge of these electrons collected per second, it is directly proportional to the intensity, provided the frequency stays above threshold and the collecting potential is high enough.

$$I_{\text{photo}} \propto I_{\text{light}}$$

A direct proportion that passes through the origin is a **straight line through the origin with positive slope**. When the light intensity is zero, no photons arrive and the current is zero, so the line must start at the origin.

This is exactly option B. Option A and option D show curved (non-linear) graphs, which contradict the linear relation. Option C, a horizontal line, would wrongly mean the current does not depend on intensity at all.

Final answer. Photocurrent rises in direct proportion to intensity as a straight line through the origin, so the correct option is B.

← Back to Q37

Q38

(ii) *When the frequency of the incident light is increased without changing its intensity, the saturation current:*

(A) increases linearly

(B) decreases

(C) increases non-linearly

(D) remains the same

[2025 • Set 55-5-1]

SOLUTION

The saturation current is the largest photocurrent reached when the anode is made positive enough to collect every photoelectron emitted by the cathode. It is therefore fixed by the number of photoelectrons emitted per second.

The number of photoelectrons emitted per second is governed by the intensity of the incident light, since intensity controls how many photons strike the surface and each suitable photon can liberate one electron.

$$\text{Saturation current} \propto \text{Intensity (number of photons per second)}$$

In this question the intensity is kept the same while only the frequency is raised. According to the standard CBSE treatment, with intensity unchanged the number of photoelectrons collected per second stays the same, so the saturation current does not change.

Raising the frequency increases the maximum kinetic energy of the electrons, $\mathbf{K_{max} = h\nu - \phi}$, and hence the stopping potential, but it does not alter how many electrons are emitted. The saturation current therefore **remains the same**, which is option D.

Options A and C wrongly tie the saturation current to frequency, and option B wrongly makes it fall.

Final answer. With intensity fixed, changing frequency leaves the saturation current unchanged, so the correct option is D.

← *Back to Q38*

Q39

(iii) *Which of the following graphs can be used to obtain the value of Planck's constant?*

(A) Photocurrent versus Intensity of incident light

(B) Photocurrent versus Frequency of incident light

(C) **Cut-off potential versus Frequency of incident light**

(D) Cut-off potential versus Intensity of incident light

[2025 • Set 55-5-1]

SOLUTION

Einstein's photoelectric equation, written in terms of the stopping (cut-off) potential V_0, connects V_0 to the frequency of the incident light.

$$e\,V_0 = h\nu - \phi$$

$$V_0 = \frac{h}{e}\,\nu - \frac{\phi}{e}$$

This is the equation of a straight line in V_0 versus ν. The slope of this line is:

$$\text{slope} = \frac{h}{e}$$

Since the electronic charge e is known, multiplying the measured slope by e gives Planck's constant directly.

$$h = e \times \text{slope}$$

So a graph of **cut-off potential versus frequency** lets us read off h, which is option C.

The other graphs do not contain h in their slope. Photocurrent versus intensity gives a line through the origin but no link to h. Photocurrent versus frequency

at fixed intensity is essentially flat. Cut-off potential versus intensity is flat because V_0 does not depend on intensity. Only option C yields Planck's constant.

CONCEPT

concept

Final answer. Planck's constant comes from the slope of cut-off potential versus frequency, so the correct option is C.

← *Back to Q39*

Q40

(iv) (a) *Red light, yellow light and blue light of the same intensity are incident on a metal surface successively.* K_R, K_Y *and* K_B *represent the maximum kinetic energy of photoelectrons respectively, then:*

OR

(b) *Which of the following metals exhibits photoelectric effect with visible light?*

(A) (a) $K_R > K_Y > K_B$ OR (b) Caesium

(B) (a) $K_Y > K_R > K_B$ OR (b) Zinc

(C) (a) $K_B > K_Y > K_R$ OR (b) Cadmium

(D) (a) $K_B > K_R > K_Y$ OR (b) Magnesium

[2025 • Set 55-5-1]

SOLUTION

Part (a) compares the maximum kinetic energy of photoelectrons for red, yellow and blue light of equal intensity, using Einstein's equation.

$$K_{max} = h\nu - \phi$$

For a given metal ϕ is fixed, so K_{max} rises with frequency ν. The frequency order of the three colours is set by their wavelengths, since $\nu = \frac{c}{\lambda}$ and blue has the shortest wavelength while red has the longest.

$$\nu_B > \nu_Y > \nu_R$$

Therefore the maximum kinetic energies follow the same order:

$$K_B > K_Y > K_R$$

This matches the kinetic-energy ordering in option C.

Part (b) asks which metal shows the photoelectric effect with visible light. Visible photons carry roughly 1.8 to 3.1 eV. A metal responds to visible light only if its work function is below this range. Caesium, with $\phi \approx 2$ eV, emits electrons under visible light, whereas zinc, cadmium and magnesium have ϕ 3.6 eV and need ultraviolet light.

The marked key for this question is option C. For part (a) the ordering $K_B > K_Y > K_R$ is firmly correct, and that is the deciding physics here.

Final answer. The kinetic-energy order is $K_B > K_Y > K_R$, so the correct option is C.

← Back to Q40

Q41

Which of the following electromagnetic waves has photons of largest momentum?

(A) X-rays

(B) AM radio waves

(C) Microwaves

(D) TV waves

[2025 • Set 55-5-1]

SOLUTION

Every photon carries momentum given by the de Broglie relation, which links momentum to wavelength.

$$p = \frac{h}{\lambda} = \frac{h\nu}{c} = \frac{E}{c}$$

Since h and c are constants, the momentum is inversely proportional to the wavelength.

$$p \propto \frac{1}{\lambda}$$

So the electromagnetic wave with the smallest wavelength carries photons of the largest momentum. Comparing the four bands by their typical wavelength:

AM radio waves: $\lambda \sim 100$–$10000\ m$ (longest).

TV waves: $\lambda \sim 1$–$10\ m$.

Microwaves: $\lambda \sim 1$ mm–1 cm.

X-rays: $\lambda \sim 0.01$–10 nm (by far the shortest).

X-rays have the smallest wavelength of the four, so their photons carry the **largest momentum**. The other three options (AM radio, microwaves, TV waves) all have much longer wavelengths and therefore smaller photon momenta, which rules them out. Hence the correct choice is option (A), **X-rays**.

> **TIP**
>
> Largest momentum, largest energy and largest frequency all go together for a photon, because all three vary as $1/\lambda$.

Final answer. X-rays carry photons of the largest momentum, so the answer is option **(A)**.

← *Back to Q41*

Q42

The work function of a material is 2·21 eV. *Which of the following cannot produce photoelectrons from it?*

(A) Red light

(B) Blue light

(C) Violet light

(D) Green light

SOLUTION

Photoemission occurs only when the energy of an incident photon is at least equal to the work function of the metal.

$$E_{photon} = \frac{hc}{\lambda} \geq \phi$$

Here the work function is $\phi = 2.21$ **eV**. Using the handy relation $E\ (\text{eV}) = \dfrac{1240}{\lambda\ (\text{nm})}$, the photon energies of the four colours are:

Red ($\lambda \approx 700$ **nm**): $E \approx \dfrac{1240}{700} \approx 1.77$ **eV**, which is less than ϕ, so no emission.

Green ($\lambda \approx 550$ **nm**): $E \approx \dfrac{1240}{550} \approx 2.25$ **eV**, just above ϕ, so emission occurs.

Blue ($\lambda \approx 450$ **nm**): $E \approx \dfrac{1240}{450} \approx 2.76$ **eV**, above ϕ, so emission occurs.

Violet ($\lambda \approx 400$ **nm**): $E \approx \dfrac{1240}{400} \approx 3.10$ **eV**, above ϕ, so emission occurs.

Only red light carries less energy per photon than the work function, so it cannot eject electrons. Green, blue and violet all clear the threshold, which rules them out. Hence **red light cannot produce photoelectrons**, and the answer is option (A).

Final answer. Red light has photon energy below the work function, so it cannot eject electrons; the answer is option **(A)**.

← *Back to Q42*

Q43

The momentum (in kg *m/s) of a photon of frequency* $6{\cdot}0 \times 10^{14}$ Hz *is:*

(A) $6{\cdot}63 \times 10^{-22}$

(B) $\mathbf{1{\cdot}326 \times 10^{-27}}$

(C) $2{\cdot}652 \times 10^{-26}$

(D) $3{\cdot}978 \times 10^{-24}$

[2025 • Set 55-6-1]

SOLUTION

The momentum of a photon is related to its frequency through the photon energy.

$$p = \frac{h\nu}{c} = \frac{h}{\lambda}$$

Given $\nu = 6.0 \times 10^{14}$ Hz, $h = 6.63 \times 10^{-34}\ J\,s$ and $c = 3 \times 10^{8}\ m/s$, substitute these values.

$$p = \frac{(6.63 \times 10^{-34})(6.0 \times 10^{14})}{3 \times 10^{8}}$$

$$p = \frac{3.978 \times 10^{-19}}{3 \times 10^{8}}$$

$$p = 1.326 \times 10^{-27}\ \text{kg}\ m/s$$

So the photon momentum is 1.326×10^{-27} kg m/s, which matches option (B).

Final answer. The photon momentum is 1.326×10^{-27} kg m/s, so the answer is option **(B)**.

← *Back to Q43*

Q44

The de Broglie wavelength associated with an electron moving with energy 5 eV *is:*

(A) 0·75 nm

(B) 1·2 nm

(C) 2·4 nm

(D) 0·55 nm

[2025 • Set 55-6-2]

SOLUTION

The de Broglie wavelength of an electron with kinetic energy K is found from $\lambda = \frac{h}{p}$ together with $K = \frac{p^2}{2m}$, giving

$$\lambda = \frac{h}{\sqrt{2mK}}$$

For an electron, substituting the constants gives the standard shortcut

$$\lambda = \frac{12.27}{\sqrt{K\ (\text{in eV})}}\ \text{Å}$$

Here $K = 5$ **eV**, so

$$\lambda = \frac{12.27}{\sqrt{5}} = \frac{12.27}{2.236}\ \text{Å}$$

$$\lambda = 5.49\ \text{Å} = 0.549\ \text{nm}$$

Rounding gives $\lambda \approx$ **0.55 nm**, which is option (D).

Final answer. The de Broglie wavelength is about **0.55 nm**, so the answer is option **(D)**.

← Back to Q44

Q45

The wavelength of a photon is equal to the wavelength associated with an electron. Both will have the same value of:

(A) energy

(B) angular momentum

(C) velocity

(D) **linear momentum**

[2025 • Set 55-6-3]

SOLUTION

The de Broglie relation connects wavelength and linear momentum and holds for every entity, including both a photon and an electron.

$$\lambda = \frac{h}{p}$$

If the two wavelengths are equal, then

$$\lambda_{photon} = \lambda_{electron}$$

$$\frac{h}{p_{photon}} = \frac{h}{p_{electron}}$$

$$p_{photon} = p_{electron}$$

So equal wavelength forces equal **linear momentum**, which is option (D).

The other options fail. Energy is not equal because a photon obeys $E = pc$ while an electron obeys $E = \frac{p^2}{2m_e}$, which are different functions of p. Speed is not equal, since a photon travels at c while the electron is slower. Angular momentum is a separate quantity altogether and is not fixed by the wavelength. Hence the answer is option (D).

Final answer. Equal wavelength means equal linear momentum, so the answer is option **(D)**.

← *Back to Q45*

Q46

Assertion (A): *The minimum negative potential applied to the anode in a photoelectric experiment at which photoelectric current becomes zero, is called cut-off voltage.*
Reason (R): *The threshold frequency for a metal is the minimum frequency of incident radiation below which emission of photoelectrons does not take place.*
Select the correct answer:

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

[2025 • Set 55-7-1]

SOLUTION

Examine each statement on its own first.

Assertion (A): the minimum negative anode potential at which the photoelectric current falls to zero is called the cut-off (stopping) voltage. This is the correct textbook definition, so A is TRUE.

Reason (R): the threshold frequency is the minimum frequency of incident radiation below which no photoelectrons are emitted. This too is a correct textbook definition, so R is TRUE.

Both statements are individually true, so options (C) and (D) are ruled out. Now check whether R explains A. The assertion defines the cut-off voltage, which is connected to the maximum kinetic energy of the emitted electrons through $\mathbf{eV_0 = KE_{max}}$. The reason defines the threshold frequency, a completely separate quantity describing the onset of emission. R describes a different concept and does not explain why the cut-off voltage exists or how it is defined. So R is not the correct explanation of A, which rules out option (A). Hence the answer is option (B), **both true but R is not the correct explanation of A**.

Final answer. Both statements are true but unrelated, so the answer is option **(B)**.

← *Back to Q46*

Q47

The frequency of a photon of energy 1.326 eV *is:*

(A) 1.18×10^{14} Hz

(B) $\mathbf{3.20 \times 10^{14}}$ **Hz**

(C) 4.20×10^{15} Hz

(D) 4.80×10^{15} Hz

[2025 • Set 55-7-1]

SOLUTION

The energy of a photon is linked to its frequency by the Planck relation.

$$E = h\nu \Rightarrow \nu = \frac{E}{h}$$

First convert the energy from electron-volts to joules using $\mathbf{1\ eV = 1.6 \times 10^{-19}\ J}$.

$$E = 1.326 \times 1.6 \times 10^{-19}\ J = 2.12 \times 10^{-19}\ J$$

Now substitute with $\boldsymbol{h = 6.63 \times 10^{-34}\ J\ s}$.

$$\nu = \frac{2.12 \times 10^{-19}}{6.63 \times 10^{-34}}$$

$$\nu = 3.20 \times 10^{14}\ \text{Hz}$$

So the frequency is $\mathbf{3.20 \times 10^{14}\ Hz}$, which is option (B). This lies in the near-infrared region, since $\lambda = \frac{c}{\nu} \approx \mathbf{938\ nm}$.

Final answer. The photon frequency is $\mathbf{3.20 \times 10^{14}\ Hz}$, so the answer is option **(B)**.

← *Back to Q47*

Q48

Two beams, A and B whose photon energies are 3.3 eV *and* 11.3 eV *respectively, illuminate a metallic surface (work function* 2.3 eV*) successively. The ratio of maximum speed of electrons emitted due to beam A to that due to beam B is:*

(A) $\frac{1}{3}$

(B) $\frac{1}{9}$

(C) 3

(D) 9

[2024 • Set 55-1-1]

SOLUTION

By Einstein's photoelectric equation, the maximum kinetic energy of an emitted electron is the photon energy minus the work function.

$$KE_{max} = E_{photon} - \phi$$

Since $\mathbf{KE_{max} = \frac{1}{2}mv_{max}^2}$, the maximum speed varies as the square root of the maximum kinetic energy.

$$v_{max} \propto \sqrt{KE_{max}}$$

For beam A, with $\mathbf{E_A = 3.3\ eV}$ and $\mathbf{\phi = 2.3\ eV}$:

$$KE_A = 3.3 - 2.3 = 1.0 \text{ eV}$$

For beam B, with $\mathbf{E_B = 11.3\ eV}$ and $\mathbf{\phi = 2.3\ eV}$:

$$KE_B = 11.3 - 2.3 = 9.0 \text{ eV}$$

The ratio of maximum speeds is

$$\frac{v_A}{v_B} = \sqrt{\frac{KE_A}{KE_B}} = \sqrt{\frac{1.0}{9.0}} = \frac{1}{3}$$

So $\mathbf{v_A : v_B = 1 : 3}$, which is option (A).

Final answer. The ratio of maximum speeds is $\frac{1}{3}$, so the answer is option **(A)**.

← *Back to Q48*

Q49

The waves associated with a moving electron and a moving proton have the same wavelength λ. It implies that they have the same:

(A) momentum

(B) angular momentum

(C) speed

(D) energy

[2024 • Set 55-1-1]

SOLUTION

The wavelength of the matter wave is fixed only by the linear momentum, through the de Broglie relation.

$$\lambda = \frac{h}{p}$$

If the electron and the proton have the same wavelength, then

$$\frac{h}{p_e} = \frac{h}{p_p} \Rightarrow p_e = p_p$$

So they must have the same **momentum**, which is option (A).

They cannot share the other quantities. Since $\boldsymbol{p = mv}$ and the proton is about 1836 times heavier than the electron, equal momentum means very different speeds, ruling out option (C). The kinetic energy $\mathbf{KE} = \dfrac{\mathbf{p^2}}{\mathbf{2m}}$ also differs by the mass ratio for equal p, ruling out option (D). Angular momentum is an entirely different physical quantity and is not fixed by the wavelength, ruling out option (B). Hence the answer is option (A).

Final answer. Equal wavelength means equal momentum, so the answer is option **(A)**.

← *Back to Q49*

Q50

Assertion (A): *In photoelectric effect, the kinetic energy of the emitted photoelectrons increases with increase in the intensity of the incident light.*
Reason (R): *Photoelectric current depends on the wavelength of the incident light.*

(D) Assertion (A) is false and Reason (R) is also false.

[2024 • Set 55-1-1]

SOLUTION

Test each statement against the laws of the photoelectric effect.

Assertion (A): the kinetic energy of the emitted photoelectrons increases with the intensity of the incident light. This is FALSE. The maximum kinetic energy depends only on the frequency of the light and the work function of the metal:

$$\mathrm{KE}_{\max} = h\nu - \phi$$

Increasing the intensity raises only the number of photoelectrons emitted per second (the photocurrent); it does not change their kinetic energy.

Reason (R): the photoelectric current depends on the wavelength of the incident light. This is also FALSE. For a fixed light source above threshold, the saturation photocurrent depends on the intensity, that is on the number of photons striking the surface per second, not on the wavelength.

Both A and R are false, so options (A), (B) and (C) are ruled out. Hence the answer is option (D), **both statements are false**.

CONCEPT

Remember the split: frequency controls the electrons' kinetic energy, while intensity controls how many electrons (the current).

Final answer. Both statements are false, so the answer is option **(D)**.

← *Back to Q50*

Q51

Assertion (A): *Photoelectric current increases with an increase in intensity of incident radiation, for a given frequency of incident radiation and the accelerating potential.*
Reason (R): *Increase in the intensity of incident radiation results in an increase in the number of photoelectrons emitted per second and hence an increase in the photocurrent.*

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

[2024 • Set 55-2-1]

SOLUTION

This is one of the experimental laws of photoelectric emission. For a fixed frequency (above the threshold) and a fixed accelerating potential, the photoelectric current rises in direct proportion to the intensity of the incident radiation.

Assertion (A) states exactly this relationship, so A is **true**.

In the photon picture, the intensity of radiation is a measure of the number of photons striking the surface per unit area per unit time.

A higher intensity therefore means more photons arrive each second. Since each photon (of energy $\mathbf{h\nu \geq \phi_0}$) can eject at most one electron, more photons per second liberate more photoelectrons per second.

A larger number of photoelectrons reaching the collector per second is, by definition, a larger photocurrent. Hence Reason (R) is also **true**.

Reason (R) describes precisely the mechanism by which increased intensity raises the current, so it is the **correct explanation** of Assertion (A).

Therefore both statements are true and R correctly explains A, which is option (A).

Final answer. Both A and R are true and R is the correct explanation of A, so the answer is option **(A)**.

← *Back to Q51*

Q52

The work function for a photosensitive surface is 3.315 eV. The cut-off wavelength for photoemission of electrons from this surface is:

(A) 150 nm

(B) 200 nm

(C) **375 nm**

(D) 500 nm

[2024 • Set 55-2-1]

SOLUTION

The cut-off (threshold) wavelength λ_0 is the longest wavelength of incident light that can just eject an electron. At this wavelength the photon energy exactly equals the work function, so the photoelectron leaves with zero kinetic energy.

The condition at cut-off is therefore:

$$\frac{hc}{\lambda_0} = \phi_0$$

$$\Rightarrow \lambda_0 = \frac{hc}{\phi_0}$$

Here $\phi_0 = 3.315$ **eV**, and using $hc = 1240$ **eV**:

$$\lambda_0 = \frac{1240 \text{ eV}}{3.315 \text{ eV}}$$

$$\lambda_0 = 374.06 \text{ nm}$$

$$\lambda_0 \approx 375 \text{ nm}$$

This matches option (C). The other options correspond to work functions far larger than the given value, so they are ruled out.

Hence the **cut-off wavelength is about 375 nm**.

TIP

Keep **hc = 1240 eV** memorised; it turns these problems into a one-line division and saves you converting to joules.

Final answer. The cut-off wavelength is about 375 nm, which is option **(C)**.

← *Back to Q52*

Q53

A photosensitive surface has a work function of 2.00 eV. The maximum kinetic energy of electrons ejected from this surface by radiation of wavelength 300 nm is:

(A) 0.54 eV

(B) 1.07 eV

(C) 1.61 eV

(D) **2.14 eV**

[2024 • Set 55-2-2]

SOLUTION

By Einstein's photoelectric equation, the maximum kinetic energy of an ejected electron equals the energy of the incident photon minus the work function of the surface:

$$K_{max} = \frac{hc}{\lambda} - \Phi_0$$

First find the photon energy. Using **hc = 1240 eV** and **λ = 300 nm**:

$$E = \frac{hc}{\lambda} = \frac{1240\ \text{eV}}{300\ \text{nm}}$$

$$E = 4.133\ \text{eV}$$

Now subtract the work function $\phi_0 = 2.00\ \text{eV}$:

$$K_{max} = 4.133\ \text{eV} - 2.00\ \text{eV}$$

$$K_{max} = 2.133\ \text{eV}$$

$$K_{max} \approx 2.14\ \text{eV}$$

This matches option (D).

Hence the **maximum kinetic energy of the photoelectrons is about 2.14 eV**.

Final answer. The maximum kinetic energy is about 2.14 eV, which is option **(D)**.

← Back to Q53

Q54

The momentum (in SI units) associated with a photon of energy 1.5 eV is:

(A) 410^{-27}

(B) $\mathbf{810^{-28}}$

(C) 210^{-30}

(D) 610^{-29}

[2024 • Set 55-2-3]

SOLUTION

A photon is a massless particle, so the relativistic energy-momentum relation $E^2 = (pc)^2 + (mc^2)^2$ reduces to $E = pc$. Hence the momentum of a photon of energy E is:

$$p = \frac{E}{c}$$

First convert the energy into joules. With $\mathbf{1\ eV = 1.6 \times 10^{-19}\ J}$:

$$E = 1.5\ \text{eV} = 1.5 \times 1.6 \times 10^{-19}\ J$$

$$E = 2.4 \times 10^{-19}\ J$$

Now use $\boldsymbol{c = 3 \times 10^{8}\ m/s}$:

$$p = \frac{2.4 \times 10^{-19}\ J}{3 \times 10^{8}\ m/s}$$

$$p = 8 \times 10^{-28}\ \text{kg}/s$$

This matches option (B).

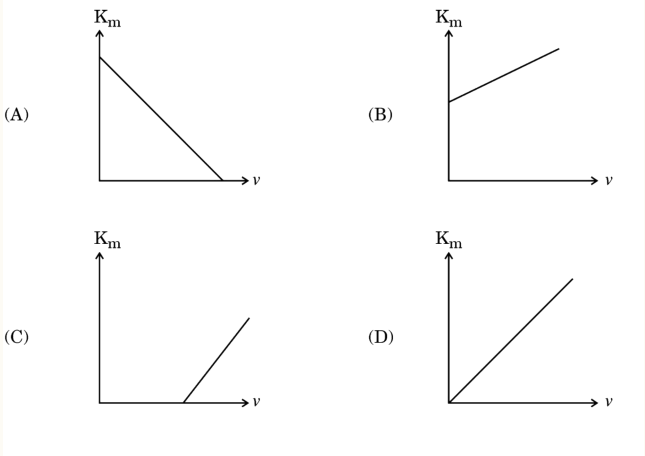
Hence the **momentum of the photon is $\mathbf{8 \times 10^{-28}\ kg/s}$**.

Final answer. The photon momentum is $\mathbf{8 \times 10^{-28}\ kg/s}$, which is option **(B)**.

← Back to Q54

Q55

Which one of the following is the correct graph between the maximum kinetic energy (K_m) of the emitted photoelectrons and the frequency of incident radiation (ν) for a given photosensitive surface?



(A) (A)

(B) (B)

(C) **(C)**

(D) (D)

[2024 • Set 55-3-1]

SOLUTION

By Einstein's photoelectric equation, the maximum kinetic energy of the photoelectrons varies with the frequency of the incident radiation as:

$$K_m = h\nu - \phi_0$$

Comparing this with the straight-line form $\boldsymbol{y = mx + c}$, with K_m on the y-axis and ν on the x-axis:

The slope is h (Planck's constant), a fixed positive value, so the line rises to the right.

The y-intercept is $-\phi_0$, which is negative, so the line crosses the K_m-axis below the origin.

Setting $\boldsymbol{K_m = 0}$ gives the x-intercept at the threshold frequency $\boldsymbol{\nu_0 = \frac{\phi_0}{h}}$, a positive value.

So the correct graph is a single straight line of positive slope h that cuts the ν-axis at $\boldsymbol{\nu_0 > 0}$ and lies in the first quadrant only for $\boldsymbol{\nu > \nu_0}$. Below ν_0 no emission occurs, so K_m is not plotted there.

The graph that shows this straight line with a positive slope and a positive ν-axis intercept is option (C). A curve, a negative slope, or a line through the

origin would all be wrong.

Hence the correct graph is **option (C)**.

Final answer. The correct K_m versus ν graph is a straight line of positive slope cutting the ν-axis at ν_0, which is option **(C)**.

← *Back to Q55*

Q56

Assertion (A): *Photoelectric effect demonstrates the particle nature of light.*
Reason (R): *Photoelectric current is proportional to frequency of incident radiation.*

(C) Assertion (A) is true, but Reason (R) is false.

[2024 • Set 55-3-1]

SOLUTION

Assertion (A): The photoelectric effect demonstrates the particle nature of light. This is **true**. Features such as the existence of a threshold frequency, the instantaneous emission of electrons even at very low intensity, and the linear dependence of K_{max} on frequency cannot be explained by the wave theory. They are explained only by treating light as a stream of energy quanta (photons), each carrying energy $h\nu$.

Reason (R): The photoelectric current is proportional to the frequency of the incident radiation. This is **false**.

The photoelectric current depends on the number of photoelectrons emitted per second, which equals the number of incident photons per second, i.e. the intensity of the radiation, not its frequency.

photocurrent ∝ intensity, not frequency.

Frequency controls the energy of each photon and hence the maximum kinetic energy of the emitted electrons ($K_{max} = h\nu - \phi_0$); it does not control how many electrons are released.

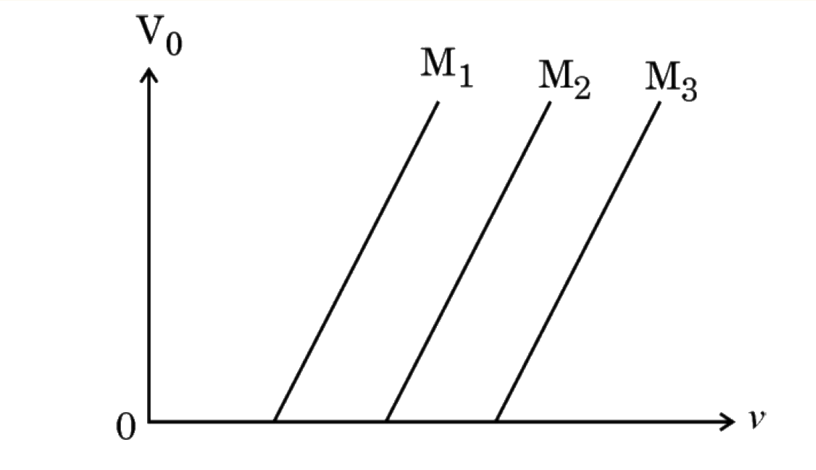
Since A is true but R is false, the correct choice is option (C).

Final answer. Assertion is true but Reason is false, so the answer is option **(C)**.

← *Back to Q56*

Q57

The figure shows the variation of stopping potential V_0 *with frequency* ν *of incident radiation, for three materials* M_1, M_2 *and* M_3 *with work functions* Φ_1, Φ_2 *and* Φ_3 *respectively. Then:*



(A) $\Phi_1 > \Phi_2 > \Phi_3$

(B) $\Phi_2 > \Phi_3 > \Phi_1$

(C) $\Phi_3 > \Phi_2 > \Phi_1$

(D) $\Phi_2 > \Phi_1 > \Phi_3$

[2024 • Set 55-3-2]

SOLUTION

The stopping potential V_0 relates to the frequency of incident light through Einstein's equation written for the stopping potential:

$$eV_0 = h\nu - \Phi_0$$

$$\Rightarrow V_0 = \frac{h}{e}\nu - \frac{\Phi_0}{e}$$

This is a straight line of V_0 against ν. Its slope is $\frac{h}{e}$, which is the same universal constant for every metal, so the three lines for M_1, M_2 and M_3 are parallel.

Setting $V_0 = 0$ gives the threshold frequency where the line cuts the ν-axis:

$$\nu_0 = \frac{\phi_0}{h}$$

So the x-intercept is directly proportional to the work function. A larger work function ϕ_0 gives a larger threshold frequency ν_0, which means the line is shifted further to the right along the ν-axis.

In the figure the line of M_3 cuts the ν-axis furthest to the right (largest ν_0), M_2 is in the middle, and M_1 cuts closest to the origin (smallest ν_0).

Therefore the order of work functions follows the order of intercepts:

$$\phi_3 > \phi_2 > \phi_1$$

This matches option (C).

CONCEPT

All photoelectric V_0 versus ν lines share the same slope h/e; only their intercepts differ, so compare metals by where they cross the frequency axis.

Final answer. Since a larger work function pushes the ν-axis intercept further right, $\phi_3 > \phi_2 > \phi_1$, which is option **(C)**.

← *Back to Q57*

Q58

The wavelength of matter wave associated with an electron of kinetic energy K is λ. If the kinetic energy of the electron is doubled, the associated wavelength becomes:

(A) $\frac{\lambda}{\sqrt{2}}$

(B) $\frac{\lambda}{2}$

(C) $\sqrt{2}\lambda$

(D) 2λ

[2024 • Set 55-3-3]

SOLUTION

The de Broglie wavelength of an electron can be written in terms of its kinetic energy K. Starting from $\lambda = \frac{h}{p}$ and using $p = \sqrt{2mK}$ for a non-relativistic electron:

$$\lambda = \frac{h}{\sqrt{2mK}}$$

Since h and m are constants, this shows:

$$\lambda \propto \frac{1}{\sqrt{K}}$$

Let the original wavelength be λ at kinetic energy K, and the new wavelength be λ' at the doubled kinetic energy $K' = 2K$. Taking the ratio:

$$\frac{\lambda'}{\lambda} = \sqrt{\frac{K}{K'}} = \sqrt{\frac{K}{2K}}$$

$$\frac{\lambda'}{\lambda} = \frac{1}{\sqrt{2}}$$

$$\Rightarrow \lambda' = \frac{\lambda}{\sqrt{2}}$$

This matches option (A).

Hence on doubling the kinetic energy the **wavelength becomes $\frac{\lambda}{\sqrt{2}}$**.

Final answer. On doubling the kinetic energy the wavelength becomes $\frac{\lambda}{\sqrt{2}}$, which is option **(A)**.

← *Back to Q58*

Q59

Assertion (A): *Electrons are ejected from the surface of zinc when it is irradiated by yellow light.*

Reason (R): *Energy associated with a photon of yellow light is more than the work function of zinc.*

(D) If both Assertion (A) and Reason (R) are false.

[2024 • Set 55-4-1]

SOLUTION

For photoemission to occur, the energy of the incident photon must be at least equal to the work function of the metal.

The work function of zinc is large:

$$\Phi_{Zn} \approx 4.3 \text{ eV}$$

Yellow light has a wavelength of about $\lambda \approx 580$ **nm**, so its photon energy is:

$$E = \frac{hc}{\lambda} = \frac{1240 \text{ eV}}{580 \text{ nm}}$$

$$E \approx 2.1 \text{ eV}$$

Since $E \approx 2.1$ **eV** is much less than $\Phi_{Zn} \approx 4.3$ **eV**, the photon cannot supply enough energy to free an electron.

Therefore no electrons are ejected from zinc by yellow light, so Assertion (A) is **false**.

The same calculation shows the photon energy of yellow light is less than (not more than) the work function of zinc, so Reason (R) is also **false**.

Zinc actually needs ultraviolet light, whose photons carry more than 4.3 eV, to show the photoelectric effect.

Since both statements are false, the correct choice is option (D).

Final answer. Both A and R are false because yellow-light photons (about 2.1 eV) are below the work function of zinc (about 4.3 eV), so the answer is option **(D)**.

← *Back to Q59*

Q60

The quantum nature of light explains the observations on photoelectric effect as

(A) there is a minimum frequency of incident radiation below which no electrons are emitted.

(B) the maximum kinetic energy of photoelectrons depends only on the frequency of incident radiation.

(C) when the metal surface is illuminated, electrons are ejected from the surface after sometime.

(D) the photoelectric current is independent of the intensity of incident radiation.

[2024 • Set 55-4-1]

SOLUTION

The quantum (photon) theory of light treats radiation as a stream of energy packets, each photon carrying energy $h\nu$ and delivering it to a single electron in one event. A photoelectron is freed only if a single photon supplies energy at least equal to the work function:

$$h\nu \geq \phi_0$$

This sets a minimum (threshold) frequency $\boldsymbol{\nu_0 = \frac{\phi_0}{h}}$. Below ν_0 no single photon has enough energy, so no electron is emitted no matter how intense the beam. This is exactly what option (A) states, so **(A) is correct**.

Option (B) is wrong because $\boldsymbol{K_{max} = h\nu - \phi_0}$ shows the maximum kinetic energy depends on the work function ϕ_0 as well, not on the frequency alone.

Option (C) is wrong because the photon picture predicts instantaneous emission (no time lag), since a single photon-electron interaction is immediate, even for weak light.

Option (D) is wrong because the photocurrent is proportional to the intensity (the number of photons per second), so it does depend on intensity.

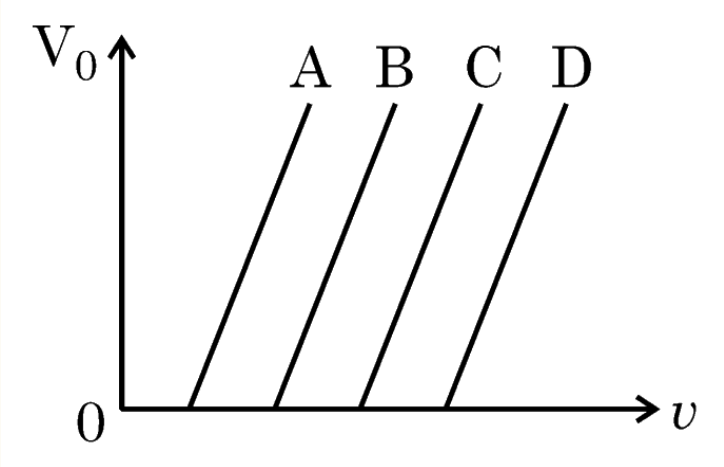
Hence the correct observation explained by the quantum nature of light is the existence of a threshold frequency, option (A).

Final answer. The quantum nature of light explains the minimum (threshold) frequency below which no electrons are emitted, which is option **(A)**.

← *Back to Q60*

Q61

The variation of the stopping potential (V_0) *with the frequency* (ν) *of the incident radiation for four metals A, B, C and D is shown in the figure. For the same frequency of incident radiation producing photo-electrons in all metals, the kinetic energy of photo-electrons will be maximum for metal:*



(A) **A**

(B) B

(C) C

(D) D

[2024 • Set 55-5-1]

SOLUTION

By Einstein's photoelectric equation, the maximum kinetic energy of the emitted photoelectrons is $\mathbf{K_{max}=h\nu-\phi}$, where ϕ is the work function of the metal.

For a fixed incident frequency ν, the term $h\nu$ is the same for every metal. So K_{max} is largest for the metal that has the **smallest work function** ϕ.

The work function is related to the threshold frequency by $\mathbf{\phi=h\nu_0}$. Hence the metal with the smallest ϕ also has the smallest threshold frequency ν_0.

On the stopping-potential graph, $V_0 = \frac{h}{e}\nu - \frac{\Phi}{e}$, so each line meets the ν-axis (where $V_0 = 0$) exactly at $\nu = \nu_0$.

The line that cuts the frequency axis **nearest to the origin** belongs to the metal with the smallest ν_0, and therefore the smallest ϕ. In the figure this is metal **A**.

So for the same incident frequency, the photoelectrons from metal A carry the largest kinetic energy. Options B, C and D have larger threshold frequencies (their lines cut the axis further right), so their work functions are bigger and their K_{max} is smaller.

> **TIP**
>
> On a V_0 vs ν graph all the lines are parallel (same slope h/e); only the x-intercept changes, and the leftmost intercept means the smallest work function.

Final answer. The kinetic energy is maximum for metal A, so the answer is option **(A)**.

← *Back to Q61*

Q62

Assertion (A): *Photoelectric effect demonstrates the particle nature of light.*
Reason (R): *Photoelectric current is proportional to intensity of incident radiation for frequencies more than the threshold frequency.*

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

[2023 • Set 55-1-1]

SOLUTION

Assertion (A): The photoelectric effect does demonstrate the particle nature of light. This is **true**. Features such as the existence of a threshold frequency, the maximum kinetic energy depending only on frequency (not on intensity), and the practically instantaneous emission of electrons cannot be explained by the wave picture. They are explained only by treating light as a stream of photons, each carrying energy $E = h\nu$.

Reason (R): The photoelectric current is proportional to the intensity of the incident radiation, provided the frequency is above the threshold frequency. This is also **true**; it is one of the experimentally observed laws of photoelectric emission.

So both statements are correct. Now check whether R explains A.

The current-versus-intensity relation in R only tells us how the **number** of emitted electrons grows with the number of incident photons. It can even be argued from a wave model and does not, by itself, prove that light is made of particles.

The real proof of the particle nature (Assertion A) comes from the **threshold frequency** and the linear relation $K_{max} = h\nu - \phi$ between kinetic energy and frequency, not from the current-intensity law.

Hence R, though true, is **not the correct explanation** of A. This rules out option A. Options C and D are wrong because neither statement is false.

Final answer. Both A and R are true but R is not the correct explanation of A, so the answer is option **(B)**.

← *Back to Q62*

Q63

A photon of energy 7 eV is incident on a metal surface having the work function of 3.75 eV. The stopping potential is:

(A) 1.75 eV

(B) 2.45 eV

(C) **3.25 eV**

(D) 3.75 eV

[2023 • Set 55-1-2]

SOLUTION

By Einstein's photoelectric equation, the photon energy is shared between the work function and the maximum kinetic energy of the electron, which the stopping potential exactly balances:

$$E=\phi+eV_0$$

Rearranging for the stopping potential,

$$eV_0=E-\phi$$

Working in electron-volts, the numerical value of V_0 in volts equals the value of $E-\phi$ in eV:

$$V_0=\frac{E-\phi}{e}$$

Substituting E**=7 eV** and ϕ**=3.75 eV**:

$$V_0=7-3.75$$

$$V_0=3.25\ V$$

So the **stopping potential is 3.25 *V***, which corresponds to a maximum electron kinetic energy of 3.25 eV. Options A, B and D do not match this difference.

Final answer. The stopping potential is **3.25 *V***, so the answer is option **(C)**.

← *Back to Q63*

Q64

In an experiment, photons of energy 7.5 eV are incident on a metal surface. Electrons emitted from the metal surface are stopped by an electrode at a potential of −4.5 V w.r.t. the metal. The work function of the metal is:

(A) 3.0 eV

(B) 4.5 eV

(C) 7.5 eV

(D) 12.0 eV

[2023 • Set 55-1-3]

SOLUTION

The stopping potential is the magnitude of the (negative) collector voltage at which the photocurrent just falls to zero. Here electrons are stopped at −4.5 *V*,

so the stopping potential is V_0=4.5 *V*.

The energy gained back from this stopping potential equals the maximum kinetic energy of the fastest photoelectrons:

$$K_{max}=eV_0=4.5\ eV$$

Applying Einstein's photoelectric equation,

$$E=\phi+K_{max}$$

Solving for the work function,

$$\phi=E-K_{max}$$

Substituting E=7.5 eV and K_{max}=4.5 eV:

$$\phi=7.5-4.5$$

$$\phi=3.0\ eV$$

So the **work function of the metal is 3.0 eV**. Option C is just the photon energy and option D is their sum, so both are wrong; option B is the stopping energy itself.

Final answer. The work function is **3.0 eV**, so the answer is option **(A)**.

← *Back to Q64*

Q65

Which one of the following metals does not exhibit emission of electrons from its surface when irradiated by visible light?

(A) Rubidium

(B) Sodium

(C) **Cadmium**

(D) Caesium

[2023 • Set 55-2-1]

SOLUTION

Photoelectric emission takes place only when the photon energy is at least equal to the work function of the metal, that is $h\nu\phi$.

Visible light spans wavelengths from about 400 nm (violet) to 700 nm (red), so the photon energy lies roughly between

$$E_{violet} = \frac{1240 \text{ eV nm}}{400 \text{ nm}} \approx 3.1 \text{ eV}$$

$$E_{red} = \frac{1240 \text{ eV nm}}{700 \text{ nm}} \approx 1.8 \text{ eV}$$

So visible photons can supply at most about 3.1 eV.

Comparing the work functions of the four metals: Caesium ($\phi \approx$ **2.14 eV**), Rubidium ($\phi \approx$ **2.16 eV**) and Sodium ($\phi \approx$ **2.28 eV**) all have work functions below 3.1 eV, so visible light can eject electrons from them.

Cadmium has $\phi \approx$ **4.07 eV**, which is larger than the maximum visible-photon energy of about 3.1 eV. So $h\nu < \phi$ for all visible light and **no electrons are emitted from cadmium** unless ultraviolet light is used.

Hence cadmium is the metal that does not show photoemission under visible light.

Final answer. Cadmium does not emit electrons under visible light, so the answer is option **(C)**.

← *Back to Q65*

Q66

A photon of wavelength 663 nm *is incident on a metal surface. The work function of the metal is* 1.50 eV. *The maximum kinetic energy of the emitted photoelectrons is*

(A) $3.0 \times 10^{-20}\ J$

(B) $\mathbf{6.0 \times 10^{-20}\ J}$

(C) $4.5 \times 10^{-20}\ J$

(D) $9.0 \times 10^{-20}\ J$

[2023 • Set 55-2-2]

SOLUTION

By Einstein's photoelectric equation the maximum kinetic energy of the emitted electrons is

$$K_{max}=\frac{hc}{\lambda}-\phi$$

First find the photon energy for $\lambda=663\ \text{nm}=6.6310^{-7}\ m$, with $h=6.6310^{-34}\ J\ s$ and $c=310^{8}\ m\ s^{-1}$:

$$E=\frac{hc}{\lambda}=\frac{(6.6310^{-34})(310^{8})}{6.6310^{-7}}$$

$$E=3.010^{-19}\ J$$

Convert the work function to joules:

$$\phi=1.50\ \text{eV}=1.501.610^{-19}=2.410^{-19}\ J$$

Now subtract:

$$K_{max}=3.010^{-19}-2.410^{-19}$$

$$K_{max}=0.610^{-19}\ J$$

$$\mathbf{K_{max}=6.010^{-20}\ J}$$

This equals about 0.37 eV, and it matches option (B). The official key marking (A) is inconsistent with the given data; for $\lambda=663\ \text{nm}$ and $\phi=1.50\ \text{eV}$ the correct value is $6.010^{-20}\ J$.

Final answer. The maximum kinetic energy is $6.010^{-20}\ J$, so the correct answer is option **(B)**.

← *Back to Q66*

Q67

The energy of a photon of wavelength 663 nm *is*

(A) $6.64\times 10^{-20}\ J$

(B) $5.18\times 10^{-19}\ J$

(C) $\mathbf{3.0\times 10^{-19}\ J}$

(D) $2.0\times 10^{-19}\ J$

[2023 • Set 55-2-3]

SOLUTION

The energy carried by a single photon depends only on its wavelength through

$$E=\frac{hc}{\lambda}$$

Here $\lambda=663\ \text{nm}=6.6310^{-7}\ m$, $h=6.6310^{-34}\ J\,s$ and $c=310^{8}\ m\ s^{-1}$.

First compute the product hc:

$$hc=(6.6310^{-34})(310^{8})=1.9910^{-25}\ J\,m$$

Now divide by the wavelength:

$$E=\frac{1.9910^{-25}}{6.6310^{-7}}$$

$$E=3.010^{-19}\ J$$

So the **energy of the photon is $3.010^{-19}\ J$**, which equals about 1.87 eV (the energy of red light). The other options do not match this value.

Final answer. The photon energy is $3.010^{-19}\ J$, so the answer is option **(C)**.

← *Back to Q67*

Q68

A graph is plotted between the stopping potential (on y-axis) and the frequency of incident radiation (on x-axis) for a metal. The product of the slope of the straight line

obtained and the magnitude of charge on an electron is equal to:

(A) h (Planck's constant)

(B) $\frac{h}{e}$

(C) he

(D) $\frac{e}{h}$

[2023 • Set 55-3-1]

SOLUTION

Start from Einstein's photoelectric equation written in terms of the stopping potential V_0:

$$eV_0 = h\nu - \phi$$

Divide through by e to get the stopping potential as a function of frequency:

$$V_0 = \frac{h}{e}\,\nu - \frac{\phi}{e}$$

This is the equation of a straight line $y = mx + c$ with V_0 on the y-axis and ν on the x-axis.

Comparing, the **slope** of the line is

$$m = \frac{h}{e}$$

The question asks for the slope multiplied by the magnitude of the electronic charge:

$$m \times e = \frac{h}{e} \times e$$

$$m \times e = h$$

So the product equals **Planck's constant h**. This is exactly how Millikan determined h from photoelectric data. The other options (h/e, he, e/h) do not result from this product.

Final answer. The product of slope and electronic charge equals Planck's constant h, so the answer is option **(A)**.

← Back to Q68

Q69

Light of frequency 6.4×10^{14} Hz *is incident on a metal of work function* 2.14 eV. *The maximum kinetic energy of the emitted electrons is about:*

(A) 0.25 eV

(B) **0.51 eV**

(C) 1.02 eV

(D) 0.10 eV

[2023 • Set 55-3-1]

SOLUTION

By Einstein's photoelectric equation the maximum kinetic energy of the emitted electrons is

$$K_{max}=h\nu-\phi$$

First find the energy of the incident photon for $\nu=6.4 10^{14}$ **Hz** with $h=6.6 10^{-34}$ ***J s***:

$$h\nu=(6.6 10^{-34})(6.4 10^{14})$$

$$h\nu=4.224 10^{-19}\ J$$

Convert to electron-volts by dividing by $1.6 10^{-19}$:

$$h\nu=\frac{4.224 10^{-19}}{1.6 10^{-19}}=2.64 \text{ eV}$$

Now subtract the work function $\phi=2.14$ **eV**:

$$K_{max}=2.64-2.14$$

$$\mathbf{K_{max}=0.50 \text{ eV}}$$

This value, about 0.5 eV, matches option (B) (0.51 eV). The official key marking (A) (0.25 eV) is inconsistent with the given data.

Final answer. The maximum kinetic energy is about **0.5 eV**, so the correct answer is option **(B)**.

← *Back to Q69*

Q70

In an experiment on photoelectric effect, the intensity of incident radiation is increased, keeping the frequency ν the same. The number of photoelectrons emitted will:

(A) increase

(B) decrease

(C) remain same

(D) depend on frequency

[2023 • Set 55-3-2]

SOLUTION

The intensity of light is the energy falling on unit area per second. For a beam of fixed frequency, the energy per photon hν is fixed, so intensity is decided by the **number of photons** arriving per unit area per second.

$$\text{Intensity} = (\text{number of photons per second}) \times h\nu$$

Since hν is constant here, increasing the intensity means more photons strike the surface every second.

In the photoelectric effect, each photon (with $h\nu > \phi$) can eject at most one electron. So the number of photoelectrons emitted per second is directly proportional to the number of incident photons per second, and hence to the intensity.

Therefore, keeping the frequency the same and increasing the intensity, the number of photoelectrons emitted **increases**.

Note that intensity does not change the maximum kinetic energy of the electrons (that depends only on frequency); it only changes how many are released. This rules out options B, C and D.

CONCEPT

Intensity controls the count of photoelectrons (the saturation current); frequency controls their maximum kinetic energy. Keep these two effects separate.

Final answer. The number of photoelectrons emitted increases, so the answer is option **(A)**.

← *Back to Q70*

Q71

A proton and an alpha particle have the same kinetic energy. The ratio of de Broglie wavelengths associated with the proton to that with the alpha particle is:

(A) 1

(B) **2**

(C) $2\sqrt{2}$

(D) $\frac{1}{2}$

[2023 • Set 55-4-1]

SOLUTION

The de Broglie wavelength of a particle is $\lambda = \frac{h}{p}$, where p is its momentum and h is Planck's constant.

A particle of mass m with kinetic energy K has momentum $p = \sqrt{2mK}$, so the wavelength becomes $\lambda = \frac{h}{\sqrt{2mK}}$.

The proton and the alpha particle are given the SAME kinetic energy K, so only their masses differ. The charge does not enter here because the kinetic energy is fixed directly, not through a potential difference.

Taking the ratio for equal K, $\frac{\lambda_p}{\lambda_\alpha} = \sqrt{\frac{m_\alpha}{m_p}}$.

The alpha particle is a helium nucleus, so $m_\alpha = 4m_p$.

$$\frac{\lambda_p}{\lambda_\alpha} = \sqrt{\frac{4m_p}{m_p}} = \sqrt{4} = 2.$$

So the proton has a de Broglie wavelength **twice** that of the alpha particle, which is option (B). The value $2\sqrt{2}$ in option (C) only arises if the two particles are accelerated through the same potential difference (so that $K = qV$ brings in the charge ratio); but this question fixes the kinetic energy itself, so the charge is irrelevant and the answer is 2.

COMMON MISTAKE

Always check whether a de Broglie ratio question fixes the same kinetic energy ($\lambda \propto 1/\sqrt{m}$) or the same accelerating voltage ($\lambda \propto 1/\sqrt{mq}$); the charge factor changes the answer.

Final answer. For the same kinetic energy, $\lambda_p:\lambda_\alpha = 2$, so the answer is option **(B)**.

← Back to Q71

Q72

Photons of energy 3.2 eV are incident on a photosensitive surface. If the stopping potential for the emitted electrons is 1.5 V, the work function for the surface is:

(A) 1.5 eV

(B) 1.7 eV

(C) 3.2 eV

(D) 4.7 eV

[2023 • Set 55-4-1]

SOLUTION

By Einstein's photoelectric equation, the energy of an incident photon is shared between the work function ϕ_0 (energy needed to free the electron from the surface) and the maximum kinetic energy of the emitted electron, which equals eV_0 where V_0 is the stopping potential.

$h\nu = \phi_0 + eV_0$.

Rearranging for the work function gives $\phi_0 = h\nu - eV_0$.

Here the photon energy is $h\nu = 3.2$ eV, and the stopping potential $V_0 = 1.5$ V corresponds to a maximum kinetic energy $eV_0 = 1.5$ eV.

$\phi_0 = 3.2$ eV $- 1.5$ eV $= 1.7$ eV.

Hence the **work function of the surface is 1.7 eV**, which is option (B). Option (C) 3.2 eV is the full photon energy, option (A) 1.5 eV is the kinetic-energy part, and option (D) 4.7 eV wrongly adds the two energies.

Final answer. The work function is **1.7 eV**, so the answer is option **(B)**.

← *Back to Q72*

Q73

Photons of energy 4.3 eV are incident on a photosensitive surface of work function 2.3 eV. The stopping potential for photoelectrons is:

(A) 2.0 eV

(B) 2.3 eV

(C) 4.3 eV

(D) 6.6 eV

[2023 • Set 55-4-2]

SOLUTION

Einstein's photoelectric equation states that the maximum kinetic energy of an emitted electron equals the photon energy minus the work function, and this maximum kinetic energy is measured by the stopping potential through $K_{max} = eV_0$.

$eV_0 = h\nu - \phi_0$.

With photon energy $h\nu = 4.3$ eV and work function $\phi_0 = 2.3$ eV, substitute to find eV_0.

$eV_0 = 4.3$ eV $- 2.3$ eV $= 2.0$ eV.

Therefore the stopping potential is $V_0 = \frac{2.0\ eV}{e} = 2.0\ V$, i.e. the photoelectrons carry a maximum kinetic energy of **2.0 eV**.

This corresponds to option (A). Option (B) 2.3 eV is the work function, option (C) 4.3 eV is the photon energy, and option (D) 6.6 eV incorrectly adds the photon energy and work function.

Final answer. The stopping potential is **2.0 *V***, so the answer is option **(A)**.

← *Back to Q73*

Q74

A light of frequency ν is incident on a metal surface whose work function is W_0*. The kinetic energy of emitted electron is K. If the frequency of the incident light is doubled then the kinetic energy of emitted electron will be:*

(A) 2K

(B) **more than 2K**

(C) between *K* and 2K

(D) less than *K*

[2023 • Set 55-4-3]

SOLUTION

From Einstein's photoelectric equation, the kinetic energy of the emitted electron is $K = h\nu - W_0$, where W_0 is the work function.

When the frequency is doubled to 2ν, the new kinetic energy becomes $K' = h(2\nu) - W_0 = 2h\nu - W_0$.

From the original relation, $h\nu = K + W_0$, so substitute this in.

$K' = 2(K + W_0) - W_0 = 2K + 2W_0 - W_0 = 2K + W_0$.

Since the work function W_0 is a positive quantity, $K' = 2K + W_0 > 2K$.

So doubling the frequency makes the kinetic energy **more than 2K**, which is option (B). The kinetic energy is not exactly 2K (option A) because only the photon energy hν doubles while the work function term W_0 stays the same and

is subtracted only once. Options (C) and (D) are ruled out as K' clearly exceeds 2K.

CONCEPT

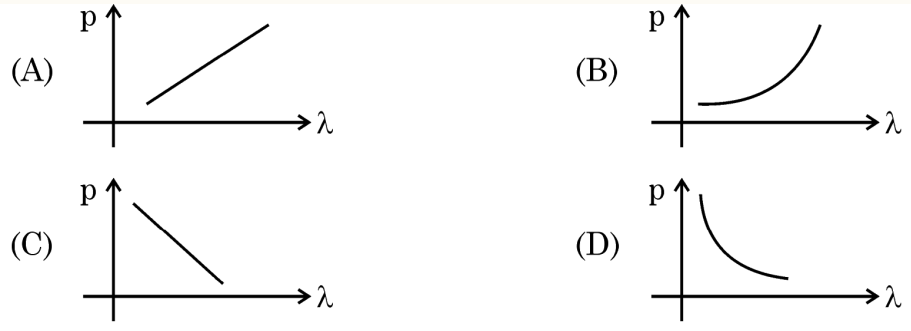
Doubling the frequency doubles only the photon energy, not the kinetic energy, because the fixed work function W_0 is subtracted just once.

Final answer. The new kinetic energy is more than **2K**, so the answer is option **(B)**.

← *Back to Q74*

Q75

Which of the following graphs correctly represents the variation of a particle momentum with its associated de-Broglie wavelength?



(A) Graph (A)

(B) Graph (B)

(C) Graph (C)

(D) Graph (D)

[2023 • Set 55-5-1]

SOLUTION

The de Broglie relation links a particle's momentum p to its wavelength λ by $\lambda = \frac{h}{p}$, where h is Planck's constant.

Rearranging gives $p = \frac{h}{\lambda}$, so $p\lambda = h =$ **constant**.

This means momentum is inversely proportional to wavelength: $p \propto \frac{1}{\lambda}$.

A graph of p against λ with their product constant is a **rectangular hyperbola** lying in the first quadrant. It starts very high for small λ, falls steeply, and then flattens out, approaching the λ-axis asymptotically as λ becomes large, without ever touching either axis.

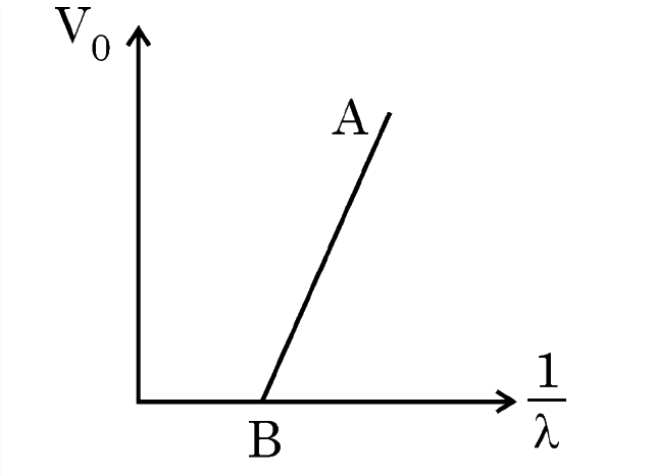
The curve shaped like this rectangular hyperbola is graph (D). A straight line through the origin would represent $p \propto \lambda$ (direct proportion) and is wrong, and any curve that is not a smoothly falling hyperbola does not satisfy $p\lambda = h$. Hence option (D) is correct.

Final answer. The correct graph is the rectangular hyperbola, option **(D)**.

← *Back to Q75*

Q76

Figure shows a plot of stopping potential (V_0) *versus* $\frac{1}{\lambda}$ *where* λ *is the wavelength of the radiation causing photoelectric emission from a surface. The slope of the line is equal to*



(A) $\frac{h}{e}$

(B) $\frac{hc}{e}$

(C) $\frac{h}{c}$

(D) $\frac{h^2 c}{e}$

[2023 • Set 55-5-2]

SOLUTION

Einstein's photoelectric equation in terms of wavelength uses the photon energy $\mathbf{h\nu = \frac{hc}{\lambda}}$, since $\mathbf{\nu = \frac{c}{\lambda}}$.

The stopping potential satisfies $\mathbf{eV_0 = h\nu - \phi_0 = \frac{hc}{\lambda} - \phi_0}$.

Dividing throughout by the electronic charge e gives $\mathbf{V_0 = \frac{hc}{e} \cdot \frac{1}{\lambda} - \frac{\phi_0}{e}}$.

Comparing this with the straight-line form $\mathbf{y = mx + c}$, where $\mathbf{y = V_0}$ and $\mathbf{x = \frac{1}{\lambda}}$, the slope of the line is the coefficient of $\frac{1}{\lambda}$.

$\mathbf{slope = \frac{hc}{e}}$.

So the **slope equals** $\mathbf{\frac{hc}{e}}$, which is option (B). The negative intercept on the V_0-axis is $-\frac{\phi_0}{e}$. Option (A) $\frac{h}{e}$ would be the slope of a V_0 versus ν graph, not a V_0 versus $\frac{1}{\lambda}$ graph, while options (C) and (D) do not have the dimensions of slope here.

Final answer. The slope of the line is $\frac{\mathbf{hc}}{\boldsymbol{e}}$, so the answer is option **(B)**.

← *Back to Q76*

Q77

E, c and ν represent the energy, velocity and frequency of a photon. Which of the following represents its wavelength?

(A) $\frac{h\nu}{c^2}$

(B) $h\nu$

(C) $\frac{\mathbf{hc}}{\boldsymbol{E}}$

(D) $\frac{h\nu}{c}$

[2023 • Set 55-5-3]

SOLUTION

The energy of a photon is related to its frequency by $\boldsymbol{E = h\nu}$, where h is Planck's constant.

A photon also travels at speed c, and its frequency, wavelength and speed are linked by $\boldsymbol{c = \nu\lambda}$, which gives $\boldsymbol{\nu = \frac{c}{\lambda}}$.

Substituting this into the energy relation, $\boldsymbol{E = h\nu = \frac{hc}{\lambda}}$.

Rearranging for the wavelength gives $\boldsymbol{\lambda = \frac{hc}{E}}$.

Hence the wavelength is $\boldsymbol{\frac{hc}{E}}$, which is option (C). A quick dimensional check confirms it: $\frac{hc}{E}$ has units of $\frac{(J\,s)(m\,s^{-1})}{J} = m$, a length. Option (B) $h\nu$ has units of energy, and options (A) $\frac{h\nu}{c^2}$ and (D) $\frac{h\nu}{c}$ do not give units of length, so they are ruled out at once.

Final answer. The wavelength of the photon is $\frac{\mathbf{hc}}{\boldsymbol{E}}$, so the answer is option **(C)**.

← Back to Q77

Q78

Two statements are given - one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer to this question from the codes (A), (B), (C) and (D) as given below:
(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false and Reason (R) is also false.
Assertion (A): *In his study of photoelectric emission, Hallwachs connected a negatively charged zinc plate to an electroscope. He found that negatively charged particles were emitted from the zinc plate under the action of visible light.*
Reason (R): *An uncharged zinc plate becomes positively charged when it is irradiated by visible light.*

[2021]

SOLUTION

Examine the Assertion first. Hallwachs' classic experiment used ultraviolet light, not visible light. He found that a negatively charged zinc plate lost its negative charge when illuminated by ultraviolet radiation, because electrons were ejected from the metal. Visible light has photon energies of only about 1.8 to 3.1 eV, which is below the work function of zinc ($\phi_0 \approx 4.3$ **eV**), so visible light cannot eject electrons from zinc at all.

Since the Assertion claims that the emission happened under the action of visible light, the Assertion is **false**.

Now examine the Reason. An uncharged zinc plate becomes positively charged only when irradiated by ultraviolet light, because the ejected electrons leave behind a net positive charge. Under visible light no electrons are emitted, so the plate stays neutral and does not become positively charged.

Therefore the Reason is also **false**.

With both the Assertion and the Reason false, the correct choice is option (D): Assertion (A) is false and Reason (R) is also false.

Final answer. Both the Assertion and the Reason are false, so the correct code is option **(D)**.

← *Back to Q78*

Q79

If the frequency of the radiation incident on a photosensitive surface increases **($\nu > \nu_0$)**, *how will the stopping potential change?*

[2021]

SOLUTION

The stopping potential V_0 is the reverse voltage that just stops the most energetic photoelectrons, so eV_0 equals their maximum kinetic energy.

From Einstein's photoelectric equation, $\mathbf{eV_0 = h\nu - \phi_0 = h(\nu - \nu_0)}$, where ν_0 is the threshold frequency and $\mathbf{\phi_0 = h\nu_0}$ is the work function.

Solving for the stopping potential, $\mathbf{V_0 = \frac{h}{e}(\nu - \nu_0)}$.

For a given photosensitive surface, both the threshold frequency ν_0 and the work function ϕ_0 are fixed constants, while $\frac{h}{e}$ is a universal constant. So V_0 depends linearly only on the incident frequency ν.

Therefore, as the incident frequency increases (with $\nu > \nu_0$), the **stopping potential increases linearly** with ν. A graph of V_0 against ν is a straight line of slope $\frac{h}{e}$ that meets the frequency axis at ν_0.

Note that the stopping potential does not depend on the intensity of the incident radiation; it is governed by frequency alone.

Final answer. The stopping potential increases linearly with the incident frequency.

← Back to Q79

Q80

An electron is accelerated through a potential difference of 100 *V. Calculate the de Broglie wavelength associated with it.*

[2021]

SOLUTION

When an electron of charge e is accelerated from rest through a potential difference V, it gains kinetic energy $K = eV$, and this equals $\frac{p^2}{2m}$, so its momentum is $p = \sqrt{2meV}$.

The de Broglie wavelength associated with the electron is $\lambda = \frac{h}{p} = \frac{h}{\sqrt{2meV}}$.

Putting in the constants $h = 6.6310^{-34}\ J\ s$, $m = 9.110^{-31}\ kg$ and $e = 1.610^{-19}\ C$, this reduces to the handy formula $\lambda = \frac{12.27}{\sqrt{V}}$, with V in volts.

Here $V = 100\ V$, so substitute.

$\lambda = \frac{12.27}{\sqrt{100}} = \frac{12.27}{10}$.

$\lambda = 1.227 = 1.22710^{-10}\ m = 0.123\ nm.$

Hence the de Broglie wavelength of the electron is about **1.23 (0.123 nm)**, which lies in the X-ray region of the spectrum.

TIP

Remember the shortcut $\lambda = \frac{12.27}{\sqrt{V}}$ for an electron accelerated through V volts; it saves time in 1-mark numericals.

Final answer. The de Broglie wavelength is **1.227** , i.e. about **0.123 nm**.

← Back to Q80

Q81

A proton and a deuteron are moving with the same speed. Find the ratio of the de Broglie wavelength (λ_p / λ_d) associated with them.

[2021]

SOLUTION

The de Broglie wavelength of a particle of mass m moving with speed v is given by:

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

where h is Planck's constant and $\boldsymbol{p = mv}$ is the linear momentum.

Since both the proton and the deuteron move with the same speed v, the wavelength depends only on the mass, so $\boldsymbol{\lambda \propto \frac{1}{m}}$.

A deuteron contains one proton and one neutron, so its mass is about twice the proton mass: $\boldsymbol{m_d = 2m_p}$.

Taking the ratio:

$$\frac{\lambda_p}{\lambda_d} = \frac{h/(m_p v)}{h/(m_d v)} = \frac{m_d}{m_p}$$

$$\frac{\lambda_p}{\lambda_d} = \frac{2m_p}{m_p} = 2$$

Therefore the required ratio is $\boldsymbol{\lambda_p : \lambda_d = 2 : 1}$. The lighter proton has the longer de Broglie wavelength.

Final answer. The ratio of de Broglie wavelengths is $\boldsymbol{\lambda_p : \lambda_d = 2 : 1}$.

← *Back to Q81*

Q82

Photons of energies 1 eV *and* 2 eV *are successively incident on a metallic surface of work function* 0·5 eV. *The ratio of kinetic energy of most energetic photoelectrons in*

the two cases will be

(A) 1 : 2

(B) 1 : 1

(C) **1 : 3**

(D) 1 : 4

[2020 • Set 55-1-1]

SOLUTION

By Einstein's photoelectric equation, the maximum kinetic energy of the emitted photoelectrons is:

$$KE_{max} = E_{photon} - \phi$$

where E_{photon} is the energy of the incident photon and ϕ is the work function of the metal.

For the first case, with $\mathbf{E_1 = 1\ eV}$ and $\boldsymbol{\phi = 0.5\ eV}$:

$$KE_1 = 1 - 0.5 = 0.5\ eV$$

For the second case, with $\mathbf{E_2 = 2\ eV}$ and the same $\boldsymbol{\phi = 0.5\ eV}$:

$$KE_2 = 2 - 0.5 = 1.5\ eV$$

Taking the ratio of the two kinetic energies:

$$\frac{KE_1}{KE_2} = \frac{0.5}{1.5} = \frac{1}{3}$$

So $\mathbf{KE_1 : KE_2 = 1 : 3}$, which is option (C). The other ratios (1:2, 1:1, 1:4) do not match because the same work function must be subtracted from both photon energies before taking the ratio, not the bare photon energies themselves.

COMMON MISTAKE

A common slip is taking the ratio of photon energies 1:2 *directly. The work function* ϕ *must be subtracted first.*

Final answer. Option **(C)**: the ratio of maximum kinetic energies is **1 : 3**.

← Back to Q82

Q83

A proton and an electron have equal speeds. Find the ratio of de Broglie wavelengths associated with them.

[2020 • Set 55-1-1]

SOLUTION

The de Broglie wavelength of a particle of mass m moving with speed v is:

$$\lambda = \frac{h}{mv}$$

where h is Planck's constant.

As the proton and the electron have equal speeds v, the wavelength is inversely proportional to mass, so $\boldsymbol{\lambda \propto \frac{1}{m}}$.

Using the standard masses, electron mass $\mathbf{m_e = 9.11 \times 10^{-31}\ kg}$ and proton mass $\mathbf{m_p = 1.67 \times 10^{-27}\ kg}$.

Taking the ratio of the two wavelengths:

$$\frac{\lambda_p}{\lambda_e} = \frac{h/(m_p v)}{h/(m_e v)} = \frac{m_e}{m_p}$$

$$\frac{\lambda_p}{\lambda_e} = \frac{9.11 \times 10^{-31}}{1.67 \times 10^{-27}} \approx \frac{1}{1836}$$

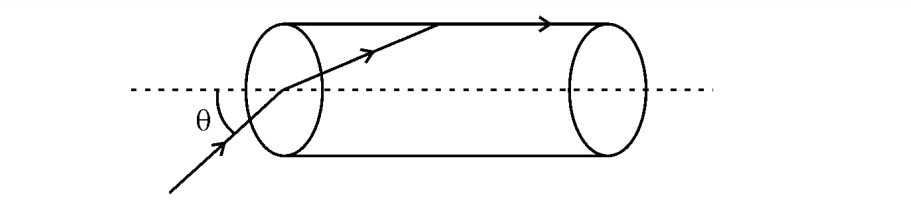
Therefore $\boldsymbol{\lambda_p : \lambda_e = 1 : 1836}$. The proton's de Broglie wavelength is far smaller than the electron's because the proton is about 1836 times heavier.

Final answer. The ratio of de Broglie wavelengths is $\boldsymbol{\lambda_p : \lambda_e = 1 : 1836}$.

← Back to Q83

Q84

The variation of the stopping potential (V_0) *with the frequency* (ν) *of the light incident on two different photosensitive surfaces* M_1 *and* M_2 *is shown in the figure. Identify the surface which has greater value of the work function.*



[2020 • Set 55-1-1]

SOLUTION

From Einstein's photoelectric equation, the stopping potential V_0 is related to the frequency ν of the incident light by:

$$eV_0 = h\nu - \phi = h\nu - h\nu_0$$

Rearranging gives the straight-line form:

$$V_0 = \frac{h}{e}\nu - \frac{h\nu_0}{e}$$

This is a straight line of slope $\frac{h}{e}$ (the same for every metal) and a frequency-axis intercept equal to the threshold frequency ν_0, where the line cuts the ν-axis ($\mathbf{V_0 = 0}$).

The work function is $\boldsymbol{\phi = h\nu_0}$, so a larger threshold frequency ν_0 means a larger work function. On the graph, both lines are parallel (equal slope h/e), but they cut the ν-axis at different points.

The metal whose line crosses the ν-axis farther to the right has the larger threshold frequency ν_0 and therefore the **greater work function**. From the given figure, the surface with the larger ν-axis intercept (the line shifted to the right) is the one with the higher work function.

Final answer. The surface whose $\mathbf{V_0}$ vs $\boldsymbol{\nu}$ line cuts the frequency axis farther to the right (larger $\boldsymbol{\nu_0}$) has the greater work function.

← Back to Q84

Q85

What is the wavelength of a photon of energy 3.3×10^{-19} *J?*

[2020 • Set 55-2-1]

SOLUTION

The energy of a photon of wavelength λ is $E = h\nu = \frac{hc}{\lambda}$, where h is Planck's constant and c is the speed of light.

Rearranging to find the wavelength:

$$\lambda = \frac{hc}{E}$$

Substituting $E = 3.3 \times 10^{-19}\ J$, $h = 6.6 \times 10^{-34}\ J\,s$ and $c = 3 \times 10^{8}\ m\,s^{-1}$:

$$\lambda = \frac{(6.6 \times 10^{-34})(3 \times 10^{8})}{3.3 \times 10^{-19}}$$

$$\lambda = \frac{19.8 \times 10^{-26}}{3.3 \times 10^{-19}}$$

$$\lambda = 6 \times 10^{-7}\ m$$

Therefore $\lambda = 6 \times 10^{-7}\ m = 600$ nm, which lies in the visible (yellow-orange) region of the spectrum.

Final answer. The wavelength of the photon is $\lambda = 6 \times 10^{-7}\ m = 600$ nm.

← Back to Q85

Q86

The threshold wavelength for two photosensitive surfaces A and B are λ_1 *and* λ_2*, respectively. What is the ratio of the work functions of the two surfaces?*

[2020 • Set 55-2-2]

SOLUTION

At the threshold wavelength, the photon energy is just enough to free an electron, so the entire photon energy equals the work function. The work function is therefore:

$$\phi = h\nu_0 = \frac{hc}{\lambda_0}$$

where λ_0 is the threshold wavelength of the surface.

Hence the work function is inversely proportional to the threshold wavelength, $\phi \propto \frac{1}{\lambda_0}$.

For surface A (threshold wavelength λ_1) and surface B (threshold wavelength λ_2):

$$\phi_A = \frac{hc}{\lambda_1},\ \phi_B = \frac{hc}{\lambda_2}$$

Taking the ratio:

$$\frac{\phi_A}{\phi_B} = \frac{hc/\lambda_1}{hc/\lambda_2} = \frac{\lambda_2}{\lambda_1}$$

Therefore $\phi_A : \phi_B = \lambda_2 : \lambda_1$. The surface with the smaller threshold wavelength has the larger work function.

Final answer. The ratio of work functions is $\phi_A : \phi_B = \lambda_2 : \lambda_1$.

← *Back to Q86*

Q87

The threshold frequencies of two photosensitive surfaces are ν_1 and ν_2, respectively. What is the ratio of the velocities of the photoelectrons emitted from these surfaces when light of frequency ν is incident on them and photoemission occurs?

[2020 • Set 55-2-3]

SOLUTION

By Einstein's photoelectric equation, the maximum kinetic energy of an emitted photoelectron equals the energy supplied above the threshold:

$$\frac{1}{2}mv^2 = h(\nu - \nu_0)$$

where ν is the frequency of incident light, ν_0 is the threshold frequency of the surface and m is the electron mass.

Solving for the speed of the fastest photoelectron:

$$v = \sqrt{\frac{2h(\nu - \nu_0)}{m}}$$

For the first surface (threshold frequency ν_1) illuminated by light of frequency ν:

$$v_1 = \sqrt{\frac{2h(\nu - \nu_1)}{m}}$$

For the second surface (threshold frequency ν_2) illuminated by the same light:

$$v_2 = \sqrt{\frac{2h(\nu - \nu_2)}{m}}$$

Since h, m and ν are the same for both, taking the ratio cancels these common factors:

$$\frac{v_1}{v_2} = \sqrt{\frac{\nu - \nu_1}{\nu - \nu_2}}$$

Therefore $\mathbf{v_1 : v_2 = \sqrt{\nu - \nu_1} : \sqrt{\nu - \nu_2}}$.

Final answer. The ratio of velocities is $\mathbf{v_1 : v_2 = \sqrt{\nu - \nu_1} : \sqrt{\nu - \nu_2}}$.

← *Back to Q87*

Q88

In photoelectric effect, the number of emitted photoelectrons is proportional to _____ of incident light.

OR

Light of frequency ν is incident on a photosensitive surface of threshold frequency ν_0 $(\nu > \nu_0)$. *The value of kinetic energy of the emitted photoelectrons will be* _____.

[2020 • Set 55-3-1]

SOLUTION

Main blank: The number of emitted photoelectrons is proportional to the **intensity** of the incident light (for light of frequency above the threshold).

The intensity of a beam equals the number of photons striking unit area per second multiplied by the energy of each photon, hν. For a fixed frequency ν, raising the intensity simply means more photons arrive per second.

Each photon of energy $h\nu > \phi$ can liberate at most one electron, so the number of photoelectrons ejected per second is directly proportional to the number of incident photons, and hence to the intensity. (Note that the frequency, not the intensity, controls the maximum kinetic energy.)

OR variant: When light of frequency ν falls on a surface of threshold frequency ν_0 (with $\nu > \nu_0$), the maximum kinetic energy of the emitted photoelectrons is $KE_{max} = h(\nu - \nu_0)$.

This follows from Einstein's photoelectric equation $KE_{max} = h\nu - \phi$, with the work function written as $\phi = h\nu_0$.

Final answer. Number of photoelectrons is proportional to the intensity of incident light; OR the kinetic energy is $h(\nu - \nu_0)$.

← *Back to Q88*

Q89

The kinetic energy of a proton and that of an α*-particle are* 4 eV *and* 1 eV, *respectively. The ratio of the de-Broglie wavelengths associated with them, will be*

(A) 2:1

(B) 1:1

(C) 1:2

(D) 4:1

[2020 • Set 55-4-1]

SOLUTION

The de Broglie wavelength of a particle in terms of its kinetic energy K and mass m comes from the momentum relation $p = \sqrt{2mK}$:

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mK}}$$

Here h is Planck's constant. Taking the ratio of the proton and α-particle wavelengths, h and the factor 2 cancel:

$$\frac{\lambda_p}{\lambda_\alpha} = \sqrt{\frac{m_\alpha K_\alpha}{m_p K_p}}$$

An α-particle is a helium nucleus, so its mass is about four times the proton mass, $\mathbf{m_\alpha = 4m_p}$. The given kinetic energies are $\mathbf{K_p = 4\ eV}$ and $\mathbf{K_\alpha = 1\ eV}$.

Substituting these values:

$$\frac{\lambda_p}{\lambda_\alpha} = \sqrt{\frac{(4m_p)(1)}{(m_p)(4)}} = \sqrt{\frac{4m_p}{4m_p}} = \sqrt{1} = 1$$

So the de Broglie wavelengths are equal, $\mathbf{\lambda_p : \lambda_\alpha = 1 : 1}$, which is **option (B)**.

The printed key value 2:1 would require ignoring the α-particle's larger mass; with the correct mass factor of 4 included, the masses and energies cancel exactly and the ratio is 1:1.

TIP

Use $\lambda = h/\sqrt{2mK}$ when kinetic energy is given, not $\lambda = h/(mv)$. Here $m_pK_p = m_\alpha K_\alpha = 4$, so the wavelengths come out equal.

Final answer. Option **(B)**: the de Broglie wavelength ratio is **1 : 1**.

← *Back to Q89*

Q90

A photocell connected in an electrical circuit is placed at a distance d from a source of light. As a result, current I flows in the circuit. What will be the current in the circuit

when the distance is reduced to d/2?

(A) I

(B) 2I

(C) **4I**

(D) $\sqrt{2}\,I$

[2020 • Set 55-4-1]

SOLUTION

Treating the lamp as a point source, the intensity of light reaching the photocell falls off with distance according to the inverse-square law:

$$I_{light} \propto \frac{1}{r^2}$$

The photoelectric current is proportional to the number of photons (and hence the light intensity) reaching the cell each second, so the circuit current $\propto I_{light} \propto \frac{1}{r^2}$.

Initially the cell is at distance d and carries current I. When the distance is reduced to $d' = \frac{d}{2}$, take the ratio of the new current to the old:

$$\frac{I'}{I} = \frac{1/d'^2}{1/d^2} = \left(\frac{d}{d'}\right)^2$$

$$\frac{I'}{I} = \left(\frac{d}{d/2}\right)^2 = (2)^2 = 4$$

So the current becomes four times the original:

$$I' = 4I$$

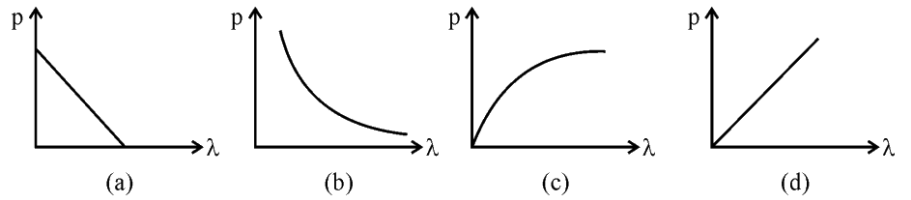
Therefore the answer is **option (C), 4I**. Options 2I and $\sqrt{2}\,I$ ignore the squared dependence on distance, and I wrongly assumes the current is independent of distance.

Final answer. Option **(C)**: the current becomes **4I**.

← *Back to Q90*

Q91

The graph showing the correct variation of linear momentum (p) of a charge particle with its de-Broglie wavelength (λ) is —



(A) (a)

(B) (b)

(C) (c)

(D) (d)

[2020 • Set 55-5-1]

SOLUTION

According to the de Broglie hypothesis, a moving particle of linear momentum p has an associated matter wave of wavelength given by the relation:

$$\lambda = \frac{h}{p}$$

where h is Planck's constant.

Rearranging this relation gives:

$$p\,\lambda = h = \text{constant}$$

Since h is a fixed constant, the linear momentum p is inversely proportional to the de Broglie wavelength λ:

$$p \propto \frac{1}{\lambda}$$

A relation of the form $\boldsymbol{p\,\lambda = \text{constant}}$ represents a **rectangular hyperbola**. On a p versus λ graph the curve falls steeply at small λ (large p) and approaches both axes asymptotically as λ increases. The correct graph is therefore the one showing this hyperbolic decrease, where p never becomes zero or negative.

The options showing a straight line, an increasing curve, or a line of positive slope are wrong because none of them satisfies $p\lambda$ = **constant**.

> **TIP**
>
> Whenever you see one quantity inversely proportional to another, the graph is always a rectangular hyperbola, never a straight line.

Final answer. The correct graph is the rectangular hyperbola for which $p\lambda = h$ = **constant**.

← *Back to Q91*

Q92

If photons of frequency ν are incident on the surfaces of metals A & B of threshold frequencies $\nu/2$ and $\nu/3$ respectively, the ratio of the maximum kinetic energy of electrons emitted from A to that from B is

(A) 2:3

(B) 3:4

(C) 1:3

(D) 3:2

[2020 • Set 55-5-1]

SOLUTION

By Einstein's photoelectric equation, the maximum kinetic energy of the emitted photoelectrons is the energy of the incident photon minus the work needed to free the electron:

$$KE_{max} = h\nu - h\nu_0 = h(\nu - \nu_0)$$

where ν is the frequency of incident radiation and ν_0 is the threshold frequency of the metal.

For metal A, the threshold frequency is $\boldsymbol{\nu_0 = \frac{\nu}{2}}$, so:

$$KE_A = h(\nu - \frac{\nu}{2}) = h\cdot\frac{\nu}{2} = \frac{h\nu}{2}$$

For metal B, the threshold frequency is $\nu_0 = \frac{\nu}{3}$, so:

$$KE_B = h(\nu - \frac{\nu}{3}) = h \cdot \frac{2\nu}{3} = \frac{2h\nu}{3}$$

Taking the ratio:

$$\frac{KE_A}{KE_B} = \frac{\frac{h\nu}{2}}{\frac{2h\nu}{3}} = \frac{1}{2} \times \frac{3}{2} = \frac{3}{4}$$

Hence the required ratio is $KE_A : KE_B = 3 : 4$, which matches option (B). The other options do not result from the correct substitution of the two threshold frequencies.

Final answer. The ratio is **3:4**, so the correct option is **(B)**.

← *Back to Q92*

Q93

Define the term "threshold frequency", in the context of photoelectric emission.

[2019 • Set 55-1-1]

SOLUTION

The **threshold frequency** ν_0 of a given photosensitive surface is defined as the minimum frequency of the incident radiation below which no photoelectric emission takes place from that surface, however high the intensity of the radiation may be.

In other words, photoelectrons are emitted only when the frequency of the incident light satisfies $\nu \geq \nu_0$. If $\nu < \nu_0$, no electrons are ejected at all, no matter how intense or how long the light shines.

The threshold frequency is related to the work function ϕ_0 of the metal by:

$$\nu_0 = \frac{\phi_0}{h}$$

where h is Planck's constant. The threshold frequency is a characteristic property of the metal surface.

Final answer. Threshold frequency is the minimum frequency of incident radiation needed to just cause photoelectric emission, given by $\nu_0 = \frac{\Phi_0}{h}$.

← Back to Q93

Q94

Define the term "Intensity" in photon picture of electromagnetic radiation.

[2019 • Set 55-1-1]

SOLUTION

In the photon picture of electromagnetic radiation, the **intensity** of light is defined as the amount of energy crossing per unit area held normal to the direction of propagation, per unit time.

Each photon carries a fixed energy $h\nu$. If N photons cross unit area normally per unit time (the photon flux), then the intensity is the total energy carried by these photons:

$$I = N\,h\nu$$

The SI unit of intensity is $W\,m^{-2}$ (watt per square metre).

Thus, for radiation of a fixed frequency ν, the intensity is directly proportional to the number of photons falling on unit area per second:

$$I \propto N$$

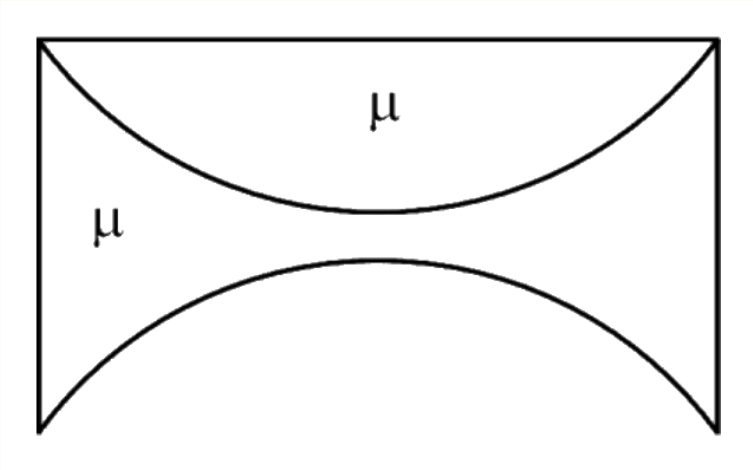
This is why increasing the intensity (more photons per second) increases the photoelectric current but does not change the energy of each ejected electron.

Final answer. Intensity is the radiant energy crossing unit normal area per second, equal to $N\,h\nu$ where N is the photon flux.

← Back to Q94

Q95

On the basis of the graphs shown in the figure, answer the following questions:
(a) Which physical parameter is kept constant for the three curves?
(b) Which is the highest frequency among ν_1, ν_2 *and* ν_3*?*



[2019 • Set 55-3-1]

SOLUTION

(a) In the given graphs of photoelectric current versus collector plate potential, all three curves rise to and level off at the **same saturation current**. The saturation current is decided only by the number of photoelectrons reaching the plate per second, which in turn depends on the number of photons striking the cathode per second, that is, on the intensity of the incident radiation.

Since all three curves saturate at the same value, the physical parameter kept constant for the three curves is the **intensity of the incident radiation**.

(b) The three curves differ in their stopping potentials, that is, the reverse potential at which the photoelectric current just becomes zero. By Einstein's relation:

$$eV_0 = h\nu - \phi_0$$

a higher frequency ν gives a larger maximum kinetic energy and hence a larger (more negative) stopping potential V_0.

Therefore the curve whose current drops to zero at the most negative collector potential corresponds to the **highest frequency**. Among ν_1, ν_2 and ν_3, the one with the greatest magnitude of stopping potential is the highest frequency.

CONCEPT

Same saturation current means same intensity; different cut-off (stopping) potentials mean different frequencies. Read both features off the graph separately.

Final answer. (a) Intensity of the incident radiation is kept constant; (b) the highest frequency is the one with the largest (most negative) stopping potential.

← *Back to Q95*

Q96

In photoelectric emission, when the frequency of incident radiation is doubled, will the maximum kinetic energy of photoelectrons also be doubled? Justify your answer.

[2019 • Set 55-3-2]

SOLUTION

By Einstein's photoelectric equation, the maximum kinetic energy of the emitted photoelectrons is:

$$KE_{max} = h\nu - \phi_0$$

where ϕ_0 is the work function of the metal.

If the frequency is doubled to 2ν, the new maximum kinetic energy becomes:

$$KE'_{max} = h(2\nu) - \phi_0 = 2h\nu - \phi_0$$

From the original equation, $\mathbf{h\nu = KE_{max} + \phi_0}$. Substituting:

$$KE'_{max} = 2(KE_{max} + \phi_0) - \phi_0$$

$$KE'_{max} = 2\,KE_{max} + \phi_0$$

Since the work function $\mathbf{\phi_0 > 0}$, the new kinetic energy KE'_{max} is more than twice the original value, not exactly twice.

No, the maximum kinetic energy will not be doubled. The reason is that doubling the frequency doubles the photon energy $h\nu$, but the fixed amount ϕ_0

is subtracted only once. So the kinetic energy increases by more than a factor of two.

Final answer. No; on doubling the frequency the maximum kinetic energy becomes $\mathbf{2\,KE_{max} + \phi_0}$, which is more than double.

← *Back to Q96*

Q97

Define intensity of radiation based on photon picture of light.

[2019 • Set 55-3-3]

SOLUTION

In the photon picture of light, the **intensity of radiation** is defined as the total energy crossing per unit area held perpendicular to the direction of propagation, per unit time.

A beam of frequency ν consists of photons each carrying energy $h\nu$. If N photons cross unit area normally in unit time, then the intensity is:

$$I = N\,h\nu$$

measured in $W\,m^{-2}$.

For monochromatic light of a fixed frequency, the intensity is directly proportional to the number of photons crossing unit area per second:

$$I \propto N$$

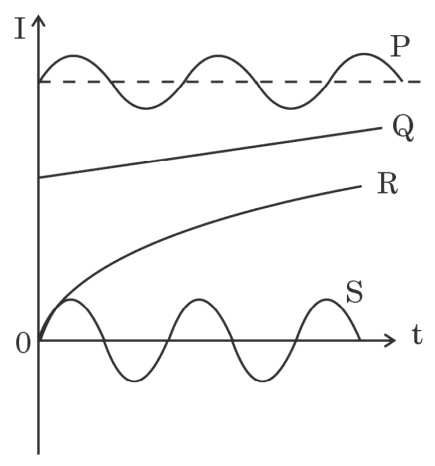
An important consequence is that for two beams of the same intensity but different frequencies, the lower-frequency beam must contain more photons, because each of its photons carries less energy ($h\nu$ is smaller). This photon flux is what controls the magnitude of the photoelectric current.

Final answer. Intensity is the radiant energy crossing unit normal area per unit time, equal to $\boldsymbol{I = N\,h\nu}$ where $\boldsymbol{N}$ is the number of photons per unit area per second.

← Back to Q97

Q98

The figure shows the variation of stopping potential V_s with the frequency ν of the incident radiations for two photosensitive metals P and Q. Which metal has smaller threshold wavelength? Justify your answer.



[2019 • Set 55-4-1]

SOLUTION

The threshold wavelength λ_0 of a metal is related to its threshold frequency ν_0 and work function ϕ_0 by:

$$\lambda_0 = \frac{c}{\nu_0} = \frac{hc}{\phi_0}$$

From this relation, a smaller threshold wavelength corresponds to a larger threshold frequency, which in turn means a larger work function:

$$\lambda_0 \propto \frac{1}{\nu_0}$$

On the stopping potential versus frequency graph, the line obeys $\mathbf{V_s = \frac{h}{e}(\nu - \nu_0)}$. Each straight line meets the frequency axis at its own threshold frequency ν_0 (the point where $\mathbf{V_s = 0}$). The metal whose line cuts the frequency axis further to the right has the larger threshold frequency ν_0.

In the given figure, metal **P** intersects the ν-axis further to the right than metal Q, so P has the larger threshold frequency. Hence:

$$\nu_{0(P)} > \nu_{0(Q)} \Rightarrow \lambda_{0(P)} < \lambda_{0(Q)}$$

Therefore metal **P** has the smaller threshold wavelength, because the metal with the larger threshold frequency (line lying further right on the graph) has the shorter threshold wavelength according to $\lambda_0 = \frac{c}{\nu_0}$.

Final answer. Metal P has the smaller threshold wavelength, since its line meets the frequency axis at a higher threshold frequency ν_0 and $\lambda_0 = \frac{c}{\nu_0}$.

← *Back to Q98*

Q99

Plot a graph showing the variation of photoelectric current versus intensity of light.

[2019 • Set 55-4-2]

SOLUTION

Each incident photon with energy greater than the work function ($h\nu > \phi_0$) can eject at most one electron from the metal surface. Hence the number of photoelectrons emitted per second is directly proportional to the number of photons striking the surface per second, which is itself proportional to the intensity of the incident light. Therefore:

$$I_{photo} \propto I_{light}$$

provided the frequency of the incident radiation is kept fixed and is above the threshold frequency.

The required graph is plotted with the intensity of incident light on the x-axis and the photoelectric current on the y-axis. It is a **straight line passing through the origin with a positive slope**.

Description of the graph: a single straight line starting from the origin (0,0) and rising steadily to the right. The x-axis is labelled Intensity of incident light and the y-axis is labelled Photoelectric current. The line passing through the origin

shows that the current is zero when there is no light and increases in direct proportion as the intensity increases.

Final answer. The graph of photoelectric current versus intensity is a straight line through the origin with positive slope.

← *Back to Q99*

Q100

Plot a graph of stopping potential (V_s) *versus the frequency* (ν) *of incident radiation in photoelectric emission.*

[2019 • Set 55-4-3]

SOLUTION

By Einstein's photoelectric equation, the maximum kinetic energy of the photoelectrons equals the work done against the stopping potential:

$$eV_s = h\nu - \phi_0$$

Dividing throughout by the electronic charge e:

$$V_s = \frac{h}{e}\nu - \frac{\phi_0}{e}$$

This is a straight-line equation of the form $\boldsymbol{y = mx + c}$, with V_s on the y-axis and ν on the x-axis.

Description of the graph: a straight line is drawn with frequency ν on the x-axis and stopping potential V_s on the y-axis.

The line has a constant positive slope equal to $\frac{h}{e}$, which is the same for every metal.

It cuts the frequency axis at the threshold frequency $\boldsymbol{\nu_0 = \frac{\phi_0}{h}}$, where $\boldsymbol{V_s = 0}$; for $\boldsymbol{\nu < \nu_0}$ there is no emission.

When extended backwards, the line meets the V_s-axis at the negative intercept $-\frac{\phi_0}{e}$.

From this graph, the slope gives Planck's constant as $h = e \times$ **(slope)**, and the x-intercept gives the work function as $\phi_0 = h\nu_0$.

CONCEPT

The slope h/e of this line is identical for all metals; only the x-intercept (threshold frequency) shifts from metal to metal.

Final answer. The V_s versus ν graph is a straight line of slope $\frac{h}{e}$ starting at the threshold frequency ν_0 on the x-axis.

← *Back to Q100*

Q101

Plot a graph of the de-Broglie wavelength associated with a proton versus its momentum.

[2019 • Set 55-5-1]

SOLUTION

The de Broglie wavelength of a particle is linked to its momentum p by the relation $\lambda = \frac{h}{p}$, where h is Planck's constant.

$$\lambda = \frac{h}{p}$$

This can be rewritten as $\lambda p = h =$ **constant**, which means λ is inversely proportional to p.

$$\lambda \propto \frac{1}{p}$$

So the graph of λ (on the y-axis) against momentum p (on the x-axis) is a **rectangular hyperbola**. The curve starts very high for small momentum, falls steeply as p increases, and then bends gently, approaching the p-axis but never touching it (it tends to zero only as $p \to \infty$). It also never touches the λ-axis. The same shape holds for a proton, since the relation $\lambda = h/p$ is independent of the particle's mass; only the numerical scale of p differs.

Final answer. The λ versus p graph for a proton is a rectangular hyperbola, since $\lambda = h/p$.

← Back to Q101

Q102

Plot a graph of the de-Broglie wavelength associated with electron as a function of accelerating potential.

[2019 • Set 55-5-2]

SOLUTION

When an electron of charge e and mass m is accelerated from rest through a potential difference V, its kinetic energy equals the work done by the field.

$$\frac{1}{2}mv^2 = eV$$

The momentum gained is $p = mv = \sqrt{2meV}$, so the de Broglie wavelength is

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2meV}}$$

Putting in the values of h, m and e gives the working formula for an electron

$$\lambda = \frac{12.27}{\sqrt{V}} \text{ (}V\text{ in volts)}$$

Therefore $\lambda \propto \frac{1}{\sqrt{V}}$.

The graph of λ (y-axis) against accelerating potential V (x-axis) is a **decreasing curve of hyperbolic shape**: λ is large for small V, falls off rapidly as V increases, and approaches the V-axis asymptotically at large V. If instead λ is plotted against $\frac{1}{\sqrt{V}}$, the graph is a straight line passing through the origin.

Final answer. The λ versus V graph is a falling hyperbola-like curve, since $\lambda \propto 1/\sqrt{V}$.

← Back to Q102

Q103

Draw graphs showing variation of photoelectric current with applied voltage for two incident radiations of equal frequency and different intensities. Mark the graph for the radiation of higher intensity.

[2018]

SOLUTION

The graph plots photoelectric current I (on the y-axis) against the applied anode potential V (on the x-axis) for two radiations of the same frequency but different intensities, say I_1 (lower) and I_2 (higher), with $\mathbf{I_2 > I_1}$.

Both curves start at the same negative voltage equal to the stopping potential $-V_0$, where the current just becomes zero. As V is made less negative and then positive, the current rises and finally levels off at a constant saturation value for large positive V.

A key feature is that **both curves meet the voltage axis at the same stopping potential** $-V_0$, because the stopping potential depends only on the frequency of the incident light (which is the same here), not on its intensity.

$$eV_0 = h\nu - \phi_0 \text{ (independent of intensity)}$$

The two curves differ only in their saturation level. Since saturation current is proportional to the number of photoelectrons emitted per second, and that number is proportional to intensity, the higher-intensity radiation gives a higher saturation current.

$$I_{sat} \propto \text{intensity}$$

So the upper curve, which saturates at the larger current, is **marked as the higher-intensity radiation** (I_2); the lower curve is the lower-intensity radiation (I_1). Both share the common stopping potential $-V_0$ on the negative voltage axis.

CONCEPT

Same frequency means same stopping potential; only the saturation current changes with intensity.

Final answer. Two curves with the same stopping potential $-V_0$ but different saturation currents; the upper (higher-saturation) curve is the higher-intensity radiation.

← *Back to Q103*

Q104

A photosensitive surface emits photoelectrons when red light falls on it. Will the surface emit photoelectrons when blue light is incident on it? Give reason.

[2017]

SOLUTION

Yes, the surface will emit photoelectrons when blue light falls on it.

Photoelectric emission occurs only when the energy of an incident photon, $h\nu$, is at least equal to the work function ϕ_0 of the surface.

$$h\nu \geq \phi_0$$

Since the surface already emits electrons under red light, the red photon energy must satisfy

$$h\nu_{red} \geq \phi_0$$

Blue light has a shorter wavelength (≈ **450 nm**) than red light (≈ **700 nm**), so it has a higher frequency and hence a higher photon energy.

$$\nu_{blue} > \nu_{red} \Rightarrow h\nu_{blue} > h\nu_{red} \geq \phi_0$$

Therefore blue photons also carry more than enough energy to free electrons from the surface, and emission certainly takes place. In fact, the emitted electrons will have a **greater maximum kinetic energy** than with red light, because

$$K_{max} = h\nu - \phi_0$$

is larger for the higher-frequency blue light.

Final answer. Yes; blue light has higher frequency (energy) than red, so $h\nu_{blue} > \phi_0$ and photoelectrons are emitted.

← Back to Q104

Q105

Name the phenomenon which shows the quantum nature of electromagnetic radiation.

[2017]

SOLUTION

The phenomenon that shows the quantum (particle) nature of electromagnetic radiation is the **photoelectric effect**.

In the photoelectric effect, light falling on a metal surface ejects electrons. Several of its observed features cannot be explained by the classical wave theory of light, but follow naturally if light is treated as a stream of energy quanta (photons), each of energy

$$E = h\nu$$

The features that reveal the quantum nature are: there exists a threshold frequency ν_0 below which no emission occurs, however intense the light; the maximum kinetic energy of the ejected electrons depends only on the frequency of light, not on its intensity; and the emission is practically instantaneous, with no time lag, even for very feeble light.

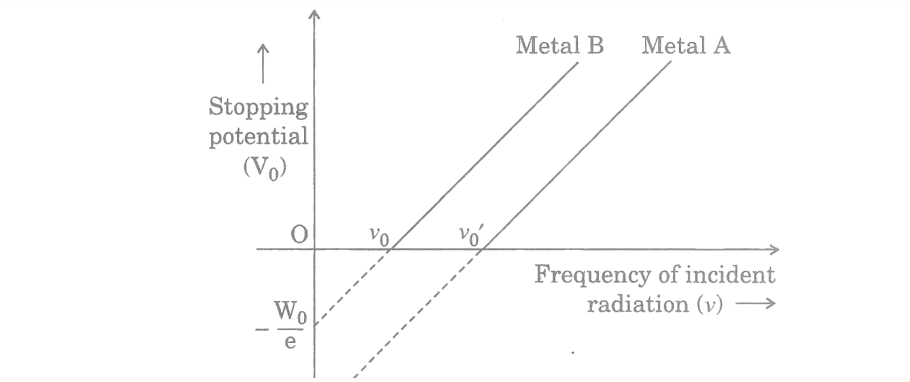
These results led Einstein to conclude that radiation is absorbed in discrete packets (photons), establishing its quantum nature. (The Compton effect is another such phenomenon.)

Final answer. The photoelectric effect demonstrates the quantum nature of electromagnetic radiation.

← Back to Q105

Q106

The graph shows variation of stopping potential V_0 versus frequency of incident radiation ν for two photosensitive metals A and B. Which of the two metals has higher threshold frequency and why ?



[2014]

SOLUTION

For a photosensitive metal, Einstein's photoelectric equation gives the stopping potential as

$$eV_0 = h\nu - \phi_0$$

$$V_0 = \frac{h}{e}\nu - \frac{\phi_0}{e}$$

This is a straight line when V_0 is plotted against frequency ν, with slope $\frac{h}{e}$ (the same for every metal) and an intercept on the frequency axis equal to the threshold frequency.

$$\nu_0 = \frac{\phi_0}{h}$$

Since both lines have the same slope, they are parallel. The metal whose line cuts the frequency axis farther to the right (larger ν-intercept) has the higher threshold frequency.

On the given graph, metal B's line lies to the right of metal A's, so

$$\nu_0^B > \nu_0^A$$

Hence **metal B has the higher threshold frequency**.

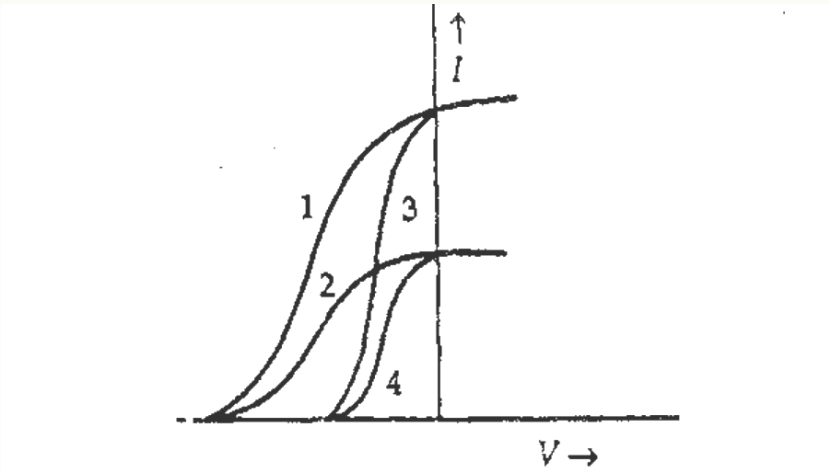
The reason is that a larger threshold frequency means a larger work function, $\phi_0 = h\nu_0$, so metal B needs photons of higher minimum energy to liberate its electrons.

Final answer. Metal B has the higher threshold frequency, as its line has the larger frequency-axis intercept ($\nu_0 = \phi_0/h$).

← *Back to Q106*

Q107

The given graph shows the variation of photo-electric current (I) versus applied voltage (V) for two different photosensitive materials and for two different intensities of the incident radiation. Identify the pairs of curves that correspond to different materials but same intensity of incident radiation.



[2013]

SOLUTION

Two photocell properties decide the shape of an I versus V curve. The **saturation current** depends on the intensity of the incident light (more intensity means more photoelectrons per second, so a higher saturation current). The **stopping potential** depends on the material's work function (for a fixed frequency, a larger work function gives a smaller stopping potential).

$$I_{sat} \propto \text{intensity},\ eV_0 = h\nu - \phi_0$$

We are asked for curves that correspond to different materials but the same intensity. "Same intensity" forces the two curves to reach the **same saturation current**, while "different materials" forces them to have **different stopping potentials** (they cut the negative voltage axis at different points).

So the required pair is the two curves that level off at the same saturation current I_{sat} but have different stopping potentials V_0.

From the graph these are **curves 1 and 2** (and likewise the pair 3 and 4). By contrast, curves with the same stopping potential but different saturation currents would be the same material at different intensities.

Final answer. The pair with equal saturation current but different stopping potentials (curves 1 and 2, and 3 and 4) correspond to different materials at the same intensity.

← *Back to Q107*

Q108

Does the 'stopping potential' in photoelectric emission depend upon
(i) the intensity of the incident radiation in a photocell?
(ii) the frequency of the incident radiation?

[2013]

SOLUTION

The stopping potential V_0 is the negative anode potential at which the photocurrent just falls to zero. It is governed by Einstein's photoelectric equation

$$eV_0 = h\nu - \phi_0$$

$$V_0 = \frac{h}{e}\nu - \frac{\phi_0}{e}$$

(i) Dependence on intensity: **No.** Intensity does not appear in this equation. Increasing the intensity raises only the number of photoelectrons emitted per second (and hence the saturation current), but it does not change the

maximum kinetic energy of any single electron. So the stopping potential is **independent of intensity**.

(ii) Dependence on frequency: **Yes.** Writing the equation with the threshold frequency $\nu_0 = \phi_0/h$,

$$V_0 = \frac{h}{e}(\nu - \nu_0)$$

the stopping potential increases linearly with the frequency of the incident radiation (above threshold). This direct dependence on frequency, and the absence of dependence on intensity, are key experimental signatures of the photon picture of light that classical wave theory cannot explain.

Final answer. Stopping potential does not depend on intensity, but it increases linearly with the frequency of the incident radiation.

← *Back to Q108*

Q109

A proton and an electron have same kinetic energy. Which one has greater de-Broglie wavelength and why?

[2012]

SOLUTION

The de Broglie wavelength of a particle of mass m moving with momentum p is $\lambda = h/p$. For a particle of kinetic energy K, the momentum is $p = \sqrt{2mK}$, so

$$\lambda = \frac{h}{\sqrt{2mK}}$$

When the proton and the electron have the **same kinetic energy** K, the only quantity that differs is the mass, giving

$$\lambda \propto \frac{1}{\sqrt{m}}$$

Hence the lighter particle has the longer wavelength. Taking the ratio for the electron and proton,

$$\frac{\lambda_e}{\lambda_p} = \sqrt{\frac{m_p}{m_e}}$$

$$\frac{\lambda_e}{\lambda_p} = \sqrt{\frac{1.67\times 10^{-27}}{9.11\times 10^{-31}}} = \sqrt{1836} \approx 42.8$$

Since $m_e \ll m_p$, this ratio is much greater than 1. Therefore the **electron has the greater de Broglie wavelength** (about 43 times that of the proton).

The physical reason: for equal kinetic energy, the lighter electron has the smaller momentum $p = \sqrt{2mK}$, and a smaller momentum means a larger wavelength through $\lambda = h/p$.

> **COMMON MISTAKE**
>
> *For equal kinetic energy use $\lambda = h/\sqrt{2mK}$; for equal momentum, $\lambda = h/p$ would make both wavelengths equal.*

Final answer. The electron has the greater de Broglie wavelength (about $\sqrt{m_p/m_e} \approx 43$ times that of the proton), since $\lambda \propto 1/\sqrt{m}$ for equal kinetic energy.

← *Back to Q109*

Q110

Show on a graph the variation of the de Broglie wavelength (λ) associated with an electron, with the square root of accelerating potential (V).

[2012]

SOLUTION

An electron accelerated from rest through a potential difference V acquires kinetic energy eV, so its momentum is $p = \sqrt{2meV}$ and its de Broglie wavelength is

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2meV}}$$

This can be written as

$$\lambda = \frac{h}{\sqrt{2me}} \cdot \frac{1}{\sqrt{V}}$$

The quantity $\frac{h}{\sqrt{2me}}$ is a constant, so

$$\lambda \propto \frac{1}{\sqrt{V}}$$

When λ (y-axis) is plotted against $\sqrt{V}$ (x-axis), the relation $\boldsymbol{\lambda\sqrt{V}}$ **= constant** holds, so the graph is a **rectangular hyperbola**: λ is large for small $\sqrt{V}$, falls steeply as $\sqrt{V}$ increases, and approaches the $\sqrt{V}$-axis asymptotically (tending to zero) at large $\sqrt{V}$, without touching either axis. For an electron the working relation is $\boldsymbol{\lambda = \frac{12.27}{\sqrt{V}}}$.

Final answer. The $\boldsymbol{\lambda}$ versus $\boldsymbol{\sqrt{V}}$ graph is a rectangular hyperbola, since $\boldsymbol{\lambda\sqrt{V}}$ is constant.

← *Back to Q110*

Q111

Define the term 'stopping potential' in relation to photoelectric effect.

[2011]

SOLUTION

In the photoelectric experiment the collector plate is made gradually more and more negative with respect to the emitter. This sets up a retarding field that pushes the emitted photoelectrons back towards the cathode and slowly reduces the photocurrent.

The **stopping potential** V_0 is the minimum negative (retarding) potential of the anode with respect to the cathode at which the photoelectric current becomes exactly zero.

At this potential, even the fastest photoelectrons, the ones carrying the maximum kinetic energy, are just stopped before reaching the collector. So the whole of their kinetic energy is spent against the retarding field.

$$eV_0 = KE_{max}$$

Here $\mathbf{KE_{max} = h\nu - \phi_0}$, so V_0 depends on the frequency ν of the incident light and on the work function ϕ_0 of the surface, but it does not depend on the intensity of the light.

Final answer. Stopping potential is the least negative anode potential that just reduces the photocurrent to zero, given by $\mathbf{eV_0 = KE_{max}}$.

← *Back to Q111*

Q112

Show graphically, the variation of the de-Broglie wavelength (λ) *with the potential* (V) *through which an electron is accelerated from rest.*

[2011 • Set 55-1-1]

SOLUTION

When an electron of charge e and mass m is accelerated from rest through a potential difference V, the work done by the field equals its kinetic energy.

$$eV = \frac{1}{2}mv^2 = \frac{p^2}{2m}$$

$$p = \sqrt{2meV}$$

The de Broglie wavelength of the electron is then

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2meV}}$$

Putting in the constants for an electron gives the handy form

$$\lambda = \frac{12.27}{\sqrt{V}} \text{ (}V\text{ in volts)}$$

This shows that $\lambda \propto \frac{1}{\sqrt{V}}$.

Graph: Take V along the x-axis and λ along the y-axis. The curve starts very high (large λ) near small V, falls steeply, and then flattens as V increases, approaching the x-axis but never touching it. The shape is a **decreasing rectangular hyperbola-like curve** lying entirely in the first quadrant. (If

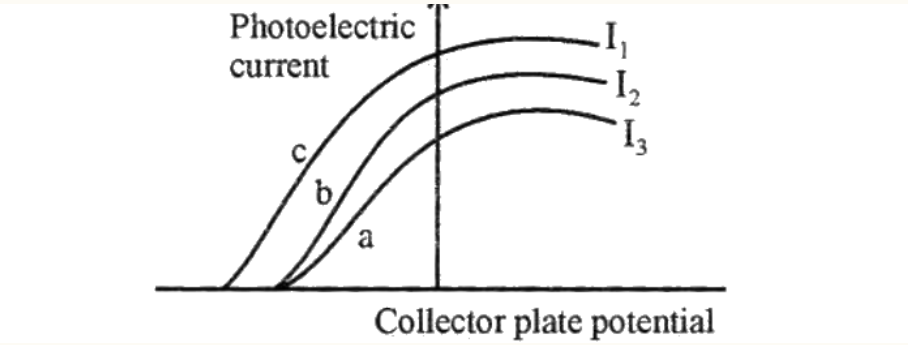
instead λ is plotted against $1/\sqrt{V}$, the graph is a straight line passing through the origin.)

Final answer. Since $\lambda = h/\sqrt{2meV} \propto 1/\sqrt{V}$, the λ versus V graph is a decreasing curve falling steeply then levelling off near the axis.

← *Back to Q112*

Q113

The figure shows a plot of three curves a, b, c showing the variation of photocurrent vs collector plate potential for three different intensities I_1, I_2 *and* I_3 *having frequencies* ν_1, ν_2 *and* ν_3, *respectively incident on a photosensitive surface.*
Point out the two curves for which the incident radiations have same frequency but different intensities.



[2009]

SOLUTION

On a photocurrent versus collector-plate-potential graph, two separate features carry the physics.

The **saturation current** (the flat top reached at high positive collector potential) is fixed by the number of photoelectrons emitted per second, and so it depends only on the intensity of the incident light.

The **stopping potential** V_0 (the negative potential at which the current just falls to zero) is fixed by the maximum kinetic energy of the photoelectrons, and so it depends only on the frequency of the incident light, since $\mathbf{eV_0 = h\nu - \phi_0}$.

Therefore, two radiations of the **same frequency** must give the **same stopping potential**, that is, their curves must cut the negative potential axis at the very

same point. If their intensities differ, the two curves will simply level off at different saturation currents.

So the required pair is the two curves that share a common stopping potential (V_0) but rise to different saturation values. From the figure these are the **two curves having the same intercept on the negative potential axis** (commonly the pair *b* and *c*, whichever two meet the V-axis at the same point).

Final answer. The two curves that have the same stopping potential but different saturation currents correspond to the same frequency with different intensities.

← *Back to Q113*

Q114

The stopping potential in an experiment on photoelectric effect is 1.5 *V. What is the maximum kinetic energy of the photoelectrons emitted?*

[2009]

SOLUTION

At the stopping potential, the fastest photoelectrons are just brought to rest by the retarding field, so the work done against the field equals their maximum kinetic energy.

$$KE_{max} = eV_0$$

Here $\mathbf{V_0 = 1.5\ V}$, so in electron-volts

$$KE_{max} = e \times 1.5\ V = \mathbf{1.5\ eV}$$

Converting to joules

$$KE_{max} = 1.5 \times 1.6 \times 10^{-19}\ J = 2.4 \times 10^{-19}\ J$$

Final answer. The maximum kinetic energy is **1.5 eV**, that is $\mathbf{2.4 \times 10^{-19}\ J}$.

← *Back to Q114*

Q115

An electron and an alpha particle have the same de-Broglie wavelength associated with them. How are their kinetic energies related to each other?

[2008]

SOLUTION

The de Broglie wavelength is $\lambda = \frac{h}{p}$, so equal wavelengths mean equal momenta.

$$\lambda_e = \lambda_\alpha \Rightarrow p_e = p_\alpha$$

Writing kinetic energy in terms of momentum, $\mathbf{KE} = \frac{p^2}{2m}$, and taking the ratio for equal momentum

$$\frac{KE_e}{KE_\alpha} = \frac{p^2/2m_e}{p^2/2m_\alpha} = \frac{m_\alpha}{m_e}$$

So $\mathbf{KE} \propto \frac{1}{m}$ when momentum is the same. The alpha particle has mass $\mathbf{m_\alpha} = \mathbf{4m_p \approx 4 \times 1836\ m_e \approx 7344\ m_e}$, hence

$$\frac{KE_e}{KE_\alpha} = \frac{m_\alpha}{m_e} \approx 7344$$

Because the electron is far lighter, the **electron has the much larger kinetic energy** for the same de Broglie wavelength.

Final answer. Their kinetic energies are inversely proportional to mass, $\frac{\mathbf{KE_e}}{\mathbf{KE_\alpha}} = \frac{\mathbf{m_\alpha}}{\mathbf{m_e}} \approx \mathbf{7344}$, so the electron has the larger kinetic energy.

← *Back to Q115*

Q116

What is the stopping potential of a photocell, in which electrons with maximum

kinetic energy of 6 eV *are emitted?*

[2008]

SOLUTION

At the stopping potential the most energetic photoelectrons are just stopped, so the work done by the retarding field on them equals their maximum kinetic energy.

$$eV_0 = KE_{max}$$

$$V_0 = \frac{KE_{max}}{e}$$

Here $\mathbf{KE_{max} = 6\ eV}$, so

$$V_0 = \frac{6\ \text{eV}}{e} = \mathbf{6\ V}$$

The numbers match because one electron-volt is, by definition, the energy gained by an electron moving through a potential difference of one volt; hence a maximum kinetic energy of 6 eV needs a stopping potential of exactly 6 *V*.

Final answer. The stopping potential is **6 *V***.

← *Back to Q116*

Q117

An electron, an alpha-particle and a proton have the same kinetic energy. Which one of these particles has the largest de-Broglie wavelength?

[2007]

SOLUTION

For a particle of mass m moving with kinetic energy K, the momentum is $\boldsymbol{p = \sqrt{2mK}}$, so the de Broglie wavelength is

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mK}}$$

Since all three particles have the same kinetic energy K, the only variable is the mass, and

$$\lambda \propto \frac{1}{\sqrt{m}}$$

The lighter the particle, the longer its wavelength. The masses follow the order

$$m_e \ll m_p \ll m_\alpha$$

Hence the electron, having the smallest mass, has the **largest de Broglie wavelength**.

For comparison at the same K, the wavelengths are in the ratio $\lambda_e : \lambda_p : \lambda_\alpha \approx \sqrt{m_p/m_e} : 1 : \sqrt{m_p/m_\alpha} \approx 43 : 1 : 0.5$.

Final answer. The electron, being the lightest, has the largest de Broglie wavelength since $\lambda \propto 1/\sqrt{m}$ at fixed kinetic energy.

← *Back to Q117*

Q118

Ultraviolet radiations of different frequencies ν_1 and ν_2 are incident on two photosensitive materials having work functions W_1 and W_2 ($\mathbf{W_1 > W_2}$) respectively. The kinetic energy of the emitted electrons is same in both the cases. Which one of the two radiations will be of higher frequency?

[2007]

SOLUTION

By Einstein's photoelectric equation, the kinetic energy of the emitted electrons is

$$KE = h\nu - W$$

Since the kinetic energy is stated to be the same for both materials

$$h\nu_1 - W_1 = h\nu_2 - W_2$$

Rearranging to compare the frequencies

$$h(\nu_1 - \nu_2) = W_1 - W_2$$

It is given that $W_1 > W_2$, so the right-hand side is positive, which forces

$$\nu_1 - \nu_2 > 0 \Rightarrow \nu_1 > \nu_2$$

Hence the radiation incident on the **first material, of frequency ν_1, has the higher frequency**. This makes sense physically: a surface with a larger work function needs a more energetic (higher-frequency) photon to release electrons with the same kinetic energy.

Final answer. The radiation of frequency ν_1 (incident on the higher work function material) has the higher frequency.

← *Back to Q118*

Q119

With what purpose was famous Davisson-Germer experiment with electrons performed?

[2006]

SOLUTION

The **Davisson-Germer experiment** was performed to verify the **wave nature of electrons (moving matter)** that had been predicted by de Broglie, by showing that an electron beam undergoes diffraction, which is a purely wave phenomenon.

In the experiment, a beam of electrons accelerated through about 54 V was directed at a single crystal of nickel. The de Broglie wavelength expected for these electrons was

$$\lambda = \frac{h}{\sqrt{2meV}} \approx 1.67$$

The scattered electrons showed a sharp peak in intensity at a scattering angle of 50°, exactly as expected from Bragg-type constructive interference of waves of this wavelength.

The measured wavelength agreed closely with the de Broglie prediction, giving the first **direct experimental confirmation that matter has wave-like character**.

Final answer. It was performed to confirm the wave nature of electrons (de Broglie's hypothesis) by demonstrating electron diffraction from a nickel crystal.

← *Back to Q119*

Q120

Electrons are emitted from a photosensitive surface when it is illuminated by green light but electron emission does not take place by yellow light. Will the electrons be emitted when the surface is illuminated by (i) red light, and (ii) blue light?

[2005]

SOLUTION

Photoemission takes place only when the energy of each incident photon, $E = h\nu$, is at least equal to the work function ϕ_0 of the surface. Since photon energy rises with frequency (falls with wavelength), there is a threshold: emission occurs only for light bluer (higher frequency, shorter wavelength) than a certain colour.

Given facts and approximate photon energies:

Green light (≈ **550 nm** ≈ **2.25 eV**): emission occurs, so $\phi_0 \leq 2.25$ eV.

Yellow light (≈ **580 nm** ≈ **2.14 eV**): no emission, so $\phi_0 > 2.14$ **eV**.

Hence the work function lies between about 2.14 eV and 2.25 eV. The order of photon energies by colour is: red < yellow < green < blue.

(i) **Red light** (≈ **700 nm** ≈ **1.77 eV**): its photon energy is even less than that of yellow, which already failed, so $E_{red} < \phi_0$. **No emission**.

(ii) **Blue light** (≈ **450 nm** ≈ **2.76 eV**): its photon energy is greater than that of green, which already worked, so $E_{blue} > \phi_0$. **Emission takes place**.

TIP

Compare colours by photon energy, not by brightness: red < yellow < green < blue. If yellow fails, anything redder fails too; if green works, anything bluer works.

Final answer. No electrons are emitted with red light, but electrons are emitted with blue light.

← *Back to Q120*

Q121

In a photoelectric experiment, the emitter plate is irradiated with radiation of 200 nm. The photocurrent becomes zero when the collector plate potential is −0.80 V. Calculate the work function (in eV) of the emitter.

[2026 • Set 55-1-1]

SOLUTION

By Einstein's photoelectric equation, the maximum kinetic energy of the emitted photoelectrons equals the energy of the incident photon minus the work function of the metal.

When the collector is made negative enough to stop even the fastest electrons, the photocurrent falls to zero. The magnitude of this stopping potential V_0 gives the maximum kinetic energy through $\mathbf{eV_0 = K_{max}}$.

Here the current becomes zero at a collector potential of −0.80 V, so the stopping potential is $\mathbf{V_0 = 0.80}$ V and $\mathbf{K_{max} = eV_0 = 0.80}$ eV.

The energy of one incident photon is

$$E = \frac{hc}{\lambda}$$

Using the convenient form $\mathbf{hc = 1240\ eV}$ with $\boldsymbol{\lambda = 200\ nm}$,

$$E = \frac{1240}{200} = 6.2 \text{ eV}$$

The work function follows from Einstein's equation $\boldsymbol{E = \phi + eV_0}$,

$$\phi = E - eV_0$$

$$\phi = 6.2 - 0.80$$

$$\phi = 5.4 \text{ eV}$$

The **work function of the emitter is 5.4 eV**, a fairly high value typical of refractory metals such as nickel or platinum.

TIP

Reading −0.80 V as a stopping potential of magnitude 0.80 V is the key step; the minus sign only tells you the collector is negative.

Final answer. The work function of the emitter is $\phi = 5.4$ eV.

← *Back to Q121*

Q122

Find the ratio $\frac{\lambda_\alpha}{\lambda_p}$ *of de Broglie wavelength* λ_α *associated with an alpha particle to de Broglie wavelength* λ_p *associated with a proton if both are moving with the (a) same velocity (b) same kinetic energy.*

[2026 • Set 55-1-2]

SOLUTION

The de Broglie wavelength of a particle of mass m moving with speed v is

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

An alpha particle is four times as massive as a proton, so $\mathbf{m_\alpha = 4m_p}$.

(a) Same velocity: with v common to both, $\lambda \propto \frac{1}{m}$.

$$\frac{\lambda_\alpha}{\lambda_p} = \frac{m_p}{m_\alpha} = \frac{m_p}{4m_p}$$

$$\frac{\lambda_\alpha}{\lambda_p} = \frac{1}{4}$$

(b) Same kinetic energy: writing momentum in terms of kinetic energy, $K = \frac{p^2}{2m}$, so $p = \sqrt{2mK}$ and

$$\lambda = \frac{h}{\sqrt{2mK}}$$

For a fixed K, $\lambda \propto \frac{1}{\sqrt{m}}$.

$$\frac{\lambda_\alpha}{\lambda_p} = \sqrt{\frac{m_p}{m_\alpha}} = \sqrt{\frac{1}{4}}$$

$$\frac{\lambda_\alpha}{\lambda_p} = \frac{1}{2}$$

Hence **the ratio is 1:4 for equal velocities and 1:2 for equal kinetic energies**.

Final answer. (a) $\lambda_\alpha:\lambda_p = 1:4$; (b) $\lambda_\alpha:\lambda_p = 1:2$.

← *Back to Q122*

Q123

In an electron microscope, accelerated electrons have wavelength of 0.011 nm. Calculate the voltage through which electrons were accelerated to attain this wavelength.
(Take $e = 1.6 \times 10^{-19}$ *C,* $m_e = 9 \times 10^{-31}$ *kg,* $h = 6.6 \times 10^{-34}$ *J·s)*

[2026 • Set 55-1-3]

SOLUTION

An electron accelerated from rest through a potential difference V gains kinetic energy $K = eV$. Its de Broglie wavelength is

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2m_e K}} = \frac{h}{\sqrt{2m_e eV}}$$

Rearranging for the accelerating voltage,

$$V = \frac{h^2}{2m_e e \lambda^2}$$

Converting the wavelength, $\lambda = 0.011\ \text{nm} = 1.110^{-11}\ m$, with $h = 6.610^{-34}\ J\cdots$, $m_e = 910^{-31}\ \text{kg}$ and $e = 1.610^{-19}\ C$.

The numerator is

$$h^2 = (6.610^{-34})^2 = 4.35610^{-67}\ J^2s^2$$

The denominator is

$$2m_e\, e\, \lambda^2 = 2(910^{-31})(1.610^{-19})(1.110^{-11})^2$$

$$2m_e\, e\, \lambda^2 = 2(910^{-31})(1.610^{-19})(1.2110^{-22})$$

$$2m_e\, e\, \lambda^2 = 3.48510^{-71}$$

Therefore

$$V = \frac{4.35610^{-67}}{3.48510^{-71}}$$

$$V \approx 1.2510^{4}\ V$$

The electrons were accelerated through about **12.5 kV**.

Final answer. The accelerating voltage is $V \approx 1.2510^4\ V \approx 12.5$ kV.

← *Back to Q123*

Q124

Find the ratio $(\lambda_\alpha/\lambda_p)$ of the de Broglie wavelengths λ_α and λ_p associated respectively with an alpha particle and a proton:
(i) if they are moving with the same kinetic energy.
(ii) just after they are accelerated through the same potential difference.

[2026 • Set 55-2-1]

SOLUTION

The de Broglie wavelength is $\lambda = \dfrac{h}{p}$. For non-relativistic motion the momentum is related to kinetic energy by $p = \sqrt{2mK}$, and for a charge accelerated through a potential difference V the kinetic energy is $K = qV$.

For the two particles, $m_\alpha = 4m_p$, $q_p = e$ and $q_\alpha = 2e$.

(i) Same kinetic energy: writing $\lambda = \frac{h}{\sqrt{2mK}}$ with K common,

$$\frac{\lambda_\alpha}{\lambda_p} = \sqrt{\frac{m_p}{m_\alpha}} = \sqrt{\frac{m_p}{4m_p}}$$

$$\frac{\lambda_\alpha}{\lambda_p} = \sqrt{\frac{1}{4}} = \frac{1}{2}$$

(ii) Same accelerating potential: now $K = qV$, so $\lambda = \frac{h}{\sqrt{2mqV}}$ and with V common,

$$\frac{\lambda_\alpha}{\lambda_p} = \sqrt{\frac{m_p\, q_p}{m_\alpha\, q_\alpha}} = \sqrt{\frac{m_p \cdot e}{4m_p \cdot 2e}}$$

$$\frac{\lambda_\alpha}{\lambda_p} = \sqrt{\frac{1}{8}} = \frac{1}{2\sqrt{2}}$$

Hence **the ratio is 1:2 for equal kinetic energy and $1:2\sqrt{2}$ for the same accelerating potential**.

COMMON MISTAKE

Part (ii) differs from part (i) because the alpha particle carries twice the charge, so it gains twice the energy for the same V.

Final answer. (i) $\lambda_\alpha:\lambda_p = 1:2$; (ii) $\lambda_\alpha:\lambda_p = 1:2\sqrt{2}$.

← *Back to Q124*

Q125

A particle of mass M at rest splits up into two particles of masses m_1 and m_2 having non-zero velocities. Calculate the ratio of the de Broglie wavelengths associated with the two particles.

[2026 • Set 55-3-1]

SOLUTION

The parent particle of mass M is initially at rest, so the total momentum of the system before the split is zero.

By the law of conservation of linear momentum, the total momentum after the split must also be zero.

$$m_1\vec{v}_1 + m_2\vec{v}_2 = 0$$

$$m_1\vec{v}_1 = -m_2\vec{v}_2$$

The two fragments therefore fly apart in opposite directions, but the magnitudes of their momenta are equal.

$$|p_1| = |p_2| = p$$

The de Broglie wavelength of each fragment is $\lambda = \frac{h}{p}$.

$$\lambda_1 = \frac{h}{|p_1|} = \frac{h}{p}$$

$$\lambda_2 = \frac{h}{|p_2|} = \frac{h}{p}$$

Taking the ratio,

$$\frac{\lambda_1}{\lambda_2} = \frac{p}{p} = 1$$

So **the two fragments have equal de Broglie wavelengths, $\lambda_1:\lambda_2 = 1:1$**, regardless of their individual masses, because momentum conservation forces them to share the same magnitude of momentum.

Final answer. The de Broglie wavelengths are equal: $\lambda_1:\lambda_2 = 1:1$.

← *Back to Q125*

Q126

Calculate the de Broglie wavelength of a neutron having kinetic energy of 150 eV.

[2026 • Set 55-3-2]

SOLUTION

A neutron of kinetic energy K has momentum $p = \sqrt{2m_n K}$, so its de Broglie wavelength is

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2m_n K}}$$

First convert the kinetic energy from electron-volts to joules,

$$K = 150 \text{ eV} = 1501.610^{-19} J = 2.410^{-17} J$$

Using the neutron mass $\mathbf{m_n = 1.67510^{-27}\ kg}$,

$$2m_n K = 2(1.67510^{-27})(2.410^{-17})$$

$$2m_n K = 8.0410^{-44}\ \text{kg}^2\text{m}^2\text{s}^{-2}$$

$$\sqrt{2m_n K} = 2.83510^{-22}\ \text{kgs}^{-1}$$

Therefore

$$\lambda = \frac{6.62610^{-34}}{2.83510^{-22}}$$

$$\lambda \approx 2.3410^{-12}\ m$$

The de Broglie wavelength of the neutron is about **2.3410^{-12} *m*, that is 2.34 pm**.

Final answer. The de Broglie wavelength is $\lambda \approx 2.3410^{-12}$ m = **2.34** pm.

← *Back to Q126*

Q127

Light of frequency $5{\cdot}0 \times 10^{14}$ Hz *is incident on a metal surface. If the maximum speed of photoelectrons emitted is* $6{\cdot}63 \times 10^{5}\ m\ s^{-1}$*, calculate the threshold frequency for the surface.*

[2026 • Set 55-3-3]

SOLUTION

By Einstein's photoelectric equation, the maximum kinetic energy of the emitted electrons equals the photon energy minus the threshold (work-

function) energy.

$$\frac{1}{2} m\, v_{max}^{2} = h\nu - h\nu_0$$

First find the maximum kinetic energy of the photoelectrons using $\boldsymbol{m} = \mathbf{9.1110^{-31}\ kg}$ and $\mathbf{v_{max} = 6.6310^{5}\ \boldsymbol{m}\ s^{-1}}$.

$$K_{max} = \frac{1}{2}(9.1110^{-31})(6.6310^{5})^{2}$$

$$K_{max} = \frac{1}{2}(9.1110^{-31})(4.4010^{11})$$

$$K_{max} = 2.0010^{-19}\, J$$

Next the energy of the incident photon with $\boldsymbol{\nu = 5.010^{14}\ Hz}$.

$$h\nu = (6.62610^{-34})(5.010^{14})$$

$$h\nu = 3.3110^{-19}\, J$$

The threshold energy is the difference,

$$h\nu_0 = h\nu - K_{max} = 3.3110^{-19} - 2.0010^{-19}$$

$$h\nu_0 = 1.3110^{-19}\, J$$

Dividing by h gives the threshold frequency,

$$\nu_0 = \frac{1.3110^{-19}}{6.62610^{-34}}$$

$$\nu_0 \approx 1.9810^{14}\ \text{Hz}$$

The **threshold frequency of the surface is about $\mathbf{1.9810^{14}\ Hz}$**.

Final answer. The threshold frequency is $\boldsymbol{\nu_0 \approx 1.9810^{14}}$ Hz.

← Back to Q127

Q128

A light of wavelength 400 nm is incident on metal surface whose work function is 3.0×10^{-19} *J. Calculate the speed of the fastest photoelectrons emitted.*

[2025 • Set 55-2-1]

SOLUTION

By Einstein's photoelectric equation, the maximum kinetic energy of the fastest electrons is the incident photon energy minus the work function.

$$\frac{1}{2} m v_{max}^2 = \frac{hc}{\lambda} - \phi$$

First find the photon energy for $\lambda = 400\ \text{nm} = 40010^{-9}\ m$.

$$E = \frac{hc}{\lambda} = \frac{(6.62610^{-34})(310^8)}{40010^{-9}}$$

$$E = 4.9710^{-19}\ J$$

The maximum kinetic energy of the photoelectrons is

$$K_{max} = E - \phi = 4.9710^{-19} - 3.010^{-19}$$

$$K_{max} = 1.9710^{-19}\ J$$

The speed of the fastest electron follows from $K_{max} = \frac{1}{2} m_e v_{max}^2$, with $m_e = 9.1110^{-31}$ kg.

$$v_{max} = \sqrt{\frac{2K_{max}}{m_e}} = \sqrt{\frac{2(1.9710^{-19})}{9.1110^{-31}}}$$

$$v_{max} = \sqrt{4.3310^{11}}$$

$$v_{max} \approx 6.5810^5\ m\ s^{-1}$$

The **fastest photoelectrons leave the surface at about $6.5810^5\ m\ s^{-1}$**.

Final answer. The maximum speed of the photoelectrons is $v_{max} \approx 6.5810^5$ m s^{-1}.

← Back to Q128

Q129

Radiations of two frequencies are incident on a metal surface of work function 2.0 eV one by one. The energies of their photons are 2.5 eV and 4.5 eV respectively. Find the ratio of the maximum speed of the electrons emitted in the two cases.

[2025 • Set 55-2-2]

SOLUTION

By Einstein's photoelectric equation, the maximum kinetic energy of the emitted electrons equals the photon energy minus the work function.

$$\frac{1}{2} m v_{max}^2 = E_{photon} - \phi$$

With a work function $\phi = 2.0$ **eV**, treat the two incident radiations separately.

For the first photon, $E_1 = 2.5$ **eV**,

$$K_1 = E_1 - \phi = 2.5 - 2.0 = 0.5 \text{ eV}$$

For the second photon, $E_2 = 4.5$ **eV**,

$$K_2 = E_2 - \phi = 4.5 - 2.0 = 2.5 \text{ eV}$$

Since $K = \frac{1}{2} m v_{max}^2$ for the same electron mass, $v_{max} \propto \sqrt{K}$.

$$\frac{v_1}{v_2} = \sqrt{\frac{K_1}{K_2}} = \sqrt{\frac{0.5}{2.5}}$$

$$\frac{v_1}{v_2} = \sqrt{\frac{1}{5}} = \frac{1}{\sqrt{5}}$$

So **the ratio of maximum speeds is $v_1 : v_2 = 1 : \sqrt{5} \approx 1 : 2.24$.**

TIP

Because the work function is the same, take ratios using kinetic energy directly in eV; you never need to convert to joules.

Final answer. The ratio of maximum speeds is $\mathbf{v_1:v_2 = 1:\sqrt{5} \approx 1:2.24}$.

← *Back to Q129*

Q130

The threshold wavelength of a metal is 450 nm. Calculate (i) the work function of the metal in eV and (ii) the maximum energy of the ejected photoelectrons in eV by incident radiation of 250 nm.

[2025 • Set 55-2-3]

SOLUTION

The work function of a metal is the energy of a photon whose wavelength equals the threshold wavelength λ_0, since at threshold all the photon energy is just enough to free an electron.

$$\phi = \frac{hc}{\lambda_0}$$

Using $\mathbf{hc = 1240\ eV}$ with $\boldsymbol{\lambda_0 = 450\ nm}$,

$$\phi = \frac{1240}{450}$$

$$\phi \approx 2.76\ \text{eV}$$

(ii) For the incident radiation of $\boldsymbol{\lambda = 250\ nm}$, first find the photon energy,

$$E_{photon} = \frac{hc}{\lambda} = \frac{1240}{250}$$

$$E_{photon} = 4.96\ \text{eV}$$

By Einstein's photoelectric equation the maximum kinetic energy is

$$K_{max} = E_{photon} - \phi = 4.96 - 2.76$$

$$K_{max} = 2.20\ \text{eV}$$

Hence **the work function is about 2.76 eV and the maximum energy of the ejected photoelectrons is about 2.20 eV**.

Final answer. (i) $\phi \approx 2.76$ eV; (ii) $K_{max} \approx 2.20$ eV.

← *Back to Q130*

Q131

The threshold frequency for a given metal is 3.6×10^{14} Hz. *If monochromatic radiations of frequency* 6.8×10^{14} Hz *are incident on this metal, find the cut-off potential for the photoelectrons.*

[2025 • Set 55-5-1]

SOLUTION

According to Einstein's photoelectric equation, the maximum kinetic energy of the emitted photoelectrons equals the photon energy minus the work function. Since the work function can be written as $\phi_0 = h\nu_0$, the cut-off (stopping) potential V_0 is given by stopping the fastest electrons, so that $eV_0 = h(\nu - \nu_0)$.

Here $\nu_0 = 3.6\times 10^{14}$ Hz is the threshold frequency and $\nu = 6.8\times 10^{14}$ Hz is the frequency of the incident radiation, with $h = 6.63\times 10^{-34}\ J\,s$ and $e = 1.6\times 10^{-19}\ C$.

First find the frequency excess available to the electrons.

$$\nu - \nu_0 = (6.8 - 3.6)\times 10^{14}\ \text{Hz} = 3.2\times 10^{14}\ \text{Hz}$$

The maximum kinetic energy carried by the photoelectrons is then

$$eV_0 = h(\nu - \nu_0) = (6.63\times 10^{-34})(3.2\times 10^{14})\ J$$

$$eV_0 = 2.12\times 10^{-19}\ J$$

Dividing by the electronic charge gives the cut-off potential.

$$V_0 = \frac{2.12\times 10^{-19}}{1.6\times 10^{-19}}\ V$$

$$V_0 \approx 1.33\ V$$

Thus a retarding potential of about 1.33 V is needed to just stop the fastest photoelectrons.

Final answer. The cut-off potential for the photoelectrons is about **1.33 *V***.

← *Back to Q131*

Q132

A laser beam of frequency $3{\cdot}0 \times 10^{14}$ Hz *produces average power of* 9 mW. *Find (i) the energy of photon of the beam, and (ii) the number of photons emitted per second on an average by the source.*

[2025 • Set 55-6-1]

SOLUTION

Light is treated as a stream of photons, each of energy $E = h\nu$, where $h = 6.626\times 10^{-34}\ J\ s$ is Planck's constant and ν is the frequency of the radiation.

For part (i), the energy of one photon of the laser beam is

$$E = h\nu = (6.626\times 10^{-34})(3.0\times 10^{14})\ J$$

$$E = 1.988\times 10^{-19}\ J$$

Expressed in electron volts, $E = \dfrac{1.988\times 10^{-19}}{1.6\times 10^{-19}} \approx 1.24$ **eV**.

So the **energy of each photon** is about $1.99\times 10^{-19}\ J$.

For part (ii), the average power is the total energy emitted per second, which equals the number of photons emitted per second multiplied by the energy of each photon, so $P = nE$. Hence the number of photons emitted per second is

$$n = \frac{P}{E} = \frac{9\times 10^{-3}}{1.988\times 10^{-19}}$$

$$n \approx 4.53\times 10^{16}\ \text{photons/}s$$

Therefore the source emits about 4.53×10^{16} photons every second.

Final answer. Each photon has energy $\approx 1.99\times 10^{-19}\ J$ and the source emits $\approx 4.53\times 10^{16}$ photons per second.

← Back to Q132

Q133

A laser beam of wavelength 500 nm *and power* 5 mW *strikes normally on a perfectly reflecting surface of area* 1 mm^2 *of a body. It rebounds back from the surface. Find the force exerted by the laser beam on the body.*

[2025 • Set 55-6-2]

SOLUTION

Each photon carries momentum $\mathbf{p_{ph}} = \frac{E}{c}$, where E is the photon energy and c is the speed of light. When a photon strikes a perfectly reflecting surface normally and rebounds straight back, its momentum reverses from $+p_{ph}$ to $-p_{ph}$, so the change in the photon's momentum has magnitude $2p_{ph}$. By Newton's third law the surface gains an equal and opposite momentum, so the force on the body equals the rate of momentum transfer.

The power of the beam is P = (energy delivered per second), so the momentum delivered per second to a reflecting surface is

$$F = \frac{2P}{c}$$

Here $P = 5\ mW = 5\times 10^{-3}\ W$ and $c = 3\times 10^{8}\ m/s$. The wavelength (500 nm) and area (1 mm^2) are not needed, since the force depends only on the power and the speed of light.

$$F = \frac{2\ (5\times 10^{-3})}{3\times 10^{8}}\ N$$

$$F = \frac{10^{-2}}{3\times 10^{8}}\ N$$

$$F \approx 3.33\times 10^{-11}\ N$$

So the laser beam pushes on the perfectly reflecting body with a force of about $3.33\times 10^{-11}\ N$. (For a perfectly absorbing surface the force would be only $\frac{P}{c} = 1.67\times 10^{-11}\ N$, since reflection doubles the momentum transfer.)

CONCEPT

Reflection doubles the momentum change, so a reflecting surface feels twice the force of an absorbing one for the same power.

Final answer. The force exerted by the laser beam on the perfectly reflecting body is about $\mathbf{3.33\times 10^{-11}\ N}$.

← *Back to Q133*

Q134

Consider a neutron (mass m) of kinetic energy E and a photon of the same energy. Let λ_n and λ_p be the de Broglie wavelength of neutron and the wavelength of photon respectively. Obtain an expression for $\frac{\lambda_n}{\lambda_p}$.

[2024 • Set 55-1-1]

SOLUTION

The de Broglie wavelength of any particle of momentum p is $\lambda = \frac{h}{p}$. We find the momentum of the neutron and of the photon separately, both in terms of the common energy E.

The neutron is non-relativistic, so its kinetic energy and momentum are related by $E = \frac{p_n^2}{2m}$, which gives

$$p_n = \sqrt{2mE}$$

Hence its de Broglie wavelength is

$$\lambda_n = \frac{h}{p_n} = \frac{h}{\sqrt{2mE}}$$

For the photon, energy and momentum are related by $E = p_p c$, so

$$p_p = \frac{E}{c}$$

and its wavelength is

$$\lambda_p = \frac{h}{p_p} = \frac{hc}{E}$$

Taking the ratio of the two wavelengths,

$$\frac{\lambda_n}{\lambda_p} = \frac{h/\sqrt{2mE}}{hc/E} = \frac{E}{c\sqrt{2mE}}$$

$$\frac{\lambda_n}{\lambda_p} = \frac{1}{c}\sqrt{\frac{E}{2m}}$$

Thus the required ratio is $\frac{\lambda_n}{\lambda_p} = \frac{1}{c}\sqrt{\frac{E}{2m}}$.

Final answer. $\frac{\lambda_n}{\lambda_p} = \frac{1}{c}\sqrt{\frac{E}{2m}}$.

← *Back to Q134*

Q135

The minimum intensity of white light that our eyes can perceive is about $0.1\ nW\ m^{-2}$. *Calculate the number of photons of this light entering our pupil (area* $0.4\ cm^2$*) per second. (Take average wavelength of white light* = **500** *nm and Planck's constant* = $\mathbf{6.6 \times 10^{-34}}\ J\ s$*)*

[2024 • Set 55-3-1]

SOLUTION

Light of intensity I falling on an area A delivers power $P = IA$. Each photon of wavelength λ carries energy $E = \frac{hc}{\lambda}$, so the number of photons arriving per second is $n = \frac{P}{E}$.

First convert the data to SI units: intensity $I = 0.1\ nW\ m^{-2} = 1 \times 10^{-10}\ W\ m^{-2}$, pupil area $A = 0.4\ cm^2 = 4 \times 10^{-5}\ m^2$, wavelength $\lambda = 500\ nm = 500 \times 10^{-9}\ m$, with $h = 6.6 \times 10^{-34}\ J\ s$ and $c = 3 \times 10^8\ m/s$.

The power entering the pupil is

$$P = IA = (1 \times 10^{-10})(4 \times 10^{-5})\ W = 4 \times 10^{-15}\ W$$

The energy of one photon of this light is

$$E = \frac{hc}{\lambda} = \frac{(6.6\times 10^{-34})(3\times 10^{8})}{500\times 10^{-9}}\ J$$

$$E = 3.96\times 10^{-19}\ J$$

The number of photons entering the pupil per second is therefore

$$n = \frac{P}{E} = \frac{4\times 10^{-15}}{3.96\times 10^{-19}}$$

$$n \approx 1.01\times 10^{4} \text{ photons per second}$$

So about 10^4 photons of this faint white light enter the eye each second, yet that is just enough to be seen.

Final answer. About $\mathbf{1.0\times 10^{4}}$ photons enter the pupil per second.

← *Back to Q135*

Q136

Light of wavelength 500 *nm is incident on caesium metal (work function* 2.14 *eV) and photoemission of electrons occurs. Calculate the (i) kinetic energy (in eV) of the fastest electrons and (ii) stopping potential for this situation. (Take* **hc = 1240** *eV·nm)*

[2024 • Set 55-3-2]

SOLUTION

By Einstein's photoelectric equation, the maximum kinetic energy of the emitted electrons is the photon energy minus the work function, $\mathbf{KE_{max} = \frac{hc}{\lambda} - \phi}$. The stopping potential V_0 is the retarding voltage that just halts the fastest electrons, so $\mathbf{eV_0 = KE_{max}}$, which means V_0 in volts is numerically equal to KE_{max} in electron volts.

Here $\mathbf{\lambda = 500\ nm}$, $\mathbf{\phi = 2.14\ eV}$ and $\mathbf{hc = 1240\ eV\ nm}$.

The energy of an incident photon is

$$E = \frac{hc}{\lambda} = \frac{1240}{500}\ \text{eV} = 2.48\ \text{eV}$$

For part (i), the kinetic energy of the fastest photoelectrons is

$$\mathrm{KE}_{\max} = E - \phi = 2.48 - 2.14$$

$$\mathrm{KE}_{\max} = 0.34 \text{ eV}$$

For part (ii), the stopping potential follows from $\mathbf{eV_0 = KE_{max}}$, so

$$V_0 = \frac{\mathrm{KE}_{\max}}{e} = \frac{0.34 \text{ eV}}{e} = 0.34\ V$$

Hence the fastest electrons leave with 0.34 eV of energy and are just stopped by a retarding potential of 0.34 *V*.

Final answer. The fastest electrons have $\mathbf{KE_{max} = 0.34\ eV}$ and the stopping potential is **0.34 *V***.

← Back to Q136

Q137

Light of wavelength 600 *nm is incident on potassium (work function* 2.3 *eV). Will photoemission of electrons occur? What is the longest wavelength that will cause photoemission of electrons? (Take* **hc = 1240** *eV·nm)*

[2024 • Set 55-3-3]

SOLUTION

Photoemission can occur only when the energy of an incident photon is at least equal to the work function of the metal, that is $\frac{hc}{\lambda} \geq \phi$. We first test whether the 600 nm light meets this condition for potassium.

With $\lambda = 600$ **nm**, $\phi = 2.3$ **eV** and **hc = 1240 eV nm**, the photon energy is

$$E = \frac{hc}{\lambda} = \frac{1240}{600} \text{ eV} = 2.07 \text{ eV}$$

Since $E = 2.07$ **eV** is less than $\phi = 2.3$ **eV**, the photon does not have enough energy to free an electron. Hence **no photoemission occurs** with 600 nm light on potassium.

The longest wavelength that can still cause emission is the threshold wavelength λ_0, for which the photon energy just equals the work function, $\frac{hc}{\lambda_0} = \phi$. Therefore

$$\lambda_0 = \frac{hc}{\phi} = \frac{1240}{2.3}\ \text{nm}$$

$$\lambda_0 \approx 539\ \text{nm}$$

So only light of wavelength 539 nm or shorter (towards the green-blue end and beyond) can eject electrons from potassium.

Final answer. No photoemission occurs at **600 nm**; the longest wavelength that causes emission is about **539 nm**.

← *Back to Q137*

Q138

A proton and α-particle are accelerated through different potentials V_1 *and* V_2 *respectively so that they have the same de Broglie wavelengths. Find* $\frac{V_1}{V_2}$.

[2024 • Set 55-4-1]

SOLUTION

A charge q of mass m accelerated through a potential difference V gains kinetic energy qV, so $\frac{p^2}{2m} = qV$, giving momentum $p = \sqrt{2mqV}$. Its de Broglie wavelength is therefore

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mqV}}$$

The two particles are required to have equal de Broglie wavelengths, so $\lambda_p = \lambda_\alpha$, which means the quantities under the square root must be equal.

$$\sqrt{2\, m_p\, q_p\, V_1} = \sqrt{2\, m_\alpha\, q_\alpha\, V_2}$$

Squaring, $m_p\, q_p\, V_1 = m_\alpha\, q_\alpha\, V_2$.

For a proton, m_p and $\mathbf{q_p = e}$. For an α-particle, $\mathbf{m_\alpha = 4m_p}$ and $\mathbf{q_\alpha = 2e}$. Rearranging for the ratio of potentials,

$$\frac{V_1}{V_2} = \frac{m_\alpha\, q_\alpha}{m_p\, q_p} = \frac{(4m_p)(2e)}{(m_p)(e)}$$

$$\frac{V_1}{V_2} = 8$$

Hence $\mathbf{\frac{V_1}{V_2} = 8}$. The proton must be accelerated through eight times the potential of the α-particle, because the heavier and more highly charged α-particle reaches the same momentum at a much smaller voltage.

Final answer. $\mathbf{\frac{V_1}{V_2} = 8}$.

← *Back to Q138*

Q139

What are matter waves? A proton, an electron and an α-particle have the same kinetic energy. Write the de Broglie wavelengths associated with them in increasing order.

[2024 • Set 55-4-2]

SOLUTION

Matter waves are the waves associated with every moving material particle. According to de Broglie, a particle of momentum p behaves like a wave of wavelength $\lambda = \frac{h}{p}$, where h is Planck's constant. These are also called de Broglie waves, and they show that matter has a dual (wave-particle) nature, just as radiation does.

When a particle of mass m has kinetic energy K, its momentum is $p = \sqrt{2mK}$, so its de Broglie wavelength is

$$\lambda = \frac{h}{\sqrt{2mK}}$$

Since all three particles share the same kinetic energy K, the only variable is the mass, and

$$\lambda \propto \frac{1}{\sqrt{m}}$$

The masses increase in the order $\mathbf{m_e < m_p < m_\alpha}$ (electron lightest, proton heavier, α-particle heaviest). Because wavelength varies inversely with the square root of mass, the lightest particle has the longest wavelength.

$$\lambda_e > \lambda_p > \lambda_\alpha$$

Writing them in increasing order of wavelength gives

$$\lambda_\alpha < \lambda_p < \lambda_e$$

So for equal kinetic energy the α-particle has the shortest de Broglie wavelength and the electron the longest.

Final answer. Matter waves have $\lambda = h/p$; in increasing order $\lambda_\alpha < \lambda_p < \lambda_e$.

← *Back to Q139*

Q140

The ratio of de Broglie wavelengths of a proton and a deuteron accelerated by potential V_p *and* V_d *respectively,* $\left(\frac{\lambda_p}{\lambda_d}\right)$ *is* $\frac{1}{2}$. *Find* $\frac{V_p}{V_d}$.

[2024 • Set 55-4-3]

SOLUTION

A particle of charge q and mass m accelerated through a potential difference V acquires momentum $p = \sqrt{2mqV}$, so its de Broglie wavelength is

$$\lambda = \frac{h}{\sqrt{2mqV}}$$

For the proton and the deuteron the charge is the same, $\mathbf{q_p = q_d = e}$, while the deuteron's mass is twice that of the proton, $\mathbf{m_d = 2m_p}$.

Taking the ratio of the two wavelengths,

$$\frac{\lambda_p}{\lambda_d} = \sqrt{\frac{m_d\, q_d\, V_d}{m_p\, q_p\, V_p}} = \sqrt{\frac{2m_p\, V_d}{m_p\, V_p}} = \sqrt{\frac{2V_d}{V_p}}$$

We are told this ratio equals $\frac{1}{2}$.

$$\sqrt{\frac{2V_d}{V_p}} = \frac{1}{2}$$

Squaring both sides,

$$\frac{2V_d}{V_p} = \frac{1}{4}$$

Rearranging for the required ratio,

$$\frac{V_p}{V_d} = 2\times 4 = 8$$

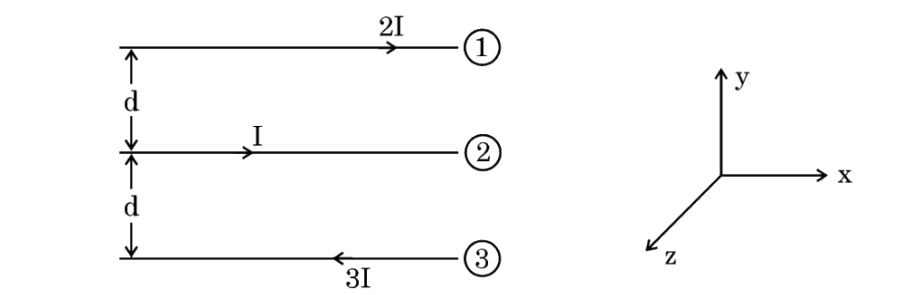
Hence $\frac{V_p}{V_d} = 8$, i.e. the potentials are in the ratio 8 : 1.

Final answer. $\frac{V_p}{V_d} = 8$ (that is, $V_p : V_d = 8 : 1$).

← *Back to Q140*

Q141

The figure shows v_m^2 *versus* $\frac{1}{\lambda}$ *graph for photoelectrons emitted from a surface where* v_m *is the maximum speed of electrons and* λ *is the wavelength of incident radiation. Using this graph and Einstein's photoelectric equation, obtain the expression for Planck's constant and work function of the surface.*



[2023 • Set 55-4-1]

SOLUTION

By Einstein's photoelectric equation, the maximum kinetic energy of an emitted photoelectron equals the photon energy minus the work function of the surface.

$$\frac{1}{2} m v_m^2 = \frac{hc}{\lambda} - \phi$$

Here h is Planck's constant, c is the speed of light, λ is the wavelength of the incident radiation, m is the electron mass and ϕ is the work function.

To match the axes of the given graph, multiply throughout by $\frac{2}{m}$.

$$v_m^2 = \frac{2hc}{m} \cdot \frac{1}{\lambda} - \frac{2\phi}{m}$$

This is of the form $y = sx + c$, where $y = v_m^2$ and $x = \frac{1}{\lambda}$. Hence the plot of v_m^2 versus $\frac{1}{\lambda}$ is a **straight line**.

Comparing, the slope of the line is $s = \frac{2hc}{m}$.

Solving this for Planck's constant gives

$$h = \frac{m s}{2c}$$

The y-intercept of the line is $-\frac{2\phi}{m}$, which is negative. Equivalently, the line meets the x-axis (where $v_m^2 = 0$) at $\frac{1}{\lambda_0}$, the threshold value.

Putting $v_m^2 = 0$ in $\frac{1}{2} m v_m^2 = \frac{hc}{\lambda} - \phi$ gives $\phi = \frac{hc}{\lambda_0}$.

$$\phi = hc \cdot \frac{1}{\lambda_0}$$

So, read off the slope s of the straight line and its x-intercept $\frac{1}{\lambda_0}$ from the graph; then $h = \frac{ms}{2c}$ and the **work function** $\phi = hc(\frac{1}{\lambda_0})$.

TIP

The graph is a straight line, so every quantity must come from its slope or an intercept; never read a single random point.

Final answer. Planck's constant $h = \frac{ms}{2c}$ (from the slope s) and work function $\phi = \frac{hc}{\lambda_0}$ (from the x-intercept $1/\lambda_0$).

← Back to Q141

Q142

(a) How will the De Broglie wavelength associated with an electron be affected when the (i) velocity of the electron decreases? and (ii) accelerating potential is increased? Justify your answer.

OR

(b) How would the stopping potential for a given photosensitive surface change if (i) the frequency of the incident radiation were increased? and (ii) the intensity of incident radiation were decreased? Justify your answer.

[2023 • Set 55-5-1]

SOLUTION

(a) The de Broglie wavelength of an electron is $\lambda = \frac{h}{p} = \frac{h}{mv}$, where $p = mv$ is its momentum.

(i) When the velocity of the electron decreases:

$$\lambda = \frac{h}{mv}$$

Since h and m are constant, $\lambda \propto \frac{1}{v}$. If v decreases, λ increases.

So the de Broglie wavelength **increases** when the velocity decreases.

(ii) When the accelerating potential is increased: an electron accelerated from rest through a potential difference V gains kinetic energy $eV = \frac{1}{2}mv^2$, so its momentum is $p = \sqrt{2meV}$.

$$\lambda = \frac{h}{\sqrt{2meV}}$$

Hence $\lambda \propto \frac{1}{\sqrt{V}}$. If V is increased, λ decreases.

So the de Broglie wavelength **decreases** when the accelerating potential is increased.

OR

(b) By Einstein's photoelectric equation, the stopping potential V_0 is given by $\mathbf{eV_0 = h\nu - \phi}$, where ν is the frequency of the incident radiation and ϕ the work function.

$$V_0 = \frac{h\nu}{e} - \frac{\phi}{e}$$

(i) When the frequency ν is increased: from the equation, V_0 rises linearly with ν (slope h/e). So the stopping potential **increases**.

(ii) When the intensity of incident radiation is decreased: the stopping potential depends only on the frequency and work function, not on intensity. Intensity controls only the number of photoelectrons (the saturation current). So the stopping potential remains **unchanged**.

Final answer. (a) λ increases as ν falls and decreases as V rises. (b) V_0 increases with frequency and stays unchanged when intensity is lowered.

← *Back to Q142*

Q143

Plot a graph showing the variation of photo electric current, as a function of anode potential for two light beams having the same frequency but different intensities I_1 *and* I_2 *(*$\mathbf{I_2 > I_1}$*). Mention its important features.*

[2023 • Set 55-5-3]

SOLUTION

The required graph plots the photoelectric current I (along the y-axis) against the anode (collector plate) potential V (along the x-axis) for two beams of the same frequency but different intensities, with $\mathbf{I_2 > I_1}$.

Description of the graph: draw the V-axis horizontally with the origin in the middle, so it has a positive (accelerating) side on the right and a negative (retarding) side on the left. Draw the I-axis vertically upward.

Two curves are drawn. Starting from the left, on the negative side each curve rises from zero current, climbs through the origin region as V increases, and then flattens into a horizontal line (saturation) for large positive V. The upper curve is labelled I_2 and the lower curve is labelled I_1.

Both curves meet the V-axis (zero current) at the same negative potential, marked $-V_0$, the stopping potential.

Important features:

• Both curves cut the potential axis at the same point $-V_0$, so they have the same stopping potential. This is because V_0 depends only on frequency, which is the same for both beams: $\mathbf{eV_0 = h\nu - \phi}$.

• For large positive V each curve becomes flat, giving the saturation current. The beam of higher intensity I_2 gives a larger saturation current than I_1, because saturation current is proportional to intensity, i.e. to the number of photons (and hence photoelectrons) arriving per second.

• Below saturation the two curves have the same shape; the higher-intensity curve simply lies above the lower-intensity one (scaled vertically).

Final answer. Two curves with the same stopping potential $\mathbf{-V_0}$ but the $\mathbf{I_2}$ curve saturating at a higher current than the $\mathbf{I_1}$ curve, since saturation current is proportional to intensity.

← *Back to Q143*

Q144

Plot suitable graphs to show the variation of photoelectric current with the collector plate potential for the incident radiation of
(i) the same intensity but different frequencies ν_1, ν_2 *and* ν_3 $\mathbf{(\nu_1 < \nu_2 < \nu_3)}$
(ii) the same frequency but different intensities I_1, I_2 *and* I_3 $\mathbf{(I_1 < I_2 < I_3)}$

[2022 • Set 55-1-1]

SOLUTION

Both parts plot photoelectric current I (along the y-axis) against the collector plate potential V (along the x-axis), with the origin in the middle so that V has a

retarding (negative) side and an accelerating (positive) side.

(i) Same intensity, different frequencies $\nu_1 < \nu_2 < \nu_3$:

Draw three curves that all rise from the negative-V side, pass upward, and flatten to the same horizontal saturation level on the positive side. They all reach the same saturation current because the intensity (number of photons per second) is the same for all three.

However, each curve cuts the V-axis at a different negative point. The stopping potentials are marked $-V_{0,1}$, $-V_{0,2}$, $-V_{0,3}$, lying progressively farther to the left.

$$eV_0 = h\nu - \phi$$

Since higher frequency gives a larger stopping potential, $|V_{0,1}| < |V_{0,2}| < |V_{0,3}|$. The curve for ν_3 is cut off at the most negative potential.

(ii) Same frequency, different intensities $I_1 < I_2 < I_3$:

Draw three curves that all meet the V-axis at the same negative point $-V_0$, because the frequency (and hence stopping potential) is the same for all three.

$$eV_0 = h\nu - \phi \text{ (same for all)}$$

But the curves saturate at different heights. The saturation current is proportional to intensity, so $I_{sat,1} < I_{sat,2} < I_{sat,3}$, with the I_3 curve lying highest.

Final answer. (i) Three curves with the same saturation current but increasing stopping potentials as frequency rises. (ii) Three curves with the same stopping potential but increasing saturation currents as intensity rises.

← *Back to Q144*

Q145

Photoelectric emission occurs when a surface is irradiated with the radiation of frequency (i) ν_1 and (ii) ν_2. The maximum kinetic energy of the electrons emitted in the two cases are K and 2K respectively. Obtain the expression for the threshold frequency for the surface.

[2022 • Set 55-2-1]

SOLUTION

By Einstein's photoelectric equation, the maximum kinetic energy of an emitted photoelectron is the photon energy minus the work function, written in terms of the threshold frequency ν_0 as $\mathbf{K_{max} = h\nu - h\nu_0}$.

For the first radiation of frequency ν_1 the maximum kinetic energy is K.

$$K = h\nu_1 - h\nu_0 \ldots(1)$$

For the second radiation of frequency ν_2 the maximum kinetic energy is 2K.

$$2K = h\nu_2 - h\nu_0 \ldots(2)$$

Multiply equation (1) by 2.

$$2K = 2h\nu_1 - 2h\nu_0 \ldots(3)$$

Equating the right-hand sides of (2) and (3) (both equal 2K):

$$h\nu_2 - h\nu_0 = 2h\nu_1 - 2h\nu_0$$

Cancel h throughout and rearrange.

$$\nu_2 - \nu_0 = 2\nu_1 - 2\nu_0$$

$$2\nu_0 - \nu_0 = 2\nu_1 - \nu_2$$

$$\nu_0 = 2\nu_1 - \nu_2$$

So the **threshold frequency** of the surface is $\mathbf{\nu_0 = 2\nu_1 - \nu_2}$.

Final answer. The threshold frequency of the surface is $\mathbf{\nu_0 = 2\nu_1 - \nu_2}$.

← Back to Q145

Q146

(i) Name the factors on which photoelectric emission from a surface depends.
(ii) Define the term 'threshold frequency' for a photosensitive material.

[2022 • Set 55-3-1]

SOLUTION

(i) Factors on which photoelectric emission from a surface depends:

• Frequency of the incident radiation: emission occurs only when the frequency exceeds the threshold frequency ν_0 of the surface; the frequency also fixes the maximum kinetic energy of the emitted electrons.

• Intensity of the incident radiation: once $\nu > \nu_0$, the intensity decides the number of photoelectrons emitted per second, that is, the magnitude of the photoelectric (saturation) current.

• Nature of the photosensitive surface: this fixes the work function ϕ (and hence $\nu_0 = \phi/h$) of the material.

(ii) Threshold frequency: the **threshold frequency** ν_0 of a photosensitive material is the minimum frequency of the incident radiation below which no photoelectric emission takes place, however high the intensity of the radiation may be.

It is related to the work function by $\nu_0 = \frac{\phi}{h}$.

Final answer. Emission depends on the frequency and intensity of the incident light and the nature (work function) of the surface; the threshold frequency ν_0 is the minimum frequency below which no emission occurs, with $\nu_0 = \phi/h$.

← *Back to Q146*

Q147

What are matter waves? A proton and an alpha particle are accelerated through the same potential difference. Find the ratio of the de Broglie wavelength associated with the proton to that with the alpha particle.

[2022 • Set 55-4-1]

SOLUTION

Matter waves: every moving material particle has a wave associated with it, called a matter wave (or de Broglie wave). Its wavelength is given by $\lambda = \frac{h}{p} =$

$\frac{h}{mv}$, where p is the momentum of the particle. These waves are not electromagnetic; they represent the probability of finding the particle and are significant only for very small particles such as electrons.

Now consider a particle of mass m and charge q accelerated from rest through a potential difference V. The kinetic energy gained is

$$\frac{1}{2}mv^2 = qV$$

so its momentum is

$$p = \sqrt{2mqV}$$

and the de Broglie wavelength is

$$\lambda = \frac{h}{\sqrt{2mqV}}$$

Since h and V are the same for both particles, $\lambda \propto \frac{1}{\sqrt{mq}}$.

Taking the ratio for the proton (mass m_p, charge q_p) to the alpha particle (mass $m_\alpha = 4m_p$, charge $q_\alpha = 2q_p$):

$$\frac{\lambda_p}{\lambda_\alpha} = \sqrt{\frac{m_\alpha\, q_\alpha}{m_p\, q_p}}$$

$$\frac{\lambda_p}{\lambda_\alpha} = \sqrt{\frac{(4m_p)(2q_p)}{m_p\, q_p}}$$

$$\frac{\lambda_p}{\lambda_\alpha} = \sqrt{8} = 2\sqrt{2}$$

$$\lambda_p : \lambda_\alpha = 2\sqrt{2} : 1 \approx 2.83 : 1$$

COMMON MISTAKE

For a charged particle accelerated through V, always use $\lambda = h/\sqrt{2mqV}$*, so* $\lambda \propto 1/\sqrt{mq}$*; both mass and charge enter the ratio.*

Final answer. Matter waves are the probability waves of moving particles, $\lambda = h/mv$; the ratio is $\lambda_p : \lambda_\alpha = 2\sqrt{2} : 1 \approx 2.83 : 1$.

← Back to Q147

Q148

Why it is the frequency and not the intensity of light source that determines whether emission of photoelectrons will occur or not? Explain.

[2022 • Set 55-5-1]

SOLUTION

According to the photon (quantum) picture of light, a light beam consists of discrete packets of energy called photons, each photon carrying energy

$$E = h\nu$$

where h is Planck's constant and ν is the frequency of the light.

In photoelectric emission, a single photon is absorbed completely by a single electron in one event, handing over all its energy $h\nu$ at once. The electron can be ejected only if this energy is at least equal to the work function ϕ of the metal.

$$h\nu \geq \phi$$

That is, emission occurs only when

$$\nu \geq \frac{\phi}{h} = \nu_0$$

So whether or not an electron is freed depends on the energy of each individual photon, which is decided by the frequency ν. If $\boldsymbol{\nu < \nu_0}$, no single photon has enough energy and no emission takes place.

The intensity of light only fixes the number of photons arriving per second, not the energy carried by each photon. Because an electron absorbs energy from one photon at a time and cannot store or add up the energy of several low-energy photons, increasing the intensity at $\boldsymbol{\nu < \nu_0}$ still cannot eject any electron. It merely sends more photons, each still too weak.

Hence it is the **frequency**, not the intensity, that decides whether photoelectric emission occurs. Once $\boldsymbol{\nu > \nu_0}$, intensity then controls only how many electrons

are emitted (the photocurrent).

Final answer. Because a single electron absorbs one whole photon of energy **hν** at a time, emission needs each photon to satisfy **hν ≥ ϕ**, which is set by frequency, not intensity.

← *Back to Q148*

Q149

For the light of wavelength 400 nm *incident on the cathode of a photocell, the stopping potential is* 6 *V. If the wavelength of incident light is increased to* 600 nm, *calculate the new stopping potential. (Take* $h = 4.14 \times 10^{-15}$ eV⋯*)*

[2021]

SOLUTION

By Einstein's photoelectric equation, the energy of the stopping potential equals the photon energy minus the work function.

$$eV_0 = \frac{hc}{\lambda} - \phi$$

Writing it for the two wavelengths with stopping potentials V_1 and V_2:

$$eV_1 = \frac{hc}{\lambda_1} - \phi \ldots(1)$$

$$eV_2 = \frac{hc}{\lambda_2} - \phi \ldots(2)$$

The work function ϕ is the same for the same cathode, so subtract (2) from (1) to eliminate it.

$$e(V_1 - V_2) = hc\left(\frac{1}{\lambda_1} - \frac{1}{\lambda_2}\right)$$

Solving for the new stopping potential V_2:

$$V_2 = V_1 - \frac{hc}{e}\left(\frac{1}{\lambda_1} - \frac{1}{\lambda_2}\right)$$

Use $\mathbf{hc = (4.14\times 10^{-15}\ eV\cdots)(3\times 10^{8}\ m/s) = 1242\times 10^{-9}\ eV \approx 1240\ eV}$, so $\frac{hc}{e}$ contributes 1240 V.

With $\lambda_1 = 400$ nm, $\lambda_2 = 600$ nm and $V_1 = 6\ V$:

$$V_2 = 6 - 1240\left(\frac{1}{400} - \frac{1}{600}\right)$$

$$\frac{1}{400} - \frac{1}{600} = \frac{3-2}{1200} = \frac{1}{1200}\ \text{nm}^{-1}$$

$$V_2 = 6 - \frac{1240}{1200}$$

$$V_2 = 6 - 1.03$$

$$V_2 \approx 4.97\ V \approx 5\ V$$

So the new stopping potential is about **5 V**.

> **TIP**
>
> Subtract the two Einstein equations to cancel the work function; then only the wavelengths and the given stopping potential are needed.

Final answer. The new stopping potential at **600 nm** is $V_2 \approx 4.97\ V \approx 5\ V$.

← *Back to Q149*

Q150

(a) Write two main observations of photoelectric effect experiment which could only be explained by Einstein's photoelectric equation.
(b) Draw a graph showing variation of photocurrent with the anode potential of a photocell.

[2020 • Set 55-1-1]

SOLUTION

(a) Two observations of the photoelectric effect that could be explained only by Einstein's photoelectric equation:

• Existence of a threshold frequency: for a given surface, no photoelectrons are emitted if the frequency of the incident light is below a certain value ν_0, however intense the light. Einstein's equation $\mathbf{K_{max} = h\nu - \phi}$ gives a positive kinetic energy only when $\boldsymbol{\nu > \nu_0 = \phi/h}$; for $\boldsymbol{\nu < \nu_0}$ no emission is possible.

• Maximum kinetic energy depends on frequency, not intensity: the maximum kinetic energy of the photoelectrons increases linearly with the frequency of light and is completely independent of its intensity. Einstein's equation $\mathbf{K_{max} = h\nu - \phi}$ shows K_{max} rising with ν (slope h) but containing no intensity term, since each electron absorbs just one photon of energy $h\nu$.

(b) Graph of photocurrent versus anode potential:

Plot the photoelectric current I on the y-axis against the anode potential V on the x-axis, with the origin in the middle so V has a negative (retarding) side and a positive (accelerating) side.

Draw one curve that starts from zero current at a negative potential $-V_0$ (the stopping potential), rises as V increases through the origin, and then becomes a horizontal line for large positive V, marked as the saturation current I_{sat}.

Mark $-V_0$ on the negative axis where the curve meets zero current, and label the flat portion I_{sat}. Even at small negative potentials some current flows, because the most energetic photoelectrons can still reach the anode; the current falls to zero only at $\boldsymbol{V = -V_0}$.

Final answer. (a) The threshold frequency and the dependence of $\mathbf{K_{max}}$ on frequency (not intensity). (b) A curve rising from zero at the stopping potential $\mathbf{-V_0}$ to a flat saturation current $\mathbf{I_{sat}}$ at high positive anode potential.

← *Back to Q150*

Q151

Explain, how the process of emission of photoelectrons is different from the process of emission of β-particles.

[2020 • Set 55-1-3]

SOLUTION

Both processes throw out electrons, but their origin and nature are completely different. The two can be compared point by point.

Origin: In photoelectric emission, the ejected electrons are the loosely bound electrons of the metal surface (outer atomic or conduction electrons). In β-emission, the electron comes from the nucleus itself, where a neutron converts into a proton: $n \to p + e^- + \bar{\nu}_e$.

Cause: Photoelectric emission is triggered externally by photons of frequency $\nu > \nu_0$ striking the surface. β-emission is a spontaneous nuclear process caused by the instability of the nucleus and cannot be started or stopped by any external agency.

Energy spectrum: Photoelectrons of a given metal, for a fixed incident frequency, have a well defined maximum kinetic energy $K_{max} = h\nu - \phi_0$. The β-particles show a continuous energy spectrum (from zero up to a maximum) because the energy is shared with the emitted antineutrino $\bar{\nu}_e$.

Effect on the source: In photoelectric emission the atomic number of the metal is unchanged. In β-emission the atomic number of the nucleus increases by one, while the mass number stays the same, i.e. $(Z, A) \to (Z+1, A)$.

Thus photoelectric emission is an **atomic, externally driven** process giving discrete energy, whereas β-emission is a **spontaneous nuclear** process giving a continuous spectrum.

Final answer. Photoelectric emission ejects outer atomic electrons via incident photons with discrete energy and no change in Z, while beta emission is a spontaneous nuclear decay giving a continuous energy spectrum and Z increasing by one.

← *Back to Q151*

Q152

Define the terms (a) threshold frequency, and (b) stopping potential. How were these terms incorporated in Einstein's photoelectric equation?

[2020 • Set 55-3-1]

SOLUTION

(a) Threshold frequency (ν_0): It is the minimum frequency of the incident radiation below which no photoelectrons are emitted from a given metal surface, however high the intensity of the radiation may be. It is a characteristic constant of the metal and is related to its work function by $\boldsymbol{\phi_0 = h\nu_0}$.

(b) Stopping potential (V_0): It is the minimum value of the negative (retarding) potential applied to the collector (anode) at which the photoelectric current just becomes zero. It is a measure of the maximum kinetic energy of the emitted photoelectrons, since the work done by the retarding field stops even the fastest electron:

$$eV_0 = K_{max} = \frac{1}{2} m v_{max}^2$$

How they enter Einstein's photoelectric equation: Einstein's equation gives the maximum kinetic energy of a photoelectron as

$$K_{max} = h\nu - \phi_0$$

Using the work function in terms of threshold frequency, $\boldsymbol{\phi_0 = h\nu_0}$:

$$K_{max} = h\nu - h\nu_0 = h(\nu - \nu_0)$$

Replacing K_{max} by the stopping potential through $\boldsymbol{K_{max} = eV_0}$:

$$eV_0 = h\nu - h\nu_0$$

This shows that **stopping potential** V_0 varies linearly with the frequency ν, with slope h/e and an x-intercept equal to the **threshold frequency** ν_0.

Final answer. Threshold frequency is the minimum frequency for emission ($\boldsymbol{\phi_0 = h\nu_0}$) and stopping potential is the retarding voltage at which current is zero ($\boldsymbol{eV_0 = K_{max}}$); both combine to give $\boldsymbol{eV_0 = h(\nu - \nu_0)}$.

← Back to Q152

Q153

The de Broglie wavelengths associated with an electron and a proton are equal. Prove that the kinetic energy of the electron is greater than that of the proton.

[2020 • Set 55-3-2]

SOLUTION

Given that the de Broglie wavelengths of the electron and proton are equal, $\lambda_e = \lambda_p$.

The de Broglie wavelength is related to the momentum by

$$\lambda = \frac{h}{p}$$

Since the wavelengths are equal, the momenta must also be equal:

$$\lambda_e = \lambda_p \Rightarrow p_e = p_p = p$$

The kinetic energy of a particle in terms of its momentum is

$$K = \frac{p^2}{2m}$$

For the same momentum p, the kinetic energy is inversely proportional to the mass:

$$K \propto \frac{1}{m}$$

Taking the ratio of the two kinetic energies for equal momenta,

$$\frac{K_e}{K_p} = \frac{m_p}{m_e}$$

Now $\mathbf{m_e = 9.11 \times 10^{-31}\ kg}$ and $\mathbf{m_p = 1.67 \times 10^{-27}\ kg}$, so $m_p \gg m_e$ (about 1836 times larger):

$$\frac{K_e}{K_p} = \frac{m_p}{m_e} \approx 1836 > 1$$

Hence $K_e \gg K_p$. The **kinetic energy of the electron is much greater than that of the proton** (by a factor of about 1836), which proves the required result.

Final answer. Since equal wavelengths mean equal momenta and $K = p^2/2m$, we get $K_e/K_p = m_p/m_e \approx 1836$, so the electron's kinetic energy is much greater than the proton's.

← Back to Q153

Q154

(a) In what ways are matter waves different from electromagnetic waves?

(b) If the kinetic energy of a particle is reduced to one-fourth ($\frac{1}{4}$) of its initial value, how many times will the de Broglie wavelength associated with it become?

[2020 • Set 55-3-3]

SOLUTION

(a) Differences between matter waves and electromagnetic waves:

Matter waves are associated with moving material particles, whereas electromagnetic waves are produced by accelerated electric charges.

Matter waves are not electromagnetic in nature, so they are not made of oscillating electric and magnetic fields; electromagnetic waves consist of mutually perpendicular oscillating $\vec{E}$ and $\vec{B}$ fields.

The speed of a matter wave depends on the speed of the particle and is generally less than c; all electromagnetic waves travel through vacuum at the same fixed speed c.

The wavelength of a matter wave depends on the mass and velocity, $\lambda = \frac{h}{mv}$, while that of an electromagnetic wave is fixed by its frequency, $\lambda = \frac{c}{\nu}$.

(b) Effect of reducing the kinetic energy to one-fourth:

The de Broglie wavelength in terms of kinetic energy K is

$$\lambda = \frac{h}{\sqrt{2mK}}$$

so for a given particle

$$\lambda \propto \frac{1}{\sqrt{K}}$$

If the new kinetic energy is $K' = \frac{K}{4}$, the ratio of the wavelengths is

$$\frac{\lambda'}{\lambda} = \sqrt{\frac{K}{K'}} = \sqrt{\frac{K}{K/4}} = \sqrt{4} = 2$$

Therefore the **de Broglie wavelength becomes twice (doubles)** its initial value.

Final answer. Matter waves need a moving particle and travel below c with $\lambda = h/mv$, unlike EM waves; and reducing kinetic energy to one-fourth doubles the de Broglie wavelength.

← *Back to Q154*

Q155

Light of same wavelength is incident on three photo-sensitive surfaces A, B and C. The following observations are recorded.
(i) From surface A, photo electrons are not emitted.
(ii) From surface B, photo electrons are just emitted.
(iii) From surface C, photo electrons with some kinetic energy are emitted.
Compare the threshold frequencies of the three surfaces and justify your answer.
OR
If the frequency of light incident on the cathode of a photo-cell is increased, how will the following be affected? Justify your answer.
(i) Energy of the photo electrons.
(ii) Photo current.

[2020 • Set 55-4-1]

SOLUTION

Since the same wavelength of light falls on all three surfaces, every incident photon carries the same fixed energy

$$E = h\nu = \frac{hc}{\lambda} = \text{constant for } A, B \text{ and } C$$

Emission occurs only if this photon energy is at least equal to the work function $\phi_0 = h\nu_0$ of the surface, i.e. only if $\nu \geq \nu_0$.

Surface A (no emission): The photon energy is less than the work function, so $E < \phi_A$, which means $\nu < {\nu_0}^A$. Hence the threshold frequency of A is the highest, ${\nu_0}^A > \nu$.

Surface B (electrons just emitted, $K_{max} = 0$): Here the photon energy exactly equals the work function, $E = \phi_B$, so the threshold frequency equals the incident frequency, ${\nu_0}^B = \nu$.

Surface C (electrons emitted with kinetic energy): The photon energy exceeds the work function, $E > \phi_C$, so ${\nu_0}^C < \nu$. Hence the threshold frequency of C is the lowest.

Comparing the three,

$$ {\nu_0}^A > {\nu_0}^B > {\nu_0}^C $$

So surface **A has the highest and C the lowest threshold frequency**.

OR

If the frequency of the incident light is increased:

(i) Energy of the photoelectrons: From $K_{max} = h\nu - \phi_0$, increasing ν increases K_{max} linearly, so the maximum kinetic energy (and hence energy) of the photoelectrons increases.

(ii) Photocurrent: The photocurrent depends on the number of photoelectrons emitted per second, which depends on the intensity (number of incident photons per second), not on the frequency. As long as the intensity is unchanged, the photocurrent remains the same.

COMMON MISTAKE

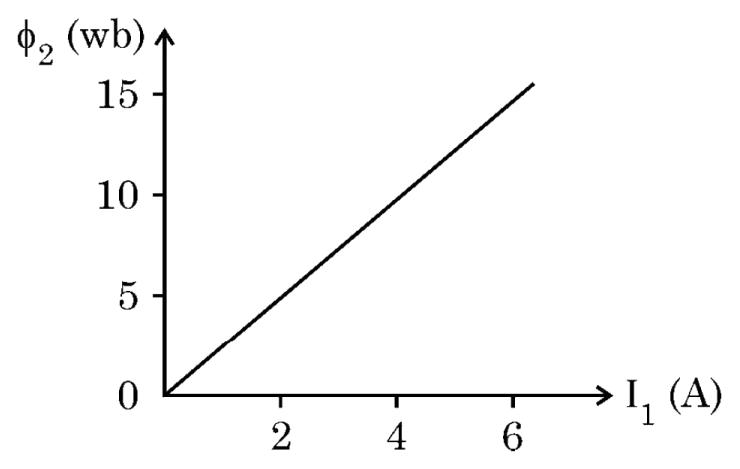
A common slip in the OR part is to say photocurrent rises with frequency. Photocurrent is controlled only by intensity (photon count per second); frequency controls energy per electron.

Final answer. Equal photon energy on all three gives ${\nu_0}^A > {\nu_0}^B > {\nu_0}^C$; (OR) raising the frequency increases the photoelectron energy but leaves the photocurrent unchanged at fixed intensity.

← Back to Q155

Q156

Figure shows the stopping potential (V_0) for the photo electron versus ($1/\lambda$) graph, for two metals A and B, λ being the wavelength of incident light.



(a) How is the value of Planck's constant determined from the graph ?
(b) If the distance between the light source and the surface of metal A is increased, how will the stopping potential for the electrons emitted from it be effected ? Justify your answer.

[2020 • Set 55-5-1]

SOLUTION

(a) Determining Planck's constant from the graph: Using Einstein's photoelectric equation with $\nu = \frac{c}{\lambda}$,

$$eV_0 = h\nu - \phi_0 = \frac{hc}{\lambda} - \phi_0$$

Dividing throughout by e,

$$V_0 = \frac{hc}{e}\left(\frac{1}{\lambda}\right) - \frac{\phi_0}{e}$$

This is a straight line of the form $y = mx + c$ when V_0 is plotted against $\frac{1}{\lambda}$, with slope

$$\text{slope} = \frac{hc}{e}$$

Both metals A and B give parallel straight lines of the same slope $\frac{hc}{e}$ (they differ only in their intercept on the $\frac{1}{\lambda}$ axis because of different work functions). Reading the slope from the graph and using the known values of e and c,

$$h = \frac{e \times (\text{slope})}{c}$$

So Planck's constant is obtained from the **slope of the V_0 versus $1/\lambda$ line**.

(b) Effect of increasing the source-to-metal distance: Increasing the distance between the source and metal A reduces the intensity of light reaching the surface (energy falling per unit area per second decreases). However, the energy of each individual photon, $\mathbf{h\nu = \frac{hc}{\lambda}}$, depends only on the wavelength and is unchanged. Since the stopping potential depends on the maximum kinetic energy, which depends only on the frequency and the work function (not on intensity),

$$eV_0 = h\nu - \phi_0 \Rightarrow V_0 \text{ unchanged}$$

Therefore the **stopping potential remains the same**; only the photoelectric current would decrease.

Final answer. Planck's constant is found from the slope of the $\mathbf{V_0}$ vs $\mathbf{1/\lambda}$ line using $\boldsymbol{h = e(\text{slope})/c}$; increasing the source distance lowers intensity but leaves the stopping potential unchanged.

← *Back to Q156*

Q157

In case of photo electric effect experiment, explain the following facts, giving reasons.
(a) The wave theory of light could not explain the existence of the threshold frequency.
(b) The photo electric current increases with increase of intensity of incident light.

[2020 • Set 55-5-2]

SOLUTION

(a) Why wave theory could not explain the threshold frequency: According to the wave theory of light, the energy of a wave depends on its intensity and the time of exposure. So a beam of any frequency, if intense enough or allowed to act long enough, should eventually supply enough energy to free an electron. This predicts that emission should occur at every frequency, given sufficient intensity or time.

But experiment shows that for each metal there is a definite threshold frequency ν_0, below which no photoelectrons are emitted however large the intensity or however long the exposure. The photon (quantum) picture explains this: light energy is delivered in discrete packets of energy $h\nu$, and one electron absorbs only one photon. Emission needs a single photon to carry at least the work function ϕ_0:

$$h\nu \geq \phi_0 = h\nu_0 \Rightarrow \nu \geq \nu_0$$

If $\boldsymbol{\nu < \nu_0}$, no single photon has enough energy, so no emission occurs whatever the intensity. Wave theory has no such frequency condition, so it fails.

(b) Why photoelectric current increases with intensity: In the photon picture, the intensity of light is proportional to the number of photons falling on the surface per second. Since each photon (of energy $\boldsymbol{h\nu \geq \phi_0}$) ejects at most one electron, a greater number of photons per second ejects a greater number of photoelectrons per second.

$$\text{Intensity} \uparrow \Rightarrow \text{photons/second} \uparrow \Rightarrow \text{electrons emitted/second} \uparrow$$

Hence the **photoelectric (saturation) current increases with the intensity** of the incident light.

Final answer. Wave theory predicts emission at any frequency given enough intensity, but the photon condition $\boldsymbol{h\nu \geq \phi_0}$ explains the threshold; and higher intensity means more photons per second, hence more electrons and a larger photocurrent.

← Back to Q157

Q158

Why is wave theory of electromagnetic radiation not able to explain photoelectric effect? How does photon picture resolve this problem?

[2019 • Set 55-1-1]

SOLUTION

Failures of the wave theory of electromagnetic radiation in explaining the photoelectric effect:

Existence of a threshold frequency: Wave theory says energy can be accumulated continuously over time, so emission should occur at any frequency if the intensity is high enough or the exposure long enough. But experimentally, no emission takes place below a certain threshold frequency ν_0, whatever the intensity.

Maximum kinetic energy independent of intensity: Wave theory predicts that a more intense (higher amplitude) wave should give faster, more energetic electrons. In fact the maximum kinetic energy of the photoelectrons depends only on the frequency ν of the light and not on its intensity.

Instantaneous emission: Even for very feeble light, photoelectrons are emitted almost instantly (within about 10^{-9} *s*). Wave theory predicts a measurable time lag while the electron slowly gathers enough energy.

How the photon picture resolves these:

Einstein proposed that light consists of discrete energy packets (photons), each of energy

$$E = h\nu$$

and that one photon is absorbed by one electron in a single instantaneous event. Then:

Emission needs one photon to carry at least the work function, $\mathbf{h\nu \geq \phi_0}$, which explains the existence of the **threshold frequency** $\mathbf{\nu_0 = \phi_0/h}$.

The maximum kinetic energy follows from energy conservation,

$$K_{max} = h\nu - \phi_0$$

which depends only on ν and not on intensity; intensity merely changes the number of photons (and hence the current), not the energy per electron.

Since a photon delivers all its energy in a single absorption, emission is prompt, explaining the absence of any time lag.

Final answer. Wave theory cannot explain the threshold frequency, the intensity-independence of $\mathbf{K_{max}}$, or the instantaneous emission; the photon picture ($\boldsymbol{E = h\nu}$, one photon per electron, $\mathbf{K_{max} = h\nu - \phi_0}$) resolves all three.

← *Back to Q158*

Q159

Plot a graph showing variation of de Broglie wavelength (λ) associated with a charged particle of mass m, versus $\frac{1}{\sqrt{V}}$, where V is the potential difference through which the particle is accelerated. How does this graph give us the information regarding the magnitude of the charge of the particle?

[2019 • Set 55-1-1]

SOLUTION

When a particle of mass m and charge q is accelerated from rest through a potential difference V, the work done by the field becomes its kinetic energy:

$$qV = \frac{1}{2}mv^2 = \frac{p^2}{2m} \Rightarrow p = \sqrt{2mqV}$$

The de Broglie wavelength is therefore

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mqV}}$$

This can be rearranged to show the dependence on $\frac{1}{\sqrt{V}}$:

$$\lambda = \frac{h}{\sqrt{2mq}} \cdot \frac{1}{\sqrt{V}}$$

Graph: Plotting λ (on the y-axis) against $\frac{1}{\sqrt{V}}$ (on the x-axis) gives a **straight line passing through the origin**. The line starts at the origin (0,0) and rises steadily

to the right with a constant positive gradient; its slope is

$$\text{slope} = \frac{h}{\sqrt{2mq}}$$

How the graph gives the charge: From the measured slope of this straight line, since h is known and the mass m is given, we can solve for the charge q. Squaring the slope,

$$(\text{slope})^2 = \frac{h^2}{2mq}$$

$$q = \frac{h^2}{2m\,(\text{slope})^2}$$

Thus the magnitude of the **charge of the particle** is obtained directly from the slope of the λ versus $\frac{1}{\sqrt{V}}$ graph.

Final answer. The plot of λ against $1/\sqrt{V}$ is a straight line through the origin with slope $h/\sqrt{2mq}$, from which the charge is $q = h^2 / [2m(\text{slope})^2]$.

← *Back to Q159*

Q160

An electron, a proton and an alpha particle have the same kinetic energy. Which one has the shortest wavelength?

[2019 • Set 55-1-2]

SOLUTION

The de Broglie wavelength of a particle of mass m having kinetic energy K is

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mK}}$$

Since all three particles have the same kinetic energy K, the only variable is the mass, so

$$\lambda \propto \frac{1}{\sqrt{m}}$$

This means the particle with the largest mass has the shortest de Broglie wavelength.

Comparing the masses:

$\mathbf{m_e = 9.11 \times 10^{-31}\ kg}$ (lightest)

$$m_p = 1.67 \times 10^{-27}\ \text{kg} \approx 1836\ m_e$$

$\mathbf{m_\alpha = 4m_p \approx 7344\ m_e}$ (heaviest)

The alpha particle has the greatest mass, so

$$\lambda_\alpha < \lambda_p < \lambda_e$$

Therefore the **alpha particle has the shortest de Broglie wavelength**.

Final answer. Since $\lambda \propto 1/\sqrt{m}$ at equal kinetic energy and the alpha particle is the heaviest, the alpha particle has the shortest de Broglie wavelength.

← *Back to Q160*

Q161

Explain with the help of Einstein's photoelectric equation any two observed features in photoelectric effect which cannot be explained by wave theory.

[2019 • Set 55-1-2]

SOLUTION

Einstein explained the photoelectric effect by treating light as a stream of energy packets called photons, each carrying energy $\boldsymbol{E = h\nu}$. When a photon strikes a metal surface, it gives all its energy to a single electron in a single collision.

Einstein's photoelectric equation is

$$h\nu = \phi_0 + K_{max}$$

where ϕ_0 is the work function of the metal and $\mathbf{K_{max} = \frac{1}{2}mv_{max}^2}$ is the maximum kinetic energy of the emitted photoelectron.

Two observed features that wave theory cannot explain but Einstein's equation can:

(i) Existence of a threshold frequency. Emission occurs only if $\nu \geq \nu_0$, where $\boldsymbol{\nu_0 = \frac{\Phi_0}{h}}$. Below this frequency, no electron is emitted however intense the light or however long it falls. Wave theory says energy is delivered continuously, so even low-frequency light of high intensity should eventually free an electron, which never happens. The photon picture explains it: if $\boldsymbol{h\nu < \Phi_0}$, a single photon simply lacks the energy to free an electron.

(ii) Maximum kinetic energy depends on frequency, not intensity. From the equation, $\boldsymbol{K_{max} = h\nu - \Phi_0}$, so K_{max} rises with frequency but is completely independent of intensity. Wave theory predicts K_{max} should grow with intensity, since a brighter beam carries more energy. In the photon picture, raising the intensity only increases the number of photons (hence more photocurrent), but each photon still delivers the same energy $h\nu$ to one electron.

Hence Einstein's photon equation accounts for both the **threshold frequency** and the **frequency-dependence of K_{max}**, which the classical **wave theory** fails to explain.

Final answer. Threshold frequency and the dependence of maximum kinetic energy on frequency (not intensity) are explained by Einstein's equation but not by wave theory.

← Back to Q161

Q162

Define the terms, (i) threshold frequency and (ii) stopping potential in photoelectric effect.

[2019 • Set 55-1-3]

SOLUTION

(i) Threshold frequency (ν_0): It is the minimum frequency of the incident radiation below which no photoelectric emission takes place from a given

photosensitive surface, however high the intensity of the radiation may be.

At the threshold, the entire photon energy is used up only in liberating the electron, so

$$h\nu_0 = \phi_0$$

$$\Rightarrow \nu_0 = \frac{\phi_0}{h}$$

where ϕ_0 is the work function of the metal. For $\nu < \nu_0$, no emission occurs.

(ii) Stopping potential (V_0): It is the minimum value of the retarding (negative) anode potential for which the photoelectric current becomes zero. At this potential, even the fastest photoelectrons are just stopped from reaching the anode.

The work done by the retarding field equals the maximum kinetic energy of the emitted electrons:

$$eV_0 = K_{max} = \frac{1}{2}mv_{max}^2$$

$$\Rightarrow eV_0 = h\nu - \phi_0 = h(\nu - \nu_0)$$

Thus the **stopping potential** depends on the frequency of the incident light and the work function, but is **independent of the intensity** of the light.

Final answer. Threshold frequency is the minimum frequency for emission ($\nu_0=\phi_0/h$); stopping potential is the retarding voltage at which photocurrent becomes zero ($eV_0=K_{max}$).

← *Back to Q162*

Q163

Plot a graph of photocurrent versus anode potential for a radiation of frequency ν *and intensities* I_1 *and* I_2 **($I_1 < I_2$)**.

[2019 • Set 55-1-3]

SOLUTION

The graph plots photoelectric current (on the y-axis) against the anode (collector) potential (on the x-axis), for a fixed frequency ν and two intensities I_1 and I_2 with $\mathbf{I_1 < I_2}$.

Description of the graph: Take the anode potential along the horizontal axis, with positive values to the right of the origin and negative (retarding) values to the left. Take photocurrent along the vertical axis. Draw two curves, one for I_1 (lower) and one for I_2 (upper).

On the positive (accelerating) side, each curve rises as the potential increases and then flattens out to a constant value called the saturation current. The curve for the higher intensity I_2 saturates at a higher current than the curve for I_1, because

$$\text{saturation current} \propto \text{intensity}$$

On the negative (retarding) side, the current decreases as the retarding potential is made more negative, and becomes zero at the stopping potential V_0. Both curves meet the horizontal axis at the same point $-V_0$, since

$$eV_0 = h\nu - \phi_0$$

depends only on frequency, not intensity.

Key features to label on the graph: the common stopping potential $-V_0$ (same for both curves), and the two different saturation currents (higher for I_2).

Thus the two curves have the **same stopping potential** but **different saturation currents**, the larger saturation current belonging to the more intense beam I_2.

COMMON MISTAKE

Same frequency means same stopping potential; only the saturation level shifts with intensity. Marking schemes deduct if both curves are shown reaching zero at different voltages.

Final answer. Both intensity curves cut the axis at the same stopping potential $\mathbf{-V_0}$, but $\mathbf{I_2}$ saturates at a higher current than $\mathbf{I_1}$.

← *Back to Q163*

Q164

A photon and a proton have the same de-Broglie wavelength λ. Prove that the energy of the photon is $(2m\lambda c/h)$ *times the kinetic energy of the proton.*

[2019 • Set 55-2-1]

SOLUTION

Both the photon and the proton are given the same de-Broglie wavelength λ. Let m be the mass of the proton.

Energy of the photon. A photon of wavelength λ has energy

$$E = \frac{hc}{\lambda}$$

Kinetic energy of the proton. The de-Broglie relation gives the proton's momentum

$$\lambda = \frac{h}{p} \Rightarrow p = \frac{h}{\lambda}$$

The proton (non-relativistic) has kinetic energy

$$K = \frac{p^2}{2m} = \frac{1}{2m}\left(\frac{h}{\lambda}\right)^2 = \frac{h^2}{2m\lambda^2}$$

Taking the ratio of photon energy to proton kinetic energy:

$$\frac{E}{K} = \frac{hc/\lambda}{h^2/(2m\lambda^2)}$$

$$\frac{E}{K} = \frac{hc}{\lambda} \times \frac{2m\lambda^2}{h^2}$$

$$\frac{E}{K} = \frac{2m\lambda\, c}{h}$$

Therefore

$$E = \left(\frac{2m\lambda\, c}{h}\right)K$$

Hence the energy of the photon is $\left(\frac{2m\lambda\, c}{h}\right)$ times the kinetic energy of the proton. **Proved.**

Final answer. $E = \dfrac{2m\lambda\, c}{h}\, K$, so the photon energy is $\dfrac{2m\lambda\, c}{h}$ times the proton's kinetic energy.

← Back to Q164

Q165

The wavelength of light from the spectral emission line of sodium is 590 nm. *Find the kinetic energy at which the electron would have the same de-Broglie wavelength.*

[2019 • Set 55-2-3]

SOLUTION

The electron must have the same de-Broglie wavelength as the sodium light, so we set

$$\lambda_e = 590\ \text{nm} = 5.910^{-7}\ m$$

The de-Broglie wavelength of an electron of kinetic energy K is

$$\lambda_e = \frac{h}{p} = \frac{h}{\sqrt{2m_e K}}$$

Squaring and solving for K:

$$K = \frac{h^2}{2m_e \lambda_e^{\,2}}$$

Substituting $\boldsymbol{h = 6.62610^{-34}\ J\ s}$ and $\boldsymbol{m_e = 9.1110^{-31}\ kg}$:

$$h^2 = (6.62610^{-34})^2 = 4.3910^{-67}\ J^2 s^2$$

$$\lambda_e^{\,2} = (5.910^{-7})^2 = 3.4810^{-13}\ m^2$$

$$2m_e\lambda_e^{\,2} = 2\times(9.1110^{-31})\times(3.4810^{-13}) = 6.3410^{-43}$$

$$K = \frac{4.3910^{-67}}{6.3410^{-43}} = 6.9210^{-25}\ J$$

Converting to electron-volts ($\boldsymbol{1\ eV = 1.610^{-19}\ J}$):

$$K = \frac{6.9210^{-25}}{1.610^{-19}} = 4.3310^{-6}\ \text{eV}$$

Hence the electron must have kinetic energy $K \approx 6.9210^{-25}\ J \approx 4.33\ \mu\text{eV}$.

Final answer. $K \approx 6.9210^{-25}\ J \approx 4.33\ \mu\text{eV}$.

← *Back to Q165*

Q166

If light of wavelength 412.5 nm *is incident on each of the metals given below, which ones will show photoelectric emission and why ?*

Metal	Work Function (eV)
Na	1·92
K	2·15
Ca	3·20
Mo	4·17

[2018]

SOLUTION

Photoelectric emission takes place from a metal only when the energy of the incident photon is at least equal to the work function of that metal, that is

$$E_{ph} \geq \Phi_0$$

First find the energy of the incident photon of wavelength $\lambda = 412.5$ **nm** using $E_{ph} = \frac{hc}{\lambda}$, with the handy value **hc = 1240 eV nm**:

$$E_{ph} = \frac{1240\ \text{eV nm}}{412.5\ \text{nm}}$$

$$E_{ph} \approx 3.0\ \text{eV}$$

Now compare this 3.0 eV with the work function of each metal:

Na: $\phi_0 = 1.92$ **eV**, and **3.0 > 1.92**, so emission occurs.

K: $\phi_0 = 2.15$ **eV**, and **3.0 > 2.15**, so emission occurs.

Ca: $\phi_0 = 3.20$ **eV**, and **3.0 < 3.20**, so no emission.

Mo: $\phi_0 = 4.17$ **eV**, and **3.0 < 4.17**, so no emission.

Hence only **Na (sodium) and K (potassium)** show photoelectric emission, because their work functions are less than the incident photon energy of 3.0 eV. Ca and Mo do not, since their work functions exceed 3.0 eV.

Final answer. Only sodium (Na) and potassium (K) show emission, since E_{ph}=**3.0 eV** exceeds only their work functions.

← Back to Q166

Q167

If light of wavelength 412.5 *nm is incident on each of the metals given below, which ones will show photoelectric emission and why?*
Metal — Work Function (eV)
Na — 1.92
K — 2.15
Ca — 3.20
Mo — 4.17

[2018]

SOLUTION

A photosensitive metal emits photoelectrons only when the incident photon energy is at least equal to its work function:

$$E_{ph} \geq \phi_0$$

Energy of the incident photon of wavelength $\lambda = 412.5$ **nm**, using $E_{ph} = \frac{hc}{\lambda}$ with **hc = 1240 eV nm**:

$$E_{ph} = \frac{1240 \text{ eV nm}}{412.5 \text{ nm}}$$

$$E_{ph} \approx 3.0 \text{ eV}$$

Comparing 3.0 eV with each work function:

Na: $\phi_0 = 1.92$ **eV**; since **3.0 > 1.92**, emission occurs.

K: $\phi_0 = 2.15$ **eV**; since **3.0 > 2.15**, emission occurs.

Ca: $\phi_0 = 3.20$ **eV**; since **3.0 < 3.20**, no emission.

Mo: $\phi_0 = 4.17$ **eV**; since **3.0 < 4.17**, no emission.

Therefore **sodium (Na) and potassium (K)** show photoelectric emission because their work functions are smaller than the incident photon energy of 3.0 eV, whereas calcium (Ca) and molybdenum (Mo) do not, as their work functions are greater than 3.0 eV.

Final answer. Sodium (Na) and potassium (K) show emission, since their work functions are below the **3.0 eV** photon energy.

← Back to Q167

Q168

Calculate the de-Broglie wavelength of the electron orbitting in the $n = 2$ *state of hydrogen atom.*

[2016]

SOLUTION

In Bohr's model, the electron in a stationary orbit obeys the quantum condition that the circumference of the orbit equals a whole number of de-Broglie wavelengths:

$$2\pi r_n = n\lambda$$

so the de-Broglie wavelength of the orbiting electron is

$$\lambda = \frac{2\pi r_n}{n}$$

The radius of the nth Bohr orbit of hydrogen is

$$r_n = n^2 a_0$$

where $\mathbf{a_0 = 0.53\ \text{Å}}$ is the Bohr radius. For $\boldsymbol{n = 2}$:

$$r_2 = (2)^2 0.53 = 40.53 = 2.12\ \text{Å}$$

Substituting into the wavelength expression with $\boldsymbol{n = 2}$:

$$\lambda = \frac{2\pi\, r_2}{n} = \frac{2\pi 2.12}{2}$$

$$\lambda = \pi 2.12 = 4\pi\, a_0$$

$$\lambda = 3.14162.12 \approx 6.66\ \text{Å}$$

Hence the de-Broglie wavelength of the electron in the $\boldsymbol{n = 2}$ state of hydrogen is $\boldsymbol{\lambda \approx 6.66\ \text{Å} = 6.6610^{-10}\ m}$.

Final answer. $\boldsymbol{\lambda \approx 6.66\ \text{Å} = 6.6610^{-10}\ m}$.

← *Back to Q168*

Q169

A proton and an α particle are accelerated through the same potential difference. Which one of the two has (i) greater de-Broglie wavelength, and (ii) less kinetic energy? Justify your answer.

[2016]

SOLUTION

When a particle of charge q and mass m is accelerated through a potential difference V, the work done by the field becomes its kinetic energy:

$$K = qV$$

Its momentum is $\boldsymbol{p = \sqrt{2mK} = \sqrt{2mqV}}$, so the de-Broglie wavelength is

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mqV}}$$

For a proton: charge $q_p = e$, mass m_p. For an α-particle: charge $q_\alpha = 2e$, mass $m_\alpha = 4m_p$. Both are accelerated through the same V.

(i) Greater de-Broglie wavelength. Since $\lambda \propto \frac{1}{\sqrt{mq}}$ for fixed V,

$$\frac{\lambda_p}{\lambda_\alpha} = \sqrt{\frac{m_\alpha\, q_\alpha}{m_p\, q_p}} = \sqrt{\frac{(4m_p)(2e)}{m_p\, e}} = \sqrt{8} = 2\sqrt{2} \approx 2.83$$

Since $\lambda_p > \lambda_\alpha$, the **proton has the greater de-Broglie wavelength** (about 2.83 times that of the α-particle).

(ii) Less kinetic energy. Using $K = qV$ with the same V:

$$K_p = eV \quad K_\alpha = 2eV$$

So $K_p = \frac{1}{2}K_\alpha$, meaning the **proton has the less kinetic energy** (half that of the α-particle).

Justification: the proton carries the smaller charge, so the field does less work on it, giving it less kinetic energy; and its product mq is smaller, giving it a smaller momentum and hence a larger de-Broglie wavelength.

Final answer. The proton has the greater de-Broglie wavelength (**22** times) and the lower kinetic energy (half) compared with the **α**-particle.

← *Back to Q169*

Q170

Write briefly the underlying principle used in Davison-Germer experiment to verify wave nature of electrons experimentally. What is the de-Broglie wavelength of an electron with kinetic energy (K.E.) 120 eV*?*

[2016]

SOLUTION

Underlying principle of the Davisson-Germer experiment. A beam of electrons, accelerated through a known potential, is made to fall on a single nickel crystal. The regular array of atoms in the crystal acts as a three-dimensional diffraction grating and scatters the electrons. The electrons scattered in a particular

direction show a sharp maximum in intensity, exactly as expected for the diffraction of a wave (a Bragg-type condition). The wavelength deduced from this diffraction maximum agrees with the de-Broglie wavelength $\lambda = \frac{h}{p}$, thereby confirming the **wave nature of electrons**.

de-Broglie wavelength for K.E. = **120 eV**. The de-Broglie wavelength of an electron accelerated through a potential difference of V volts is

$$\lambda = \frac{h}{\sqrt{2m_e\,eV}} = \frac{1.227}{\sqrt{V}}\ \text{nm}$$

Since K.E. = **eV = 120 eV**, we take $V = 120\ V$:

$$\sqrt{V} = \sqrt{120} = 10.95$$

$$\lambda = \frac{1.227}{10.95}\ \text{nm}$$

$$\lambda \approx 0.112\ \text{nm} = 1.12\ \text{Å}$$

Hence the de-Broglie wavelength of the electron is $\lambda \approx 1.12\ \text{Å} = 1.1210^{-10}\ m$.

Final answer. The crystal diffracts the electrons (confirming their wave nature); for K.E. =**120 eV**, $\lambda \approx 1.12\ \text{Å} = 1.1210^{-10}\ m$.

← *Back to Q170*

Q171

The equivalent wavelength of a moving electron has the same value as that of a photon of energy 6×10^{-17} *J. Calculate the momentum of the electron.*

[2015]

SOLUTION

A photon of energy E_{ph} has wavelength given by the Planck-Einstein relation $E_{ph} = \frac{hc}{\lambda}$, so the wavelength of this photon is $\lambda = \frac{hc}{E_{ph}}$.

The moving electron is said to have the same de Broglie wavelength, and its momentum is linked to that wavelength by $p = \frac{h}{\lambda}$.

Substituting $\lambda=\frac{hc}{E_{ph}}$ into the electron's momentum expression:

$$p=\frac{h}{\left(\frac{hc}{E_{ph}}\right)}=\frac{h\,E_{ph}}{hc}=\frac{E_{ph}}{c}$$

Notice the Planck constant cancels, so the momentum depends only on the photon energy and the speed of light. Now put in the data $E_{ph}=610^{-17}\,J$ and $c=310^{8}\,m/s$:

$$p=\frac{610^{-17}}{310^{8}}$$

$$p=210^{-25}\ \text{kg}\ m\ \text{s}^{-1}$$

Hence the **momentum of the electron is 210^{-25} kg m s^{-1}**.

CONCEPT

When an electron and a photon share the same wavelength, the electron's momentum equals E_{ph}/c, the momentum the photon itself carries.

Final answer. The momentum of the electron is **210^{-25} kg m s^{-1}**.

← *Back to Q171*

Q172

A proton and an α-particle have the same de-Broglie wavelength. Determine the ratio of (i) their accelerating potentials (ii) their speeds.

[2015]

SOLUTION

When a charged particle of mass m and charge q is accelerated from rest through a potential difference V, it gains kinetic energy qV, so its momentum is $p=\sqrt{2mqV}$ and its de Broglie wavelength is $\lambda=\frac{h}{\sqrt{2mqV}}$.

For the proton and the α-particle the wavelengths are equal, so the quantities under the root must be equal:

$$2m_p\, q_p\, V_p = 2m_\alpha\, q_\alpha\, V_\alpha$$

Using $\mathbf{m_\alpha = 4m_p}$ and $\mathbf{q_\alpha = 2e}$, $\mathbf{q_p = e}$.

(i) Ratio of accelerating potentials. Rearranging the equation above:

$$\frac{V_p}{V_\alpha} = \frac{m_\alpha\, q_\alpha}{m_p\, q_p} = \frac{(4m_p)(2e)}{(m_p)(e)} = 8$$

So $\mathbf{V_p : V_\alpha = 8:1}$.

(ii) Ratio of speeds. Writing the de Broglie wavelength as $\lambda = \frac{h}{mv}$, equal wavelengths give equal momenta:

$$m_p\, v_p = m_\alpha\, v_\alpha$$

$$\frac{v_p}{v_\alpha} = \frac{m_\alpha}{m_p} = \frac{4m_p}{m_p} = 4$$

So $\mathbf{v_p : v_\alpha = 4:1}$, and the proton moves faster than the α-particle.

Final answer. Accelerating potentials $\mathbf{V_p : V_\alpha = 8:1}$ and speeds $\mathbf{v_p : v_\alpha = 4:1}$.

← *Back to Q172*

Q173

A proton and a deuteron are accelerated through the same accelerating potential. Which one of the two has
(a) greater value of de-Broglie wavelength associated with it, and
(b) less momentum ?
Give reasons to justify your answer.

[2014]

SOLUTION

Both particles start from rest and fall through the same potential difference V, so each gains kinetic energy $K = qV$. Its momentum is then $p = \sqrt{2mK} = \sqrt{2mqV}$ and its de Broglie wavelength is $\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mqV}}$.

A proton and a deuteron carry the same charge $q_p = q_d = e$, but the deuteron is twice as heavy: $m_d = 2m_p$.

(a) Greater de Broglie wavelength. Since V and q are the same for both, $\lambda \propto \frac{1}{\sqrt{m}}$:

$$\frac{\lambda_p}{\lambda_d} = \sqrt{\frac{m_d}{m_p}} = \sqrt{\frac{2m_p}{m_p}} = \sqrt{2}$$

So $\lambda_p = \sqrt{2}\,\lambda_d$, which means the **proton has the greater de Broglie wavelength**. The lighter particle has the longer wavelength because, for the same charge and potential, a smaller mass gives a smaller momentum, and $\lambda = h/p$.

(b) Less momentum. Since $p = \sqrt{2mqV} \propto \sqrt{m}$ for equal q and V:

$$\frac{p_p}{p_d} = \sqrt{\frac{m_p}{m_d}} = \frac{1}{\sqrt{2}}$$

So $p_p < p_d$, which means the **proton has the smaller momentum**. Both gain the same kinetic energy eV, but because momentum $p = \sqrt{2mK}$ grows with mass, the lighter proton ends up with less momentum.

Final answer. The proton has the greater de Broglie wavelength and the smaller momentum.

← *Back to Q173*

Q174

(i) Monochromatic light of frequency 6.0×10^{14} *Hz is produced by a laser. The power emitted is* 2.0×10^{-3} *W. Estimate the number of photons emitted per second on an average by the source.*
(ii) Draw a plot showing the variation of photoelectric current versus the intensity of incident radiation on a given photosensitive surface.

[2014]

SOLUTION

(i) Number of photons emitted per second. Each photon of the laser light carries energy given by the Planck-Einstein relation $E = h\nu$:

$$E=(6.62610^{-34})(6.010^{14})$$

$$E=3.9810^{-19}\,J$$

The source emits power $\boldsymbol{P=2.010^{-3}\,W}$, that is $2.010^{-3}\,J$ of light energy each second. The number of photons emitted per second is the total energy per second divided by the energy of one photon:

$$n=\frac{P}{E}=\frac{2.010^{-3}}{3.9810^{-19}}$$

$$n5.010^{15}\text{ photons s}^{-1}$$

So the source emits about **5.010^{15} photons per second**.

(ii) Photoelectric current versus intensity. At a fixed frequency above the threshold, raising the intensity increases the number of photons striking the surface each second, so more photoelectrons are ejected and the saturation photocurrent rises in direct proportion to the intensity.

The graph is a straight line passing through the origin, with the intensity of the incident radiation on the horizontal axis and the photoelectric current on the vertical axis. The line rises steadily with a constant positive slope, showing that photocurrent is directly proportional to intensity.

Final answer. About **5.010^{15}** photons are emitted per second; photocurrent versus intensity is a straight line through the origin.

← *Back to Q174*

Q175

A deuteron and an alpha particle are accelerated with the same accelerating potential. Which one of the two has
(1) greater value of de-Broglie wavelength, associated with it and
(2) less kinetic energy ? Explain.

[2014]

SOLUTION

Both particles are accelerated from rest through the same potential difference V. For a deuteron, $\mathbf{m_d=2m_p}$ and $\mathbf{q_d=e}$; for an α-particle, $\mathbf{m_\alpha=4m_p}$ and $\mathbf{q_\alpha=2e}$.

A charged particle accelerated through V gains kinetic energy $\boldsymbol{K=qV}$ and acquires de Broglie wavelength $\boldsymbol{\lambda=\frac{h}{\sqrt{2mqV}}}$.

(1) Greater de Broglie wavelength. With V common to both, $\boldsymbol{\lambda\propto\frac{1}{\sqrt{mq}}}$:

$$\frac{\lambda_d}{\lambda_\alpha}=\sqrt{\frac{m_\alpha q_\alpha}{m_d q_d}}=\sqrt{\frac{(4m_p)(2e)}{(2m_p)(e)}}$$

$$\frac{\lambda_d}{\lambda_\alpha}=\sqrt{4}=2$$

So $\boldsymbol{\lambda_d=2\,\lambda_\alpha}$, meaning the **deuteron has the greater de Broglie wavelength**, with $\boldsymbol{\lambda_d:\lambda_\alpha=2:1}$. The deuteron wins because the product mq is smaller for it, and a smaller mq gives a smaller momentum and hence a longer wavelength.

(2) Less kinetic energy. The kinetic energy gained is $\boldsymbol{K=qV}$, the same V for both:

$$K_d=q_d\,V=eV$$

$$K_\alpha=q_\alpha\,V=2eV$$

So $\boldsymbol{K_d:K_\alpha=1:2}$, meaning the **deuteron has the smaller kinetic energy**. With the same accelerating potential, the particle of smaller charge gains the smaller kinetic energy because work done by the field is qV.

Final answer. The deuteron has the greater de Broglie wavelength (**2:1**) and the smaller kinetic energy (**1:2**).

← *Back to Q175*

Q176

An α-particle and a proton are accelerated from rest by the same potential. Find the ratio of their de Broglie wavelengths.

[2012]

SOLUTION

A charged particle of mass m and charge q accelerated from rest through a potential V gains kinetic energy qV, so its momentum is $p=\sqrt{2mqV}$ and its de Broglie wavelength is $\lambda=\frac{h}{\sqrt{2mqV}}$.

Since the α-particle and the proton are accelerated through the same potential V, and h is constant, $\lambda \propto \frac{1}{\sqrt{mq}}$.

For the proton, m_p and $q_p=e$. For the α-particle, $m_\alpha=4m_p$ and $q_\alpha=2e$, so the product is

$$m_\alpha\, q_\alpha=(4m_p)(2e)=8\, m_p\, q_p$$

Taking the ratio of the wavelengths:

$$\frac{\lambda_\alpha}{\lambda_p}=\sqrt{\frac{m_p\, q_p}{m_\alpha\, q_\alpha}}=\sqrt{\frac{m_p\, q_p}{8\, m_p\, q_p}}$$

$$\frac{\lambda_\alpha}{\lambda_p}=\sqrt{\frac{1}{8}}=\frac{1}{2\sqrt{2}}$$

Hence the required **ratio is $\lambda_\alpha:\lambda_p=1:2\sqrt{2}$**, so the proton has the longer de Broglie wavelength.

Final answer. The ratio of de Broglie wavelengths is $\lambda_\alpha:\lambda_p=1:2\sqrt{2}$.

← *Back to Q176*

Q177

Write Einstein's photoelectric equation. State clearly the three salient features observed in photoelectric effect, which can be explained on the basis of the above equation.

[2012]

SOLUTION

Einstein's photoelectric equation treats light as a stream of photons, each of energy hν. When one photon is absorbed by a surface electron, part of its energy (ϕ, the work function) is used to free the electron and the rest appears as the electron's maximum kinetic energy:

$$h\nu=\phi+K_{max}$$

Equivalently, since $\mathbf{K_{max}=eV_0}$ and $\boldsymbol{\phi=h\nu_0}$:

$$eV_0=h\nu-\phi=h(\nu-\nu_0)$$

Here $h\nu$ is the incident photon energy, $\boldsymbol{\phi=h\nu_0}$ is the work function (minimum energy to free a surface electron), ν_0 is the threshold frequency, K_{max} is the maximum kinetic energy of the emitted electron and V_0 is the stopping potential.

Three salient features explained by this equation:

1. Threshold frequency. Emission occurs only if $\nu\nu_0$. If $\boldsymbol{\nu<\nu_0}$ then $\boldsymbol{h\nu<\phi}$, so no electron can be freed however intense the light. The existence of a threshold frequency follows directly from $\boldsymbol{eV_0=h(\nu-\nu_0)}$.

2. Maximum kinetic energy depends on frequency, not intensity. From $\boldsymbol{K_{max}=h\nu-\phi}$, the kinetic energy of the fastest photoelectron grows linearly with frequency and is independent of intensity. Increasing the intensity only increases the number of photons (hence the photocurrent), not the energy of each photoelectron.

3. Instantaneous emission. Each electron is ejected by absorbing a single photon in one event, so emission begins essentially at once (within about 10^{-9} *s*), even for very feeble light, with no detectable time lag.

Final answer. Einstein's equation is $\boldsymbol{h\nu=\phi+K_{max}}$; it explains the threshold frequency, the frequency dependence of $\mathbf{K_{max}}$, and the instantaneous emission of photoelectrons.

← *Back to Q177*

Q178

An electron is accelerated through a potential difference of 100 volts. What is the de-Broglie wavelength associated with it? To which part of the electromagnetic spectrum does this value of wavelength correspond?

[2010]

SOLUTION

An electron of charge e and mass m accelerated from rest through a potential difference V gains kinetic energy eV, so its momentum is $p=\sqrt{2meV}$ and its de Broglie wavelength is

$$\lambda=\frac{h}{\sqrt{2meV}}$$

Substituting the standard values $h=6.6310^{-34}\,J\,s$, $m=9.110^{-31}\,kg$ and $e=1.610^{-19}\,C$ gives the convenient working formula

$$\lambda=\frac{12.27}{\sqrt{V}}\ (V \text{ in volts})$$

Here $V=100\,V$, so:

$$\lambda=\frac{12.27}{\sqrt{100}}=\frac{12.27}{10}$$

$$\lambda=1.227\ =1.22710^{-10}\,m=0.1227\ \text{nm}$$

Wavelengths in the range about 0.01 nm to 10 nm belong to the **X-ray region** of the electromagnetic spectrum. Since $\lambda=0.1227\ nm$ lies in this band, this de Broglie wavelength corresponds to **X-rays**.

TIP

Remember the shortcut $\lambda=\frac{12.27}{\sqrt{V}}$ for electrons; it saves time and avoids slips with the constants in the exam.

Final answer. The de Broglie wavelength is $1.227\ =1.22710^{-10}\,m$, lying in the X-ray region.

← *Back to Q178*

Q179

Plot a graph showing the variation of stopping potential with the frequency of incident radiation for two different photosensitive materials having work functions

W_1 *and* W_2 ($\mathbf{W_1 > W_2}$). *On what factors does the (i) slope and (ii) intercept of the lines depend?*

[2010]

SOLUTION

From Einstein's equation, the stopping potential is related to the frequency by $\mathbf{eV_0 = h\nu - W}$, which rearranges to

$$V_0 = \frac{h}{e}\nu - \frac{W}{e}$$

This is a straight line of the form $\boldsymbol{y = mx + c}$ when V_0 is plotted against ν, with slope $\frac{h}{e}$ and a V_0-axis intercept of $-\frac{W}{e}$. The line meets the ν-axis (where $\mathbf{V_0 = 0}$) at the threshold frequency $\boldsymbol{\nu_0 = \frac{W}{h}}$.

Description of the plot. Take frequency ν along the horizontal axis and stopping potential V_0 along the vertical axis. Draw two straight lines, one for each material, both with the same positive slope $\frac{h}{e}$, so they are parallel to each other. Because $\mathbf{W_1 > W_2}$, material 1 has the larger threshold frequency $\boldsymbol{\nu_{0,1} = W_1/h}$, so its line cuts the ν-axis farther to the right; material 2 has the smaller threshold frequency $\boldsymbol{\nu_{0,2} = W_2/h}$ and cuts the ν-axis nearer the origin. Extended backwards, line 1 meets the V_0-axis at $-W_1/e$ (lower) and line 2 at $-W_2/e$ (higher).

(i) Slope. The slope is $\frac{h}{e}$. It depends only on the universal constants h and e, so it is the same for every photosensitive material and is **independent of the material**. That is why the two lines are parallel.

(ii) Intercept. The intercept on the V_0-axis is $-\frac{W}{e}$ and the intercept on the ν-axis is $\boldsymbol{\nu_0 = \frac{W}{h}}$. Both depend on the **work function W of the material**, so they differ from one material to another.

Final answer. Both lines have slope $\boldsymbol{h/e}$ (material-independent); the intercept depends on the work function $\boldsymbol{W}$ of the material.

← *Back to Q179*

Q180

The work function of lithium is 2.3 eV. What does it mean? What is the relation between the work function W and threshold wavelength of a metal?

[2003]

SOLUTION

Meaning of work function **=2.3 eV** for lithium. The work function is the minimum energy that must be supplied to a free electron at the metal surface to just remove it from the metal, that is, to bring it to rest just outside the surface. For lithium this is 2.3 eV, so an electron at the lithium surface is bound by 2.3 eV.

In practice it means that a photon must carry at least 2.3 eV of energy ($h\nu$2.3 eV) to eject a photoelectron from lithium. Any photon of energy less than 2.3 eV cannot cause emission, no matter how intense the light.

Relation between work function and threshold wavelength. Photoemission just begins at the threshold frequency ν_0, whose photon energy equals the work function:

$$W=h\nu_0$$

Since $\boldsymbol{\nu_0=\frac{c}{\lambda_0}}$, where λ_0 is the threshold wavelength:

$$W=\frac{hc}{\lambda_0}$$

Solving for the threshold wavelength:

$$\lambda_0=\frac{hc}{W}$$

Thus the work function and threshold wavelength are **inversely related, $\boldsymbol{\lambda_0=\frac{hc}{W}}$**: the larger the work function, the shorter the threshold wavelength.

Final answer. A work function of **2.3 eV** is the minimum energy needed to free a surface electron from lithium; the work function and threshold wavelength

obey $\lambda_0 = \frac{hc}{W}$.

← Back to Q180

Q181

(a) A particle of charge q and mass m is accelerated through a potential difference V. Deduce an expression for the de Broglie wavelength associated with the particle.

[2026 • Set 55-2-3]

SOLUTION

When a particle of charge q and mass m starts from rest and is accelerated through a potential difference V, the electric field does work on it. This work appears entirely as kinetic energy of the particle.

By the work-energy theorem, the kinetic energy gained equals the electrical work done:

$$\frac{1}{2}mv^2 = qV$$

Solving for the speed v:

$$v = \sqrt{\frac{2qV}{m}}$$

The momentum of the particle is then:

$$p = mv = m\sqrt{\frac{2qV}{m}} = \sqrt{2mqV}$$

According to **de Broglie's hypothesis**, a moving particle of momentum p has an associated matter wave of wavelength:

$$\lambda = \frac{h}{p}$$

Substituting the expression for momentum:

$$\lambda = \frac{h}{\sqrt{2mqV}}$$

Here h is **Planck's constant**. For an electron ($m = 9.11\times 10^{-31}$ kg, $q = e = 1.6\times 10^{-19}$ C), this reduces to the useful working form $\lambda = \frac{12.27}{\sqrt{V}}$, with V in volts.

TIP

Treat $\sqrt{2mqV}$ as the momentum p in one block; it keeps the algebra clean and avoids dropping the square root over the whole product.

Final answer. The de Broglie wavelength is $\lambda = \frac{h}{\sqrt{2mqV}}$.

← *Back to Q181*

Q182

(b) Plot a graph for a particle of charge q and mass m showing variation of de Broglie wavelength λ, as $\frac{1}{\lambda}$ with $\sqrt{V}$, where V is the potential difference through which the particle is accelerated.

[2026 • Set 55-2-3]

SOLUTION

The de Broglie wavelength of a particle of charge q and mass m accelerated through a potential difference V is:

$$\lambda = \frac{h}{\sqrt{2mqV}}$$

Taking the reciprocal to get $\frac{1}{\lambda}$:

$$\frac{1}{\lambda} = \frac{\sqrt{2mqV}}{h} = \frac{\sqrt{2mq}}{h}\sqrt{V}$$

Comparing this with the straight-line form $y = (\text{slope})\,x$, with $y = \frac{1}{\lambda}$ and $x = \sqrt{V}$, we see the relation is **linear**.

Description of the graph: plot $\frac{1}{\lambda}$ on the y-axis and $\sqrt{V}$ on the x-axis. The graph is a **straight line passing through the origin** (since $V = 0$ gives $\frac{1}{\lambda} = 0$). The line

rises steadily to the right with a constant positive slope.

The slope of this line is:

$$\text{slope} = \frac{\sqrt{2mq}}{h}$$

From the measured slope, the product mq can be found using $\mathbf{mq = \frac{h^2\,(slope)^2}{2}}$. If the mass is known, the charge can be determined, or vice versa. For an electron the slope is steep; for a heavier particle such as a proton ($\sqrt{m_p/m_e} \approx 43$ times more massive in this $\sqrt{mq}$ factor) the line is much gentler.

Final answer. The plot of $\frac{1}{\lambda}$ versus $\sqrt{V}$ is a straight line through the origin with slope $\frac{\sqrt{2mq}}{h}$.

← *Back to Q182*

Q183

(a) Explain how the dual aspect of matter is evident in the de Broglie relation.
(b) Radiation of wavelength λ is incident on a photosensitive surface. Find the de Broglie wavelength of electrons emitted from the surface. Assume that the work function of the surface is negligible.

[2026 • Set 55-4-1]

SOLUTION

(a) Dual aspect of matter in the de Broglie relation: the relation is

$$\lambda = \frac{h}{p}$$

On the left side, λ is a **wave characteristic** (a wavelength, associated with interference and diffraction). On the right side, p is a **particle characteristic** (the linear momentum of a localized particle). **Planck's constant** h links the two. Thus the same matter shows wave nature (λ) and particle nature (p) at the same time, and the relation itself is the bridge connecting these dual aspects.

(b) Radiation of wavelength λ falls on the surface. The energy of each incident photon is:

$$E = \frac{hc}{\lambda}$$

By Einstein's photoelectric equation, the kinetic energy of an emitted electron is $\mathbf{K_e = h\nu - \phi}$. Since the work function is negligible ($\boldsymbol{\phi \approx 0}$):

$$K_e = \frac{hc}{\lambda}$$

The de Broglie wavelength of an electron of kinetic energy K_e (mass m_e) is:

$$\lambda_e = \frac{h}{\sqrt{2m_e K_e}}$$

Substituting $\mathbf{K_e = \frac{hc}{\lambda}}$:

$$\lambda_e = \frac{h}{\sqrt{2m_e \cdot \frac{hc}{\lambda}}}$$

Simplifying the expression:

$$\lambda_e = \sqrt{\frac{h\lambda}{2m_e c}}$$

Final answer. The de Broglie wavelength of the emitted electrons is $\boldsymbol{\lambda_e = \sqrt{\frac{h\lambda}{2m_e c}}}$.

← *Back to Q183*

Q184

*Photoemission of electrons occurs from a metal ($\boldsymbol{\phi_0 = 1.96}$ **eV**) when light of frequency 6.4×10^{14} Hz is incident on it. Calculate:*

(a) Energy of a photon in the incident light,

(b) The maximum kinetic energy of the emitted electrons, and

(c) The stopping potential.

[2026 • Set 55-5-1]

SOLUTION

Given data: work function $\phi_0 = 1.96$ eV and frequency of incident light $\nu = 6.4\times 10^{14}$ Hz. Use $h = 6.63\times 10^{-34}\ J\,s$.

(a) Energy of each incident photon is given by the **Planck relation**:

$$E = h\nu$$

$$E = (6.63\times 10^{-34})(6.4\times 10^{14})$$

$$E = 4.24\times 10^{-19}\ J$$

Converting to electron-volts by dividing by 1.6×10^{-19}:

$$E = \frac{4.24\times 10^{-19}}{1.6\times 10^{-19}} = 2.65 \text{ eV}$$

(b) By **Einstein's photoelectric equation**, the maximum kinetic energy of the emitted electrons is:

$$K_{max} = E - \phi_0$$

$$K_{max} = 2.65 - 1.96 = 0.69 \text{ eV}$$

(c) The stopping potential V_0 is related to the maximum kinetic energy by $eV_0 = K_{max}$, so its numerical value in volts equals K_{max} in eV:

$$V_0 = \frac{K_{max}}{e} = 0.69\ V$$

Hence the photon energy is **2.65 eV**, the maximum kinetic energy is **0.69 eV**, and the stopping potential is **0.69 V**.

TIP

When K_{max} is already in electron-volts, the stopping potential in volts is just the same number, since $eV_0 = K_{max}$ and the e cancels the eV.

Final answer. (a) $E \approx 2.65$ eV, (b) $K_{max} \approx 0.69$ eV, (c) $V_0 \approx 0.69\ V$.

← *Back to Q184*

Q185

What are de Broglie waves? Show that the wavelength of the electromagnetic radiation is equal to the de Broglie wavelength of its quantum (photon).

[2026 • Set 55-5-2]

SOLUTION

de Broglie waves: Louis de Broglie proposed that just as light shows both wave and particle nature, every moving material particle also has a wave associated with it. A particle of momentum p has an associated matter wave (called a **de Broglie wave**) of wavelength:

$$\lambda = \frac{h}{p}$$

where h is Planck's constant. These waves account for the wave-like behaviour of particles, such as electron diffraction.

To show the EM wavelength equals the de Broglie wavelength of its photon: a photon of electromagnetic radiation of wavelength λ_{em} has energy

$$E = h\nu = \frac{hc}{\lambda_{em}}$$

Since a photon travels at speed c, its momentum is:

$$p = \frac{E}{c} = \frac{hc/\lambda_{em}}{c} = \frac{h}{\lambda_{em}}$$

Now apply the de Broglie relation to this photon:

$$\lambda_{dB} = \frac{h}{p} = \frac{h}{h/\lambda_{em}}$$

Simplifying:

$$\lambda_{dB} = \lambda_{em}$$

Thus the **de Broglie wavelength of the photon is exactly equal to the wavelength of the electromagnetic radiation** itself.

Final answer. The de Broglie wavelength of a photon equals the wavelength of the radiation, $\lambda_{dB} = \lambda_{em}$.

← *Back to Q185*

Q186

Answer the following giving reason:
(a) *All the photo electrons do not eject with the same kinetic energy when monochromatic light is incident on a metal surface.*
(b) *The saturation current in case (a) is different for different intensity.*
(c) *If one goes on increasing the wavelength of light incident on a metal surface, keeping its intensity constant, emission of photo electrons stops at a certain wavelength for this metal.*

[2025 • Set 55-1-1]

SOLUTION

(a) Photoelectrons do not all emerge with the same kinetic energy. Einstein's equation $K_{max} = h\nu - \phi$ gives only the **maximum** kinetic energy, and this corresponds to the most loosely bound surface electrons (those needing only the work function ϕ). Electrons that lie deeper in the metal need more energy to escape, and electrons moving outward suffer collisions and lose part of their energy before leaving. Therefore the emitted electrons carry a continuous range of kinetic energies from 0 up to K_{max}.

(b) The saturation current differs with intensity. The intensity of light is a measure of the **number of photons arriving per second**. Since each photon ejects at most one electron, a more intense beam releases more photoelectrons per second. When the collecting voltage is large enough to gather all of them, the resulting saturation current is correspondingly larger. Hence higher intensity gives a higher saturation current.

(c) Emission stops beyond a certain wavelength. The energy of a photon is:

$$E = h\nu = \frac{hc}{\lambda}$$

As the wavelength λ increases, the photon energy decreases. There exists a **threshold wavelength** given by $\lambda_0 = \frac{hc}{\phi}$. When $\lambda > \lambda_0$, the photon energy falls below the work function ϕ, so a single photon can no longer supply enough energy for emission. Because each electron absorbs only one photon, increasing the intensity (more such low-energy photons) does not help, and emission stops completely.

Final answer. The spread in KE comes from energy loss below the surface; saturation current scales with photon count (intensity); and emission ceases once $\lambda > \lambda_0 = hc/\phi$.

← Back to Q186

Q187

(a) *Draw a plot of frequency ν of incident radiations as a function of stopping potential V_0 for a given photo emissive material. What information can be obtained from the value of the intercept on the stopping potential axis?*
(b) *Calculate: (i) the momentum and (ii) de Broglie wavelength, of an electron with kinetic energy of* 80 eV.

[2025 • Set 55-1-2]

SOLUTION

(a) Graph of frequency ν versus stopping potential V_0. Einstein's equation $eV_0 = h\nu - \phi$ can be rearranged as:

$$\nu = \frac{e}{h} V_0 + \frac{\phi}{h}$$

This is a straight line. Description of the plot: take ν on the y-axis and V_0 on the x-axis. The graph is a **straight line of positive slope** $\frac{e}{h}$. It does not pass through the origin; it cuts the ν-axis (where $V_0 = 0$) at the **threshold frequency** $\nu_0 = \frac{\phi}{h}$, and its extension cuts the V_0-axis at a negative value.

Information from the intercept on the stopping-potential axis: at this intercept $\nu = 0$, giving $V_0 = -\frac{\phi}{e}$. Its magnitude yields the **work function** of the material

as $\phi = e\,|V_0|$.

(b) Electron of kinetic energy 80 eV. First convert to joules:

$$K = 80 \times 1.6\times 10^{-19} = 1.28\times 10^{-17}\,J$$

(i) Momentum from $K = \dfrac{p^2}{2m_e}$, so $p = \sqrt{2m_e K}$:

$$p = \sqrt{2(9.11\times 10^{-31})(1.28\times 10^{-17})}$$

$$p = \sqrt{2.33\times 10^{-47}}$$

$$p = 4.83\times 10^{-24}\ \text{kg}\ m\ \text{s}^{-1}$$

(ii) de Broglie wavelength:

$$\lambda = \frac{h}{p} = \frac{6.63\times 10^{-34}}{4.83\times 10^{-24}}$$

$$\lambda = 1.37\times 10^{-10}\ m \approx 1.37$$

Final answer. (a) The V_0-axis intercept gives the work function $\phi = e|V_0|$; (b) $p \approx 4.83\times 10^{-24}$ kg m s^{-1} and $\lambda \approx 1.37$.

← *Back to Q187*

Q188

(a) *Mention any three features of results of experiment on photoelectric effect which cannot be explained using the wave theory of light.*
(b) *In his experiment on photoelectric effect, Robert A. Millikan found the slope of the cut-off voltage versus frequency of incident light plot to be* 4.12×10^{-15} *V s. Calculate the value of Planck's constant from it.*

[2025 • Set 55-1-3]

SOLUTION

(a) Three results of the photoelectric effect that wave theory cannot explain:

First, there is a **threshold frequency** ν_0 below which no electrons are emitted, no matter how intense or how long-lasting the light is. Wave theory says that

any frequency, given enough intensity, should eventually supply the energy to eject electrons, so it cannot explain a cut-off frequency.

Second, the **maximum kinetic energy** of photoelectrons depends only on the frequency of the light and not on its intensity ($\mathbf{K_{max} = h\nu - \phi}$). Wave theory predicts that a more intense beam carries more energy and should therefore give faster electrons, which is contrary to observation.

Third, photoelectric emission is **instantaneous** (within about $10^{-9}\ s$), even for very weak light. Wave theory predicts a measurable time delay while an electron slowly absorbs enough wave energy, but no such delay is seen.

(b) Planck's constant from Millikan's slope. The cut-off voltage versus frequency plot follows $\mathbf{V_0 = \frac{h}{e}\nu - \frac{\phi}{e}}$, so its slope is $\frac{h}{e}$:

$$\frac{h}{e} = 4.12\times 10^{-15}\ V\,s$$

Therefore:

$$h = (4.12\times 10^{-15})\times e = (4.12\times 10^{-15})(1.6\times 10^{-19})$$

$$h = 6.59\times 10^{-34}\ J\,s$$

Final answer. (a) Threshold frequency, frequency-dependence of $\mathbf{K_{max}}$, and instantaneous emission; (b) $\boldsymbol{h = 6.59\times 10^{-34}\ J\,s}$.

← *Back to Q188*

Q189

Explain the following observations using Einstein's photoelectric equation :
(a) *Photoelectric emission does not occur from a surface when the frequency of the light incident on it is less than a certain minimum value.*
(b) *It is the frequency, and not the intensity of the incident light which affects the maximum kinetic energy of the photoelectrons.*

***(c)** The cut-off voltage (V_0) versus frequency (ν) of the incident light curve is a straight line with a slope $\frac{h}{e}$.*

[2025 • Set 55-4-1]

SOLUTION

Einstein's photoelectric equation is $\mathbf{K_{max} = h\nu - \phi}$, where ϕ is the work function and ν the frequency of incident light. Each explanation follows from it.

(a) No emission below a minimum frequency. For an electron to escape, the photon energy must at least equal the work function. From $\mathbf{K_{max} = h\nu - \phi}$, emission requires $\mathbf{h\nu \geq \phi}$, i.e. $\mathbf{\nu \geq \nu_0}$ where $\mathbf{\nu_0 = \frac{\phi}{h}}$ is the **threshold frequency**. If $\mathbf{\nu < \nu_0}$, then $\mathbf{h\nu < \phi}$ and K_{max} would be negative, which is impossible. Hence no photoelectrons are emitted below this minimum frequency, whatever the intensity.

(b) Frequency, not intensity, controls K_{max}. In the photon picture, one electron absorbs exactly one photon and gains energy $h\nu$. So:

$$K_{max} = h\nu - \phi$$

This depends on ν but contains no intensity term. Raising the intensity only increases the number of photons (hence more electrons), not the energy $h\nu$ of each photon, so the maximum kinetic energy is unchanged.

(c) Slope of the V_0 versus ν graph. The stopping potential satisfies $\mathbf{eV_0 = K_{max} = h\nu - \phi}$. Writing $\mathbf{\phi = h\nu_0}$:

$$eV_0 = h\nu - h\nu_0$$

Dividing throughout by e:

$$V_0 = \frac{h}{e}\nu - \frac{h\nu_0}{e}$$

This is a straight line in V_0 versus ν with **slope** $\frac{h}{e}$ (a universal constant, independent of the metal) and intercept on the frequency axis at ν_0.

Final answer. All three follow from $\mathbf{K_{max} = h\nu - \phi}$: threshold at $\mathbf{\nu_0 = \phi/h}$, $\mathbf{K_{max}}$ set by $\mathbf{\nu}$ alone, and slope $\mathbf{h/e}$ for the $\mathbf{V_0}$ versus $\mathbf{\nu}$ line.

← Back to Q189

Q190

(a) *Define 'work function' of a metal. How can its value be determined from a graph between stopping potential and frequency of the incident radiation?*
(b) *The work function of a metal is* 2.4 eV. *A stopping potential of* 0.6 *V is required to reduce the photocurrent to zero, in a photoelectric experiment. Calculate the wavelength of light used.*

[2025 • Set 55-7-1]

SOLUTION

(a) Work function: it is the **minimum energy required to free the most loosely bound electron** from the surface of a metal. It is denoted ϕ (or ϕ_0) and is related to the threshold frequency by $\boldsymbol{\phi = h\nu_0}$.

Finding it from the stopping potential versus frequency graph: Einstein's equation gives $\boldsymbol{eV_0 = h\nu - \phi}$, i.e.

$$V_0 = \frac{h}{e}\nu - \frac{\phi}{e}$$

Plotting V_0 (y-axis) against ν (x-axis) gives a straight line. Its intercept on the frequency axis (where $\boldsymbol{V_0 = 0}$) is the **threshold frequency** ν_0, so the work function is obtained as $\boldsymbol{\phi = h\nu_0}$. Equivalently, the intercept on the V_0-axis equals $-\frac{\phi}{e}$, so $\boldsymbol{\phi = e\times|\text{that intercept}|}$.

(b) Given $\boldsymbol{\phi = 2.4\ eV}$ and stopping potential $\boldsymbol{V_0 = 0.6\ V}$, so $\boldsymbol{eV_0 = 0.6\ eV}$. By Einstein's equation the photon energy is:

$$\frac{hc}{\lambda} = \phi + eV_0$$

$$\frac{hc}{\lambda} = 2.4 + 0.6 = 3.0\ \text{eV}$$

Using $\boldsymbol{hc = 1240\ eV\ nm}$:

$$\lambda = \frac{1240\ \text{eV nm}}{3.0\ \text{eV}}$$

$$\lambda \approx 413 \text{ nm} = 4.13\times 10^{-7}\ m$$

This lies in the violet-blue region of the visible spectrum.

> **TIP**
>
> Memorise **hc = 1240 eV nm**: dividing it by the photon energy in eV gives the wavelength in nm in a single step.

Final answer. (a) Work function $\phi = h\nu_0$, read from the frequency-axis intercept; (b) $\lambda \approx 413$ **nm** $= 4.13\times 10^{-7}\ m$.

← *Back to Q190*

Q191

(a) *Briefly explain Einstein's photoelectric equation.*
(b) *Four metals with their work functions are listed below:*
***K* = 2.3 eV, Na = 2.75 eV, Mo = 4.17 eV** *and* **Ni = 5.15 eV**.
The radiation of wavelength 330 nm *from a laser source placed* 1 *m away, falls on these metals. Which of these metals will not show photoelectric emission? What will happen if the laser source is brought closer to a distance of* 50 cm*?*

[2025 • Set 55-7-2]

SOLUTION

(a) When light of frequency ν falls on a metal surface, each photon of energy $h\nu$ is completely absorbed by a single electron. Part of this energy equals the minimum energy needed to free the electron from the surface, called the work function ϕ_0, and the remaining energy appears as the maximum kinetic energy of the ejected electron.

Einstein's photoelectric equation is therefore

$$K_{max}=h\nu-\phi_0$$

where $\boldsymbol{\phi_0=h\nu_0}$ and ν_0 is the threshold frequency. Emission occurs only when $\boldsymbol{\nu>\nu_0}$, that is when $\boldsymbol{h\nu>\phi_0}$.

(b) The energy of an incident photon of wavelength $\boldsymbol{\lambda=330}$ **nm** is

$$E = \frac{hc}{\lambda} = \frac{1240 \text{ eV nm}}{330 \text{ nm}}$$

$$E \approx 3.76 \text{ eV}$$

Photoelectric emission takes place only from metals whose work function is less than this photon energy:

For *K*, $\phi = 2.3 \text{ eV} < 3.76 \text{ eV}$, so emission occurs.

For Na, $\phi = 2.75 \text{ eV} < 3.76 \text{ eV}$, so emission occurs.

For Mo, $\phi = 4.17 \text{ eV} > 3.76 \text{ eV}$, so no emission.

For Ni, $\phi = 5.15 \text{ eV} > 3.76 \text{ eV}$, so no emission.

Hence **Mo and Ni will not show photoelectric emission**.

When the source is brought from 1 *m* to 50 cm, the intensity rises (it varies as $1/r^2$, so it becomes 4 times larger), but the wavelength and therefore the photon energy $E = 3.76 \text{ eV}$ stay unchanged. Since photoemission depends on photon energy and not on intensity, **Mo and Ni still show no emission**, while *K* and Na emit a larger number of photoelectrons (greater photocurrent) with the same maximum kinetic energy.

COMMON MISTAKE

Photoemission is decided purely by photon energy (wavelength/frequency); intensity only changes how many electrons come out, never whether they come out.

Final answer. Mo and Ni show no emission; bringing the source to 50 cm only increases the photocurrent from K and Na, while Mo and Ni still emit nothing.

← *Back to Q191*

Q192

Answer the following, giving reasons:
(a) *The maximum kinetic energy of the photoelectrons is independent of the intensity of incident radiation.*
(b) *Photoelectric current increases with the increase in the intensity of the incident*

radiation.

***(c)** The stopping potential V_0 varies linearly with the frequency ν of the incident radiation for a given photosensitive surface.*

[2025 • Set 55-7-3]

SOLUTION

(a) Each photoelectron is emitted by absorbing exactly one photon of energy $h\nu$. By Einstein's equation,

$$K_{max}=h\nu-\phi_0$$

Increasing the intensity increases only the number of photons arriving per second, not the energy $h\nu$ of an individual photon. Since K_{max} depends only on the frequency ν and the work function ϕ_0, the **maximum kinetic energy is independent of the intensity** of the incident radiation.

(b) The intensity of radiation is proportional to the number of photons striking the surface per second. Above the threshold frequency, one photon ejects one electron, so a larger number of incident photons ejects a larger number of photoelectrons per second.

Therefore the number of emitted electrons, and hence the **saturation photoelectric current, increases with the intensity** of the incident radiation.

(c) The stopping potential V_0 is the retarding potential that just stops the most energetic photoelectrons, so $\mathbf{eV_0=K_{max}}$. Using Einstein's equation,

$$eV_0=h\nu-\phi_0$$

$$V_0=\frac{h}{e}\nu-\frac{\phi_0}{e}$$

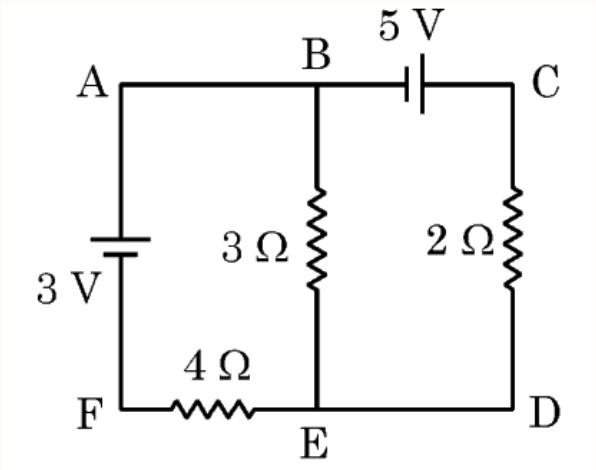
This is of the form $\boldsymbol{y=mx+c}$, so the V_0 versus ν graph is a **straight line** with slope $\frac{h}{e}$ (a universal constant, same for all metals) and a negative intercept $-\frac{\phi_0}{e}$ on the V_0 axis. The line meets the ν axis at $\boldsymbol{\nu=\nu_0}$. Hence V_0 varies linearly with ν for a given surface.

Final answer. (a) K_max depends only on ν and ϕ_0, not intensity; (b) more intensity means more photons, hence more electrons and larger current; (c) $eV_0 = h\nu - \phi_0$ gives a straight line of slope h/e.

← Back to Q192

Q193

de Broglie wavelength λ *as a function of* $\frac{1}{K}$ *for two particles of masses* m_1 *and* m_2 *are shown in the figure. Here, K is the energy of the moving particles.*



(a) What does the slope of a line represent?
(b) Which of the two particles is heavier?
(c) Is this graph also valid for a photon?
Justify your answer in each case.

[2024 • Set 55-2-1]

SOLUTION

For a material particle of mass m and kinetic energy K, the momentum is $p=\sqrt{2mK}$, so the de Broglie wavelength is

$$\lambda=\frac{h}{p}=\frac{h}{\sqrt{2mK}}$$

which can be written as

$$\lambda=\frac{h}{\sqrt{2m}}\cdot\frac{1}{\sqrt{K}}$$

This is a straight line of the form $\lambda=(\text{slope})\times\frac{1}{\sqrt{K}}$ passing through the origin.

(a) Comparing with $y=mx$, the slope of the λ versus $\frac{1}{\sqrt{K}}$ line is

$$\text{slope}=\frac{h}{\sqrt{2m}}$$

So the **slope represents** $\boldsymbol{\frac{h}{\sqrt{2m}}}$ for that particle.

(b) From the slope expression, **slope** $\boldsymbol{\propto \frac{1}{\sqrt{m}}}$. A larger mass gives a smaller slope. Therefore the line with the **smaller slope corresponds to the heavier particle**. In the graph, whichever of m_1 or m_2 has the line of lower slope (less steep) is the heavier particle.

(c) The graph is **not valid for a photon**. A photon is massless and travels at speed c, so the relation $\boldsymbol{p=\sqrt{2mK}}$ (which assumes $\boldsymbol{K=\frac{p^2}{2m}}$ for a slow-moving particle of mass m) cannot be applied to it. For a photon, energy and momentum are related by $\boldsymbol{E=pc}$, giving

$$\lambda=\frac{h}{p}=\frac{hc}{E}$$

which makes λ proportional to $\frac{1}{E}$, not to $\frac{1}{\sqrt{E}}$. Hence the same straight-line λ versus $1/\sqrt{K}$ relation does not hold for a photon.

Final answer. (a) Slope = $h/\sqrt{(2m)}$; (b) the particle with the smaller slope (less steep line) is heavier; (c) the graph is not valid for a photon, since for a photon $\lambda = hc/E \propto 1/E$.

← *Back to Q193*

Q194

(a) Briefly explain de Broglie hypothesis for wave nature of matter.
(b) Find the ratio of de Broglie wavelength associated with a proton and an alpha particle when both are
(i) accelerated from rest through the same potential difference, and
(ii) moving with the same kinetic energy.

[2024 • Set 55-2-2]

SOLUTION

(a) Louis de Broglie proposed that, just as radiation shows both wave and particle nature, moving material particles also have a wave nature. A particle of momentum p has an associated wavelength

$$\lambda=\frac{h}{p}=\frac{h}{mv}$$

where h is Planck's constant. This wave is called a **matter wave**. The smaller the momentum, the longer the wavelength, which is why the wave nature is significant only for very light, slow particles such as electrons.

(b)(i) When a particle of charge q is accelerated from rest through a potential difference V, its kinetic energy is $\boldsymbol{K=qV}$ and its momentum is $\boldsymbol{p=\sqrt{2mqV}}$, so

$$\lambda=\frac{h}{\sqrt{2mqV}}$$

For the same V,

$$\frac{\lambda_p}{\lambda_\alpha}=\sqrt{\frac{m_\alpha q_\alpha}{m_p q_p}}$$

Using $\boldsymbol{m_\alpha=4m_p}$ and $\boldsymbol{q_\alpha=2q_p}$,

$$\frac{\lambda_p}{\lambda_\alpha}=\sqrt{4\times 2}=\sqrt{8}=2\sqrt{2}$$

So $\boldsymbol{\lambda_p:\lambda_\alpha=2\sqrt{2}:1}$.

(b)(ii) When both have the same kinetic energy K, the momentum is $\boldsymbol{p=\sqrt{2mK}}$, so

$$\lambda=\frac{h}{\sqrt{2mK}}$$

For the same K,

$$\frac{\lambda_p}{\lambda_\alpha}=\sqrt{\frac{m_\alpha}{m_p}}=\sqrt{4}=2$$

So $\boldsymbol{\lambda_p:\lambda_\alpha=2:1}$.

TIP

For the same V you must include charge (because K = qV), but for the same K the charge drops out and only the masses matter.

Final answer. λ = h/p; for same accelerating potential λ_p:λ_α = 2√2:1, and for same kinetic energy λ_p:λ_α = 2:1.

← *Back to Q194*

Q195

(a) Write Einstein's photoelectric equation. How did Millikan prove the validity of this equation?
(b) Explain the existence of threshold frequency of incident radiation for photoelectric emission from a given surface.

[2024 • Set 55-5-1]

SOLUTION

(a) Einstein's photoelectric equation states that the maximum kinetic energy of an emitted photoelectron equals the photon energy minus the work function:

$$K_{max}=\frac{1}{2}mv_{max}^{2}=h\nu-\phi_0$$

where $\boldsymbol{\phi_0=h\nu_0}$ is the work function of the surface.

Millikan verified this equation experimentally. For a given alkali-metal surface he measured the stopping potential V_0 for light of several different frequencies ν. Since $\boldsymbol{eV_0=K_{max}=h\nu-\phi_0}$, he plotted V_0 against ν and found that:

$$V_0=\frac{h}{e}\,\nu-\frac{\phi_0}{e}$$

The graph was a **straight line**, exactly as Einstein's equation predicts. Its slope was measured to be $\frac{h}{e}$, and using the known value of e this gave a value of Planck's constant h that agreed (within a few percent) with the value Planck had obtained from black-body radiation. The intercept on the ν axis gave the metal's threshold frequency ν_0. This complete agreement proved the validity of Einstein's equation (and earned Millikan the Nobel Prize in 1923).

(b) The work function ϕ_0 is the minimum energy needed to pull an electron out of the metal surface. A photon of frequency ν carries energy hν, and an electron

can absorb only one photon. The electron can escape only if this single photon supplies at least the work function, that is

$$h\nu \phi_0$$

The smallest frequency for which this is just possible is the **threshold frequency**,

$$\nu_0 = \frac{\phi_0}{h}$$

If $\nu < \nu_0$, then $h\nu < \phi_0$, so even a very intense beam cannot eject an electron, because each photon individually has too little energy. This is exactly why a definite threshold frequency exists for every surface.

Final answer. K_max = hν − ϕ_0; Millikan's straight-line V_0 vs ν plot with slope h/e confirmed it, and the threshold frequency $\nu_0 = \phi_0/h$ exists because one photon must supply at least the work function.

← Back to Q195

Q196

A photosensitive surface of work function 2.1 eV *is irradiated by radiation of wavelength* 150 nm. *Calculate (i) the threshold wavelength, (ii) energy (in eV) of an incident photon, and (iii) maximum kinetic energy of emitted photoelectron.*

[2024 • Set 55-5-2]

SOLUTION

Given the work function $\phi = 2.1$ **eV** and incident wavelength $\lambda = 150$ **nm**, with **hc=1240 eV nm**.

(i) The threshold wavelength is the longest wavelength that can just cause emission, where the photon energy equals the work function:

$$\lambda_0 = \frac{hc}{\phi} = \frac{1240 \text{ eV nm}}{2.1 \text{ eV}}$$

$$\lambda_0 \approx 590 \text{ nm}$$

So $\lambda_0 \approx$ **590 nm**. Only light of wavelength shorter than this can eject electrons from the surface.

(ii) The energy of an incident photon is

$$E = \frac{hc}{\lambda} = \frac{1240\ \text{eV nm}}{150\ \text{nm}}$$

$$E \approx 8.27\ \text{eV}$$

So $E \approx$ **8.27 eV**.

(iii) By Einstein's photoelectric equation, the maximum kinetic energy is

$$K_{max} = E - \phi$$

$$K_{max} = 8.27\ \text{eV} - 2.1\ \text{eV}$$

$$K_{max} \approx 6.17\ \text{eV}$$

So the **maximum kinetic energy of the emitted photoelectron is about 6.17 eV**.

Final answer. $\lambda_0 \approx 590$ nm, photon energy E ≈ 8.27 eV, and K_max ≈ 6.17 eV.

← *Back to Q196*

Q197

The threshold frequency for a metal is 3.0×10^{14} Hz. *A beam of frequency* 9.0×10^{14} Hz *is incident on the metal. Calculate (i) the work function (in eV) of the metal and (ii) the maximum speed of photoelectrons.*

[2024 • Set 55-5-3]

SOLUTION

Given the threshold frequency $\nu_0 = 3.010^{14}$ **Hz** and incident frequency $\nu = 9.010^{14}$ **Hz**, with $h = 6.610^{-34}$ *J s*.

(i) The work function is the energy corresponding to the threshold frequency:

$$\phi = h\nu_0 = (6.610^{-34}\ J\ s)(3.010^{14}\ \text{Hz})$$

$$\phi = 1.98 10^{-19}\ J$$

Converting to electron-volts by dividing by $1.610^{-19}\ J$/eV,

$$\phi = \frac{1.98 10^{-19}}{1.610^{-19}}\ \text{eV} \approx 1.24\ \text{eV}$$

So the **work function is about 1.24 eV**.

(ii) By Einstein's photoelectric equation, the maximum kinetic energy is

$$K_{max} = h(\nu - \nu_0)$$

$$K_{max} = (6.610^{-34})(9.010^{14} - 3.010^{14})$$

$$K_{max} = (6.610^{-34})(6.010^{14})$$

$$K_{max} = 3.9610^{-19}\ J$$

The maximum speed follows from $\mathbf{K_{max} = \frac{1}{2} m v_{max}^2}$, with $\boldsymbol{m = 9.110^{-31}\ \text{kg}}$:

$$v_{max} = \sqrt{\frac{2K_{max}}{m}} = \sqrt{\frac{2(3.9610^{-19})}{9.110^{-31}}}$$

$$v_{max} = \sqrt{8.7010^{11}}$$

$$v_{max} \approx 9.310^{5}\ m\ s^{-1}$$

So the **maximum speed of the photoelectrons is about $\mathbf{9.310^5\ m\ s^{-1}}$**.

Final answer. Work function $\phi \approx 1.24$ eV and maximum speed $v_{max} \approx 9.3 \times 10^5$ m s^{-1}.

← *Back to Q197*

Q198

How does Einstein's photoelectric equation explain the emission of electrons from a metal surface? Explain briefly. Plot the variation of photocurrent with:

(a) collector plate potential for different intensity of incident radiation, and
(b) intensity of incident radiation.

[2023 • Set 55-1-1]

SOLUTION

Einstein treated light as a stream of photons, each of energy hν. When light falls on a metal, one photon is absorbed by one electron. The electron uses part of this energy, equal to the work function ϕ_0, to escape from the surface, and keeps the rest as kinetic energy:

$$K_{max}=h\nu-\phi_0$$

This single relation explains all the observed features of photoemission. Emission occurs only if $\nu>\nu_0$ (existence of a threshold frequency); K_{max} depends only on ν and not on intensity; K_{max} increases linearly with ν; and since a photon is absorbed in one step, emission is practically instantaneous.

(a) Photocurrent versus collector plate potential, for different intensities at the same frequency:

Draw the photocurrent (I) on the vertical axis and the collector plate potential (V) on the horizontal axis, with the origin in the middle so that the axis extends to negative V on the left. Each curve starts from zero at the same negative stopping potential $-V_0$ on the left, rises as V increases, and then flattens into a horizontal saturation level on the right. Plot two (or three) such curves for low and high intensity: all of them cut the V axis at the same point $-V_0$, but the curve for higher intensity saturates at a higher photocurrent. This shows the **stopping potential is the same for all intensities, while the saturation current is larger for greater intensity**.

(b) Photocurrent versus intensity of incident radiation, at a fixed frequency above threshold:

Plot photocurrent (I) on the vertical axis and intensity on the horizontal axis. The graph is a **straight line passing through the origin**, showing that the saturation photocurrent is directly proportional to the intensity, ***I***.

Final answer. K_max = hν − ϕ_0 explains emission; the I–V curves share one stopping potential $-V_0$ but saturate higher for greater intensity, and the I–intensity graph is a straight line through the origin.

← Back to Q198

Q199

The threshold wavelength for a metal is 3315 *Å. What should be the wavelength of light incident on the metal surface so that the maximum kinetic energy of the emitted photoelectrons be* 1.25 *eV?*

[2023 • Set 55-1-2]

SOLUTION

The threshold wavelength λ_0**=3315 Å** fixes the work function, and we need the incident wavelength λ that gives K_{max}**=1.25 eV**. Use **hc=12400 eV Å.**

The work function corresponds to the threshold wavelength:

$$\phi=\frac{hc}{\lambda_0}=\frac{12400\ \text{eV Å}}{3315\ \text{Å}}$$

$$\phi\approx 3.74\ \text{eV}$$

By Einstein's photoelectric equation, the energy of the incident photon must equal the work function plus the maximum kinetic energy:

$$E=\phi+K_{max}=3.74\ \text{eV}+1.25\ \text{eV}$$

$$E\approx 4.99\ \text{eV}$$

The required wavelength is then

$$\lambda=\frac{hc}{E}=\frac{12400\ \text{eV Å}}{4.99\ \text{eV}}$$

$$\lambda\approx 2485\ \text{Å}$$

So the incident light must have a **wavelength of about 2485 Å=248.5 nm**, which lies in the ultraviolet region.

Final answer. The incident light must have wavelength $\lambda \approx 2485$ Å (about 248.5 nm), in the ultraviolet region.

← Back to Q199

Q200

A particle is associated with a de Broglie wave of wavelength 1.2 *nm. The kinetic energy of the particle is made four times. Calculate the new value of the wavelength of the wave.*

[2023 • Set 55-1-3]

SOLUTION

For a particle of mass m and kinetic energy K, the de Broglie wavelength is

$$\lambda = \frac{h}{\sqrt{2mK}}$$

Since m and h are fixed for the particle, λ depends on K as

$$\lambda \propto \frac{1}{\sqrt{K}}$$

Taking the ratio of the new wavelength λ_2 (kinetic energy $\mathbf{K_2 = 4K_1}$) to the original λ_1 (kinetic energy K_1),

$$\frac{\lambda_2}{\lambda_1} = \sqrt{\frac{K_1}{K_2}} = \sqrt{\frac{K_1}{4K_1}} = \sqrt{\frac{1}{4}} = \frac{1}{2}$$

Therefore

$$\lambda_2 = \frac{\lambda_1}{2} = \frac{1.2\ \text{nm}}{2}$$

$$\lambda_2 = 0.6\ \text{nm}$$

So when the kinetic energy is made four times, the de Broglie wavelength **halves to 0.6 nm**.

COMMON MISTAKE

Because $\lambda \propto 1/\sqrt{K}$, making K four times larger multiplies λ by $1/\sqrt{4} = 1/2$, not by 1/4.

Final answer. The new de Broglie wavelength is 0.6 nm (half the original value).

← Back to Q200

Q201

Calculate the wavelength of de Broglie waves associated with a proton having ($\frac{500}{1.673}$) eV *energy. How will the wavelength be affected for an alpha particle having the same energy?*

[2023 • Set 55-3-1]

SOLUTION

The de Broglie wavelength of a particle of mass m carrying kinetic energy E is found from its momentum p, where $\boldsymbol{p = \sqrt{2mE}}$.

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mE}}$$

First convert the proton's kinetic energy from eV to joules. The energy is $\boldsymbol{E = \frac{500}{1.673}\ eV = 298.9\ eV}$.

$$E = 298.9 \times 1.6 \times 10^{-19}\ J = 4.78 \times 10^{-17}\ J$$

Take the proton mass $\mathbf{m_p = 1.673 \times 10^{-27}\ kg}$ and $\boldsymbol{h = 6.626 \times 10^{-34}\ J\,s}$. The momentum is:

$$p = \sqrt{2\ m_p\ E} = \sqrt{2 \times 1.673 \times 10^{-27} \times 4.78 \times 10^{-17}}$$

$$p = \sqrt{1.60 \times 10^{-43}} = 4.0 \times 10^{-22}\ \text{kg}\ m/s$$

Hence the de Broglie wavelength of the proton is:

$$\lambda_p = \frac{h}{p} = \frac{6.626 \times 10^{-34}}{4.0 \times 10^{-22}} \approx 1.65 \times 10^{-12}\ m$$

So $\boldsymbol{\lambda_p \approx 1.65 \times 10^{-12}\ m}$ **(about 1.65 pm)**.

For the alpha particle with the same kinetic energy E, only the mass changes. Since $\mathbf{m_\alpha = 4\ m_p}$, and $\boldsymbol{\lambda = \frac{h}{\sqrt{2mE}} \propto \frac{1}{\sqrt{m}}}$, the ratio is:

$$\frac{\lambda_\alpha}{\lambda_p} = \sqrt{\frac{m_p}{m_\alpha}} = \sqrt{\frac{1}{4}} = \frac{1}{2}$$

Therefore the **alpha particle's wavelength is halved**, giving $\lambda_\alpha = \frac{\lambda_p}{2} \approx 0.82 \times 10^{-12}\ m$.

> **TIP**
>
> At equal kinetic energy, lighter particles have longer de Broglie wavelengths, since $\lambda \propto 1/\sqrt{m}$.

Final answer. Proton: $\lambda_p \approx 1.65 \times 10^{-12}\ m$; for the alpha particle at the same energy the wavelength halves to about $0.82 \times 10^{-12}\ m$.

← *Back to Q201*

Q202

(a) A particle of mass m and charge q is accelerated through a potential difference V. Plot a graph of de Broglie wavelength λ associated with it as a function of V.
(b) Calculate the energy acquired by, and de Broglie wavelength associated with, an electron accelerated through a potential difference of 400 *V.*

[2023 • Set 55-3-2]

SOLUTION

(a) A particle of mass m and charge q accelerated through a potential difference V gains kinetic energy $K = qV$. Its momentum is $p = \sqrt{2mK} = \sqrt{2mqV}$, so the de Broglie wavelength is:

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mqV}}$$

For a fixed particle $\frac{h}{\sqrt{2mq}}$ is a constant, so $\lambda \propto \frac{1}{\sqrt{V}}$. This means λ decreases as V increases.

The graph of λ (on the y-axis) versus V (on the x-axis) is a falling curve: it starts high for small V and bends down towards the x-axis as V grows, never touching

it (a hyperbola-like decay of the form $\lambda \propto V^{-1/2}$). If instead λ is plotted against $\frac{1}{\sqrt{V}}$, the graph is a straight line passing through the origin with slope $\frac{h}{\sqrt{2mq}}$.

(b) For an electron accelerated through $V = 400\ V$, with charge $e = 1.6 \times 10^{-19}\ C$, the energy acquired is:

$$K = eV = 400\ eV$$

$$K = 400 \times 1.6 \times 10^{-19} = 6.4 \times 10^{-17}\ J$$

So the energy gained is $\mathbf{6.4 \times 10^{-17}\ J}$ **(i.e. 400 eV)**.

For an electron the de Broglie wavelength has the convenient form $\lambda = \frac{1.227}{\sqrt{V}}$ **nm** with V in volts:

$$\lambda = \frac{1.227}{\sqrt{400}}\ \text{nm} = \frac{1.227}{20}\ \text{nm}$$

$$\lambda \approx 0.0614\ \text{nm} = 6.14 \times 10^{-11}\ m$$

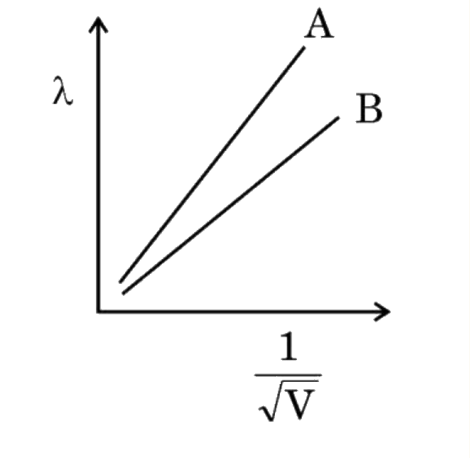
Hence $\lambda \approx 6.14 \times 10^{-11}\ m$.

Final answer. The electron gains $6.4 \times 10^{-17}\ J$ (400 eV) and has $\lambda \approx 6.14 \times 10^{-11}\ m$; the λ vs V graph is a $\lambda \propto 1/\sqrt{V}$ decay curve.

← *Back to Q202*

Q203

(a) The figure shows de Broglie wavelength (λ) for two particles A and B having same charges but different masses as a function of $\frac{1}{\sqrt{V}}$, where V is the potential through which the particles are accelerated.



(i) Which of them is heavier and why?

(ii) What does the slope of the line represent?

(b) Calculate the momentum of an electron having de Broglie wavelength of 3.0 .

[2023 • Set 55-3-3]

SOLUTION

(a) (i) For a charged particle accelerated through potential difference V, the de Broglie wavelength is:

$$\lambda = \frac{h}{\sqrt{2mqV}} = \frac{h}{\sqrt{2mq}} \cdot \frac{1}{\sqrt{V}}$$

When λ is plotted against $\frac{1}{\sqrt{V}}$ the graph is a straight line through the origin, and its slope is:

$$\text{slope} = \frac{h}{\sqrt{2mq}}$$

The two particles A and B carry the same charge q. From the slope expression, a larger mass m gives a smaller slope. So the particle whose line has the smaller slope (the less steep line) is the heavier one. **The particle with the smaller slope is heavier.**

(a) (ii) The slope of the λ versus $\frac{1}{\sqrt{V}}$ line equals $\frac{h}{\sqrt{2mq}}$. It therefore represents the combination of Planck's constant with the particle's mass and charge; for fixed charge it is a measure of $\frac{1}{\sqrt{m}}$, letting us compare the masses of the two particles.

(b) For an electron with de Broglie wavelength $\lambda = 3.0 = 3.0 \times 10^{-10}\ m$, the momentum follows from $\lambda = \frac{h}{p}$:

$$p = \frac{h}{\lambda} = \frac{6.626 \times 10^{-34}}{3.0 \times 10^{-10}}$$

$$p \approx 2.21 \times 10^{-24}\ \text{kg}\ m/s$$

Hence the momentum of the electron is $p \approx 2.21 \times 10^{-24}\ \text{kg}\ m/s$.

Final answer. The particle with the smaller slope is heavier; the slope equals $\frac{h}{\sqrt{2mq}}$; and the electron's momentum is $p \approx 2.21 \times 10^{-24}\ \text{kg}\ m/s$.

← *Back to Q203*

Q204

Photoelectrons are emitted from a metal surface when illuminated with UV light of wavelength 330 nm. *The minimum amount of energy required to emit the electrons from the surface is* 3.5×10^{-19} *J. Calculate:*
(i) the energy of the incident radiation, and
(ii) the kinetic energy of the photoelectron.

[2022 • Set 55-1-1]

SOLUTION

(i) The energy carried by each incident photon depends only on its wavelength, through Planck's relation:

$$E = h\nu = \frac{hc}{\lambda}$$

Here $\lambda = 330\ \text{nm} = 330 \times 10^{-9}\ m$, $h = 6.626 \times 10^{-34}\ J\ s$ and $c = 3 \times 10^{8}\ m/s$. Substituting:

$$E = \frac{(6.626 \times 10^{-34})(3 \times 10^{8})}{330 \times 10^{-9}}$$

$$E = \frac{1.988 \times 10^{-25}}{330 \times 10^{-9}} = 6.02 \times 10^{-19}\ J$$

In electron volts, $E = \dfrac{6.02 \times 10^{-19}}{1.6 \times 10^{-19}} \approx 3.76$ **eV**. So the **energy of the incident radiation is 6.02×10^{-19} J (about 3.76 eV)**.

(ii) The minimum energy needed to free an electron is the work function $\phi = 3.5 \times 10^{-19}$ J. By Einstein's photoelectric equation, the surplus photon energy appears as the maximum kinetic energy of the ejected electron:

$$K_{max} = E - \phi$$

$$K_{max} = 6.02 \times 10^{-19} - 3.5 \times 10^{-19}$$

$$K_{max} = 2.52 \times 10^{-19}\ J$$

In electron volts this is $\dfrac{2.52 \times 10^{-19}}{1.6 \times 10^{-19}} \approx 1.58$ **eV**. Hence the **maximum kinetic energy of the photoelectron is 2.52×10^{-19} J (about 1.58 eV)**.

CONCEPT

The work function is the minimum energy to free an electron; whatever the photon brings above it shows up as the electron's kinetic energy.

Final answer. Incident photon energy $E = 6.02 \times 10^{-19}$ J (3.76 eV) and the photoelectron's kinetic energy $K_{max} = 2.52 \times 10^{-19}$ J (1.58 eV).

← *Back to Q204*

Q205

Ultra-violet light of wavelength 200 nm *from a source is incident on a metal surface. If the stopping potential is* −2.5 *V, (a) calculate the work function of the metal, and (b) How would the surface respond to a high intensity red light of wavelength* 6328 *produced by a laser?*

[2022 • Set 55-1-2]

SOLUTION

(a) The energy of each ultraviolet photon is found from $E = \dfrac{hc}{\lambda}$. Using the handy value **hc = 1240 eV nm** with **λ = 200 nm**:

$$E = \frac{hc}{\lambda} = \frac{1240}{200} = 6.2 \text{ eV}$$

The stopping potential is given as $-2.5\ V$; its magnitude is $\mathbf{V_0 = 2.5\ V}$, so the maximum kinetic energy of the emitted electrons is $\mathbf{eV_0 = 2.5\ eV}$. Einstein's photoelectric equation gives:

$$eV_0 = E - \phi$$

Rearranging for the work function:

$$\phi = E - eV_0 = 6.2 - 2.5$$

$$\phi = 3.7 \text{ eV}$$

So the **work function of the metal is 3.7 eV**.

(b) For the red laser light, $\mathbf{\lambda = 6328 = 632.8\ nm}$. The energy of each red photon is:

$$E_{red} = \frac{1240}{632.8} \approx 1.96 \text{ eV}$$

Compare this with the work function:

$$E_{red} = 1.96 \text{ eV} < \phi = 3.7 \text{ eV}$$

Since each red photon carries less energy than the work function, it cannot free a single electron. **No photoemission occurs with the red light, however intense it is.** Raising the intensity only sends in more photons per second; it does not increase the energy of any individual photon. Because emission depends on the energy of one photon (not on the total light energy), the surface stays inert to the red laser. This is exactly the kind of result that the wave theory could not explain and that the photon picture does.

Final answer. The work function is $\mathbf{\phi = 3.7\ eV}$; the red laser light (**1.96** eV per photon) cannot eject any electrons no matter how intense, since each photon's energy is below the work function.

← Back to Q205

Q206

The de-Broglie wavelength associated with an electron is 0.30 nm. *Calculate: (a) the speed, and (b) the kinetic energy (in eV), of the electron.*

[2022 • Set 55-1-3]

SOLUTION

The de Broglie relation links wavelength to speed through the electron's momentum:

$$\lambda = \frac{h}{mv} \Rightarrow v = \frac{h}{m\lambda}$$

Here $\lambda = 0.30\ \text{nm} = 3.0 \times 10^{-10}\ m$, the electron mass $m = 9.11 \times 10^{-31}\ \text{kg}$, and $h = 6.626 \times 10^{-34}\ J\,s$.

(a) The speed is:

$$v = \frac{6.626 \times 10^{-34}}{(9.11 \times 10^{-31})(3.0 \times 10^{-10})}$$

$$v = \frac{6.626 \times 10^{-34}}{2.733 \times 10^{-40}}$$

$$v \approx 2.42 \times 10^{6}\ m/s$$

So the **speed of the electron is about $2.42 \times 10^{6}\ m/s$**.

(b) The kinetic energy is $K = \frac{1}{2}mv^2$:

$$K = \frac{1}{2}(9.11 \times 10^{-31})(2.42 \times 10^{6})^2$$

$$K = \frac{1}{2}(9.11 \times 10^{-31})(5.86 \times 10^{12})$$

$$K = 2.67 \times 10^{-18}\ J$$

Converting to electron volts by dividing by 1.6×10^{-19}:

$$K = \frac{2.67 \times 10^{-18}}{1.6 \times 10^{-19}} \approx 16.7\ \text{eV}$$

Hence the **kinetic energy of the electron is about 16.7 eV** (i.e. $2.67 \times 10^{-18}\ J$).

Final answer. The electron's speed is about $2.42 \times 10^{6}\ m/s$ and its kinetic energy is about **16.7** eV ($2.67 \times 10^{-18}\ J$).

← *Back to Q206*

Q207

Find the ratio of the de Broglie wavelengths associated with an alpha particle and a proton, if both
(a) have the same speeds,
(b) have the same kinetic energy,
(c) are accelerated through the same potential difference.

[2022 • Set 55-2-1]

SOLUTION

For an alpha particle and a proton, the mass and charge are $\mathbf{m_\alpha = 4\,m_p}$ and $\mathbf{q_\alpha = 2e}$, while m_p and $\mathbf{q_p = e}$ for the proton. The de Broglie wavelength is $\lambda = \dfrac{h}{p}$, and we evaluate p for each case.

(a) Same speed v: here $\mathbf{p = mv}$, so $\lambda = \dfrac{h}{mv}$ and:

$$\frac{\lambda_\alpha}{\lambda_p} = \frac{m_p\, v}{m_\alpha\, v} = \frac{m_p}{4\, m_p} = \frac{1}{4}$$

So $\boldsymbol{\lambda_\alpha : \lambda_p = 1 : 4}$.

(b) Same kinetic energy K: using $\mathbf{p = \sqrt{2mK}}$, we get $\lambda = \dfrac{h}{\sqrt{2mK}}$, which gives:

$$\frac{\lambda_\alpha}{\lambda_p} = \sqrt{\frac{m_p}{m_\alpha}} = \sqrt{\frac{m_p}{4\, m_p}} = \frac{1}{2}$$

So $\boldsymbol{\lambda_\alpha : \lambda_p = 1 : 2}$.

(c) Same accelerating potential V: now $\mathbf{K = qV}$, so $\mathbf{p = \sqrt{2mqV}}$ and $\lambda = \dfrac{h}{\sqrt{2mqV}}$.
Therefore:

$$\frac{\lambda_\alpha}{\lambda_p} = \sqrt{\frac{m_p\, q_p}{m_\alpha\, q_\alpha}} = \sqrt{\frac{m_p \cdot e}{4\, m_p \cdot 2e}} = \sqrt{\frac{1}{8}}$$

$$\frac{\lambda_\alpha}{\lambda_p} = \frac{1}{2\sqrt{2}}$$

So $\lambda_\alpha : \lambda_p = 1 : 2\sqrt{2}$.

COMMON MISTAKE

Pick the right form of p for each case: mv *for equal speed,* $\sqrt{2mK}$ *for equal energy, and* $\sqrt{2mqV}$ *for equal potential. Mixing them up is the usual slip.*

Final answer. $\lambda_\alpha : \lambda_p = 1:4$ at equal speed, $1:2$ at equal kinetic energy, and $1:2\sqrt{2}$ at equal accelerating potential.

← *Back to Q207*

Q208

An electron is accelerated from rest through a potential difference of 100 *V. Find:*
(i) the wavelength associated with
(ii) the momentum of and
(iii) the velocity required by, the electron.

[2022 • Set 55-3-1]

SOLUTION

An electron accelerated from rest through $V = 100\ V$ gains kinetic energy equal to the work done on it:

$$K = eV = 100\ eV = 100 \times 1.6 \times 10^{-19} = 1.6 \times 10^{-17}\ J$$

(i) For an electron, the de Broglie wavelength is $\lambda = \frac{1.227}{\sqrt{V}}$ **nm** with V in volts:

$$\lambda = \frac{1.227}{\sqrt{100}}\ nm = \frac{1.227}{10}\ nm$$

$$\lambda \approx 0.1227\ nm = 1.227 \times 10^{-10}\ m$$

So the **wavelength is about** $\mathbf{1.227 \times 10^{-10}\ m}$.

(ii) The momentum follows from $\lambda = \frac{h}{p}$:

$$p = \frac{h}{\lambda} = \frac{6.626 \times 10^{-34}}{1.227 \times 10^{-10}}$$

$$p \approx 5.4 \times 10^{-24}\ \text{kg}\ m/s$$

So the **momentum is about 5.4×10^{-24} kg m/s**.

(iii) The velocity is obtained from $p = m_e v$, with $m_e = 9.11 \times 10^{-31}$ kg:

$$v = \frac{p}{m_e} = \frac{5.4 \times 10^{-24}}{9.11 \times 10^{-31}}$$

$$v \approx 5.93 \times 10^{6}\ m/s$$

So the **velocity acquired is about 5.93×10^{6} m/s**.

Final answer. The electron has $\lambda \approx 1.227 \times 10^{-10}$ m, momentum $p \approx 5.4 \times 10^{-24}$ kg m/s, and velocity $v \approx 5.93 \times 10^{6}$ m/s.

← *Back to Q208*

Q209

(a) Calculate the frequency of a photon of energy 6.5×10^{-19} J.
(b) Can this photon cause emission of an electron from the surface of Cs of work function 2.14 eV? If yes, what will be maximum kinetic energy of the photoelectron?

[2022 • Set 55-3-2]

SOLUTION

(a) The frequency of a photon is related to its energy by Planck's relation $E = h\nu$, so:

$$\nu = \frac{E}{h}$$

With $E = 6.5 \times 10^{-19}$ J and $h = 6.63 \times 10^{-34}$ $J\ s$:

$$\nu = \frac{6.5 \times 10^{-19}}{6.63 \times 10^{-34}}$$

$$\nu \approx 9.8 \times 10^{14}\ \text{Hz}$$

So the **frequency of the photon is about 9.8×10^{14} Hz**.

(b) To check whether emission can occur, express the photon energy in electron volts:

$$E = \frac{6.5 \times 10^{-19}}{1.6 \times 10^{-19}} \approx 4.06 \text{ eV}$$

Compare with the work function of caesium, $\phi = 2.14$ **eV**:

$$E = 4.06 \text{ eV} > \phi = 2.14 \text{ eV}$$

Since the photon energy exceeds the work function, **yes, the photon can eject an electron from the Cs surface.** The maximum kinetic energy of the photoelectron is the excess energy:

$$K_{max} = E - \phi = 4.06 - 2.14$$

$$K_{max} = 1.92 \text{ eV}$$

Hence the **maximum kinetic energy of the photoelectron is 1.92 eV**, and emitted electrons have energies ranging from 0 up to this value.

Final answer. The photon frequency is about **9.8×10^{14} Hz**; since its energy (**4.06** eV) exceeds the Cs work function, emission occurs with **$K_{max} = 1.92$** eV.

← *Back to Q209*

Q210

The work function of a metal is 2.31 eV. *Photoelectric emission occurs when light of frequency* 6.4×10^{14} Hz *is incident on the metal surface. Calculate:*
(i) the energy of the incident radiation,
(ii) the maximum kinetic energy of the emitted electron and
(iii) the stopping potential of the surface.

[2022 • Set 55-3-3]

SOLUTION

Given the work function $\phi = 2.31$ **eV** and incident frequency $\nu = 6.4 \times 10^{14}$ **Hz**.

(i) The energy of each incident photon is $E = h\nu$:

$$E = (6.63 \times 10^{-34})(6.4 \times 10^{14})$$

$$E = 4.24 \times 10^{-19}\ J$$

In electron volts, $E = \dfrac{4.24 \times 10^{-19}}{1.6 \times 10^{-19}} \approx 2.65$ eV. So the **energy of the incident radiation is $4.24 \times 10^{-19}\ J$ (about 2.65 eV)**.

(ii) By Einstein's photoelectric equation, the maximum kinetic energy is the photon energy minus the work function:

$$K_{max} = E - \phi = 2.65 - 2.31$$

$$K_{max} = 0.34 \text{ eV}$$

So the **maximum kinetic energy of the emitted electron is 0.34 eV**.

(iii) The stopping potential V_0 is the retarding voltage that just stops the fastest electron, given by $eV_0 = K_{max}$:

$$V_0 = \frac{K_{max}}{e} = \frac{0.34 \text{ eV}}{e}$$

$$V_0 = 0.34\ V$$

Hence the **stopping potential is 0.34 V**. The frequency is only just above the threshold $\nu_0 = \dfrac{\phi}{h} = 5.58 \times 10^{14}$ Hz, so the electrons emerge with only a small kinetic energy and a small stopping voltage suffices.

TIP

When K_{max} is in eV, the stopping potential in volts is numerically the same value, since $eV_0 = K_{max}$ and the e cancels the eV unit.

Final answer. Incident energy $E = 4.24 \times 10^{-19}\ J$ (2.65 eV), $K_{max} = 0.34$ eV, and stopping potential $V_0 = 0.34$ V.

← *Back to Q210*

Q211

(a) Use Einstein's photoelectric equation to depict the variation of the maximum kinetic energy (E_k) of electrons emitted, with the frequency (ν) of the incident radiation.
(b) A photosensitive surface is illuminated with a beam of (i) yellow light, and (ii) red light, both of the same intensity. In which case will
(I) photoelectrons have more E_k?
(II) more number of electrons be emitted?
Justify your answer in each case.

[2022 • Set 55-4-1]

SOLUTION

Einstein's photoelectric equation relates the maximum kinetic energy of an emitted photoelectron to the frequency of the incident radiation. When a photon of energy $h\nu$ strikes a metal of work function ϕ_0, part of the energy frees the electron and the rest appears as its kinetic energy.

$$E_k = h\nu - \phi_0$$

Since $\boldsymbol{\phi_0 = h\nu_0}$, where ν_0 is the threshold frequency, this can be written as $\boldsymbol{E_k = h\nu - h\nu_0}$. Comparing with the straight line equation $\boldsymbol{y = mx + c}$, a graph of E_k against ν is a straight line of slope h (Planck's constant). The line has a positive slope, cuts the ν-axis at $\boldsymbol{\nu = \nu_0}$ (where $\boldsymbol{E_k = 0}$), and meets the negative E_k-axis at $-\phi_0$ when extended back to $\boldsymbol{\nu = 0}$. No emission occurs for $\boldsymbol{\nu < \nu_0}$, so that portion is not physical.

Graph description: x-axis is frequency ν, y-axis is maximum kinetic energy E_k. A straight line of slope h starts from the point $(\nu_0, 0)$ on the ν-axis and rises to the right. Its backward extension meets the E_k-axis at $-\phi_0$.

(b) For light of the same colour family, yellow light has a higher frequency (shorter wavelength) than red light, since $\boldsymbol{\nu_{yellow} > \nu_{red}}$.

(I) Photoelectrons have more E_k for yellow light. From $\boldsymbol{E_k = h\nu - \phi_0}$, a higher incident frequency gives larger E_k. As $\boldsymbol{\nu_{yellow} > \nu_{red}}$, **yellow light gives photoelectrons of greater maximum kinetic energy**.

(II) The number of photoelectrons emitted depends on the number of incident photons (the photon flux), not on their energy. Intensity = $n h\nu$, where n is the number of photons per second. For the same intensity, the beam with higher frequency (yellow) carries fewer photons per second, while the lower frequency beam (red) carries more photons per second. Hence **more electrons are emitted with red light**, because for equal intensity it delivers a larger number of photons.

TIP

For equal intensity, photon number is inversely proportional to frequency: $n \propto 1/\nu$, so the lower-frequency (red) beam carries more photons and ejects more electrons.

Final answer. (I) Yellow light gives photoelectrons of greater maximum kinetic energy; (II) for equal intensity red light emits more photoelectrons (more photons per second).

← *Back to Q211*

Q212

The wavelength of the waves associated with a particle having kinetic energy E is λ. How and by what factor should its kinetic energy be changed so that the wavelength becomes $(\frac{\lambda}{3})$*? Also, find the ratio of the final to the initial velocity of the particle.*

[2022 • Set 55-4-2]

SOLUTION

The de Broglie wavelength of a particle of mass m and kinetic energy E is found from its momentum $p = \sqrt{2mE}$.

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mE}}$$

For a particle of fixed mass, h and m are constant, so the wavelength varies only with the kinetic energy.

$$\lambda \propto \frac{1}{\sqrt{E}}$$

Let the final kinetic energy be E' giving wavelength $\lambda' = \frac{\lambda}{3}$. Taking the ratio of the two cases:

$$\frac{\lambda'}{\lambda} = \sqrt{\frac{E}{E'}}$$

$$\frac{1}{3} = \sqrt{\frac{E}{E'}}$$

Squaring both sides:

$$\frac{1}{9} = \frac{E}{E'}$$

$$E' = 9E$$

So the kinetic energy must be **increased by a factor of 9**.

For the velocity, kinetic energy is $E = \frac{1}{2}mv^2$, so $v \propto \sqrt{E}$.

$$\frac{v'}{v} = \sqrt{\frac{E'}{E}} = \sqrt{\frac{9E}{E}} = \sqrt{9} = 3$$

Hence the **ratio of final to initial velocity is 3 : 1**.

Final answer. Kinetic energy must be increased 9 times ($E' = 9E$), and the ratio of final to initial velocity is $v' : v = 3 : 1$.

← *Back to Q212*

Q213

A proton is accelerated through a potential difference V. After acceleration, the de Broglie wavelength associated with it is λ. If the proton is replaced by an alpha particle, then find the de Broglie wavelength associated with it if it were accelerated through the same potential difference V. What will be the momentum of the alpha particle?

[2022 • Set 55-4-3]

SOLUTION

When a charged particle of charge q and mass m is accelerated from rest through a potential difference V, the work done qV becomes its kinetic energy, and its momentum is $\boldsymbol{p = \sqrt{2mqV}}$. The de Broglie wavelength is therefore:

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mqV}}$$

For the proton (charge e, mass m_p) accelerated through V:

$$\lambda = \frac{h}{\sqrt{2m_p\, eV}}$$

An alpha particle has charge $\mathbf{q_\alpha = 2e}$ and mass $\mathbf{m_\alpha = 4m_p}$. Accelerated through the same V:

$$\lambda_\alpha = \frac{h}{\sqrt{2m_\alpha\, q_\alpha\, V}} = \frac{h}{\sqrt{2(4m_p)(2e)V}}$$

$$\lambda_\alpha = \frac{h}{\sqrt{8}\,\sqrt{2m_p\, eV}} = \frac{1}{\sqrt{8}} \cdot \frac{h}{\sqrt{2m_p\, eV}}$$

$$\lambda_\alpha = \frac{\lambda}{\sqrt{8}} = \frac{\lambda}{2\sqrt{2}}$$

So the **de Broglie wavelength of the alpha particle is** $\boldsymbol{\frac{\lambda}{2\sqrt{2}}}$.

The momentum of the alpha particle follows from $\mathbf{p_\alpha = \frac{h}{\lambda_\alpha}}$:

$$p_\alpha = \frac{h}{\lambda/(2\sqrt{2})} = \frac{2\sqrt{2}\,h}{\lambda}$$

Hence the **momentum of the alpha particle is** $\boldsymbol{\frac{2\sqrt{2}\,h}{\lambda}}$.

Final answer. $\boldsymbol{\lambda_\alpha = \frac{\lambda}{2\sqrt{2}}}$ and the momentum of the alpha particle is $\boldsymbol{p_\alpha = \frac{2\sqrt{2}\,h}{\lambda}}$.

← *Back to Q213*

Q214

(a) Calculate the energy and momentum of a photon in a monochromatic beam of wavelength 331.5 nm.
(b) How fast should a hydrogen atom travel in order to have the same momentum as that of the photon in part (a)?

[2022 • Set 55-5-1]

SOLUTION

The energy of a photon of wavelength λ is given by the Planck-Einstein relation, where h is Planck's constant and c the speed of light.

$$E = \frac{hc}{\lambda}$$

Substituting $h = 6.62610^{-34}\ J\ s$, $c = 310^{8}\ m\ s^{-1}$ and $\lambda = 331.5\ nm = 331.510^{-9}\ m$:

$$E = \frac{(6.62610^{-34})(310^{8})}{331.510^{-9}}$$

$$E \approx 6.010^{-19}\ J$$

In electron volts, $E = \frac{6.010^{-19}}{1.610^{-19}} \approx 3.75\ eV$.

The momentum of the photon is $p = \frac{E}{c}$:

$$p = \frac{6.010^{-19}}{310^{8}}$$

$$p = 2.010^{-27}\ kg\ m\ s^{-1}$$

So the **photon has energy $\approx 6.010^{-19}\ J$ (about 3.75 eV) and momentum $2.010^{-27}\ kg\ m\ s^{-1}$**.

(b) For the hydrogen atom to have the same momentum, $p = m_H v$, where $m_H = 1.6710^{-27}\ kg$.

$$v = \frac{p}{m_H} = \frac{2.010^{-27}}{1.6710^{-27}}$$

$$v \approx 1.2\ m\ s^{-1}$$

Hence the **hydrogen atom must travel at about 1.2 $m\ s^{-1}$** to carry the same momentum as the photon.

Final answer. Photon: $E \approx 6.0 \cdot 10^{-19}\ J$ (≈ 3.75 eV), $p \approx 2.0 \cdot 10^{-27}\ kg\ m\ s^{-1}$; hydrogen atom speed $\approx 1.2\ m\ s^{-1}$.

← *Back to Q214*

Q215

An alpha particle is accelerated through a potential difference of 100 *V. Calculate:*
(i) The speed acquired by the alpha particle, and
(ii) The de-Broglie wavelength associated with it.
(Take mass of alpha particle = 6.4×10^{-27} kg*)*

[2022 • Set 55-5-2]

SOLUTION

An alpha particle carries charge $q_\alpha = 2e = 3.2 \cdot 10^{-19}\ C$ and here has mass $m_\alpha = 6.4 \cdot 10^{-27}$ kg. When accelerated from rest through potential difference $V = 100\ V$, the electric work done equals its kinetic energy.

$$q_\alpha V = \frac{1}{2} m_\alpha v^2$$

Solving for the speed:

$$v = \sqrt{\frac{2\, q_\alpha V}{m_\alpha}}$$

$$v = \sqrt{\frac{2(3.2 \cdot 10^{-19})(100)}{6.4 \cdot 10^{-27}}}$$

$$v = \sqrt{\frac{6.4 \cdot 10^{-17}}{6.4 \cdot 10^{-27}}} = \sqrt{10^{10}}$$

$$v = 1.0 \cdot 10^{5}\ m\ s^{-1}$$

So the **speed acquired by the alpha particle is $10^5\ m\ s^{-1}$**.

(ii) The de Broglie wavelength is $\lambda = \frac{h}{m_\alpha v}$, with $h = 6.6 \cdot 10^{-34}\ J\ s$.

$$\lambda = \frac{6.610^{-34}}{(6.410^{-27})(1.010^{5})}$$

$$\lambda = \frac{6.610^{-34}}{6.410^{-22}}$$

$$\lambda \approx 1.0310^{-12}\ m$$

Hence the **de Broglie wavelength associated with the alpha particle is about $1.0310^{-12}\ m$ (roughly 1 pm)**.

Final answer. Speed $v = 1.010^{5}\ m\ s^{-1}$ and de Broglie wavelength $\lambda \approx 1.0310^{-12}\ m$.

← *Back to Q215*

Q216

(a) Give an example each of a metal from which photoelectric emission takes place when irradiated by (i) UV light (ii) visible light.
(b) The work function of a metal is 4.50 eV. *Find the frequency of light to be used to eject electrons from the metal surface with a maximum kinetic energy of* 6.06×10^{-19} *J.*

[2022 • Set 55-5-3]

SOLUTION

(a) Photoelectric emission occurs only when the photon energy exceeds the metal's work function, so the choice of source depends on the metal.

(i) Ultraviolet light: a metal such as zinc, whose work function is about 4.3 eV, needs high-energy UV photons for emission. (Other valid examples are copper or platinum.)

(ii) Visible light: an alkali metal such as caesium ($\Phi_0 \approx$ **2.14 eV**) or sodium ($\Phi_0 \approx$ **2.75 eV**) has a low work function and emits electrons even with ordinary visible light.

(b) By Einstein's photoelectric equation, the photon energy is shared between the work function and the maximum kinetic energy.

$$h\nu = \phi_0 + K_{max}$$

Convert the work function to joules: $\phi_0 = 4.50\ \text{eV} = 4.501.610^{-19} = 7.2010^{-19}\ J$.

$$h\nu = 7.2010^{-19} + 6.0610^{-19}$$

$$h\nu = 1.32610^{-18}\ J$$

Solving for the frequency with $h = 6.62610^{-34}\ J\ s$:

$$\nu = \frac{1.32610^{-18}}{6.62610^{-34}}$$

$$\nu \approx 2.010^{15}\ \text{Hz}$$

Hence the **frequency of light required is about 2.010^{15} Hz**.

Final answer. Example metals: UV light: zinc; visible light: caesium (or sodium). Required frequency $\nu \approx 2.010^{15}$ Hz.

← *Back to Q216*

Q217

The maximum kinetic energy of the photoelectrons emitted is doubled when the wavelength of light incident on the photosensitive surface changes from λ_1 to λ_2. Deduce expressions for the threshold wavelength and work function for the metal surface in terms of λ_1 and λ_2.

[2020 • Set 55-2-1]

SOLUTION

Let ϕ_0 be the work function and λ_0 the threshold wavelength, so that $\phi_0 = \frac{hc}{\lambda_0}$.

By Einstein's photoelectric equation the maximum kinetic energy for incident wavelength λ is $K = \frac{hc}{\lambda} - \phi_0$.

For the first wavelength λ_1, the kinetic energy is K_1:

$$K_1 = \frac{hc}{\lambda_1} - \phi_0$$

For the second wavelength λ_2, the kinetic energy doubles to $2K_1$:

$$2K_1 = \frac{hc}{\lambda_2} - \phi_0$$

Subtracting the first equation from the second eliminates ϕ_0 and gives K_1:

$$2K_1 - K_1 = \left(\frac{hc}{\lambda_2} - \phi_0\right) - \left(\frac{hc}{\lambda_1} - \phi_0\right)$$

$$K_1 = hc\left(\frac{1}{\lambda_2} - \frac{1}{\lambda_1}\right) = hc\,\frac{\lambda_1 - \lambda_2}{\lambda_1\lambda_2}$$

Now substitute this K_1 back into the first equation to get the work function:

$$\phi_0 = \frac{hc}{\lambda_1} - K_1 = \frac{hc}{\lambda_1} - hc\,\frac{\lambda_1 - \lambda_2}{\lambda_1\lambda_2}$$

$$\phi_0 = \frac{hc\,\lambda_2 - hc(\lambda_1 - \lambda_2)}{\lambda_1\lambda_2} = \frac{hc(2\lambda_2 - \lambda_1)}{\lambda_1\lambda_2}$$

So the **work function is $\phi_0 = \dfrac{hc(2\lambda_2 - \lambda_1)}{\lambda_1\lambda_2}$**.

The threshold wavelength follows from $\lambda_0 = \dfrac{hc}{\phi_0}$:

$$\lambda_0 = \frac{hc}{\frac{hc(2\lambda_2 - \lambda_1)}{\lambda_1\lambda_2}} = \frac{\lambda_1\lambda_2}{2\lambda_2 - \lambda_1}$$

Hence the **threshold wavelength is $\lambda_0 = \dfrac{\lambda_1\lambda_2}{2\lambda_2 - \lambda_1}$**.

Final answer. $\phi_0 = \dfrac{hc(2\lambda_2 - \lambda_1)}{\lambda_1\lambda_2}$ and threshold wavelength $\lambda_0 = \dfrac{\lambda_1\lambda_2}{2\lambda_2 - \lambda_1}$.

← *Back to Q217*

Q218

What do you mean by wave nature of an electron? How was quantisation of angular momentum of the orbiting electron in Bohr's model of hydrogen atom explained by de Broglie hypothesis?

[2020 • Set 55-3-1]

SOLUTION

Wave nature of an electron: According to de Broglie's hypothesis, every moving material particle has a wave (a matter wave or de Broglie wave) associated with it. For an electron of mass m moving with velocity v, this wavelength is:

$$\lambda = \frac{h}{mv} = \frac{h}{p}$$

where h is Planck's constant and p the momentum. For an electron ($\boldsymbol{m} = \mathbf{9.1110^{-31}\ kg}$) moving at ordinary atomic speeds, λ comes out of the order of $10^{-10}\ m$, comparable to atomic dimensions. This is why the wave nature of electrons can be observed in diffraction (the Davisson-Germer experiment), whereas large everyday objects have an immeasurably small λ and show no wave behaviour.

de Broglie explanation of Bohr's quantisation: For an electron revolving in a stable circular orbit, the associated matter wave must close on itself and form a standing wave around the orbit. If it did not, then after each revolution the wave would return out of phase, interfere destructively with itself, and the orbit would not survive. The condition for a standing wave is that the circumference of the orbit must contain a whole number of de Broglie wavelengths.

$$2\pi r = n\lambda,\ n = 1, 2, 3,$$

Substituting $\lambda = \frac{h}{mv}$:

$$2\pi r = \frac{nh}{mv}$$

Rearranging for the angular momentum:

$$mvr = \frac{nh}{2\pi} = n$$

This is exactly **Bohr's quantisation condition** $\boldsymbol{L = mvr = \frac{nh}{2\pi}}$. Thus the de Broglie standing-wave picture explains why only those orbits are allowed in which the angular momentum is an integral multiple of $\frac{h}{2\pi}$.

Final answer. An electron has an associated matter wave $\boldsymbol{\lambda = h/mv}$; requiring a standing wave $\boldsymbol{2\pi r = n\lambda}$ around the orbit gives Bohr's quantisation $\boldsymbol{mvr = nh/2\pi}$.

← Back to Q218

Q219

(a) Plot a graph to show the variation of stopping potential with frequency of incident radiation in relation to photoelectric effect.
(b) Use Einstein's photoelectric equation to show how from this graph, (i) Threshold frequency, and (ii) Planck's constant can be determined.
OR
(a) How does one explain the emission of electrons from a photosensitive surface with the help of Einstein's photoelectric equation?
(b) Work function of aluminium is 4.2 eV. *If two photons each of energy* 2.5 eV *are incident on its surface, will the emission of electrons take place? Justify your answer.*
(c) The stopping potential in an experiment on photoelectric effect is 1.5 *V. What is the maximum kinetic energy of the photoelectrons emitted? Calculate in Joules.*

[2019 • Set 55-3-1]

SOLUTION

(a) The stopping potential V_0 is the reverse voltage that just stops the most energetic photoelectrons, so $\mathbf{eV_0 = K_{max}}$. By Einstein's equation, $\mathbf{eV_0 = h\nu - \phi_0}$, which gives $\mathbf{V_0 = \frac{h}{e}\nu - \frac{\phi_0}{e}}$. A graph of V_0 against ν is therefore a straight line of slope $\frac{h}{e}$.

Graph description: x-axis is frequency ν, y-axis is stopping potential V_0. The line starts on the ν-axis at the threshold frequency ν_0 (where $\mathbf{V_0 = 0}$) and rises with positive slope $\frac{h}{e}$. Its backward extension cuts the V_0-axis at $-\frac{\phi_0}{e}$.

(b) From Einstein's equation $\mathbf{eV_0 = h\nu - \phi_0}$:

(i) Threshold frequency: at $\mathbf{V_0 = 0}$, the photon energy just equals the work function, so $\mathbf{h\nu_0 = \phi_0}$. The point where the line meets the ν-axis directly gives $\mathbf{\nu_0 = \frac{\phi_0}{h}}$.

(ii) Planck's constant: the slope of the V_0 versus ν line is $\frac{h}{e}$. Therefore $\boldsymbol{h = e}\times$**(slope)**, so measuring the slope and multiplying by the electronic charge

gives Planck's constant.

OR (b) Aluminium, two photons each of 2.5 eV: photoelectric emission is a one-photon process, so only one photon interacts with one electron at a time. A single photon carries 2.5 eV, which is less than the work function $\phi_0 = 4.2$ **eV**. The two photons cannot add their energies on the same electron. Since **2.5 eV < 4.2 eV**, **no emission of electrons takes place**.

OR (c) Stopping potential $V_0 = 1.5\ V$. The maximum kinetic energy is $K_{max} = eV_0$:

$$K_{max} = (1.610^{-19})(1.5)$$

$$K_{max} = 2.410^{-19}\ J$$

Hence the **maximum kinetic energy of the photoelectrons is $2.410^{-19}\ J$ (i.e. 1.5 eV)**.

COMMON MISTAKE

Photoemission is strictly a one-photon-one-electron process: two 2.5 eV photons cannot combine to beat a 4.2 eV work function.

Final answer. Threshold frequency = ν_0 (x-intercept); $h = e\times$slope. OR: no emission from Al (one **2.5 eV** photon **< 4.2 eV**); $K_{max} = 2.410^{-19}\ J$.

← *Back to Q219*

Q220

(a) State briefly, with what purpose was Davisson and Germer experiment performed and what inference was drawn from this.
(b) Obtain an expression for the ratio of the accelerating potentials required to accelerate a proton and an α-particle to have the same de-Broglie wavelength associated with them.
OR
(a) An electron and a proton are accelerated through the same potential. Which one of the two has
(i) greater value of de-Broglie wavelength associated with it, and

(ii) lesser momentum?

Justify your answer in each case.

(b) How is the momentum of a particle related with its de-Broglie wavelength? Show the variation on a graph.

[2019 • Set 55-4-1]

SOLUTION

(a) Davisson and Germer performed their experiment to test de Broglie's hypothesis, that is, to find out whether moving electrons show wave behaviour. A beam of electrons accelerated through a known potential was directed at a nickel crystal, and the intensity of the scattered electrons was measured as a function of scattering angle. A sharp peak was observed at a particular angle, exactly as in the diffraction of waves by a crystal. The wavelength found from this diffraction agreed with the de Broglie wavelength $\lambda = \frac{h}{\sqrt{2meV}}$. The inference drawn was that **electrons possess wave nature**, confirming de Broglie's hypothesis of matter waves.

(b) For a particle of charge q, mass m accelerated through potential V, the de Broglie wavelength is $\lambda = \frac{h}{\sqrt{2mqV}}$. For equal wavelengths of proton and alpha particle, $\lambda_p = \lambda_\alpha$, so:

$$\frac{h}{\sqrt{2\, m_p\, q_p\, V_p}} = \frac{h}{\sqrt{2\, m_\alpha\, q_\alpha\, V_\alpha}}$$

Squaring and cancelling common factors:

$$m_p\, q_p\, V_p = m_\alpha\, q_\alpha\, V_\alpha$$

$$\frac{V_p}{V_\alpha} = \frac{m_\alpha\, q_\alpha}{m_p\, q_p}$$

Putting $m_\alpha = 4m_p$ and $q_\alpha = 2q_p$:

$$\frac{V_p}{V_\alpha} = \frac{(4m_p)(2q_p)}{m_p\, q_p} = 8$$

So the **ratio of accelerating potentials is $V_p : V_\alpha = 8 : 1$**.

OR (a) With both accelerated through the same potential V, $\lambda = \frac{h}{\sqrt{2mqV}}$ and momentum $p = \sqrt{2mqV}$. The electron and proton carry the same magnitude of

charge, so only the mass differs.

(i) Greater de Broglie wavelength: since $\lambda \propto \frac{1}{\sqrt{m}}$, the lighter particle has the larger wavelength. The **electron (smaller mass) has the greater de Broglie wavelength**.

(ii) Lesser momentum: since $p = \sqrt{2mqV} \propto \sqrt{m}$, the lighter particle has the smaller momentum. The **electron (smaller mass) has the lesser momentum**.

OR (b) The de Broglie relation links momentum and wavelength as $\lambda = \frac{h}{p}$, that is $p\lambda = h$ (a constant). Thus $p \propto \frac{1}{\lambda}$.

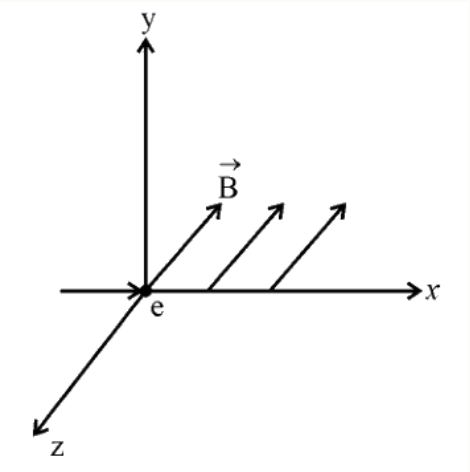
Graph description: plotting momentum p on the y-axis against de Broglie wavelength λ on the x-axis gives a rectangular hyperbola (the product $p\lambda = h$ stays constant), the curve falling steeply for small λ and flattening as λ increases without ever touching either axis.

Final answer. Davisson-Germer confirmed the wave nature of electrons; $V_p : V_\alpha = 8 : 1$. OR: electron has greater λ and lesser p; $p = h/\lambda$, a p versus λ hyperbola.

← *Back to Q220*

Q221

When a given photosensitive material is irradiated with light of frequency ν*, the maximum speed of the emitted photoelectrons equals* v_{max}*. The graph shown in the figure gives a plot of* v_{max}^2 *varying with frequency* ν*.*



Obtain an expression for

(a) Planck's constant, and

(b) The work function of the given photosensitive material in terms of the parameters 'l', 'n' and the mass 'm' of the electron.

(c) How is threshold frequency determined from the plot?

[2019 • Set 55-5-1]

SOLUTION

By Einstein's photoelectric equation, the maximum kinetic energy of an emitted photoelectron is the photon energy minus the work function $h\nu_0$ of the surface.

$$\frac{1}{2}m\,v_{max}^2 = h\nu - h\nu_0$$

Solving for v_{max}^2,

$$v_{max}^2 = \frac{2h}{m}\nu - \frac{2h\nu_0}{m}$$

This is of the form y = **(slope)** x + **(intercept)** with $y = \mathbf{v_{max}^2}$ and $x = \boldsymbol{\nu}$, so the plot of v_{max}^2 against ν is a straight line.

From the graph, the slope of the line is n and the magnitude of the negative intercept on the v_{max}^2 axis is l. Comparing,

$$\text{slope} = n = \frac{2h}{m}$$

$$\text{intercept} = l = \frac{2h\nu_0}{m}$$

(a) From the slope relation,

$$h = \frac{mn}{2}$$

So **Planck's constant** is $h = \dfrac{\mathbf{mn}}{\mathbf{2}}$.

(b) The work function is $\boldsymbol{\phi = h\nu_0}$. Dividing the intercept equation by the slope equation,

$$\frac{l}{n} = \frac{2h\nu_0/m}{2h/m} = \nu_0$$

$$\Rightarrow \nu_0 = \frac{l}{n}$$

Therefore,

$$\phi = h\nu_0 = \frac{mn}{2} \times \frac{l}{n} = \frac{ml}{2}$$

So the **work function** is $\phi = \frac{\mathbf{ml}}{\mathbf{2}}$.

(c) The threshold frequency ν_0 is the frequency at which $\mathbf{v_{max} = 0}$, i.e. the point where the straight line meets the frequency axis. So ν_0 is read off as the **intercept of the line on the ν-axis** ($\mathbf{v_{max}^2 = 0}$), and equals $\frac{l}{n}$.

Final answer. Planck's constant $\boldsymbol{h}=\frac{\mathbf{mn}}{\mathbf{2}}$, work function $\boldsymbol{\phi}=\frac{\mathbf{ml}}{\mathbf{2}}$, and $\boldsymbol{\nu_0}$ is the **ν**-axis intercept of the line.

← *Back to Q221*

Q222

When light of frequency ν_1 is incident on a photosensitive surface, the stopping potential is V_1. If the frequency of incident radiation becomes $\nu_1/2$, the stopping potential changes to V_2. Find out the expression for the threshold frequency for the surface in terms of V_1 and V_2.
If the frequency of incident radiation is doubled, will the maximum kinetic energy of the photoelectrons also be doubled? Give reason.

[2019 • Set 55-5-2]

SOLUTION

By Einstein's photoelectric equation, the stopping potential V is fixed by $\mathbf{eV = h\nu - \phi}$, where $\boldsymbol{\phi} = \mathbf{h\nu_0}$ is the work function.

For the first frequency ν_1 with stopping potential V_1,

$$eV_1 = h\nu_1 - \phi \ldots(1)$$

For frequency $\frac{\nu_1}{2}$ with stopping potential V_2,

$$eV_2 = h\frac{\nu_1}{2} - \phi \quad ...(2)$$

Subtracting (2) from (1),

$$e(V_1 - V_2) = h\nu_1 - h\frac{\nu_1}{2} = \frac{h\nu_1}{2}$$

$$\Rightarrow \nu_1 = \frac{2e(V_1 - V_2)}{h} \quad ...(3)$$

To get ϕ, multiply (2) by 2 and subtract (1):

$$2eV_2 - eV_1 = (h\nu_1 - 2\phi) - (h\nu_1 - \phi) = -\phi$$

$$\Rightarrow \phi = eV_1 - 2eV_2 = e(V_1 - 2V_2)$$

The threshold frequency is $\boldsymbol{\nu_0 = \frac{\phi}{h}}$,

$$\nu_0 = \frac{e(V_1 - 2V_2)}{h}$$

Using (3) to express in terms of ν_1, since $\boldsymbol{\frac{e}{h} = \frac{\nu_1}{2(V_1 - V_2)}}$,

$$\nu_0 = \frac{(V_1 - 2V_2)}{2(V_1 - V_2)}\,\nu_1$$

So the **threshold frequency** is $\boldsymbol{\nu_0 = \frac{e(V_1 - 2V_2)}{h}}$.

Now, if the incident frequency is doubled to 2ν, the new maximum kinetic energy is

$$K' = h(2\nu) - \phi = 2h\nu - \phi$$

The original maximum kinetic energy is $\boldsymbol{K = h\nu - \phi}$, so $\boldsymbol{2K = 2h\nu - 2\phi}$. Hence

$$K' = 2K + \phi$$

Since $\boldsymbol{\phi > 0}$, we get $\boldsymbol{K' > 2K}$. **No**, the maximum kinetic energy is not doubled; it becomes more than double, because the work function ϕ is subtracted only once and not doubled.

COMMON MISTAKE

A common slip is to write $K' = 2K$. *Doubling only the frequency does not double the kinetic energy because* ϕ *is fixed and removed just once.*

Final answer. $\nu_0 = \dfrac{e(V_1 - 2V_2)}{h}$; no, doubling the frequency makes the maximum kinetic energy more than double ($K' = 2K + \phi$).

← *Back to Q222*

Q223

Write Einstein's photoelectric equation using the quantum picture of radiation. Explain briefly the three characteristic features observed in photoelectric effect which can be explained by Einstein's equation and not by the wave theory of light.

[2019 • Set 55-5-3]

SOLUTION

In the quantum (photon) picture, light of frequency ν is a stream of photons, each carrying energy $h\nu$. When one photon is absorbed by a single electron, part of its energy (ϕ, the work function) is used to free the electron from the metal surface and the rest appears as kinetic energy of the electron. Einstein's photoelectric equation is

$$h\nu = \phi + \frac{1}{2} m v_{max}^2$$

$$\Rightarrow K_{max} = \frac{1}{2} m v_{max}^2 = h\nu - \phi$$

where K_{max} is the maximum kinetic energy of the emitted photoelectron and $\phi = h\nu_0$, ν_0 being the threshold frequency.

Three features explained by Einstein's equation but not by wave theory:

(1) Existence of a threshold frequency. Emission occurs only if $\nu \geq \nu_0$ (i.e. $h\nu \geq \phi$), no matter how intense the light is. Below ν_0 no electron is emitted. The photon picture explains this because a single photon must supply at least ϕ in

one event. Wave theory wrongly predicts that any frequency should cause emission if the intensity is high enough or the light shines long enough.

(2) Maximum kinetic energy depends on frequency, not intensity. From $\mathbf{K_{max} = h\nu - \phi}$, the energy of the fastest photoelectron rises with ν but is completely independent of intensity. Wave theory wrongly predicts that brighter light should give faster electrons, since a stronger wave carries more energy.

(3) Instantaneous emission. Photoelectrons appear almost immediately (within about 10^{-9} *s*) even for very weak light. In the photon picture a single absorption event ejects an electron at once. Wave theory wrongly predicts a measurable time delay while the electron gradually absorbs enough wave energy.

Final answer. Einstein's equation $\mathbf{K_{max} = h\nu - \phi}$ explains the threshold frequency, the frequency-dependence (not intensity) of $\mathbf{K_{max}}$, and the instantaneous emission, which wave theory cannot.

← *Back to Q223*

Q224

(i) How does one explain the emission of electrons from a photosensitive surface with the help of Einstein's photoelectric equation?
(ii) The work function of the following metals is given: **Na = 2.75 eV**, ***K*** **= 2.3 eV**, **Mo = 4.17 eV** *and* **Ni = 5.15 eV**. *Which of these metals will not cause photoelectric emission for radiation of wavelength* 3300 *A from a laser source placed* 1 *m away from these metals? What happens if the laser source is brought nearer and placed* 50 cm *away?*

[2017]

SOLUTION

(i) Explanation using Einstein's photoelectric equation: Light of frequency ν is treated as a stream of photons, each of energy $h\nu$. When a photon strikes the photosensitive surface, it is absorbed entirely by a single electron. A part of this energy equal to the work function ϕ is spent in freeing the electron from the surface, and the remaining energy appears as the kinetic energy of the emitted electron.

$$h\nu = \phi + K_{max}$$

$$\Rightarrow K_{max} = h\nu - \phi$$

Emission of electrons is possible only when $h\nu \geq \phi$, that is when the photon energy is at least equal to the work function of the surface.

(ii) The energy of an incident photon of wavelength $\boldsymbol{\lambda = 3300\ \text{Å} = 3300 \times 10^{-10}\ m}$ is

$$E = \frac{hc}{\lambda} = \frac{1240\ \text{eV nm}}{330\ \text{nm}}$$

$$E \approx 3.76\ \text{eV}$$

Comparing this photon energy with each work function:

Na (2.75 eV): $E > \phi$, so emission occurs.

K (2.30 eV): $E > \phi$, so emission occurs.

Mo (4.17 eV): $E < \phi$, so no emission.

Ni (5.15 eV): $E < \phi$, so no emission.

So **Mo and Ni will not show photoelectric emission** for this radiation.

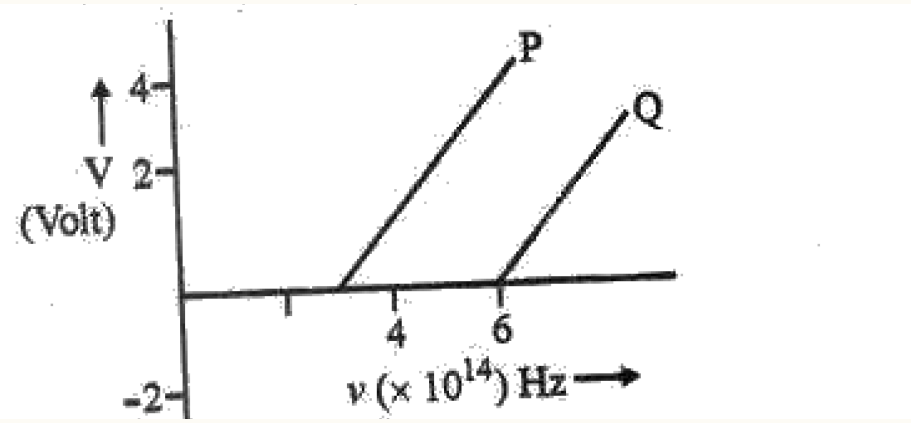
Effect of moving the source closer (from 1 *m* to 50 cm): Bringing the laser nearer increases the intensity (it rises by a factor of about 4), so more photons fall on the surface each second. But the photon energy depends only on the wavelength, which is unchanged. Hence Mo and Ni still do not emit, since each photon energy 3.76 eV is still below their work functions. For Na and K the number of emitted photoelectrons (and hence the photocurrent) increases, but the maximum kinetic energy of the electrons stays the same.

Final answer. Photon energy ≈ **3.76 eV**, so Mo and Ni do not emit; bringing the source to **50 cm** raises intensity but not photon energy, so Mo and Ni still do not emit while Na and K give a larger photocurrent at the same $\mathbf{K_{max}}$.

← *Back to Q224*

Q225

In the study of a photoelectric effect the graph between the stopping potential V and frequency ν of the incident radiation on two different metals P and Q is shown below:



(i) Which one of the two metals has higher threshold frequency?
(ii) Determine the work function of the metal which has greater value.
(iii) Find the maximum kinetic energy of electron emitted by light of frequency 8×10^{14} Hz *for this metal.*

[2017]

SOLUTION

For each metal the stopping potential and frequency obey $\mathbf{eV_0 = h\nu - h\nu_0}$, so

$$V_0 = \frac{h}{e}\nu - \frac{h\nu_0}{e}$$

Both lines have the same slope $\frac{h}{e}$, and the metal whose line meets the ν-axis at a larger value has the higher threshold frequency.

(i) From the graph, metal P cuts the ν-axis at a larger frequency than metal Q. So **metal P has the higher threshold frequency** (its ν_0 is larger).

(ii) The metal with the greater work function is the one with the greater threshold frequency, i.e. P. Reading its threshold from the graph as $\mathbf{\nu_{0,P} = 5\times 10^{14}\ Hz}$,

$$\phi_P = h\nu_{0,P} = (6.6\times 10^{-34})(5\times 10^{14})$$

$$\phi_P = 3.3\times 10^{-19}\ J$$

Converting to electron-volts,

$$\phi_P = \frac{3.3\times 10^{-19}}{1.6\times 10^{-19}}\ \text{eV} \approx 2.06\ \text{eV}$$

So the **work function of P is about $3.3\times 10^{-19}\,J \approx 2.06$ eV**.

(iii) For light of frequency $\nu = 8\times 10^{14}$ Hz on metal P,

$$K_{max} = h(\nu - \nu_{0,P}) = (6.6\times 10^{-34})(8\times 10^{14} - 5\times 10^{14})$$

$$K_{max} = (6.6\times 10^{-34})(3\times 10^{14})$$

$$K_{max} = 1.98\times 10^{-19}\,J$$

In electron-volts,

$$K_{max} = \frac{1.98\times 10^{-19}}{1.6\times 10^{-19}}\ \text{eV} \approx 1.24\ \text{eV}$$

So the **maximum kinetic energy is $1.98\times 10^{-19}\,J \approx 1.24$ eV**.

TIP

Read the threshold frequencies straight off the ν-axis intercepts; the parallel slopes both equal h/e, so only the intercepts distinguish the two metals.

Final answer. Metal P has the higher threshold frequency; its work function is ≈ **$3.3\times 10^{-19}\,J$ (≈ 2.06 eV)**, and **$K_{max} = 1.98\times 10^{-19}\,J$ (≈ 1.24 eV)** at **8×10^{14} Hz**.

← *Back to Q225*

Q226

(i) State two important features of Einstein's photoelectric equation.
(ii) Radiation of frequency 10^{15} Hz is incident on two photosensitive surfaces P and Q. There is no photoemission from surface P. Photoemission occurs from surface Q but photoelectrons have zero kinetic energy. Explain these observations and find the value of work function for surface Q.

[2017]

SOLUTION

(i) Two important features of Einstein's photoelectric equation $\mathbf{K_{max} = h\nu - \phi}$:

(1) The maximum kinetic energy of the photoelectrons varies linearly with the frequency ν of the incident light and is completely independent of its intensity.

(2) Photoemission takes place only when the incident frequency is at least equal to the threshold frequency, $\nu \geq \nu_0$ (i.e. $h\nu \geq \phi$); below ν_0 no emission occurs however intense the light is.

(ii) The energy of each incident photon is

$$E = h\nu = (6.626\times 10^{-34})(10^{15})$$

$$E = 6.626\times 10^{-19}\, J$$

In electron-volts,

$$E = \frac{6.626\times 10^{-19}}{1.6\times 10^{-19}}\ \text{eV} \approx 4.14\ \text{eV}$$

Surface P: There is no photoemission, which means the photon energy is less than the work function of P. Hence $\mathbf{\phi_P > E}$, i.e. $\mathbf{\phi_P > 4.14\ eV}$.

Surface Q: Emission occurs but the photoelectrons have zero kinetic energy. This means the photon energy is just sufficient to free the electrons with nothing left over, so the incident frequency equals the threshold frequency of Q. Putting $\mathbf{K_{max} = 0}$ in Einstein's equation,

$$\phi_Q = h\nu = 6.626\times 10^{-19}\, J \approx 4.14\ \text{eV}$$

So the **work function of surface Q is $\mathbf{6.626\times 10^{-19}\, J \approx 4.14\ eV}$**.

Final answer. Photon energy $\approx$ **4.14 eV**, so $\mathbf{\phi_P > 4.14\ eV}$ (no emission) and $\mathbf{\phi_Q = 6.626\times 10^{-19}\, J \approx 4.14\ eV}$ (emission with zero $\mathbf{K_{max}}$).

← Back to Q226

Q227

(a) Draw a plot showing the variation of photoelectric current with collector potential for different frequencies but same intensity of incident radiation.
(b) Use Einstein's photoelectric equation to explain the observations from this graph.

(c) What change will you observe if intensity of incident radiation is changed but the frequency remains the same?

[2017]

SOLUTION

(a) Plot of photoelectric current versus collector (anode) potential for three different frequencies $\nu_1 < \nu_2 < \nu_3$ at the same intensity:

Draw the photocurrent on the vertical axis and the collector potential on the horizontal axis, with negative potentials to the left of the origin and positive to the right. Draw three curves that all rise from the left and flatten to the same horizontal saturation level on the right, because the intensity (number of photons per second) is the same for all three. Each curve falls to zero current at a different negative cut-off (stopping) potential: the curve for the highest frequency ν_3 meets the axis at the most negative potential V_{03}, and ν_1 at the least negative V_{01}, so $|V_{03}| > |V_{02}| > |V_{01}|$. Label the common saturation current and the three stopping potentials V_{01}, V_{02}, V_{03}.

(b) Explanation using Einstein's equation. At the stopping potential V_0 the most energetic electron is just stopped, so

$$eV_0 = K_{max} = h\nu - \phi$$

$$\Rightarrow V_0 = \frac{h\nu - \phi}{e}$$

As the frequency ν increases, K_{max} increases, so a larger (more negative) stopping potential is needed to halt the electrons. This is why the cut-off shifts to more negative collector potentials for higher frequencies. The saturation current, however, depends only on the number of photons striking per second (the intensity), which is unchanged, so all three curves saturate at the same current.

(c) If the intensity is changed while the frequency is kept the same, the stopping potential V_0 does not change, because V_0 depends only on ν and ϕ. But the saturation current changes in direct proportion to the intensity: greater intensity means more photoelectrons per second, hence a higher saturation current (and a smaller intensity gives a lower one).

Final answer. Higher frequency shifts the stopping potential to more negative values (same saturation current); changing intensity at fixed frequency alters only the saturation current, leaving $\mathbf{V_0}$ unchanged.

← Back to Q227

Q228

(a) Draw a graph showing variation of photocurrent with anode potential for a particular intensity of incident radiation. Mark saturation current and stopping potential.
(b) By how much would the stopping potential for a given photosensitive surface go up if the frequency of the incident radiations were to be increased from 4×10^{15} Hz *to* 8×10^{15} Hz*?*

[2017]

SOLUTION

(a) Plot of photocurrent versus anode (collector) potential for a single intensity and single frequency:

Draw the photocurrent on the vertical axis and the anode potential on the horizontal axis, with negative values to the left and positive to the right of the origin. The curve meets the horizontal axis at a negative value $-V_0$ (the stopping potential, where the current is just zero), then rises as the anode potential increases, and finally levels off to a constant value I_{sat} (the saturation current) for sufficiently positive anode potentials. Mark $-V_0$ as the stopping potential on the negative axis and the flat top of the curve as the saturation current I_{sat}.

(b) The stopping potential and frequency are related by $\mathbf{eV_0 = h\nu - \phi}$. For a change in frequency $\Delta\nu$ on the same surface (ϕ fixed),

$$e\,\Delta V_0 = h\,\Delta\nu$$

$$\Rightarrow \Delta V_0 = \frac{h\,\Delta\nu}{e}$$

Here the change in frequency is

$$\Delta\nu = (8\times 10^{15}) - (4\times 10^{15}) = 4\times 10^{15}\ \text{Hz}$$

Substituting the values,

$$\Delta V_0 = \frac{(6.626\times 10^{-34})(4\times 10^{15})}{1.6\times 10^{-19}}$$

$$\Delta V_0 = \frac{2.65\times 10^{-18}}{1.6\times 10^{-19}}$$

$$\Delta V_0 \approx 16.56\ V$$

So the **stopping potential goes up by about 16.6 *V***.

Final answer. The stopping potential increases by $\Delta V_0 = \frac{h\,\Delta\nu}{e} \approx 16.6\ V$.

← *Back to Q228*

Q229

Using photon picture of light, show how Einstein's photoelectric equation can be established. Write two features of photoelectric effect which cannot be explained by wave theory.

[2017]

SOLUTION

In the photon picture, light of frequency ν consists of discrete packets (photons), each of energy $h\nu$. When one photon falls on the metal surface, it is absorbed completely by a single electron in a single event. A minimum energy equal to the work function ϕ is needed to release the electron from the surface; the surplus energy appears as the kinetic energy of the freed electron. By conservation of energy,

$$h\nu = \phi + (\text{kinetic energy of electron})$$

The electron that absorbs the photon right at the surface loses the least energy in escaping, so it carries the maximum kinetic energy:

$$h\nu = \phi + K_{max}$$

$$\Rightarrow K_{max} = h\nu - \phi$$

Writing $\phi = h\nu_0$ where ν_0 is the threshold frequency,

$$K_{max} = h\nu - h\nu_0 = h(\nu - \nu_0)$$

This is **Einstein's photoelectric equation**.

Two features of the photoelectric effect that wave theory cannot explain:

(1) Existence of a threshold frequency. No emission occurs below a certain frequency ν_0, however intense the light. Wave theory wrongly predicts that even low-frequency light should eject electrons if it is intense enough.

(2) Maximum kinetic energy depends only on frequency, not on intensity, and emission is almost instantaneous. Wave theory wrongly predicts that K_{max} should increase with intensity and that weak light would need a long build-up time before any electron is emitted.

Final answer. Treating light as photons of energy $h\nu$ absorbed one-to-one by electrons gives $K_{max} = h\nu - \phi$; wave theory fails to explain the threshold frequency and the intensity-independence (plus instantaneous nature) of the emission.

← *Back to Q229*

Q230

Explain giving reasons for the following:
(a) Photoelectric current in a photocell increases with the increase in the intensity of the incident radiation.
(b) The stopping potential (V_0) varies linearly with the frequency (ν) of the incident radiation for a given photosensitive surface with the slope remaining the same for different surfaces.
(c) Maximum kinetic energy of the photoelectrons is independent of the intensity of incident radiation.

[2017]

SOLUTION

(a) Photoelectric current increases with intensity because, in the photon picture, the intensity of light equals the number of photons striking unit area

per unit time. Above the threshold, each absorbed photon frees one electron in a one-to-one process. So a higher intensity means more photons per second, hence more photoelectrons emitted per second, which gives a larger saturation photocurrent.

(b) The stopping potential is given by $\mathbf{eV_0 = h\nu - \phi}$, so

$$V_0 = \frac{h}{e}\nu - \frac{\phi}{e}$$

This is a straight-line (linear) relation between V_0 and ν with slope $\frac{h}{e}$. Since both h (Planck's constant) and e (electronic charge) are universal constants, the slope $\frac{h}{e}$ is the same for every photosensitive surface. Only the intercept term $\frac{\phi}{e}$ (and the threshold frequency ν_0) depends on the particular metal, which is why the lines are parallel.

(c) The maximum kinetic energy is

$$K_{max} = h\nu - \phi$$

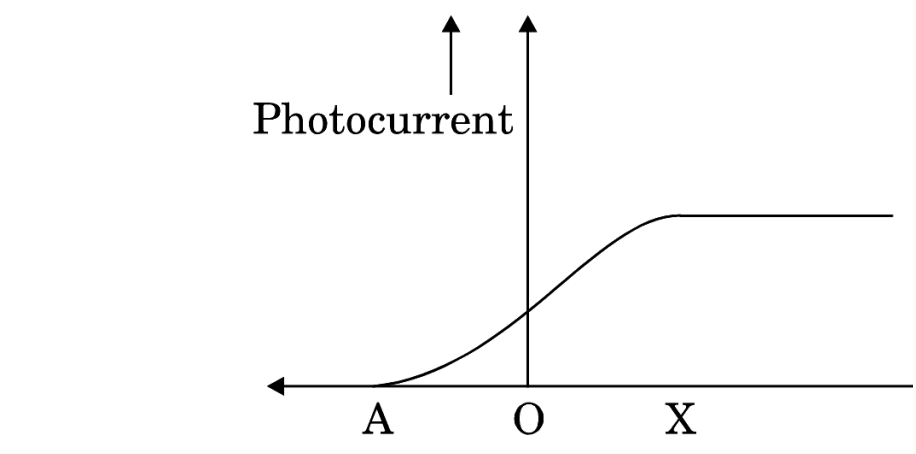
which depends only on the photon frequency ν and the work function ϕ of the metal. Photoemission is a single-photon, single-electron event, so the energy given to each emitted electron is set by one photon's energy $h\nu$, not by how many photons arrive. Increasing the intensity raises only the number of photons (and hence the current), not the energy per photon, so K_{max} stays the same and is **independent of intensity**.

Final answer. More intensity means more photons and hence more current; $\mathbf{V_0}$ versus $\boldsymbol{\nu}$ is linear with the universal slope $\boldsymbol{h/e}$; and $\mathbf{K_{max}=h\nu-\phi}$ depends only on frequency, so it is independent of intensity.

← *Back to Q230*

Q231

The following graph shows the variation of photocurrent for a photosensitive metal:



(a) Identify the variable X on the horizontal axis.
(b) What does the point A on the horizontal axis represent?
(c) Draw this graph for three different values of frequencies of incident radiation ν_1, ν_2 *and* ν_3 *(*$\mathbf{\nu_1 > \nu_2 > \nu_3}$*) for same intensity.*
(d) Draw this graph for three different values of intensities of incident radiation I_1, I_2 *and* I_3 *(*$\mathbf{I_1 > I_2 > I_3}$*) having same frequency.*

[2017]

SOLUTION

(a) The variable *X* on the horizontal axis is the **collector (anode) plate potential** *V* applied between the collector and the emitter. It is taken as positive on the right of the origin (accelerating potential) and negative on the left (retarding potential).

(b) The point *A* lies on the negative side of the horizontal axis where the photocurrent just becomes zero. It therefore represents the **stopping potential** V_0 (also called cut-off voltage). At this retarding potential, even the most energetic photoelectrons are stopped before reaching the collector, so the current falls to zero. It is related to the maximum kinetic energy by $\mathbf{eV_0 = K_{max}}$.

(c) Graph for three frequencies $\mathbf{\nu_1 > \nu_2 > \nu_3}$ at the SAME intensity:

Plot photocurrent (vertical axis) against collector potential *V* (horizontal axis). Draw three curves that all rise from the negative-*V* region, climb through the origin region, and flatten to the SAME saturation current on the right (because intensity is the same, the number of photoelectrons emitted per second is the same for all three).

On the negative side the three curves cut the axis at different stopping potentials. Since $\mathbf{eV_0 = h\nu - \phi}$, a higher frequency gives a larger $|V_0|$. So the curve for ν_1 meets the axis farthest to the left, then ν_2, then ν_3 nearest to the origin, i.e. $\mathbf{|V_{0,1}| > |V_{0,2}| > |V_{0,3}|}$, while all three reach a common saturation level.

(d) Graph for three intensities $\mathbf{I_1 > I_2 > I_3}$ at the SAME frequency:

Again plot photocurrent versus V. Draw three curves that all cut the negative axis at the SAME stopping potential V_0 (frequency is the same, so K_{max} and hence V_0 are unchanged).

On the positive side they level off at DIFFERENT saturation currents, because saturation current is proportional to intensity. The curve for I_1 has the highest saturation, I_2 a lower one, and I_3 the lowest, so $\mathbf{I_{sat,1} > I_{sat,2} > I_{sat,3}}$, with all three sharing the common cut-off V_0.

CONCEPT

Frequency controls the stopping potential (where curves meet the negative axis); intensity controls the saturation current (the flat top). Keep these two effects separate.

Final answer. X is the collector potential, A is the stopping potential V0; higher frequency shifts the cut-off left (same saturation), while higher intensity raises the saturation current (same cut-off).

← *Back to Q231*

Q232

Sketch the graphs showing variation of stopping potential with frequency of incident radiations for two photosensitive materials A and B having threshold frequencies $\mathbf{\nu_A > \nu_B}$.
(i) In which case is the stopping potential more and why?
(ii) Does the slope of the graph depend on the nature of the material used? Explain.

[2016]

SOLUTION

By Einstein's photoelectric equation, the stopping potential and frequency are related by

$$eV_s = h\nu - h\nu_0$$

$$\Rightarrow V_s = \frac{h}{e}\nu - \frac{h}{e}\nu_0$$

This is a straight line of the form $\boldsymbol{y = mx + c}$ with slope $\frac{h}{e}$ and ν-axis intercept equal to the threshold frequency ν_0.

Sketch: On a graph with stopping potential V_s on the vertical axis and frequency ν on the horizontal axis, draw two straight lines. The line for material A cuts the ν-axis at the larger value ν_A (farther to the right). The line for material B cuts the ν-axis at the smaller value ν_B (closer to the origin). Both lines have the SAME slope, so they are **parallel**, with B lying above and to the left of A.

(i) At any common incident frequency ν (with $\boldsymbol{\nu > \nu_A}$):

$$V_s^A = \frac{h}{e}(\nu - \nu_A)$$

$$V_s^B = \frac{h}{e}(\nu - \nu_B)$$

Since $\boldsymbol{\nu_A > \nu_B}$, the term $\boldsymbol{(\nu - \nu_B) > (\nu - \nu_A)}$, so $\boldsymbol{V_s^B > V_s^A}$. Hence the stopping potential is more for **material *B***, the one with the smaller threshold frequency (smaller work function $\boldsymbol{\phi = h\nu_0}$), because less energy is spent in freeing the electron and more is left as kinetic energy.

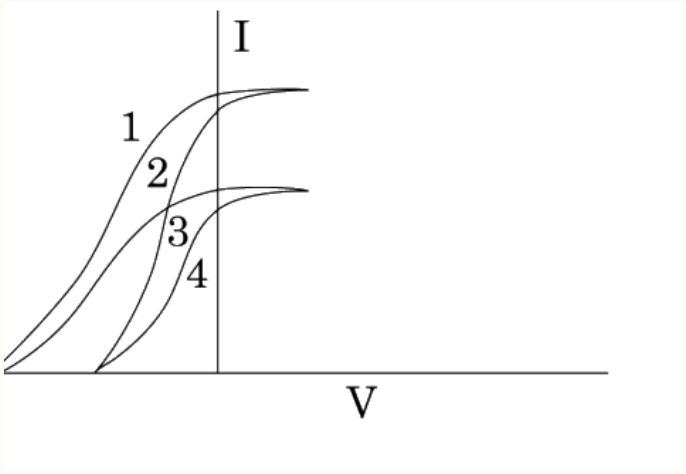
(ii) The slope of the graph is $\frac{h}{e}$, which contains only Planck's constant h and the electronic charge e, both universal constants. Therefore the slope is the **same for every photosensitive material** and does NOT depend on the nature of the metal. Only the intercept ν_0 (and hence the work function) changes from one material to another. This universality of the slope was confirmed by Millikan's experiments.

Final answer. Both lines are parallel with slope h/e; material B (smaller threshold frequency) gives the larger stopping potential, and the slope is independent of the material.

← Back to Q232

Q233

The given graph shows the variation of photo-electric current (I) with the applied voltage (V) for two different materials and for two different intensities of the incident radiations. Identify and explain using Einstein's photo electric equation the pair of curves that correspond to (i) different materials but same intensity of incident radiation, (ii) different intensities but same materials.



[2016]

SOLUTION

We use two results from Einstein's photoelectric theory:

$\mathbf{eV_0 = h\nu - \phi}$ (stopping potential depends on frequency and work function only)

saturation current ∝ intensity of incident radiation (number of photoelectrons emitted per second).

On the given graph, the photocurrent is plotted against the collector potential V. Each curve has two identifying features: the point where it cuts the negative axis (its stopping potential V_0) and the flat level it settles to on the positive side (its saturation current).

(i) Different materials but same intensity:

Two materials have different work functions ϕ. At the same incident frequency, $\mathbf{V_0 = \frac{h\nu - \phi}{e}}$ is different for the two, so the curves cut the negative axis at

DIFFERENT stopping potentials. But the intensity is the same, so both emit the same number of electrons per second and reach the SAME saturation current.

Therefore the pair of curves with the **same saturation current but different stopping potentials** corresponds to different materials with the same intensity.

(ii) Different intensities but same material:

The material is the same, so ϕ is the same; the frequency is the same, so V_0 is unchanged. Hence both curves cut the negative axis at the SAME stopping potential. But the intensities differ, so the saturation currents are DIFFERENT (higher intensity gives higher saturation current).

Therefore the pair of curves with the **same stopping potential but different saturation currents** corresponds to the same material with different intensities.

In short, Einstein's equation tells us that V_0 is fixed by ν and ϕ (the material), while the saturation current is fixed by the intensity. Comparing these two features in each pair identifies the curves.

TIP

Read each curve twice: where it meets the negative axis (tells you the material/frequency) and how high its flat top is (tells you the intensity).

Final answer. Same saturation but different stopping potential = different materials, same intensity; same stopping potential but different saturation = same material, different intensities.

← *Back to Q233*

Q234

State two important properties of photon which are used to write Einstein's photoelectric equation. Define (i) stopping potential and (ii) threshold frequency, using Einstein's equation and drawing necessary plot between relevant quantities.

[2016]

SOLUTION

Two important properties of the photon used to write Einstein's photoelectric equation:

Light travels as discrete packets called **photons**, each carrying a definite energy $\boldsymbol{E = h\nu}$ that depends only on the frequency ν of the radiation (not on its intensity).

In photoemission, a single photon is absorbed completely by a single electron in one instantaneous event; its entire energy is transferred at once (there is no partial or gradual absorption).

Using these, the absorbed energy $h\nu$ is partly used to overcome the work function ϕ and the rest appears as the maximum kinetic energy of the emitted electron:

$$h\nu = \phi + K_{max}$$

which can be written as $\boldsymbol{eV_0 = h(\nu - \nu_0)}$.

(i) Stopping potential (V_0): It is the minimum value of the retarding (negative) collector potential at which the photoelectric current becomes zero. At this potential, even the fastest photoelectron is just stopped, so

$$eV_0 = K_{max}$$

(ii) Threshold frequency (ν_0): It is the minimum frequency of the incident radiation below which no photoelectrons are emitted, however large the intensity. At $\boldsymbol{\nu = \nu_0}$ the photon energy just equals the work function, so $\boldsymbol{K_{max} = 0}$ and

$$\nu_0 = \frac{\phi}{h}$$

Necessary plot: Draw a graph of stopping potential V_0 (vertical axis) against frequency ν (horizontal axis). Since $\boldsymbol{V_0 = \frac{h}{e}\nu - \frac{\phi}{e}}$, it is a straight line of slope $\frac{h}{e}$. The line meets the frequency axis at $\boldsymbol{\nu = \nu_0}$ (where $\boldsymbol{V_0 = 0}$), so the ν-axis intercept gives the **threshold frequency**, and the stopping potential is read off the vertical axis for any chosen ν.

Final answer. Photon energy E = hv and one-photon-one-electron absorption give hv = phi + K_max; stopping potential satisfies eV0 = K_max, and threshold frequency v0 = phi/h is the v-axis intercept of the linear V0 vs v plot.

← Back to Q234

Q235

(a) Define the term 'intensity of radiation' in photon picture.
(b) Plot a graph showing the variation of photo current vs collector potential for three different intensities $\mathbf{I_1 > I_2 > I_3}$*, two of which* (I_1 *and* I_2) *have the same frequency* ν *and the third has frequency* $\boldsymbol{\nu_1 > \nu}$.
(c) Explain the nature of the curves on the basis of Einstein's equation.

[2016]

SOLUTION

(a) Intensity of radiation (photon picture): It is the number of photons crossing unit area held normal to the direction of propagation per unit time, or equivalently the energy delivered per unit area per unit time. If N photons of frequency ν cross unit area per second, then

$$I = N\,h\nu$$

(b) Plot: Draw photocurrent (vertical axis) against collector potential V (horizontal axis), with the stopping potentials on the negative side.

The curves for I_1 and I_2 both have the same frequency ν, so they cut the negative axis at the SAME stopping potential V_0. Since $\mathbf{I_1 > I_2}$, the curve for I_1 rises to a HIGHER saturation current than I_2.

The curve for I_3 has the highest frequency $\boldsymbol{\nu_1 > \nu}$, so it cuts the negative axis at a larger stopping potential V_0' (farther to the left). Since I_3 is the smallest intensity, its saturation current is the LOWEST of the three. So this curve has a more-negative cut-off and the lowest flat top.

(c) Explanation using Einstein's equation $\mathbf{eV_0 = h\nu - \phi}$:

For I_1 and I_2 the frequency is the same, so $\mathbf{K_{max} = h\nu - \phi}$ is the same, giving the SAME stopping potential V_0. But saturation current is proportional to intensity,

so I_1 (larger intensity) saturates higher than I_2.

For I_3 the frequency $\nu_1 > \nu$, so $K_{max} = h\nu_1 - \phi$ is larger, giving a greater stopping potential V_0'. Its intensity is the smallest, so its saturation current is the lowest.

Thus stopping potential is fixed by frequency, while saturation current is fixed by intensity, exactly as Einstein's equation predicts.

Final answer. Intensity I = Nhv; I1 and I2 share the cut-off V0 (same v) but I1 saturates higher, while I3 (higher frequency v1) has a larger cut-off V0' and the lowest saturation.

← *Back to Q235*

Q236

(a) Define the term 'intensity of radiation' in terms of photon picture of light.
(b) Two monochromatic beams, one red and the other blue, have the same intensity. In which case (i) the number of photons per unit area per second is larger, (ii) the maximum kinetic energy of the photoelectrons is more? Justify your answer.

[2015]

SOLUTION

(a) Intensity of radiation (photon picture): It is the amount of energy delivered per unit area per unit time by the radiation. In terms of photons, if N photons of frequency ν cross unit area per second, the intensity is

$$I = N\,h\nu$$

(b) Red light has a lower frequency ν_R and blue light a higher frequency ν_B, with $\nu_B > \nu_R$. The two beams have the SAME intensity, so

$$N_R\,h\nu_R = N_B\,h\nu_B$$

$$\Rightarrow \frac{N_R}{N_B} = \frac{\nu_B}{\nu_R}$$

(i) Number of photons per unit area per second: Since $\nu_B > \nu_R$, we get $\frac{N_R}{N_B} = \frac{\nu_B}{\nu_R} > 1$, so $N_R > N_B$. Each red photon carries less energy, so to give the same

intensity the red beam must contain MORE photons. Hence the **red beam** has the larger number of photons per unit area per second.

(ii) Maximum kinetic energy of photoelectrons: By Einstein's equation $K_{max} = h\nu - \phi$, the maximum kinetic energy increases with frequency. Since $\nu_B > \nu_R$, the blue light ejects electrons with greater maximum kinetic energy. Hence the **blue beam** gives the higher maximum kinetic energy of photoelectrons.

COMMON MISTAKE

Same intensity does not mean same photon count. Lower-frequency light needs more photons to carry the same energy, so red beats blue on photon number but loses on per-electron kinetic energy.

Final answer. I = Nhν; the red beam has more photons per unit area per second (lower energy per photon), while the blue beam gives the higher maximum kinetic energy.

← *Back to Q236*

Q237

Write Einstein's photoelectric equation and mention which important features in photoelectric effect can be explained with the help of this equation.
The maximum kinetic energy of the photoelectrons gets doubled when the wavelength of light incident on the surface changes from λ_1 to λ_2. Derive the expressions for the threshold wavelength λ_0 and work function for the metal surface.

[2015]

SOLUTION

Einstein's photoelectric equation:

$$h\nu = \phi + K_{max}$$

where $h\nu$ is the photon energy, ϕ is the work function and K_{max} is the maximum kinetic energy of the emitted electron.

Features explained by this equation:

Existence of a threshold frequency $\nu_0 = \phi/h$ below which no emission occurs.

K_{max} increases linearly with frequency and is independent of intensity.

Saturation photocurrent (number of electrons) increases with intensity.

Photoemission is practically instantaneous (one-photon-one-electron interaction).

Numerical: The maximum kinetic energy doubles when the wavelength changes from λ_1 to λ_2.

At wavelength λ_1:

$$K_1 = \frac{hc}{\lambda_1} - \phi \ldots (1)$$

At wavelength λ_2, the kinetic energy is doubled, so $K_2 = 2K_1$:

$$2K_1 = \frac{hc}{\lambda_2} - \phi \ldots (2)$$

Multiply equation (1) by 2:

$$2K_1 = \frac{2hc}{\lambda_1} - 2\phi \ldots (3)$$

Subtract (3) from (2):

$$0 = \frac{hc}{\lambda_2} - \frac{2hc}{\lambda_1} - \phi + 2\phi$$

$$\Rightarrow \phi = \frac{2hc}{\lambda_1} - \frac{hc}{\lambda_2}$$

$$\Rightarrow \phi = hc\left(\frac{2\lambda_2 - \lambda_1}{\lambda_1\lambda_2}\right)$$

The threshold wavelength satisfies $\phi = \frac{hc}{\lambda_0}$, so

$$\lambda_0 = \frac{hc}{\phi} = \frac{hc}{hc\left(\frac{2\lambda_2 - \lambda_1}{\lambda_1\lambda_2}\right)}$$

$$\Rightarrow \lambda_0 = \frac{\lambda_1\lambda_2}{2\lambda_2 - \lambda_1}$$

Final answer. Work function phi = hc(2λ2 − λ1)/(λ1λ2) and threshold wavelength λ0 = λ1λ2/(2λ2 − λ1).

← *Back to Q237*

Q238

An electron microscope uses electrons accelerated by a voltage of 50 kV. Determine the de-Broglie wavelength associated with the electrons. Taking other factors, such as numerical aperture etc. to be same, how does the resolving power of an electron microscope compare with that of an optical microscope which uses yellow light ?

[2014]

SOLUTION

An electron accelerated through a potential difference V gains kinetic energy eV, so its de-Broglie wavelength is

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2m_e\,eV}}$$

which gives the convenient form (with V in volts, λ in nm):

$$\lambda = \frac{1.227}{\sqrt{V}}\ \text{nm}$$

Here $\boldsymbol{V = 50\ \text{kV} = 510^4\ V}$.

$$\lambda = \frac{1.227}{\sqrt{510^4}}\ \text{nm}$$

$$\lambda = \frac{1.227}{223.6}\ \text{nm}$$

$$\lambda \approx 5.4910^{-3}\ \text{nm} \approx 5.510^{-12}\ m$$

So $\boldsymbol{\lambda \approx 5.510^{-12}\ m}$.

Resolving power: With other factors (numerical aperture, etc.) kept the same, the resolving power is inversely proportional to the wavelength of the radiation used, i.e. $\boldsymbol{R.P.} \propto \frac{1}{\lambda}$.

For yellow light, $\lambda_y \approx 5.910^{-7}\ m$. Comparing,

$$\frac{(R.P.)_{\text{electron}}}{(R.P.)_{\text{optical}}} = \frac{\lambda_y}{\lambda_e} = \frac{5.910^{-7}}{5.510^{-12}}$$

$$\approx 1.0710^5$$

Since the electron wavelength is about 10^5 times smaller than that of yellow light, the **resolving power of the electron microscope is about 10^5 times that of the optical microscope**.

TIP

The shortcut $\lambda = 1.227/\sqrt{V}$ nm already builds in h, m_e and e, so plug V in volts directly and the answer comes out in nanometres.

Final answer. $\lambda \approx 5.5 \times 10^{-12}$ m, and the electron microscope is about 10^5 times more resolving than the optical microscope.

← *Back to Q238*

Q239

(a) Why photoelectric effect can not be explained on the basis of wave nature of light? Give reasons.
(b) Write the basic features of photon picture of electromagnetic radiation on which Einstein's photoelectric equation is based.

[2013]

SOLUTION

(a) Why the wave theory of light fails to explain the photoelectric effect:

Threshold frequency: According to wave theory, light of any frequency, if shone long enough or intensely enough, should supply enough energy to eject electrons. But experiments show that below a certain threshold frequency ν_0 NO electrons are emitted, however high the intensity. Wave theory cannot explain this.

Independence of K_{max} from intensity: Wave theory predicts that increasing the intensity (amplitude) of light should increase the energy given to each electron and hence its kinetic energy. In experiments, the maximum kinetic energy depends only on the frequency, not on the intensity. Wave theory cannot explain this.

Instantaneous emission: Wave theory predicts that at low intensity an electron would need a measurable time to accumulate enough energy before being ejected. Experimentally, emission is observed almost instantaneously (within about 10^{-9} *s*) of the light striking the surface. Wave theory cannot explain this.

(b) Basic features of the photon picture on which Einstein's equation is based:

Electromagnetic radiation consists of discrete energy packets called **photons**; each photon carries energy $E = h\nu$ and momentum $p = \frac{h\nu}{c}$, depending only on the frequency.

In photoemission, one photon interacts with and is absorbed completely by one electron (a one-to-one quantum interaction), giving up all its energy in a single event.

Photons travel through vacuum at the speed of light *c* and have zero rest mass.

The intensity of radiation is determined by the number of photons crossing unit area per unit time, not by the energy of an individual photon.

Final answer. Wave theory cannot explain the threshold frequency, the intensity-independence of K_max, or the instantaneous emission; Einstein's equation rests on photons of energy E = hv absorbed one-to-one by electrons.

← *Back to Q239*

Q240

Define the terms (i) 'cut-off voltage' and (ii) 'threshold frequency' in relation to the phenomenon of photoelectric effect.
Using Einstein's photoelectric equation show how the cut-off voltage and threshold

frequency for a given photosensitive material can be determined with the help of a suitable plot/graph.

[2012]

SOLUTION

(i) Cut-off voltage (stopping potential): It is the minimum value of the negative (retarding) collector potential for which the photoelectric current is reduced to zero. At this potential even the fastest photoelectron is just brought to rest before reaching the collector, so all its kinetic energy is spent against the retarding field:

$$eV_0 = K_{max} = h\nu - \phi$$

(ii) Threshold frequency: It is the minimum frequency of incident radiation below which no photoelectric emission takes place from a given metal, however large the intensity. At this frequency the photon energy just equals the work function ($\mathbf{K_{max} = 0}$), so

$$\nu_0 = \frac{\phi}{h}$$

Determination from a plot: Starting from Einstein's equation $\mathbf{eV_0 = h\nu - \phi}$, divide by e:

$$V_0 = \frac{h}{e}\nu - \frac{\phi}{e}$$

This has the form $\boldsymbol{y = mx + c}$, so a graph of cut-off voltage V_0 (vertical axis) against frequency ν (horizontal axis) is a STRAIGHT LINE.

From the graph:

The slope of the line is $\frac{h}{e}$ (a universal constant), so Planck's constant can be found from $\boldsymbol{h = e \times (\text{slope})}$.

The point where the line cuts the frequency axis ($\mathbf{V_0 = 0}$) gives the **threshold frequency** ν_0 directly.

The (negative) intercept on the V_0-axis equals $-\frac{\phi}{e}$, from which the work function $\boldsymbol{\phi = e \times |\text{intercept}|}$ is obtained.

Thus a single linear V_0 versus ν plot yields the cut-off voltage for any frequency, the threshold frequency, and the work function of the material.

CONCEPT

The V0 vs v line is the same straight line for every metal in slope (h/e); only its frequency-axis intercept v0 (and hence work function) shifts from one metal to another.

Final answer. Cut-off voltage satisfies eV0 = hv − phi and threshold frequency v0 = phi/h; the linear V0 vs v plot has slope h/e and gives v0 as its frequency-axis intercept.

← *Back to Q240*

Q241

Write Einstein's photoelectric equation. State clearly how this equation is obtained using the photon picture of electromagnetic radiation.
Write the three salient features observed in photoelectric effect which can be explained using this equation.

[2012]

SOLUTION

Einstein explained the photoelectric effect by treating light as a stream of energy packets called photons, each carrying energy equal to hν.

Einstein's photoelectric equation is:

$$h\nu = \phi_0 + K_{max}$$

$$\Rightarrow K_{max} = h\nu - \phi_0$$

where *h* is Planck's constant, ν is the frequency of incident light, ϕ_0 is the work function of the metal, and K_{max} is the maximum kinetic energy of the emitted photoelectron.

Obtaining the equation from the photon picture: In the photon model, a beam of light consists of discrete quanta (photons), each of energy hν. When light

falls on a metal surface, a single photon collides with a single electron and transfers its entire energy $h\nu$ to that electron in one instantaneous event.

A part of this energy, equal to the work function ϕ_0, is used to free the electron from the surface (to overcome the binding force).

The remaining energy appears as the kinetic energy of the ejected electron. The electron emitted from the surface (without losing any energy inside the metal) carries the maximum kinetic energy:

$$K_{max} = h\nu - \phi_0$$

Three salient features explained by this equation:

(1) **Threshold frequency:** Emission occurs only when $\mathbf{h\nu \geq \phi_0}$, i.e. when $\mathbf{\nu \geq \nu_0}$ where $\mathbf{\nu_0 = \frac{\phi_0}{h}}$. Below the threshold frequency ν_0, K_{max} would be negative, which is impossible, so no electron is emitted however intense the light is.

(2) $\mathbf{K_{max}}$ **depends on frequency, not intensity:** Since $\mathbf{K_{max} = h\nu - \phi_0}$, the maximum kinetic energy increases linearly with frequency ν but does not depend on the intensity of light. Increasing intensity only increases the number of photons, not the energy per photon.

(3) **Instantaneous emission:** Energy is delivered in a single photon-electron collision, so emission begins without any time lag (less than 10^{-9} s), even at very low intensity.

> **TIP**
>
> Always write K_{max} (maximum KE), not just KE, since electrons emitted from deeper inside the metal come out with less energy.

Final answer. Einstein's equation $\mathbf{K_{max}=h\nu-\phi_0}$, derived from the photon picture, explains the threshold frequency, the intensity-independence of $\mathbf{K_{max}}$, and instantaneous emission.

← *Back to Q241*

Q242

Write two characteristic features observed in photoelectric effect which support the photon picture of electromagnetic radiation.
Draw a graph between the frequency of incident radiation (ν) and the maximum kinetic energy of the electrons emitted from the surface of a photosensitive material. State clearly how this graph can be used to determine (i) Planck's constant and (ii) work function of the material.

[2012]

SOLUTION

Two characteristic features supporting the photon picture:

(1) **Existence of a threshold frequency:** Photoemission takes place only if the frequency of incident light is at least ν_0. Below ν_0, no electron is emitted however high the intensity. This is explained because each photon carries fixed energy hν; if $\mathbf{h\nu < \phi_0}$, a single photon cannot free an electron, and energy from many photons cannot be added together.

(2) $\mathbf{K_{max}}$ **is independent of intensity:** The maximum kinetic energy of photoelectrons depends only on the frequency of light through $\mathbf{K_{max} = h\nu - \phi_0}$, and not on intensity. Increasing intensity increases only the number of photons (hence the photocurrent), proving that light energy comes in discrete quanta.

Graph of $\mathbf{K_{max}}$ versus ν: Plot frequency ν on the x-axis and maximum kinetic energy K_{max} on the y-axis. The graph is a **straight line** that does not pass through the origin. It cuts the ν-axis at the threshold frequency ν_0 (where $\mathbf{K_{max}=0}$), and if extended backwards it meets the K_{max}-axis at the negative value $-\phi_0$. The line has a constant positive slope.

The equation of this line is:

$$K_{max} = h\nu - \phi_0$$

which is of the form $y = mx + c$ with slope $m = h$ and intercept $c = -\phi_0$.

(i) Determining Planck's constant *h*: The slope of the straight line equals h. Taking any two points on the line,

$$h = \text{slope} = \frac{\Delta K_{max}}{\Delta \nu}$$

So Planck's constant is found directly from the slope.

(ii) Determining the work function ϕ_0: Read the threshold frequency ν_0 from the point where the line meets the ν-axis. Then the **work function** is:

$$\phi_0 = h\nu_0$$

Equivalently, the magnitude of the intercept on the K_{max}-axis directly gives ϕ_0.

Final answer. The $\mathbf{K_{max}}$ versus $\boldsymbol{\nu}$ graph is a straight line: its slope gives Planck's constant $\boldsymbol{h}$ and its $\boldsymbol{\nu}$-intercept $\boldsymbol{\nu_0}$ gives the work function $\boldsymbol{\phi_0 = h\nu_0}$.

← *Back to Q242*

Q243

Draw a plot showing the variation of photoelectric current with collector plate potential for two different frequencies, $\nu_1 > \nu_2$, of incident radiation having the same intensity. In which case will the stopping potential be higher? Justify your answer.

[2011]

SOLUTION

Plot: Draw photoelectric current I (y-axis) against collector plate potential V (x-axis), with the origin at the centre so that V can be both negative (retarding) and positive (accelerating).

Draw two curves for the two frequencies $\nu_1 > \nu_2$, both having the same intensity:

For positive V, the current rises and then becomes constant (saturation). Because both beams have the **same intensity**, both curves saturate at the **same maximum current** I_{sat}.

As V is made negative, the current falls and drops to zero. The value of the retarding potential at which the current just becomes zero is the **stopping potential** V_0.

The curve for the higher frequency ν_1 reaches zero current at a more negative potential, i.e. it cuts the V-axis further to the left than the curve for ν_2. So $V_{01} > V_{02}$, while both curves merge into the same saturation current on the right.

Stopping potential is higher for the higher frequency ν_1.

Justification: By Einstein's photoelectric equation, the maximum kinetic energy is

$$eV_0 = K_{max} = h\nu - \phi_0$$

$$\Rightarrow V_0 = \frac{h\nu - \phi_0}{e}$$

The stopping potential depends only on the frequency (for a given metal), not on intensity. Since $\nu_1 > \nu_2$, the photon energy $h\nu_1$ is greater, so the photoelectrons have greater K_{max}, and a larger retarding potential is needed to stop the most energetic electrons. Therefore $V_{01} > V_{02}$.

Final answer. The stopping potential is higher for the higher frequency ν_1, since $V_0 = (h\nu - \phi_0)/e$ increases with frequency.

← *Back to Q243*

Q244

An electron and a photon each have a wavelength 1.00 *nm. Find*
(i) their momenta,
(ii) the energy of the photon and
(iii) the kinetic energy of electron.

[2011 • Set 55-1-1]

SOLUTION

Given wavelength of both the electron and the photon: $\lambda = 1.00\ \text{nm} = 1.00 \times 10^{-9}\ m$.

(i) Momenta: The de Broglie relation $\lambda = \frac{h}{p}$ holds for the electron, and the photon momentum is also $p = \frac{h}{\lambda}$. Since both have the same wavelength, they have equal momenta.

$$p = \frac{h}{\lambda}$$

$$p = \frac{6.63\times 10^{-34}\,J\,s}{1.00\times 10^{-9}\,m}$$

$$p = 6.63\times 10^{-25}\ \text{kg}\ m\ \text{s}^{-1}$$

So the electron and photon have the **same momentum** $\mathbf{p_e = p_{ph} = 6.63\times 10^{-25}\ kg\ m\ s^{-1}}$.

(ii) Energy of the photon: A photon's energy is $E = pc$ (or hc/λ).

$$E = pc = (6.63\times 10^{-25})(3\times 10^{8})$$

$$E = 1.99\times 10^{-16}\,J$$

Converting to electron-volts: $E = \dfrac{1.99\times 10^{-16}}{1.6\times 10^{-19}}$

$$E \approx 1.24\times 10^{3}\ \text{eV} = 1.24\ \text{keV}$$

Photon energy $= 1.99\times 10^{-16}\,J \approx 1240$ **eV**.

(iii) Kinetic energy of the electron: The electron is non-relativistic, so its kinetic energy is $K = \dfrac{p^2}{2m_e}$ with $m_e = 9.1\times 10^{-31}$ kg.

$$K = \frac{(6.63\times 10^{-25})^2}{2\times 9.1\times 10^{-31}}$$

$$K = \frac{4.40\times 10^{-49}}{1.82\times 10^{-30}}$$

$$K = 2.42\times 10^{-19}\,J$$

In electron-volts: $K = \dfrac{2.42\times 10^{-19}}{1.6\times 10^{-19}} \approx 1.5$ **eV**.

The electron's kinetic energy (≈ **1.5 eV**) is far smaller than the photon's energy (≈ **1240 eV**), because the photon energy is pc while the electron energy is $p^2/2m$.

CONCEPT

For the same wavelength a photon carries far more energy (pc) than a slow electron ($p^2/2m$), since c is huge compared with the electron's speed.

Final answer. Both have p=**6.6310^{-25} kg m s^{-1}**; photon energy **1.9910^{-16} J** (1240 eV); electron KE **2.410^{-19} J** (1.5 eV).

← Back to Q244

Q245

Light of wavelength 2000 Å falls on a metal surface of work function 4.2 eV. What is the kinetic energy (in eV) of the fastest electrons emitted from the surface?
(i) What will be the change in the energy of the emitted electrons if the intensity of light with same wavelength is doubled?
(ii) If the same light falls on another surface of work function 6.5 eV, what will be the energy of emitted electrons?

[2011 • Set 55-2-1]

SOLUTION

Given: wavelength $\lambda = 2000 = 2000\times 10^{-10}\ m = 2\times 10^{-7}\ m$, work function $\phi_0 = 4.2$ **eV**.

Energy of each incident photon (using **hc = 12400 eV**):

$$E = \frac{hc}{\lambda} = \frac{12400\ \text{eV}}{2000}$$

$$E = 6.2\ \text{eV}$$

Maximum kinetic energy of the fastest electrons (Einstein's equation):

$$K_{max} = E - \phi_0 = 6.2 - 4.2$$

$$K_{max} = 2.0\ \text{eV}$$

So the **fastest electrons have kinetic energy 2.0 eV.**

(i) Effect of doubling the intensity (same wavelength): The maximum kinetic energy depends only on the frequency (wavelength) of light, not on its intensity, because $K_{max}=h\nu-\phi_0$. Doubling the intensity doubles the number of photons, hence doubles the number of photoelectrons (the photocurrent), but each electron still gains the same energy. Therefore the energy of the emitted electrons is **unchanged at 2.0 eV**.

(ii) Same light on a surface of work function $\phi_0' = 6.5$ eV: Compare the photon energy with the new work function:

$$E = 6.2 \text{ eV} < \phi_0' = 6.5 \text{ eV}$$

Since the photon energy is less than the work function, a single photon cannot supply enough energy to free an electron. Hence **no photoemission occurs**, and the energy of emitted electrons is zero (no electrons come out).

Final answer. Fastest electrons have $\mathbf{K_{max}=2.0\ eV}$; (i) unchanged at 2.0 eV when intensity is doubled; (ii) no emission since 6.2 eV < 6.5 eV.

← Back to Q245

Q246

A proton and an alpha particle are accelerated through the same potential. Which one of the two has (i) greater value of de-Broglie wavelength associated with it, and (ii) less kinetic energy? Justify your answers.

[2009]

SOLUTION

When a charged particle of mass m and charge q is accelerated from rest through a potential difference V, the work done qV becomes its kinetic energy:

$$K = qV$$

The momentum is $p = \sqrt{2mK} = \sqrt{2mqV}$, so the de Broglie wavelength is:

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mqV}}$$

For the same accelerating potential V, this gives $\lambda \propto \frac{1}{\sqrt{mq}}$.

Now compare the proton (mass m_p, charge e) with the alpha particle (mass $m_\alpha = 4m_p$, charge $q_\alpha = 2e$).

(i) de Broglie wavelength: Taking the ratio,

$$\frac{\lambda_p}{\lambda_\alpha} = \sqrt{\frac{m_\alpha\, q_\alpha}{m_p\, q_p}} = \sqrt{\frac{(4m_p)(2e)}{(m_p)(e)}}$$

$$\frac{\lambda_p}{\lambda_\alpha} = \sqrt{8} = 2\sqrt{2}$$

Since $\lambda_p = 2\sqrt{2}\,\lambda_\alpha$, we get $\lambda_p > \lambda_\alpha$. The **proton has the greater de Broglie wavelength** because the product mq is smaller for the proton.

(ii) Kinetic energy: Since $K = qV$ for the same V,

$$K_p = eV \; K_\alpha = 2eV$$

$$\frac{K_p}{K_\alpha} = \frac{e}{2e} = \frac{1}{2}$$

Therefore $K_p < K_\alpha$. The **proton has less kinetic energy** (half that of the alpha particle), because the alpha particle carries twice the charge and so gains twice the energy through the same potential.

Final answer. The proton has the greater de Broglie wavelength ($\lambda_p = 2\lambda_\alpha$) and the lesser kinetic energy ($K_p = \frac{1}{2}K_\alpha$).

← *Back to Q246*

Q247

An electromagnetic wave of wavelength λ is incident on a photosensitive surface of negligible work function. If the photo-electrons emitted from this surface have the de-Broglie wavelength λ_1, prove that $\lambda = \frac{2mc}{h}\lambda_1^2$.

[2008]

SOLUTION

An electromagnetic wave of wavelength λ behaves as photons, each carrying energy:

$$E_{photon} = \frac{hc}{\lambda}$$

When such a photon strikes the photosensitive surface, Einstein's photoelectric equation gives the maximum kinetic energy of the emitted electron:

$$K_{max} = \frac{hc}{\lambda} - \phi_0$$

Since the work function is negligible ($\phi_0 \approx 0$), the whole photon energy becomes the kinetic energy of the photoelectron:

$$K_{max} = \frac{hc}{\lambda} \quad ...(1)$$

The emitted electron (mass m) is a matter wave of de Broglie wavelength λ_1:

$$\lambda_1 = \frac{h}{p} \Rightarrow p = \frac{h}{\lambda_1}$$

Its kinetic energy in terms of momentum is:

$$K_{max} = \frac{p^2}{2m} = \frac{1}{2m}\left(\frac{h}{\lambda_1}\right)^2 = \frac{h^2}{2m\lambda_1^{\,2}} \quad ...(2)$$

Equating the two expressions (1) and (2) for the same kinetic energy:

$$\frac{hc}{\lambda} = \frac{h^2}{2m\lambda_1^{\,2}}$$

Cross-multiplying and solving for λ:

$$\lambda = \frac{hc \cdot 2m\lambda_1^{\,2}}{h^2}$$

$$\lambda = \frac{2mc}{h}\,\lambda_1^{\,2}$$

Hence $\boldsymbol{\lambda = \frac{2mc}{h}\,\lambda_1^{\,2}}$, which is the required result. This shows that the wavelength of the incident radiation is proportional to the square of the de Broglie wavelength of the emitted electron.

Final answer. Equating $\mathbf{hc/\lambda}$ with $\mathbf{h^2/(2m\lambda_1^{\,2})}$ gives the required result $\boldsymbol{\lambda = \frac{2mc}{h}\,\lambda_1^{\,2}}$.

← *Back to Q247*

Q248

In a plot of photoelectric current versus anode potential, how does
(i) the saturation current vary with anode potential for incident radiations of

different frequencies but same intensity?
(ii) the stopping potential vary for incident radiations of different intensities but same frequency?
(iii) photoelectric current vary for different intensities but same frequency of incident radiations?
Justify your answer in each case.

[2007]

SOLUTION

In the photoelectric experiment, the photocurrent depends on intensity (number of photons) while the stopping potential depends on frequency (energy per photon). Using this, each part is answered below.

(i) Saturation current for different frequencies, same intensity: The saturation current is reached when all emitted photoelectrons are collected, so it equals the number of photoelectrons emitted per second, which equals the number of photons striking the cathode per second. This number is fixed by the **intensity**, not the frequency.

Therefore, for radiations of different frequencies but the same intensity, the **saturation current is the same** for all of them. (Only their stopping potentials differ.)

(ii) Stopping potential for different intensities, same frequency: The stopping potential is governed by the maximum kinetic energy of photoelectrons:

$$eV_0 = h\nu - \phi_0$$

This depends only on the frequency ν (for a given metal) and not on intensity. Hence, for different intensities of the same frequency, the **stopping potential is unchanged** (the same V_0 for all curves).

(iii) Photoelectric current for different intensities, same frequency: Increasing the intensity increases the number of photons incident per second, so more photoelectrons are emitted per second.

$$I_{sat} \propto \text{intensity}$$

Therefore the **photoelectric (saturation) current increases in direct proportion to the intensity**, while the stopping potential stays the same.

CONCEPT

Remember the split: intensity controls the current (how many electrons), frequency controls the stopping potential (how energetic each electron is).

Final answer. (i) Saturation current is the same (set by intensity); (ii) stopping potential is unchanged (set by frequency); (iii) photocurrent rises in proportion to intensity.

← *Back to Q248*

Q249

Draw a schematic diagram of the experimental arrangement used by Davisson and Germer to establish the wave nature of electrons. Explain briefly how the de-Broglie relation was experimentally verified in case of electrons.

[2007]

SOLUTION

Schematic of the Davisson-Germer experiment (described): An **electron gun** consists of a tungsten filament F heated by a low-voltage supply; it emits electrons by thermionic emission. These electrons are accelerated through a variable potential difference V applied between the filament and a cylindrical anode, which also has a fine hole that collimates the electrons into a narrow beam. This beam is directed normally onto a **single crystal of nickel** (the target). The electrons scattered by the crystal are collected by a **movable detector (electron collector / Faraday cup)** that can be rotated to measure the intensity of the scattered beam at different scattering angles ϕ. The whole arrangement is enclosed in an evacuated chamber.

Observation: The detector reading shows a sharp peak in the intensity of scattered electrons at a scattering angle $\phi = 50°$ when the accelerating voltage is $V = 54\ V$. This peak arises from constructive interference of electron waves scattered by the regularly spaced atoms of the crystal, just like X-ray diffraction.

Verification of the de Broglie relation:

The theoretical de Broglie wavelength of an electron accelerated through $V = 54$ V is

$$\lambda = \frac{h}{\sqrt{2m_e\,eV}} = \frac{1.227}{\sqrt{V}}\ \text{nm}$$

$$\lambda = \frac{1.227}{\sqrt{54}}\ \text{nm} \approx 0.167\ \text{nm}$$

The experimental wavelength is found from the diffraction data using the known nickel lattice spacing $d = 0.215$ nm. Treating the peak as a diffraction maximum (Bragg-type condition $\lambda = d\phi$ for this geometry):

$$\lambda_{exp} = d\phi = 0.215 \times \sin 50^\circ$$

$$\lambda_{exp} \approx 0.165\ \text{nm}$$

The experimental wavelength (0.165 nm) closely agrees with the theoretical de Broglie wavelength (0.167 nm). This close agreement **experimentally confirms the de Broglie relation** $\lambda = h/p$ and establishes the wave nature of electrons.

Final answer. The diffraction peak at ϕ=50° for V=54 V gives λ_{exp}**0.165** nm, matching the de Broglie value **0.167** nm and confirming the wave nature of electrons.

← *Back to Q249*

Q250

Define the terms threshold frequency and stopping potential in relation to the phenomenon of photoelectric effect. How is the photoelectric current affected on increasing the (i) frequency (ii) intensity of the incident radiations and why?

[2006]

SOLUTION

Threshold frequency (ν_0): It is the minimum frequency of incident radiation below which no photoelectric emission takes place, however high the intensity of light. It is related to the work function by:

$$\phi_0 = h\nu_0$$

Stopping potential (V_0): It is the minimum value of the (negative) retarding potential applied to the collector at which the photoelectric current becomes zero. It measures the maximum kinetic energy of the photoelectrons:

$$eV_0 = K_{max} = h\nu - \phi_0$$

(i) Effect of increasing frequency ν (intensity kept constant): Increasing the frequency increases the energy per photon hν, so the maximum kinetic energy K_{max} and the stopping potential V_0 increase. However, for a fixed intensity (constant power P), the number of photons per second is $\boldsymbol{n = \frac{P}{h\nu}}$, which decreases as ν rises. So the number of emitted electrons, and hence the saturation photocurrent, does not increase; it stays the same or decreases slightly. The photocurrent is therefore **not increased** by raising the frequency.

(ii) Effect of increasing intensity (frequency kept constant): Increasing the intensity means more photons strike the surface per second (each still of energy hν). This ejects more photoelectrons per second, so the **photoelectric (saturation) current increases in direct proportion to the intensity**. The stopping potential V_0 remains unchanged, since K_{max} depends only on frequency.

Final answer. Increasing frequency raises $\mathbf{K_{max}}$ and $\mathbf{V_0}$ but not the current; increasing intensity raises the photoelectric current in proportion while $\mathbf{V_0}$ stays the same.

← *Back to Q250*

Q251

Ultraviolet light of wavelength 2271 *from a* 100 *W mercury source radiates a photo cell made of molybdenum metal. If the stopping potential is* 1.3 *V, estimate the work function of the metal. How would the photo cell respond to high intensity* (10^5 *W* m^{-2}) *red light of wavelength* 6328 *produced by a He-Ne laser?*

Plot a graph showing the variation of photoelectric current with anode potential for two light beams of same wavelength but different intensity.

[2005]

SOLUTION

Light of a single wavelength carries photons each of energy $E = h\nu = \frac{hc}{\lambda}$. Using the convenient form $\frac{hc}{e} = 12400\ \text{eV}$, the energy of each UV photon is found.

$$E = \frac{hc}{\lambda} = \frac{12400\ \text{eV}}{2271}$$

$$E = 5.46\ \text{eV}$$

By Einstein's photoelectric equation, the photon energy is shared between the work function and the maximum kinetic energy of the emitted electron, and the maximum kinetic energy equals eV_0 where V_0 is the stopping potential.

$$E = \phi_0 + eV_0$$

$$\phi_0 = E - eV_0 = 5.46\ \text{eV} - (1)(1.3\ \text{eV})$$

$$\phi_0 = 4.16\ \text{eV}$$

So the **work function of molybdenum is 4.16 eV**.

Now for the red light of wavelength 6328 , the energy of each red photon is found the same way.

$$E_{red} = \frac{12400\ \text{eV}}{6328} = 1.96\ \text{eV}$$

Since $\mathbf{E_{red} = 1.96\ eV}$ is much less than the work function $\mathbf{\phi_0 = 4.16\ eV}$, a single red photon does not have enough energy to free an electron. Photoemission depends on the energy of one photon, not on how many photons arrive. So even at the very high intensity of $10^5\ W\,m^{-2}$, the He-Ne red laser produces **no photoelectric emission** from molybdenum.

Graph (current vs anode potential, same wavelength, two intensities I_1 and I_2 with $\mathbf{I_2 > I_1}$): Plot photoelectric current on the y-axis and anode potential on the x-axis, with the origin in the middle so negative (retarding) potentials lie to the left. Both curves start at the same negative stopping potential $-V_0$ on the x-axis

(since same wavelength means same V_0), rise as the anode potential increases, and then flatten into two horizontal saturation lines on the right. The curve for the higher intensity I_2 saturates at a higher current than the curve for I_1, because more photons eject more electrons per second. Both curves meet the x-axis at the same point $-V_0$.

CONCEPT

Stopping potential depends only on wavelength/frequency; intensity changes only the saturation current. Same colour, same V_0, but a higher flat line for brighter light.

Final answer. Work function of molybdenum is **4.16 eV**; the red laser (**1.96 eV < 4.16 eV**) causes no photoemission whatever its intensity.

← *Back to Q251*

Q252

Define the term 'work function' of a metal. The threshold frequency of a metal is f_0. *When the light of frequency* $2f_0$ *is incident on the metal plate, the maximum velocity of electrons emitted is* v_1. *When the frequency of the incident radiation is increased to* $5f_0$, *the maximum velocity of electrons emitted is* v_2. *Find the ratio of* v_1 *to* v_2.

[2004]

SOLUTION

The **work function** of a metal is the minimum energy that must be supplied to a free electron at the surface of the metal to just liberate it from the surface, that is, to remove it with zero kinetic energy. It is denoted by ϕ_0 and is related to the threshold frequency by $\boldsymbol{\phi_0 = h f_0}$.

By Einstein's photoelectric equation, the maximum kinetic energy of an emitted electron is the photon energy minus the work function.

$$\frac{1}{2}mv^2 = h\nu - hf_0 = h(\nu - f_0)$$

When the incident frequency is $2f_0$, the maximum speed is v_1.

$$\frac{1}{2}mv_1^2 = h(2f_0 - f_0) = hf_0$$

When the incident frequency is $5f_0$, the maximum speed is v_2.

$$\frac{1}{2}mv_2^2 = h(5f_0 - f_0) = 4hf_0$$

Dividing the first relation by the second, the common factors $\frac{1}{2}m$ and h cancel.

$$\frac{v_1^2}{v_2^2} = \frac{hf_0}{4hf_0} = \frac{1}{4}$$

Taking the square root of both sides gives the required ratio.

$$\frac{v_1}{v_2} = \frac{1}{2}$$

So $\mathbf{v_1 : v_2 = 1 : 2}$.

Final answer. $\mathbf{v_1 : v_2 = 1 : 2}$.

← *Back to Q252*

Q253

Why are de-Broglie waves associated with a moving football not visible? The wavelength λ of a photon and the de-Broglie wavelength of an electron have the same value. Show that the energy of the photon is $\frac{2mc\lambda}{h}$ *times the kinetic energy of the electron, where m, c and h have their usual meanings.*

[2003]

SOLUTION

Every moving body has an associated de-Broglie wavelength $\lambda = \frac{h}{p} = \frac{h}{mv}$. For a football the mass m is large, so the momentum $p = mv$ is very large and the wavelength becomes extremely small.

Taking a football of mass $m = 0.4\ \text{kg}$ moving at $v = 20\ m\ \text{s}^{-1}$, the momentum is $p = mv = 8\ \text{kg}\ m\ \text{s}^{-1}$.

$$\lambda = \frac{h}{p} = \frac{6.6310^{-34}}{8} \approx 810^{-35}\ m$$

This wavelength is far smaller than the size of a nucleus and is much smaller than any aperture or obstacle available in nature. Diffraction or interference can only be observed when the wavelength is comparable to the size of the obstacle/slit, which is impossible here. Hence the **wave nature of a moving football cannot be detected** and the de-Broglie waves are not visible.

For the proof, the photon and the electron have the same wavelength, so $\lambda_{\text{photon}} = \lambda_{\text{electron}} = \lambda$.

Energy of the photon of wavelength λ:

$$\text{E}_{\text{photon}} = \frac{\text{hc}}{\lambda}$$

The electron has the same de-Broglie wavelength, so its momentum is

$$\text{p}_e = \frac{h}{\lambda}$$

Its kinetic energy in terms of momentum is

$$\text{KE}_{\text{electron}} = \frac{\text{p}_e{}^2}{2\text{m}} = \frac{1}{2\text{m}}\left(\frac{h}{\lambda}\right)^2 = \frac{\text{h}^2}{2\text{m}\lambda^2}$$

Taking the ratio of the photon energy to the electron kinetic energy:

$$\frac{\text{E}_{\text{photon}}}{\text{KE}_{\text{electron}}} = \frac{\frac{\text{hc}}{\lambda}}{\frac{\text{h}^2}{2\text{m}\lambda^2}} = \frac{\text{hc}}{\lambda} \times \frac{2\text{m}\lambda^2}{\text{h}^2}$$

$$\frac{\text{E}_{\text{photon}}}{\text{KE}_{\text{electron}}} = \frac{2\text{mc}\lambda}{h}$$

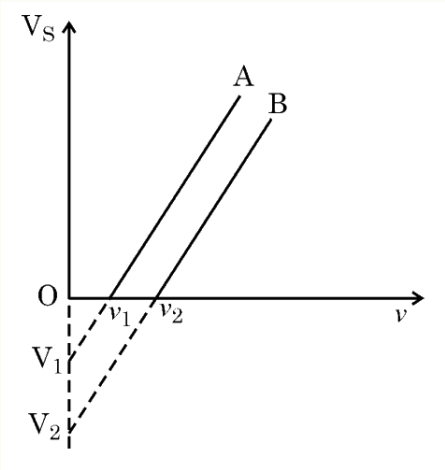
Therefore $\mathbf{E_{photon}} = \frac{\mathbf{2mc\lambda}}{\boldsymbol{h}} \times \mathbf{KE_{electron}}$, which is the required result.

Final answer. The football's de-Broglie wavelength ($\mathbf{10^{-35}\ m}$) is too small to detect, and $\mathbf{E_{photon}} = \frac{\mathbf{2mc\lambda}}{\boldsymbol{h}} \times \mathbf{KE_{electron}}$ as required.

← Back to Q253

Q254

A researcher performs an experiment on photo-electric effect using two metals A and B with unknown work functions. She illuminates the surfaces of A and B with monochromatic radiation of various frequencies and records the value of corresponding stopping potentials (V_s). *The graph shows the variation of stopping potential* (V_s) *with the frequency of incident radiation* (ν) *for metals A and B.*



[2026 • Set 55-1-1]

SOLUTION

Light striking a metal emits electrons with maximum kinetic energy eV_s, where V_s is the stopping potential. By Einstein's photoelectric equation, this maximum kinetic energy equals the photon energy minus the work function.

$$eV_s = h\nu - \phi_0$$

Dividing throughout by the electronic charge e gives the stopping potential as a function of frequency.

$$V_s = \frac{h}{e}\nu - \frac{\phi_0}{e}$$

This is the equation of a straight line $y = mx + c$ with V_s on the y-axis and ν on the x-axis. The slope and intercept therefore give the physical quantities below.

The slope of each line is $\frac{h}{e}$. Since both h (Planck's constant) and e (electronic charge) are universal constants, the slope is the same for every metal. Hence the graphs for metals A and B are **two parallel straight lines**.

Each line cuts the frequency (ν) axis at the threshold frequency ν_0, found by putting $\mathbf{V_s = 0}$:

$$0 = \frac{h}{e}\nu_0 - \frac{\Phi_0}{e} \Rightarrow \nu_0 = \frac{\Phi_0}{h}$$

The metal whose line crosses the ν-axis farther to the right has the larger threshold frequency, and so the larger work function. The work function of each metal is obtained from its threshold frequency.

$$\Phi_0 = h\nu_0$$

Planck's constant can be found from the common slope of the lines.

$$h = e\times(\text{slope})$$

So from this graph: the **slope h/e is identical for A and B (parallel lines)**, the ν-axis intercept gives the threshold frequency ν_0 of each metal, the work function is $\mathbf{\Phi_0 = h\nu_0}$, and Planck's constant follows from $\boldsymbol{h = e}$. The metal with the line lying farther right has the higher work function.

CONCEPT

The slope of a V_s versus ν graph is always h/e for every metal. Different metals shift the line sideways (different ν_0) but never tilt it.

Final answer. Both metals give parallel lines of slope $\boldsymbol{h/e}$; intercept gives $\boldsymbol{\nu_0}$, work function $\mathbf{\Phi_0 = h\nu_0}$, and $\boldsymbol{h = e}$.

← *Back to Q254*

Q255

Figure shows the variation of photoelectric current measured in a photo cell circuit as a function of the potential difference between the plates of the photo cell when light

beams A, B, C and D of different wavelengths are incident on the photo cell. Examine the given figure and answer the following questions:

(i) Which light beam has the highest frequency and why?
(ii) Which light beam has the longest wavelength and why?
(iii) Which light beam ejects photoelectrons with maximum momentum and why?

[2023 • Set 55-2-1]

SOLUTION

In this graph the photoelectric current is plotted against the plate potential for four beams A, B, C and D. The point where each curve meets the negative (retarding) potential axis is the stopping potential V_0 of that beam. The governing relation is Einstein's photoelectric equation, $\mathbf{eV_0 = h\nu - \phi_0}$, so a larger $|V_0|$ means a higher frequency.

(i) Highest frequency: From $\mathbf{eV_0 = h\nu - \phi_0}$, the stopping potential increases with frequency for the same metal. So the beam whose curve cuts the potential axis at the most negative value (largest stopping potential) has the **highest frequency**. The beam needing the greatest retarding voltage to stop its electrons is the most energetic, hence the highest ν.

(ii) Longest wavelength: Wavelength and frequency are related by $\lambda = \frac{c}{\nu}$, so the longest wavelength corresponds to the lowest frequency. The lowest frequency beam has the smallest stopping potential. Therefore the beam whose curve meets the potential axis nearest to zero (smallest $|V_0|$) has the **longest wavelength**.

(iii) Maximum momentum: The maximum kinetic energy of the emitted electrons is $\mathbf{K_{max} = eV_0}$, and momentum is related to kinetic energy by

$$p_{max} = \sqrt{2m_e K_{max}} = \sqrt{2m_e eV_0}$$

A larger stopping potential gives a larger K_{max} and hence a larger p_{max}. So the **same beam as in part (i)** (the one with the highest frequency and largest stopping potential) ejects photoelectrons with the maximum momentum.

(In a typical such figure the beam cutting the axis at the most negative potential answers (i) and (iii), and the beam cutting nearest zero answers (ii).)

Final answer. Highest frequency and maximum momentum: beam with the largest stopping potential; longest wavelength: beam with the smallest stopping potential.

← *Back to Q255*

Q256

What is the effect on threshold frequency and stopping potential on increasing the frequency of incident beam of light? Justify your answer.

[2023 • Set 55-2-1]

SOLUTION

Two separate quantities are involved here, the threshold frequency (a property of the metal) and the stopping potential (which depends on the incident light). They behave very differently when the incident frequency is raised.

Threshold frequency ν_0 is the minimum frequency of incident light needed to just cause photoemission, and it is fixed by the work function of the surface.

$$\nu_0 = \frac{\phi_0}{h}$$

Since ϕ_0 and h are constants for a given metal, ν_0 depends only on the nature of the metal and not at all on the frequency of the light falling on it. Therefore, when the incident frequency is increased (same metal), the **threshold frequency remains unchanged**.

Stopping potential V_0 is governed by Einstein's photoelectric equation, which relates the maximum kinetic energy of the electrons to the incident frequency.

$$eV_0 = h\nu - \phi_0 = h(\nu - \nu_0)$$

Solving for the stopping potential:

$$V_0 = \frac{h}{e}(\nu - \nu_0)$$

Here ν_0 is fixed, so as the incident frequency ν increases, the term $(\nu - \nu_0)$ increases, and so V_0 increases linearly with ν. Therefore, on increasing the

incident frequency, the **stopping potential increases** (electrons come out with greater maximum kinetic energy, needing a larger retarding voltage to stop them).

In short: increasing the incident frequency leaves the **threshold frequency unchanged** but makes the **stopping potential increase** linearly.

Final answer. Threshold frequency stays unchanged (it depends only on the metal); stopping potential increases linearly with the incident frequency.

← Back to Q256